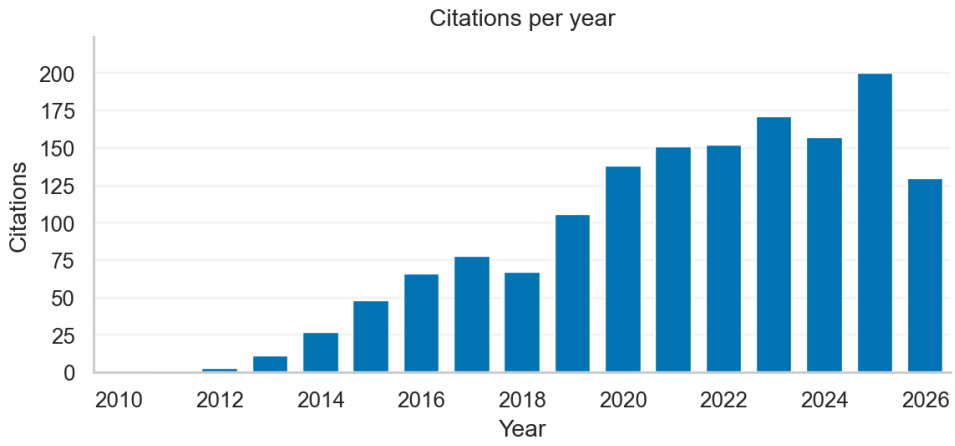
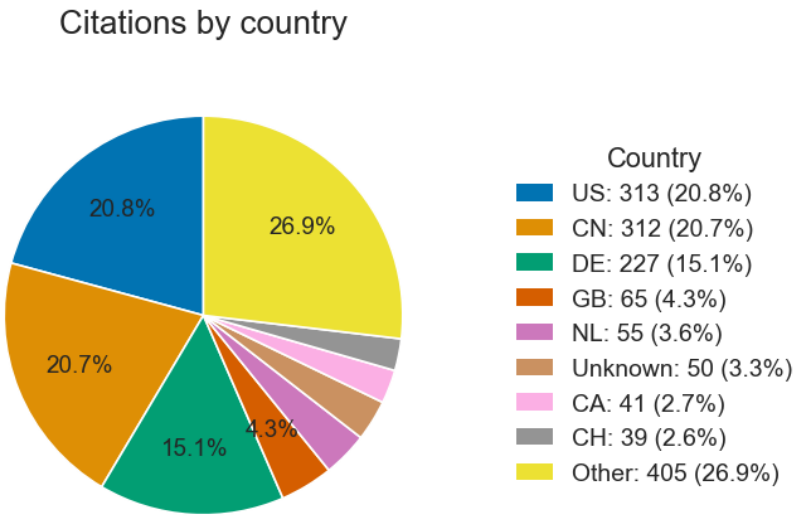
Author Citation Report

Author: Markus Aswendt
ORCID: https://orcid.org/0000-0003-1423-0934
Total citations: 1507
Generated: 2026-10-16 14:17

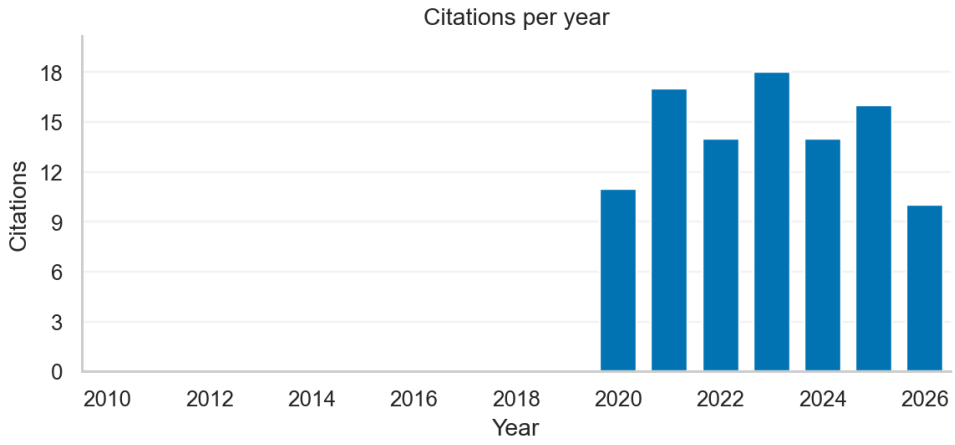


Summary
Total citations: **1507**
Average/year: **94.19**
Period: 2011-2026

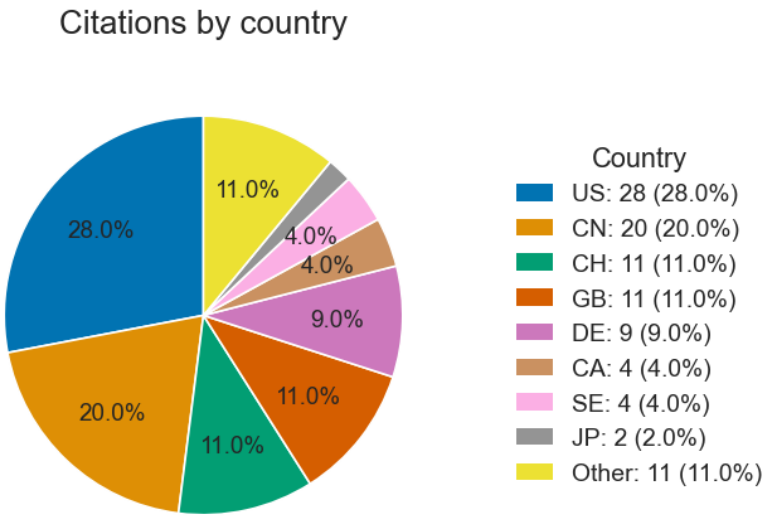


Publication

Goubran M, Leuze CW, Hsueh B, Aswendt M, Li Y, Tian Q, Cheng M, Crow A, Steinberg GK, McNab JA, Deisseroth KA, Zeineh M. Multimodal image registration and connectivity analysis for integration of connectomic data from microscopy to MRI. *Nature Communications*. 2019 Dec 3;10(1):5504. doi: 10.1038/s41467-019-13374-0. PMID: 31796741.



Citation summary
Total citations: **100**
Average/year: **12.50**
Period: 2019-2026
SCImago cites/doc 2y (2025): **16.6**
SJR 2025: **4.9** (Q1)



Cited by

1. Hsueh B, Chen R, Jo Y, Tang DD, Raffiee M, Kim YS, Inoue M, Randles S, Ramakrishnan C, Patel S, Kim DK, Liu TX, Kim SH, Tan L, Mortazavi L, Cordero A, Shi J, Zhao M, Ho TT, Crow A, Yoo AW, Raja C, Evans KE, Bernstein DJ, Zeineh M, Goubran M, Deisseroth KA. Cardiogenic control of affective behavioural state. *Nature*. 2023 Mar 1;615.0(7951):292-299. doi: 10.1038/s41586-023-05748-8. PMID: 36859543.

2. Hoffmann M, Billot B, Greve DN, Iglesias JE, Fischl BR, Dalca AV. SynthMorph: Learning Contrast-Invariant Registration Without Acquired Images. *IEEE Transactions on Medical Imaging*. 2021 Sep 29;41.0(3):543-558. doi: 10.1109/tmi.2021.3116879. PMID: 34587005.

3. Claudi F, Tyson AL, Petrucco L, Margrie TW, Portugues R, Branco T. Visualizing anatomically registered data with brainrender. *eLife*. 2021 Mar 19;10.0. doi: 10.7554/elife.65751. PMID: 33739286.

4. Tyson AL, Rousseau CV, Niedworok CJ, Keshavarzi S, Tsitoura C, Cossell L, Strom M, Margrie TW. A deep learning algorithm for 3D cell detection in whole mouse brain image datasets. *PLoS Computational Biology*. 2021 May 28;17.0(5):e1009074. doi: 10.1371/journal.pcbi.1009074. PMID: 34048426.

5. Yang JY, Yeh C, Poupon C, Calamante F. Diffusion MRI tractography for neurosurgery: the basics, current state, technical reliability and challenges. *Physics in Medicine and Biology*. 2021 Jun 22;66.0(15):15TR01. doi: 10.1088/1361-6560/ac0d90. PMID: 34157706.

6. Qu L, Li Y, Xie P, Liu L, Wang Y, Wu J, Liu Y, Wang T, Li L, Guo K, Wan W, Ouyang L, Xiong F, Kolstad AC, Wu Z, Xu F, Zheng Y, Gong H, Luo Q, Bi G, Dong H, Hawrylycz M, Zeng H, Peng H. Cross-modal coherent registration of whole mouse brains. *Nature Methods*. 2021 Dec 9;19.0(1):111-118. doi: 10.1038/s41592-021-01334-w. PMID: 34887551.

7. Xu WL, Lin J, Gao MY, Chen Y, Cao J, Pu J, Huang L, Zhao J, Qian K. Rapid Computer-Aided Diagnosis of Stroke by Serum Metabolic Fingerprint Based Multi-Modal Recognition. *Advanced Science*. 2020 Sep 23;7.0(21):2002021. doi: 10.1002/advs.202002021. PMID: 33173737.

8. Chiaruttini N, Castoldi C, Requie LM, Camarena-Delgado C, Bianco BD, Gräff J, Seitz A, Silva BA. ABBA+BraiAn, an integrated suite for whole-brain mapping, reveals brain-wide differences in immediate-early genes induction upon learning. *Cell Reports*. 2025 Jun 23;44.0(7):115876. doi: 10.1016/j.celrep.2025.115876. PMID: 40553651.

9. Johnson GA, Tian Y, Ashbrook DG, Cofer GP, Cook JJH, Gee JC, Hall A, Hornburg KJ, Kaczorowski CC, Qi YP, Yeh FF, Wang N, WHITE LE, Williams RW. Merged magnetic resonance and light sheet microscopy of the whole mouse brain. *Proceedings of the National Academy of Sciences*. 2023 Apr 17;120.0(17):e2218617120. doi: 10.1073/pnas.2218617120. PMID: 37068254.

10. Cho N, Squair JW, Aureli V, James ND, Bole-Feysot L, Dewany I, Hankov N, Baud L, Leonhartsberger A, Sveistyte K, Skinnider MA, Gautier M, Laskaratos A, Galan K, Goubran M, Ravier J, Merlos F, Batti L, Pagès S, Bérard N, Intering N, Varescon C, Watrin A, Duguet L, Carda S, Bartholdi K, Hutson TH, Kathe C, Hodara M, Anderson MA, Draganski B, Demesmaeker R, Asboth L, Barraud Q, Bloch J, Courtine G. Hypothalamic deep brain stimulation augments walking after spinal cord injury. *Nature Medicine*. 2024 Dec 1;30.0(12):3676-3686. doi: 10.1038/s41591-024-03306-x. PMID: 39623087.

11. Aswendt M, Pallast N, Wieters F, Baues M, Hoehn M, Fink GR. Lesion Size- and Location-Dependent Recruitment of Contralesional Thalamus and Motor Cortex Facilitates Recovery after Stroke in Mice. *Translational Stroke Research*. 2020 Mar 12;12.0(1):87-97. doi: 10.1007/s12975-020-00802-3. PMID: 32166716.

12. Rijsketic DR, Casey AB, Barbosa DAN, Zhang X, Hietamies TM, Ramirez-Ovalle G, Pomrenze MB, Halpern CH, Williams LM, Malenka RC, Heifets BD. UNRAVELing the synergistic effects of psilocybin and environment on brain-wide immediate early gene expression in mice. *Neuropsychopharmacology*. 2023 May 29;48.0(12):1798-1807. doi: 10.1038/s41386-023-01613-4. PMID: 36865251.

13. Luo N, Shi W, Yang Z, Song M, Jiang T. Multimodal Fusion of Brain Imaging Data: Methods and Applications. *Machine Intelligence Research*. 2024 Jan 15;21.0(1):136-152. doi: 10.1007/s11633-023-1442-8.

14. Li Z, Shang Z, Liu J, Zhen H, Zhu E, Zhong S, Sturgess RN, Zhou Y, Hu X, Zhao X, Wu Y, Li P, Lin R, Ren J. D-LMBmap: a fully automated deep-learning pipeline for whole-brain profiling of neural circuitry. *Nature Methods*. 2023 Sep 28;20.0(10):1593-1604. doi: 10.1038/s41592-023-01998-6. PMID: 37770711.

15. Evans AK, Park HH, Woods CE, Lam RK, Rijsketic DR, Xu CL, Chu EK, Ciari P, Blumenfeld SE, Vidano LM, Saw NL, Heifets BD, Shamloo M. Impact of noradrenergic inhibition on neuroinflammation and pathophysiology in mouse models of Alzheimer's disease. *Journal of Neuroinflammation*. 2024 Dec 18;21.0(1):322. doi: 10.1186/s12974-024-03306-1. PMID: 39574895.

16. Han X, Chen J, Zhao Y, Kang R, Wei Y, Zhou S. Dual antibody-guided drug delivery systems using MOF-PQDs nanocomposites for precise tumor diagnosis and combination therapy. *Chemical Engineering Journal*. 2025 Jan 5;505.0:159275. doi: 10.1016/j.cej.2025.159275.

17. Otomo K, Omura T, Nozawa Y, Edwards S, Sato Y, Saito Y, Yagishita S, Uchida H, Watakabe Y, Naitou K, Yanai R, Sahara N, Takagi S, Katayama R, Iwata Y, Shiokawa T, Hayakawa Y, Otsuka K, Watanabe-Takano H, Haneda Y, Fukuhara S, Fujiwara M, Nii T, Meno C, Takeshita N, Yashiro K, Rocabado JMR, Kaku M, Yamada T, Oishi Y, Koike H, Cheng Y, Sekine K, Koga J, Sugiyama K, Kimura K, Karube F, Kim H, Manabe I, Nemoto T, Tainaka K, Hamada A, Brismar H, Susaki EA. descSPIM: an affordable and easy-to-build light-sheet microscope optimized for tissue clearing techniques. *Nature Communications*. 2024 Jun 12;15.0(1):4941. doi: 10.1038/s41467-024-49131-1. PMID: 38866781.

18. Attarpour A, Osmann J, Rinaldi A, Qi T, Lal N, Patel SB, Rozak MW, Yu F, Cho N, Squair JW, McLaurin J, Raffiee M, Deisseroth KA, Courtine G, Li Y, Stefanovic B, Goubran M. A deep learning pipeline for three-dimensional brain-wide mapping of local neuronal ensembles in teravoxel light-sheet microscopy. *Nature Methods*. 2025 Jan 27;22.0(3):600-611. doi: 10.1038/s41592-024-02583-1. PMID: 39870865.

19. Mancini M, Casamitjana A, Peter L, Robinson E, Crampsie S, Thomas DL, Holton JL, Jaunmuktane Z, Iglesias JE. A multimodal computational pipeline for 3D histology of the human brain. *Scientific Reports*. 2020 Aug 14;10.0(1):13839. doi: 10.1038/s41598-020-69163-z. PMID: 32796937.

20. Liu B, Hu Y, Ma G, Xiao Y, Zhang B, Liang Y, Zhong P, Zeng X, Lin Z, Kong H, Wu G, Du Z, Fang Y, Huang M, Wang L, Yang X, Yu H. Reduced Retinal Microvascular Perfusion in Patients With Stroke Detected by Optical Coherence Tomography Angiography. *Frontiers in Aging Neuroscience*. 2021 Apr 13;13.0:628336. doi: 10.3389/fnagi.2021.628336. PMID: 33927607.

21. Tyson AL, Margrie TW. Mesoscale microscopy and image analysis tools for understanding the brain. *Progress in Biophysics and Molecular Biology*. 2021 Jul 1;168.0:81-93. doi: 10.1016/j.pbiomolbio.2021.06.013. PMID: 34216639.

22. Huang R, Zhou X, Chen G, Su L, Liu Z, Zhou P, Weng J, Min Y. Advances of functional nanomaterials for magnetic resonance imaging and biomedical engineering applications. *Wiley Interdisciplinary Reviews Nanomedicine and Nanobiotechnology*. 2022 Apr 20;14.0(4):e1800. doi: 10.1002/wnan.1800. PMID: 35445588.

23. Tran MQ, Do T, Tran H, Tjiputra E, Tran QD, Nguyen A. Light-Weight Deformable Registration Using Adversarial Learning With Distilling Knowledge. *IEEE Transactions on Medical Imaging*. 2022 Jan 6;41.0(6):1443-1453. doi: 10.1109/tmi.2022.3141013. PMID: 34990354.

24. Pallast N, Wieters F, Nill M, Fink GR, Aswendt M. Graph theoretical quantification of white matter reorganization after cortical stroke in mice. *NeuroImage*. 2020 May 4;217.0:116873. doi: 10.1016/j.neuroimage.2020.116873. PMID: 32380139.

25. Newmaster KT, Kronman FN, Wu Y, Kim Y. Seeing the Forest and Its Trees Together: Implementing 3D Light Microscopy Pipelines for Cell Type Mapping in the Mouse Brain. *Frontiers in Neuroanatomy*. 2022 Jan 14;15.0:787601.

doi: 10.3389/fnana.2021.787601. PMID: 35095432.

26. Dong Y, Li C, Zhang Y, Li P, Qi FZ. Exascale image processing for next-generation beamlines in advanced light sources. *Nature Reviews Physics*. 2022 May 18;4.0(7):427-428. doi: 10.1038/s42254-022-00465-z.

27. Monte B, Constantinou S, Koundal S, Lee H, Dai F, Gursky ZH, Van Nostrand WE, Darbinyan A, Zloković BV, Wardlaw JM, Benveniste H. Characterization of perivascular space pathology in a rat model of cerebral small vessel disease by in vivo magnetic resonance imaging. *Journal of Cerebral Blood Flow & Metabolism*. 2022 Jun 8;42.0(10):1813-1826. doi: 10.1177/0271678x221105668. PMID: 35673963.

28. Wehrli PM, Ge J, Michno W, Koutarapu S, Dreos A, Jha D, Zetterberg H, Blennow K, Hanrieder J. Correlative Chemical Imaging and Spatial Chemometrics Delineate Alzheimer Plaque Heterogeneity at High Spatial Resolution. *JACS Au*. 2023 Mar 7;3.0(3):762-774. doi: 10.1021/jacsau.2c00492. PMID: 37006756.

29. Silveira ACC, Antunes ASLM, Athié MCP, da Silva BF, Santos JVRD, Canateli C, Fontoura MA, Pinto ADS, Pimentel-Silva LR, Avansini SH, Carvalho MDO. Between neurons and networks: investigating mesoscale brain connectivity in neurological and psychiatric disorders. *Frontiers in Neuroscience*. 2024 Feb 20;18.0:1340345. doi: 10.3389/fnins.2024.1340345. PMID: 38445254.

30. Abe P, Lavalley A, Morassut I, Santinha AJ, Roig-Puiggros S, Javed A, Klingler E, Baumann N, Prados J, Platt RJ, Jabaudon D. Molecular programs guiding arealization of descending cortical pathways. *Nature*. 2024 Sep 11;634.0(8034):644-651. doi: 10.1038/s41586-024-07895-y. PMID: 39261725.

31. García-Saldivar P, Garimella AH, Garza-Villarreal EA, Salcido FAM, Concha L, Merchant H. PREEMACS: Pipeline for preprocessing and extraction of the macaque brain surface. *NeuroImage*. 2020 Dec 24;227.0:117671. doi: 10.1016/j.neuroimage.2020.117671. PMID: 33359348.

32. Perens J, Salinas CG, Roostalu U, Skytte JL, Gundlach C, Hecksher-Sørensen J, Dahl AB, Dyrby TB. Multimodal 3D Mouse Brain Atlas Framework with the Skull-Derived Coordinate System. *Neuroinformatics*. 2023 Feb 21;21.0(2):269-286. doi: 10.1007/s12021-023-09623-9. PMID: 36809643.

33. Zhu J, Liu X, Liu Z, Deng Y, Xu J, Liu K, Zhang R, Meng X, Fei P, Yu T, Zhu DY. SOLID: minimizing tissue distortion for brain-wide profiling of diverse architectures. *Nature Communications*. 2024 Sep 27;15.0(1):8303. doi: 10.1038/s41467-024-52560-7. PMID: 39333107.

34. Li X, Zhang Y, Liu Y, Li P, Hu H, Wang L, He P, Dong Y, Zhang C. A high-throughput big-data orchestration and processing system for the High Energy Photon Source. *Journal of Synchrotron Radiation*. 2023 Sep 1;30.0(6):1086-1091. doi: 10.1107/s1600577523006951. PMID: 37729071.

35. Tang W, Li Z, Li Z, Zou Y, Liao J, Li B. A multimodal pipeline for image correction and registration of mass spectrometry imaging with microscopy. *Analytica Chimica Acta*. 2023 Oct 27;1283.0:341969. doi: 10.1016/j.aca.2023.341969. PMID: 37977791.

36. Santoro M, Lam RK, Blumenfeld SE, Tan W, Ciari P, Chu EK, Saw NL, Rijsketic DR, Lin JS, Heifets BD, Shamloo M. Mapping of catecholaminergic denervation, neurodegeneration, and inflammation in 6-OHDA-treated Parkinson's disease mice. *npj Parkinson s Disease*. 2025 Feb 11;11.0(1):28. doi: 10.1038/s41531-025-00872-w. PMID: 39483924.

37. Menzel M, Gräßel D, Rajković I, Zeineh M, Georgiadis M. Using light and X-ray scattering to untangle complex neuronal orientations and validate diffusion MRI. *eLife*. 2023 May 11;12.0. doi: 10.7554/elife.84024. PMID: 37166005.

38. Koukalova L, Chmelová M, Amlerova Z, Vargová L. Out of the core: the impact of focal ischemia in regions beyond the penumbra. *Frontiers in Cellular Neuroscience*. 2024 Mar 5;18.0:1336886. doi: 10.3389/fncel.2024.1336886. PMID: 38504666.

39. Mansour H, Azrak R, Cook JJH, Hornburg KJ, Qi YP, Tian Y, Williams RW, Yeh FF, WHITE LE, Johnson GA. The Duke Mouse Brain Atlas: MRI and light sheet microscopy stereotaxic atlas of the mouse brain. *Science Advances*. 2025 May 1;11.0(18):eadq8089. doi: 10.1126/sciadv.adq8089. PMID: 40305623.

40. Pang Z, Leung VH, Wang CC, Attarpour A, Rinaldi A, Shen H, Moya-Garzón MD, Sigua LH, Rammel C, Selke A, Glynn C, Yender M, Xu S, Moslehi JJ, Wu P, Long JZ, Goubran M, Cravatt BF, Li Y. Mapping cellular targets of covalent cancer drugs in the entire mammalian body. *Cell*. 2025 Dec 22;189.0(3):725-738.e15. doi: 10.1016/j.cell.2025.11.030. PMID: 40236136.

41. Feng H, Zhao Y, Li Y, Qi X, Shen S, Zhou S. Multi-Armed Anti-CD40-Mediated Dual Drug Delivery System Based on Mesoporous Silica/Au Nanorod Nanocomposites for Multimodality Imaging and Combination Therapy. *ACS Applied Nano Materials*. 2023 Jul 17;6.0(14):13001-13012. doi: 10.1021/acsanm.3c01722.

42. Elhaminia B, Alsalemi A, Nasir ES, Jahanifar M, Awan R, Young LS, Rajpoot N, Minhas F, Raza SEA. From traditional to deep learning approaches in whole slide image registration: A methodological review. *Journal of Pathology Informatics*. 2025 Sep 17;19.0:100512. doi: 10.1016/j.jpi.2025.100512. PMID: 41126805.

43. Hahn JD, Swanson LW, Bowman I, Foster NN, Zingg B, Bienkowski MS, Hintiryan H, Dong H. An open access mouse brain flatmap and upgraded rat and human brain flatmaps based on current reference atlases. *The Journal of Comparative Neurology*. 2020 Jun 8;529.0(3):576-594. doi: 10.1002/cne.24966. PMID: 32511750.

44. Wang X, Zeng W, Yang X, Zhang Y, Fang C, Zeng S, Han Y, Fei P. Bi-channel image registration and deep-learning segmentation (BIRDS) for efficient, versatile 3D mapping of mouse brain. *eLife*. 2021 Jan 18;10.0. doi: 10.7554/elife.63455. PMID: 33459255.

45. Scharwächter L, Schmitt FJ, Pallast N, Fink GR, Aswendt M. Network analysis of neuroimaging in mice. *NeuroImage*. 2022 Mar 17;253.0:119110. doi: 10.1016/j.neuroimage.2022.119110. PMID: 35311664.

46. Giampetraglia M, Weigelin B. Recent advances in intravital microscopy for preclinical research. *Current Opinion in Chemical Biology*. 2021 Jul 15;63.0:200-208. doi: 10.1016/j.cbpa.2021.05.010. PMID: 34274700.

47. Shafique A, Babaie M, Sajadi MSS, Batten AJ, Skdar S, Tizhoosh HR. Automatic Multi-Stain Registration of Whole Slide Images in Histopathology. *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society*. 2021 Nov 1;2021.0:3622-3625. doi: 10.1109/embc46164.2021.9629970. PMID: 34892022.

48. Chandrashekhar V, Tward DJ, Crowley D, Crow A, Wright MA, Hsueh B, Gore F, Machado TA, Branch A, Rosenblum JS, Deisseroth KA, Vogelstein JT. CloudReg: automatic terabyte-scale cross-modal brain volume registration. *Nature Methods*. 2021 Jul 12;18.0(8):845-846. doi: 10.1038/s41592-021-01218-z. PMID: 34253927.

49. Takata N, Sato N, Komaki Y, Okano H, Tanaka KF. Flexible annotation atlas of the mouse brain: combining and dividing brain structures of the Allen Brain Atlas while maintaining anatomical hierarchy. *Scientific Reports*. 2021 Mar 18;11.0(1):6234. doi: 10.1038/s41598-021-85807-0. PMID: 33737651.

50. Ruthig P, Schönwiesner M. Common principles in the lateralization of auditory cortex structure and function for vocal communication in primates and rodents. *European Journal of Neuroscience*. 2022 Jan 4;55.0(3):827-845. doi: 10.1111/ejn.15590. PMID: 34984748.

51. Li Y, Wu J, Lu D, Xu C, Zheng Y, Peng H, Qu L. mBrainAligner-Web: a web server for cross-modal coherent registration of whole mouse brains. *Bioinformatics*. 2022 Aug 11;38.0(19):4654-4655. doi: 10.1093/bioinformatics/btac549. PMID: 35951750.

52. Cabeen RP, Toga AW, Allman JM. Frontoinsular cortical microstructure is linked to life satisfaction in young adulthood. *Brain Imaging and Behavior*. 2021 Apr 6;15.0(6):2775-2789. doi: 10.1007/s11682-021-00467-y. PMID: 33825124.

53. Choi YK, Feng L, Jeong W, Kim J. Connecto-informatics at the mesoscale: current advances in image processing and analysis for mapping the brain connectivity. *Brain Informatics*. 2024 Jun 4;11.0(1):15. doi: 10.1186/s40708-024-00228-9. PMID: 38833195.

54. Ma X, Xing Y, Zhai R, Du Y, Yan H. Development and advancements in rodent MRI-based brain atlases. *Heliyon*. 2024 Mar 1;10.0(6):e27421. doi: 10.1016/j.heliyon.2024.e27421. PMID: 38510053.

55. Shamim M, Faridi AR, Masood F. Analysis of Emerging Trends and Challenges in Multimodal Data. *International Conference on Computing for Sustainable Global Development*. 2025 Apr 2;:1-7. doi: 10.23919/indiacom66777.2025.11115680.

56. Scarpelli M, Healey DR, Mehta SV, Kodibagkar VD, Quarles CC. A practical method for multimodal registration and assessment of whole-brain disease burden using PET, MRI, and optical imaging. *Scientific Reports*. 2020 Oct 14;10.0(1):17324. doi: 10.1038/s41598-020-74459-1. PMID: 33057180.

57. Moldovanu S, Pană L, Biswas A, Moraru L. Combining Sparse and Dense Features to Improve Multi-Modal Registration for Brain DTI Images. *Entropy*. 2020 Nov 14;22.0(11):1299. doi: 10.3390/e22111299. PMID: 33287067.

58. Stouffer KM, Witter MP, Tward DJ, Miller MI. Projective diffeomorphic mapping of molecular digital pathology with tissue MRI. *Communications Engineering*. 2022 Dec 13;1.0(1). doi: 10.1038/s44172-022-00044-1. PMID: 37284027.

59. Willekens SMA, Morini F, Mediavilla T, Nilsson E, Orädd G, Hahn M, Chotiwan N, Visa M, Berggren P, Ilegems E, Överby AK, Ahlgren U, Marcellino DJ. An MR-based brain template and atlas for optical projection tomography and light sheet fluorescence microscopy in neuroscience. *Frontiers in Neuroscience*. 2024 Mar 27;18.0:1328815. doi: 10.3389/fnins.2024.1328815. PMID: 38601090.

60. Piluso S, Verasztó C, Carey H, Delattre É, L'Yvonnet T, Colnot É, Romani A, Bjaalie JG, Markram H, Keller DX. An extended and improved CCFv3 annotation and Nissl atlas of the entire mouse brain. *Imaging Neuroscience*. 2025 Jan 1;3.0. doi: 10.1162/imag_a_00565. PMID: 40800985.

61. Vihma H, James L, Nourie HC, Smith AL, Liang S, Friar CA, Vulli T, Xing L, Cowley DO, Burette AC, Philpot BD. A dual-reporter mouse for therapeutic discovery in Angelman syndrome. *JCI Insight*. 2026 Feb 3;11.0(5). doi: 10.1172/jci.insight.197028. PMID: 41632831.

62. Jung F, Cao X, Heymans L, Carlén M. DMC-BrainMap is an open-source, end-to-end tool for multi-feature brain mapping in different species. *Cell Reports Methods*. 2026 Feb 1;6.0(2):101302. doi: 10.1016/j.crmeth.2026.101302. PMID: 41666930.

63. Fan P, Ruan Y, HU K, Wang H, Ye J, Wang S, Yuan R, Yang G, Liu F, He W, Li G, Yan X, CHU S, Zhang Z, Chen N. Ginsenoside Rg1 alleviates post-ischemic stroke neuroinflammation by inhibiting CKLF1-mediated suppression of dead/dying neuron clearance. *Acta Pharmaceutica Sinica B*. 2026 Apr 9;16.0(6):3632-3654. doi: 10.1016/j.apsb.2026.04.003. PMID: 42368585.

64. Tian Y, Cook JJH, Johnson GA. Restoring morphology of light sheet microscopy data based on magnetic resonance histology. *Frontiers in Neuroscience*. 2023 Jan 4;16.0:1011895. doi: 10.3389/fnins.2022.1011895. PMID: 36685227.

65. Duarte FT, Ramosaj M, Hasanovic E, Regio S, Sipion M, Rey MJ, Déglon N. Semi-automated workflows to quantify AAV transduction in various brain areas and predict gene editing outcome for neurological disorders. *Molecular Therapy - Methods & Clinical Development*. 2023 Mar 27;29.0:254-270. doi: 10.1016/j.omtm.2023.03.013. PMID: 37090478.

66. Aswendt M, Wieters F. Structural integrity and remodeling underlying functional recovery after stroke. *Neural Regeneration Research*. 2020 Dec 17;16.0(7):1423. doi: 10.4103/1673-5374.301004. PMID: 33318437.

67. Liang Z, Zhang J. Mouse brain MR super-resolution using a deep learning network trained with optical imaging data. *Frontiers in Radiology*. 2023 May 15;3.0:1155866. doi: 10.3389/fradi.2023.1155866. PMID: 37492378.

68. Malovani C, Friedman N, Ben-Eliezer N, Tavor I. Tissue Probability Based Registration of Diffusion-Weighted Magnetic Resonance Imaging. *Journal of Magnetic Resonance Imaging*. 2021 Apr 24;54.0(4):1066-1076. doi: 10.1002/jmri.27654. PMID: 33894095.

69. Mahani FSN, Kalantari A, Fink GR, Hoehn M, Aswendt M. A systematic review of the relationship between magnetic resonance imaging based resting-state and structural networks in the rodent brain. *Frontiers in Neuroscience*. 2023 Jul 24;17.0:1194630. doi: 10.3389/fnins.2023.1194630. PMID: 37554291.

70. Mittal S, Kim J, Bhargava R. Statistical Considerations and Tools to Improve Histopathologic Protocols with Spectroscopic Imaging. *Applied Spectroscopy*. 2022 Mar 16;76.0(4):428-438. doi: 10.1177/00037028211066327. PMID: 35296146.

71. Sun R, Wu J, Miao Y, Ouyang L, Qu L. Progressive 3D biomedical image registration network based on deep self-calibration. *Frontiers in Neuroinformatics*. 2022 Sep 21;16.0:932879. doi: 10.3389/fninf.2022.932879. PMID: 36213548.

72. Jiang P, Wu S, Qin W, Xie Y. Complex Large-Deformation Multimodality Image Registration Network for Image-Guided Radiotherapy of Cervical Cancer. *Bioengineering*. 2024 Dec 23;11.0(12):1304. doi: 10.3390/bioengineering11121304. PMID: 39768121.

73. Suliman MA, Williams LZJ, Fawaz A, Robinson EC. Unsupervised multimodal surface registration with geometric deep learning. *Medical Image Analysis*. 2025 Oct 4;107.0(Pt B):103821. doi: 10.1016/j.media.2025.103821. PMID: 41101194.

74. Bao L, Luo Z, Zhu X, Xu Y. Automated identification of protein expression intensity and classification of protein cellular locations in mouse brain regions from immunofluorescence images. *Medical & Biological Engineering & Computing*. 2023 Dec 27;62.0(4):1105-1119. doi: 10.1007/s11517-023-02985-x. PMID: 38150111.

75. Wu C, Gao Y, Huang G, Ma X. Dual-branch Bayesian adversarial learning for multi-view clustering. *Knowledge-Based Systems*. 2026 Apr 6;342.0:115942. doi: 10.1016/j.knosys.2026.115942.

76. Xhima K, McMahon D, Ntiri EE, Goubran M, Hynynen K, Aubert I. Intravenous and Non-invasive Drug Delivery to the Mouse Basal Forebrain Using MRI-guided Focused Ultrasound. *BIO-PROTOCOL*. 2021 Jan 1;11.0(12):e4056. doi: 10.21769/bioprotoc.4056. PMID: 34262999.

77. Claudi F, Tyson AL, Petrucco L, Margrie TW, Portugues R, Branco T. Brainrender : a python-based software for visualizing anatomically registered data. *bioRxiv (Cold Spring Harbor Laboratory)*. 2020 Feb 25;. doi: 10.1101/2020.02.23.961748.

78. Johnson GA, Tian Y, Cofer GP, Cook JC, Gee JC, Hall A, Hornburg KJ, Qi YP, Yeh FF, Wang N, WHITE LE, Williams RW. HiDiver: A Suite of Methods to Merge Magnetic Resonance Histology, Light Sheet Microscopy, and Complete Brain Delineations. *bioRxiv (Cold Spring Harbor Laboratory)*. 2022 Feb 10;. doi: 10.1101/2022.02.10.479607.

79. Tyson AL, Margrie TW. Mesoscale microscopy for micromammals: image analysis tools for understanding the rodent brain. *arXiv (Cornell University)*. 2021 Feb 23;. doi: 10.48550/arxiv.2102.11812.

80. Schmitt O, Eipert P, Ruß F, Beier J, Kadir K, Horn AKE. The brainstem connectome database. *Scientific Data*. 2022 Apr 12;9.0(1):168. doi: 10.1038/s41597-022-01219-3. PMID: 35414055.

81. Mansour H, Azrak R, Cook JJH, Hornburg KJ, Qi YP, Tian Y, Williams RW, Yeh FF, WHITE LE, Johnson GA. An Open Resource: MR and light sheet microscopy stereotaxic atlas of the mouse brain. *bioRxiv (Cold Spring Harbor Laboratory)*. 2024 Mar 31;. doi: 10.1101/2024.03.28.587246. PMID: 38586051.

82. Wehrli PM, Michno W, Guérard L, Fernández-Rodríguez J, Bergh ARJ, Blennow K, Zetterberg H, Hanrieder J. Spatial Chemometrics and Comprehensive Chemical Imaging based Molecular Histopathology Delineates Anatomical Heterogeneity at Single Pixel Resolution. *ChemRxiv*. 2020 Feb 5;. doi: 10.26434/chemrxiv.11800209.v1.

83. Stouffer KM, Witter MP, Tward DJ, Miller MI. Projective LDDMM: Mapping Molecular Digital Pathology with Tissue MRI. *bioRxiv (Cold Spring Harbor Laboratory)*. 2022 Apr 24;. doi: 10.1101/2022.04.22.489163.

84. Oz S, Saar G, Olszakier S, Heinrich R, Kompanets M, Berlin S. Tissue expansion of ‘cleared’ whole brains reduces contrast in ex vivo MRI. *Research Square*. 2021 Sep 14;. doi: 10.21203/rs.3.rs-877945/v1.

85. Attarpour A, Raffiee M, Xu T, Osmann J, Patel SB, Yu F, Au B, Clappison AS, Biparva M, Zhu R, Crow A, Gharagoz BE, Rozak MW, Aubert I, McLaurin J, Deisseroth KA, Stefanovic B, Goubran M. Quantitative profiling of whole-brain connectomes at single-axon resolution using deep learning and high-resolution light sheet microscopy. *bioRxiv (Cold Spring Harbor Laboratory)*. 2025 Nov 16;. doi: 10.1101/2025.11.14.688340. PMID: 41292859.

86. Abe P, Lavalley A, Morassut I, Klingler E, Santinha AJ, Platt RJ, Jabaudon D. Developmental molecular controls over arealization of descending cortical motor pathways. *bioRxiv (Cold Spring Harbor Laboratory)*. 2023 Jun 30;. doi: 10.1101/2023.06.29.546438.

87. Ciarrusta J, Arichi T. Neurobiology and the connectome. *Elsevier eBooks*. 2023 Jan 1;:3-23. doi: 10.1016/b978-0-323-85280-7.00012-9.

88. Marques FC, Wehrle E, Müller R. Correlative Multimodal Imaging for Multiscale Analysis of Complex Biological Systems: Validation Methods and Performance Analysis. *Reference series in biomedical engineering*. 2023 Jan 1;:1-36. doi: 10.1007/978-3-030-85569-7_17-1.

89. Ali LM, Yim AK, Gerbi E, Nguyen T, Tu N, Ikede FP, Sampaleanu R, Grigore D, Waligorski JE, Kremitzki CL, Yuan L, Dong W, Mitra RD, Milbrandt J, Buchser W. Biophysical Simulation Enables Multi-Scale Segmentation and Atlas Mapping for Top-Down Spatial Omics of the Nervous System. *bioRxiv (Cold Spring Harbor Laboratory)*. 2024 Aug 3;. doi: 10.1101/2024.08.02.605366.

90. Aggarwal M. Preclinical and ex vivo tractography: Techniques and applications at high field. *Elsevier eBooks*. 2025 Jan 1;:653-667. doi: 10.1016/b978-0-12-818894-1.00025-2.

91. Chau GR, Weber EMM, Batra A, Ni L, Zeineh M, Chaudhari AS, Adeli E, Knowles JK, McNab JA. Non-parametric prediction of brain MRI microstructure using transfer learning. *Imaging Neuroscience*. 2025 Jan 1;3.0. doi: 10.1162/imag_a_00548. PMID: 40800932.

92. Marques FC, Mathavan N, Ohs N, Goenczoel C, Kendall JJ, Singh A, Yilmaz D, Günther D, Kuhn GA, Schulte FA, Wehrle E, Müller R. Spatial µProBe: a correlative multimodal imaging approach for spatial profiling of biological micro-environments. *bioRxiv (Cold Spring Harbor Laboratory)*. 2025 Apr 26;. doi: 10.1101/2025.04.23.650252.

93. Marques FC, Wehrle E, Müller R. Correlative Multimodal Imaging for Multiscale Analysis of Complex Biological Systems: Validation Methods and Performance Analysis. *Reference series in biomedical engineering*. 2025 Jan 1;:387-422. doi: 10.1007/978-3-031-91544-4_17.

94. Ali LM, Yim AK, Gerbi E, Nguyen T, Tu N, Ikede F, Sampaleanu R, Grigore D, Waligorski JE, Kremitzki CL, Yuan L, Dong W, Mitra RD, Milbrandt J, Buchser W. Biophysical simulation enables segmentation and nervous system atlas mapping for image first spatial omics. *npj Systems Biology and Applications*. 2025 Dec 14;12.0(1):3. doi: 10.1038/s41540-025-00627-6. PMID: 41392269.

95. Zhu S, Dinsdale NK, Jbabdi S, Miller KL, Howard AF. Microscopy-informed structural connectivity mapping in the in vivo human brain via domain adaptation. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Jun 18;. doi: 10.64898/2026.06.14.732211. PMID: 42367983.

96. Feng AY, Barbosa DAN, Casey AB, Rijsketic DR, Salgado JS, Huang H, Malenka RC, Hermes D, Miller KJ, Halpern CH, Heifets BD. Cross-species brain-wide mapping reveals convergent mesocorticolimbic engagement by nucleus accumbens deep brain stimulation. *Molecular Psychiatry*. 2026 Aug 18;. doi: 10.1038/s41380-026-03828-5. PMID: 42613417.

97. Darzi F, Guerrero RED, Bocklitz TW. A Multistep Registration Framework for Multimodal Histology Across Same and Consecutive Sections. *Journal of Imaging Informatics in Medicine*. 2026 Sep 10;. doi: 10.1007/s10278-026-02278-3. PMID: 42722955.

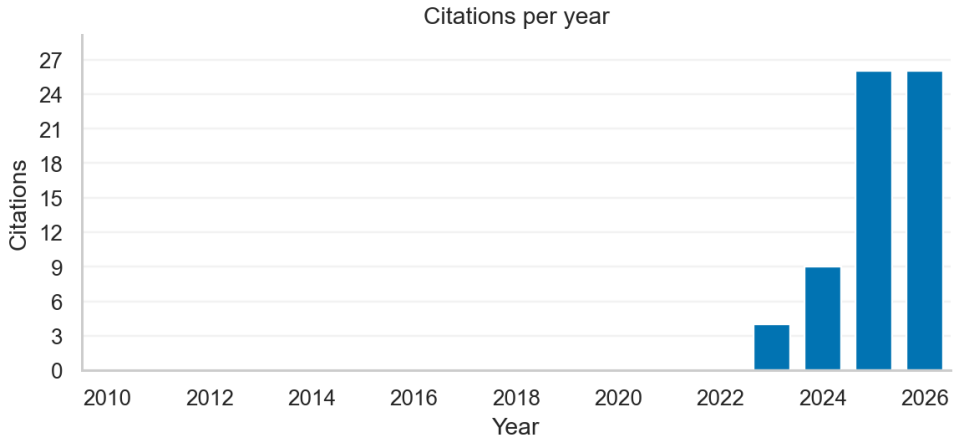
98. Li Y, Huang H, Ling N, Qu L, Zhou Q. DGRSFReg: A domain-generalized representation similarity framework for multimodal brain image registration. *Expert Systems with Applications*. 2026 Sep 22;334.0:134472. doi: 10.1016/j.eswa.2026.134472.

99. Policet-Bétend H, Jordá-Siquier T, Cuenu G, Badre M, Schranz S, Da Costa J, Brockmann C, Senn P, Huber D, Lamy CM. Whole human organ clearing and multimodal mapping. *eLife*. 2026 Oct 12;. doi: 10.7554/elife.112713.1.

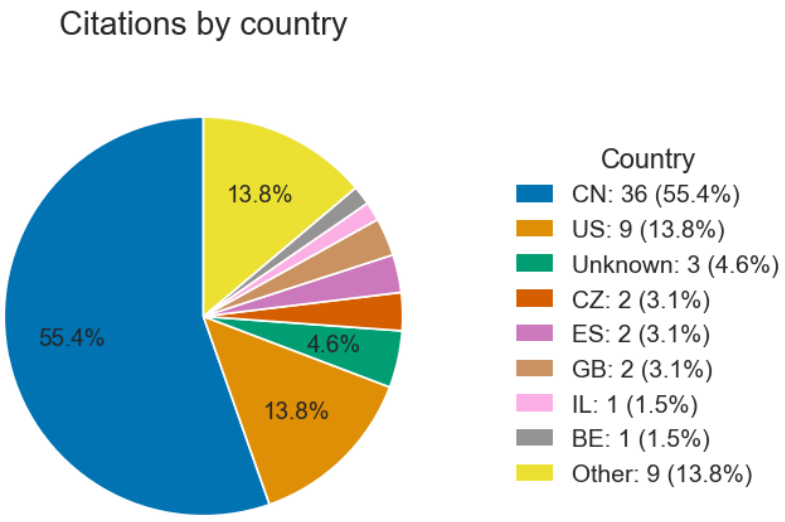
100. Policet-Bétend H, Jordá-Siquier T, Cuenu G, Badre M, Schranz S, Da Costa J, Brockmann C, Senn P, Huber D, Lamy CM. Whole human organ clearing and multimodal mapping. *eLife*. 2026 Oct 12;. doi: 10.7554/elife.112713.

Publication

Stokowska A, Aswendt M, Žucha D, Lohmann S, Wieters F, Suarez JM, Atkins AL, Li Y, Miteva M, Lewin J, Wiedermann D, Diedenhofen M, Naluai ÅT, Abaffy P, Valihrach L, Kubista M, Hoehn M, Pekny M, Pekna M. Complement C3a treatment accelerates recovery after stroke via modulation of astrocyte reactivity and cortical connectivity. *Journal of Clinical Investigation*. 2023 Mar 30;133(10). doi: 10.1172/jci162253. PMID: 36995772.



Citation summary
Total citations: **65**
Average/year: **16.25**
Period: 2023-2026
SCImago cites/doc 2y (2025): **9.84**
SJR 2025: **4.81** (Q1)



Cited by

1. Mastellos DC, Hajishengallis G, Lambris JD. A guide to complement biology, pathology and therapeutic opportunity. *Nature reviews. Immunology*. 2023 Sep 5;24.0(2):118-141. doi: 10.1038/s41577-023-00926-1. PMID: 37670180.

2. Petr V, Thurman JM. The role of complement in kidney disease. *Nature Reviews Nephrology*. 2023 Sep 21;19.0(12):771-787. doi: 10.1038/s41581-023-00766-1. PMID: 37735215.

3. Tenner AJ, Petrisko TJ. Knowing the enemy: strategic targeting of complement to treat Alzheimer disease. *Nature Reviews Neurology*. 2025 Mar 24;21.0(5):250-264. doi: 10.1038/s41582-025-01073-y. PMID: 40128350.

4. Li XX, Woodruff TM. The complement system: Biology, pathology, and therapeutic interventions. *Pharmacological Reviews*. 2025 Jun 26;77.0(5):100079. doi: 10.1016/j.pharmr.2025.100079. PMID: 40683108.

5. Li D, Huo X, Shen L, Qian M, Wang J, Mao S, Chen W, Li RQ, Zhu T, Zhang B, Liu K, Wu F, Bai Y. Astrocyte heterogeneity in ischemic stroke: Molecular mechanisms and therapeutic targets. *Neurobiology of Disease*. 2025 Mar 25;209.0:106885. doi: 10.1016/j.nbd.2025.106885. PMID: 40139279.

6. Zelek WM, Tenner AJ. Complement therapeutics in neurodegenerative diseases. *Immunobiology*. 2025 Jun 16;230.0(4):153089. doi: 10.1016/j.imbio.2025.153089. PMID: 40544661.

7. Zhong J, Peng Y, Zhang L, Xiao B, Zhang M. Astrocyte Lipid Droplet Dynamics Orchestrate Neurological Disorders and Therapeutic Horizons. *Small Science*. 2025 Jun 8;5.0(9):2500152. doi: 10.1002/smsc.202500152. PMID: 40917390.

8. Wang D, Huang W, Shi J, Liu F, Jiang W, Chen K, Zhang S, Li X, Lin L. FGF21, a modulator of astrocyte reactivity, protects against ischemic brain injury through anti-inflammatory and neurotrophic pathways. *Acta Pharmacologica Sinica*. 2025 Feb 28;46.0(7):1834-1851. doi: 10.1038/s41401-024-01462-x. PMID: 40021824.

9. Li Y, Zeng H, Zhong X. Complement C3 in panvascular disease: a central integrator of immune signaling and vascular remodeling. *Clinical Science*. 2025 Nov 1;139.0(21):1373-1403. doi: 10.1042/cs20257865. PMID: 41187099.

10. Akhter N, Lambay A, Almotairi R, Hamadi A, Bhatia K, Ahmad S, Ducruet AF. The Complex Role of the Complement C3a Receptor (C3aR) in Cerebral Injury and Recovery Following Ischemic Stroke. *Cells*. 2025 Sep 15;14.0(18):1440. doi: 10.3390/cells14181440. PMID: 41002405.

11. Zhang S, Yuan M, Zhou J, Yuan ZK, Wang L, Gong C, Lu H, Cheng X, Wang X, He Q, Hu L, Wang B, He C, Fang Y, Lin S, Zi W, He Y, Zhao C, Zheng H, Niu J, Mei F, Sun B, Xie Q, Yang Q. BST2 expression at astrocyte borders promotes microglial recruitment via the C3/C3aR signaling. *Neuron*. 2025 Oct 24;114.0(1):67-85.e8. doi: 10.1016/j.neuron.2025.09.038. PMID: 41138733.

12. Park J, Riew T. Nanoparticle-Mediated Nose-to-Brain Delivery for Ischemic Stroke Therapy: Preclinical Insights. *Pharmaceutics*. 2025 Nov 9;17.0(11):1447. doi: 10.3390/pharmaceutics17111447. PMID: 41304786.

13. Phan NV, Rathbun EM, Ouyang Y, Carmichael ST, Segura T. Biology-driven material design for ischaemic stroke repair. *Nature Reviews Bioengineering*. 2023 Oct 27;2.0(1):44-63. doi: 10.1038/s44222-023-00117-6.

14. Schartz ND, Aroor A, Li Y, Pinzón-Hoyos N, Brewster AL. Mice deficient in complement C3 are protected against recognition memory deficits and astrogliosis induced by status epilepticus. *Frontiers in Molecular Neuroscience*. 2023 Nov 14;16.0:1265944. doi: 10.3389/fnmol.2023.1265944. PMID: 38035266.

15. Yang L, Wu J, Zhang F, Zhang L, Zhang X, Zhou J, Pang J, Xie B, Xie H, Jiang YQ, Peng J. Microglia aggravate white matter injury via C3/C3aR pathway after experimental subarachnoid hemorrhage. *Experimental Neurology*. 2024 Jun 10;379.0:114853. doi: 10.1016/j.expneurol.2024.114853. PMID: 38866102.

16. Liu R, Yu Y, Ge Q, Feng R, Zhong G, Luo L, Han Z, Wang T, Huang C, Xue J, Huang Z. Genistein-3′-sodium sulfonate promotes brain functional rehabilitation in ischemic stroke rats by regulating astrocytes polarization through NF-κB signaling pathway. *Chemico-Biological Interactions*. 2024 Jul 25;400.0:111159. doi: 10.1016/j.cbi.2024.111159. PMID: 39059603.

17. Garton TP, Smith MD, Kesharwani A, Gharagozloo M, Oh S, Na CH, Absinta M, Reich DS, Zack DJ, Calabresi PA. Myeloid lineage C3 induces reactive gliosis and neuronal stress during CNS inflammation. *Nature Communications*. 2025 Apr 11;16.0(1):3481. doi: 10.1038/s41467-025-58708-3. PMID: 40216817.

18. Liu H, Jiang M, Chen Z, Li C, Yin X, Zhang X, Wu M. The Role of the Complement System in Synaptic Pruning after Stroke. *Aging and Disease*. 2024 Jan 1;16.0(3):0. doi: 10.14336/ad.2024.0373. PMID: 39012667.

19. Zhang Y, Zhu Z, Li Z, Feng J, Long J, Deng Y, Ahmed W, Khan AA, Huang S, Fu Q, Chen L. Sbno1 mediates cell-cell communication between neural stem cells and microglia through small extracellular vesicles. *Cell & Bioscience*. 2024 Sep 29;14.0(1):125. doi: 10.1186/s13578-024-01296-4. PMID: 39343943.

20. Mu Y, Yang X, Feng Y, Zhang L, Xu J, Li M, Wu R, Li S, He X, Zuo Z, Hu X. Physical exercise promotes white matter repair after ischemic stroke. *Neural Regeneration Research*. 2025 Apr 29;21.0(6):2397-2406. doi: 10.4103/nrr.nrr-d-24-00861. PMID: 40313089.

21. Gleichman AJ, Kawaguchi R, Rathbun EM, Sofroniew MV, Carmichael ST. Regionally mapped astrocytic responses to cortical and white matter stroke show differential roles in astrocyte-induced vascular remodeling. *Neuron*. 2025 Oct 10;113.0(24):4178-4198.e9. doi: 10.1016/j.neuron.2025.09.023. PMID: 41075786.

22. Rao DP, Hammad AK, Al-Hetty HRAK, Kareem AK, Hjazi A, Ahmad I, Abbas Z, albaz RY, Mustafa YF, Abdulmoaty MAZ. Targeting Reactive Astrocytes with Flavonoids to Alleviate Neuroinflammation and Promote Synaptic Function. *Molecular Neurobiology*. 2026 Jan 1;63.0(1):414. doi: 10.1007/s12035-026-05670-w. PMID: 41627560.

23. Chen X, Li K, Liu S, Zhao J, Tiwari S, Zeng F, ZHAO Y, Zhang P, Tang H, Zhao H, Kettenmann H, Xiang X, Zhu X. Trem2 exacerbates ischemic brain injury through Gpnmb in a photothrombotic stroke model. *Proceedings of the National Academy of Sciences*. 2026 Apr 14;123.0(16):e2523148123. doi: 10.1073/pnas.2523148123. PMID: 41980098.

24. Zhong X, Li Y, Xie Y, Jiang H, Liu Z, Dong W, Wang H, Zeng H. Complement C3/C3a-CCL9 feedback loop orchestrates inflammatory crosstalk to accelerate aortic dissection. *Biochemical Pharmacology*. 2025 Oct 13;243.0(Pt 1):117421. doi: 10.1016/j.bcp.2025.117421. PMID: 41093012.

25. Wang Y, Wang D, Li D, Ma Y. The role of astrocytes in ischemic stroke - mechanisms, functions and treatment. *Frontiers in Cell and Developmental Biology*. 2025 Dec 10;13.0:1700564. doi: 10.3389/fcell.2025.1700564. PMID: 41450734.

26. Verkhratsky A, Lee CJ, Chun H, Göritz C, Harkany T, Lee J, Lee S, Lindskog M, Koh W, Mulder J, Nam M, Ottersen OP, Pekna M, Pekny M, Pękowska A, Ryu H, Sohn CH, Tretiakov EO, Untiet V, Viney TJ, Youn W, Yi C, Zorec R, Yun MR, Cheong E, Nordberg A. Curing the brain: in search for new astrocyte-specific therapies. *Experimental & Molecular Medicine*. 2026 Apr 24;58.0(4):1086-1127. doi: 10.1038/s12276-026-01712-4. PMID: 42032208.

27. Jia Y, Song Y, Xue H, Li X, Zhang Y, Fan S, Xu Y, Ding Z, Qiu Y, Wu Z, Zhao P. Sevoflurane postconditioning mitigates neuronal hypoxic-ischemic injury via regulating reactive astrocytic STAT3 protein modification. *Chemico-Biological Interactions*. 2024 Nov 12;405.0:111308. doi: 10.1016/j.cbi.2024.111308. PMID: 39536892.

28. Magnusen AF, Pandey MK. Complement System and Adhesion Molecule Skirmishes in Fabry Disease: Insights into Pathogenesis and Disease Mechanisms. *International Journal of Molecular Sciences*. 2024 Nov 14;25.0(22):12252. doi: 10.3390/ijms252212252. PMID: 39596318.

29. Xu B, Bai Y, Liu Z, Li J, Chen K, Yan Z, Wang Y, Xu J, Wang H, Wang H, Shi F, Wang J, Wang H, Wang H. Harnessing patient-derived antibodies-induced microglial complement 1q expression: Novel therapy for anti-NMDAR encephalitis.

Molecular Therapy. 2025 Aug 6;33.0(11):5721-5740. doi: 10.1016/j.ymthe.2025.08.001. PMID: 40776448.

30. Pang N, Liu Y, Yu S, Li Y, Shi K. CircFGFR2 induces astrocyte pyroptosis to promote ischemic stroke. *Neuroscience*. 2025 Aug 19;584.0:356-366. doi: 10.1016/j.neuroscience.2025.08.030. PMID: 40840702.

31. Liang Q, Wang Y, Li Z. Comprehensive bioinformatics analysis identifies metabolic and immune-related diagnostic biomarkers shared between diabetes and COPD using multi-omics and machine learning. *Frontiers in Endocrinology*. 2025 Jan 8;15.0:1475958. doi: 10.3389/fendo.2024.1475958. PMID: 39845878.

32. Berkowitz SH, Goldberg Z, Golderman V, Dori A, Maggio N, Devaux JJ, Stein ES. Complement C3a and C5a Receptors Are Presented in Mouse Sciatic and Human Sural Nerves and Selectively Modulate the Neuronal Function of Large-Caliber Fibers in Mice. *Journal of Neurochemistry*. 2025 Jun 1;169.0(6):e70129. doi: 10.1111/jnc.70129. PMID: 40539483.

33. Xiong S, Sun M, Zhang Y, Kong P, Gan L, Gao L, Xu K, Wu H, Zhu D, Lin Y, Li R, Luo C. Astrocytic BEST1 can serve as a target for functional recovery after ischemic stroke. *Molecular Therapy*. 2025 Mar 20;33.0(7):3209-3230. doi: 10.1016/j.ymthe.2025.03.022. PMID: 40119514.

34. Xu L, Wang X, Xu H, Chen S, Jiang W, Li H, Afridi SK, Wang Y, Luo W, Yang H, Tang C, Qiu W. Targeting C3aR/STAT3 signaling attenuates neuroinflammation and reverses neurological impairment in Neuromyelitis optica. *Fundamental Research*. 2025 Nov 8;6.0(5):3491-3506. doi: 10.1016/j.fmre.2025.11.001. PMID: 42846538.

35. Sun L, Wang YL, Liu Y, Chang X, Mao X, Wu X, Chu M, Niu H, Shi M, Guo D, Zhu Z, Zhao J. Associations of Serum Complement Biomarkers With Adverse Clinical Outcomes Among Patients With Ischemic Stroke. *Journal of the American Heart Association*. 2026 Jan 14;15.0(2):e046136. doi: 10.1161/jaha.125.046136. PMID: 41532542.

36. Gao S, Su R, Gao J, Zhang G, Xu Z, Han B, Ying Z, Zhang Z, Zhang J, Liu W. Fujian Tablets promote corticospinal tract remodeling and improve motor function in MCAO rats by regulating BDNF shearing enzyme-mediated proBDNF/mBDNF balance. *Journal of Ethnopharmacology*. 2026 Jan 17;361.0:121235. doi: 10.1016/j.jep.2026.121235. PMID: 41548621.

37. Gaudet AD, Fonken LK. Phagocytosis and neuroinflammation: orchestrating central nervous system homeostasis, repair, and the resolution of inflammation. *Trends in Neurosciences*. 2026 Mar 1;49.0(3):198-216. doi: 10.1016/j.tins.2026.01.002. PMID: 41771723.

38. Shang Y, Kuang B, Zheng J, Du Y, Shang T, Luo B, Sun Y, Zheng L, Zhang B, Liu L, Zou W. The Interaction of Neutrophil Extracellular Traps and Complement in Intracerebral Hemorrhage. *Cellular and Molecular Neurobiology*. 2026 Mar 19;46.0(1). doi: 10.1007/s10571-026-01675-0. PMID: 41857255.

39. Castillo-González J, Vargas-Rodríguez P, Serrano-Martínez I, Cuenca-Martagón A, Forte-Lago I, Price M, Buscemi L, Hirt L, Ruiz JL, González-Rey E. Lack of cortistatin drives neuroimmune and vascular dysfunction in brain ischemia. *bioRxiv (Cold Spring Harbor Laboratory)*. 2024 Sep 14;. doi: 10.1101/2024.09.11.611691.

40. Chen H, Frank JA, Tan C, Lee AG, Kopchock R, Chiang T, Kim AY, Galvan MD, Fraser JF, Dornbos D, Nour HA, Millson N, Tomlinson S, McCullough LD, Pennypacker KR, Cheng M, Bliss TM, Steinberg GK. Luminal Vascular Dysfunction Drives Rapid Blood Brain Barrier Injury in Hyperglycemic Stroke: Key Roles for Luminal Glycocalyx and Complement. *bioRxiv (Cold Spring Harbor Laboratory)*. 2025 Jun 30;. doi: 10.1101/2025.06.26.661858. PMID: 40667062.

41. References. *Elsevier eBooks*. 2024 Nov 8;:471-578. doi: 10.1016/b978-0-443-23565-8.09996-8.

42. Anisman H, Hayley SP, Kusnecov A. Illness Comorbidities in Relation to Inflammatory Processes. *Elsevier eBooks*. 2024 Nov 8;:443-469. doi: 10.1016/b978-0-443-23565-8.00003-x.

43. Wang C, Meng Y, Zhu Y. Human Umbilical Cord Mesenchymal Stem Cells Alleviate White Matter Injury in Preterm Rats by Modulating Complement C3-Mediated Microglial Polarization. *SSRN Electronic Journal*. 2025 Jan 1;. doi: 10.2139/ssrn.5132844.

44. Marchetti V, Turečková J, Kriška J, Heřmanová Z, Anděrová M. Studying the Role of Astrocytes in Ischemic Acute Brain Slices. *Springer eBooks*. 2025 Jan 1;:77-109. doi: 10.1007/978-3-031-94215-0_3.

45. Hua J, Chen D, Sun Y, Liu Z, Fang X, Sun H, Chen L, Zhang L, Cao Y, Zhang L, Chen Q. Structural-functional connectivity coupling in motor-brain networks following acute ischemic stroke. *Journal of NeuroEngineering and Rehabilitation*. 2026 Apr 7;23.0(1). doi: 10.1186/s12984-026-01981-0. PMID: 41947201.

46. Sun W, Wang J, Yao Z, Pan F, Gan Z, Yang Y, Lu J, Zhou L, Xu G. Causal effects of lung function and COPD on functional outcome following ischemic stroke: a univariable and multivariable Mendelian randomization study. *medRxiv*. 2025 Sep 5;. doi: 10.1101/2025.09.03.25335062.

47. Hamer D, Butorac A, Petrinec D, Berecki M, Mendes VM, Manadas BJ, Kelava V, Hackenberger BK, Glasnović A, Lovrić M, Gajović S, Radmilović MD. Proteomic Analysis of Mouse Cerebral Cortex Following Experimental Ischemic Stroke: Identifying Novel Biomarkers of Damage and Repair. *Cellular and Molecular Neurobiology*. 2025 Dec 4;46.0(1):8. doi: 10.1007/s10571-025-01645-y. PMID: 41342963.

48. Bracke L, Delaere J, Haspeslagh E, De Winter K, Driege Y, Bilgraer R, Delahaye T, Hack CE, Van de Walle I. Development of an anti-rat complement C2 antibody that improves renal outcome in a rat kidney transplant model. *Frontiers in Immunology*. 2026 Jan 15;16.0:1668376. doi: 10.3389/fimmu.2025.1668376. PMID: 41624835.

49. Suleiman N, Sanusi F, Bulama I, Saidu UF, Idris Y, Bilbis LS. Neuroinflammatory mechanisms in ischemic stroke and phytotherapeutic interventions for enhanced recovery: A review. *American Journal of Pharmacotherapy and Pharmaceutical Sciences*. 2026 Jan 21;5.0:2. doi: 10.25259/ajpps_2026_002.

50. Zou A, You H, Xiao J. The immunothrombosis pathway linking pulmonary disease to stroke and neurodegeneration: mechanisms and therapeutic implications. *Frontiers in Pharmacology*. 2026 Mar 31;17.0:1811794. doi: 10.3389/fphar.2026.1811794. PMID: 41988520.

51. Bo X, Wang M, Liu Y, Zhong Y. The Role of the C3a-C3aR Pathway in Diseases: Latest Research Advances. *Mediators of Inflammation*. 2026 Jan 1;2026.0(1):e1754881. doi: 10.1155/mi/1754881. PMID: 41999083.

52. (8449) SL, Suo L, Chen H, Wang Q, Mao D, Cai Y, Qin B. C3a targets the Hippo-YAP signaling pathway to promote Müller cells glial-mesenchymal transition in a high-glucose in vitro model. *European journal of medical research*. 2026 Apr 29;31.0(1). doi: 10.1186/s40001-026-04304-0.

53. Zhu S, Zhong S, Lin XP, Zhao C, Li Y. Emerging Roles of Adenosine Metabolism in Astrocytes During Brain Injury. *CNS Neuroscience & Therapeutics*. 2026 May 1;32.0(5):e70889. doi: 10.1002/cns.70889. PMID: 42136261.

54. Song Q, Zhao M, Sun W, Xie Q, Zhang Y, Tang W, Li M. Electroacupuncture Promotes Synaptic Recovery After Cerebral Ischemia-Reperfusion by Activating SIRT1 to Inhibit the NF-κB Pathway and Regulate Astrocyte Phenotypic Transformation. *Neurochemical Research*. 2026 May 18;51.0(3). doi: 10.1007/s11064-026-04775-x. PMID: 42149357.

55. Castillo-González J, Vargas-Rodríguez P, Serrano-Martínez I, Cuenca-Martagón A, Forte-Lago I, Price M, Buscemi L, Hirt L, Ruiz JL, González-Rey E. Lack of cortistatin drives neuroimmune and vascular dysfunction in brain ischemia. *Journal of Biomedical Science*. 2026 Jun 20;33.0(1). doi: 10.1186/s12929-026-01271-w. PMID: 42321803.

56. Li X, Tian X, Zhang J, Pan M, Xu B, Wei J, Yan D, Fang W. The ubiquitin E3 ligase TRIM59 in myeloid cell regulates complement C3 through degrading CCAAT/enhancer binding protein β to promote neuron survival after ischemic stroke. *Cell & Bioscience*. 2026 Jun 30;16.0(1). doi: 10.1186/s13578-026-01605-z. PMID: 42380918.

57. Huang P, Jiang S, Chen Z, Liu N, Li H, Wang A, Luo L, Huang W, Chen S, Deng Y. Complement C3a induces synaptic elimination in the hippocampus and neurobehavioral deficits in later life of septic neonatal rats triggered by lipopolysaccharide. *Free Radical Biology and Medicine*. 2026 Jul 2;254.0:579-594. doi: 10.1016/j.freeradbiomed.2026.07.003. PMID: 42385800.

58. Zhang Y, Pu J, Shen Z, Zhang Y, Liu J, Di Zhang, Deng K, Yao Y. Global Trends and Evolving Frontiers in Intranasal Delivery for CNS Diseases (2000-2025): A Bibliometric Analysis and Systematic Review. *World Neurosurgery*. 2026 Jul 2;212.0:125172. doi: 10.1016/j.wneu.2026.125172. PMID: 42392306.

59. Yin Q, Chen S, Yang D, Li R, Hang H, Xu C, Wang X, Li J, Zhao J, Wang J, Chen J, Sun PH, Zhang T, Hu Q, Xu J, Xue L, Liu C, Huang S, Zhao H, Wang Y. IL7R depletion mitigates neuroinflammation and ischemic stroke via AKT dephosphorylation dependent LCP1 suppression in mice. *Neurotherapeutics*. 2026 Jul 1;23.0(4):e00966. doi: 10.1016/j.neurot.2026.e00966. PMID: 42447806.

60. Zhang Y, Xia Y, Chen Z, Hou S, Li J, Pang M, Wang Y. C3aR inhibition alleviates cerebral ischemia-reperfusion injury via STING-mediated ferroptosis through SLC7A11/GPX4. *iScience*. 2026 Jul 31;29.0(8):116982. doi: 10.1016/j.isci.2026.116982. PMID: 42572629.

61. Liu C, Liu Q, Jing W, Liu L, Wei L, Li X, Du X. Microglial Immunometabolic Remodeling After Ischemic Stroke: The Interplay Between Mitochondrial Stress and the Ischemic Microenvironment. *Translational Stroke Research*. 2026 Aug 22;17.0(5). doi: 10.1007/s12975-026-01492-z. PMID: 42631879.

62. Song L, Sun Z, Zhang X, Shang G, Du S, Cheng F, Xu W. Inflammation-centered neurovascular-immune-metabolic remodeling in ischemic stroke: stage-dependent mechanisms, regulated cell death, and therapeutic translation. *Frontiers in Immunology*. 2026 Sep 3;17.0:1918327. doi: 10.3389/fimmu.2026.1918327. PMID: 42755742.

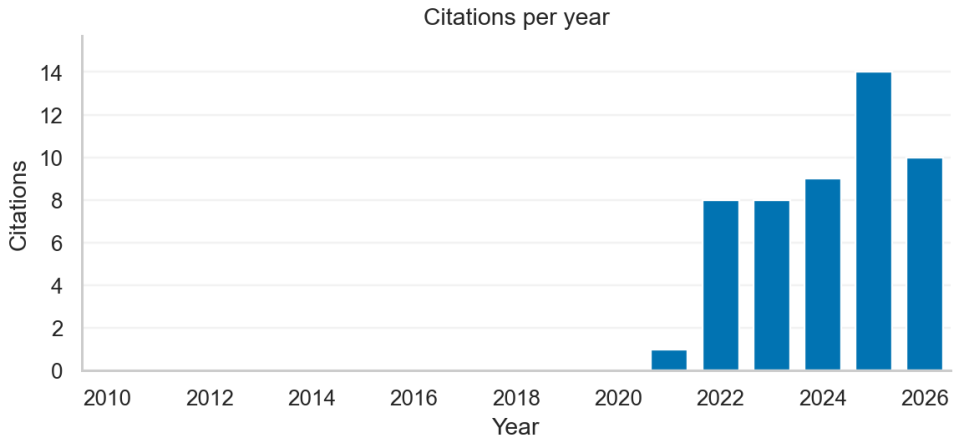
63. Thapa R, Bhat AA, Fuloria NK, Varghese N, Narain K, Biswas A, Biswas S, Bhatia S, Gan SH, Fuloria S. Canonical complement C3a-C3AR1 signalling in ageing and age-related disease: Cellular mechanisms, tissue-specific effects and translational challenges. *Ageing Research Reviews*. 2026 Sep 9;122.0:103367. doi: 10.1016/j.arr.2026.103367. PMID: 42716442.

64. Shu H, Luo Q, Hu P, Xu Y, Zhu W, Yang L, Lei K, Luo M, Zhang S, Zhu X, Zhu X, Liao Q, Chen Z, Zeng Q, Zhu X, Zhu X, Zhu Z, Yan T. TRIM28 SUMOylates IRF3 in astrocytes to drive neutrophils infiltration and ischemic cerebral injury. *Cell Death and Differentiation*. 2026 Sep 15;. doi: 10.1038/s41418-026-01871-x. PMID: 42745034.

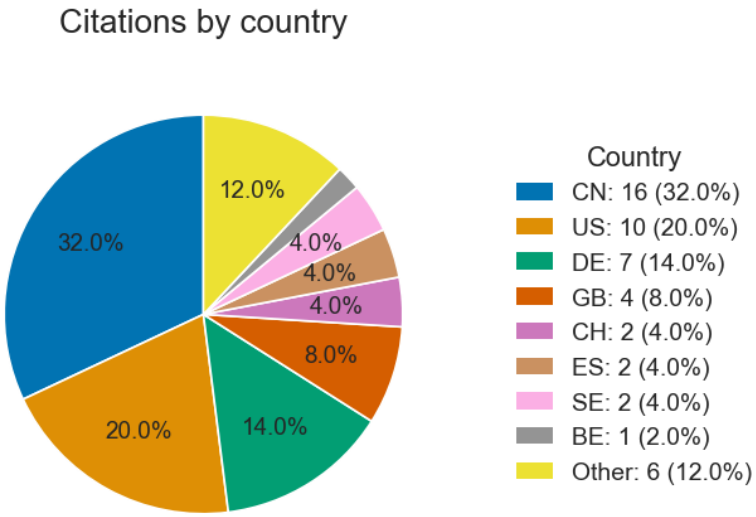
65. Ramos AJ, Villarreal A. “Stand by me”: astrocyte-microglia crosstalk in central nervous system development, homeostasis, injury, and disease. *Frontiers in Network Physiology*. 2026 Sep 15;6.0:1895622. doi: 10.3389/fnetp.2026.1895622. PMID: 42824483.

Publication

Aswendt M, Wilhelmsson U, Wieters F, Stokowska A, Schmitt FJ, Pallast N, de Pablo Y, Mohammed L, Hoehn M, Pekna M, Pekny M. Reactive astrocytes prevent maladaptive plasticity after ischemic stroke. *Progress in Neurobiology*. 2021 Dec 15;209:102199. doi: 10.1016/j.pneurobio.2021.102199. PMID: 34921928.



Citation summary
Total citations: **50**
Average/year: **8.33**
Period: 2021-2026
SCImago cites/doc 2y (2025): **4.74**
SJR 2025: **2.4** (Q1)



Cited by

1. Verkhratsky A, Butt AM, Li B, Illés P, Zorec R, Semyanov AV, Tang Y, Sofroniew MV. Astrocytes in human central nervous system diseases: a frontier for new therapies. *Signal Transduction and Targeted Therapy*. 2023 Oct 12;8.0(1):396. doi: 10.1038/s41392-023-01628-9. PMID: 37828019.

2. Luo J. TGF-β as a Key Modulator of Astrocyte Reactivity: Disease Relevance and Therapeutic Implications. *Biomedicines*. 2022 May 23;10.0(5):1206. doi: 10.3390/biomedicines10051206. PMID: 35625943.

3. Wang T, Qiu X, Wu H, Chen K, Liu S, Li Y, He T, Zhao J. Vimentin as a potential target for diverse nervous system diseases. *Neural Regeneration Research*. 2022 Oct 27;18.0(5):969. doi: 10.4103/1673-5374.355744. PMID: 36254976.

4. Stokowska A, Aswendt M, Žucha D, Lohmann S, Wieters F, Suarez JM, Atkins AL, Li Y, Miteva M, Lewin J, Wiedermann D, Diedenhofen M, Naluai ÅT, Abaffy P, Valihrach L, Kubista M, Hoehn M, Pekny M, Pekna M. Complement C3a treatment accelerates recovery after stroke via modulation of astrocyte reactivity and cortical connectivity. *Journal of Clinical Investigation*. 2023 Mar 30;133.0(10). doi: 10.1172/jci162253. PMID: 36995772.

5. He L, Duan X, Li S, Zhang R, Dai X, Lu M. Unveiling the role of astrocytes in postoperative cognitive dysfunction. *Ageing Research Reviews*. 2024 Feb 5;95.0:102223. doi: 10.1016/j.arr.2024.102223. PMID: 38325753.

6. Chen W, Su G, Chai M, An Y, Song J, Zhang Z. Astrogliosis and glial scar in ischemic stroke - focused on mechanism and treatment. *Experimental Neurology*. 2024 Dec 27;385.0:115131. doi: 10.1016/j.expneurol.2024.115131. PMID: 39733853.

7. Kim J, Trzebanski S, Shin S, Schori L, Friedman GRF, Chapal-Ilani N, Kadam A, Vicario R, Aust O, Bugaeva P, Piątek S, Ismajli LK, Hoffmann CJ, Scheller M, Boura-Halfon S, Kaushansky N, Golani O, Solomon A, Liu Z, Amann L, Boehm-Sturm P, Koch S, Wenger N, Ginhoux F, Prinz M, Avraham R, Harms C, Geissmann F, Müller-Tidow C, Uderhardt S, Milenković I,

Shlush LI, Jung S. Clonal hematopoiesis-associated motoric deficits caused by monocyte-derived microglia accumulating in aging mice. *Cell Reports*. 2025 Apr 23;44.0(5):115609. doi: 10.1016/j.celrep.2025.115609. PMID: 40279248.

8. Yang D, Su J, Chen Y, Chen G. The NF-κB pathway: Key players in neurocognitive functions and related disorders. *European Journal of Pharmacology*. 2024 Oct 4;984.0:177038. doi: 10.1016/j.ejphar.2024.177038. PMID: 39369877.

9. Hou J, Klippel K, Bose P. Glutamate Excitotoxicity: A Key Secondary Injury Mechanism of Traumatic Brain Injury and Spinal Cord Injury. *Frontiers in Bioscience-Landmark*. 2025 Oct 31;30.0(10):38904. doi: 10.31083/fbl38904. PMID: 41198536.

10. Collyer E, Blanco-Suárez E. Astrocytes in stroke-induced neurodegeneration: a timeline. *Frontiers in Molecular Medicine*. 2023 Sep 7;3.0:1240862. doi: 10.3389/fmmed.2023.1240862. PMID: 39086680.

11. Li Y, Li X, Zhao Y, Chen HH, Wang C, Wu A, Guo X, Huang Y, Wang Q, Hao L, Li X, Ji Y, Ban J, Wang G, Cao J, Pan Z. NAT10 inhibition alleviates astrocyte autophagy by impeding ac4C acetylation of Timp1 mRNA in ischemic stroke. *Acta Pharmaceutica Sinica B*. 2025 Mar 21;15.0(5):2575-2592. doi: 10.1016/j.apsb.2025.03.042. PMID: 40487658.

12. Luo X, Zhu Z, Zheng T, Li D, Wei Y, Yang H, Yu H, Han J, Wang Y, Wang L, Huang Z. Astrocytic YAP prevents the glutamate neurotoxicity by upregulation of EAAT2 expression and promotes the gain of stemness in astrocytes in ischemic stroke mice. *Cell Death and Disease*. 2025 Jul 30;16.0(1):577. doi: 10.1038/s41419-025-07806-7. PMID: 40739083.

13. Akhter N, Lambay A, Almotairi R, Hamadi A, Bhatia K, Ahmad S, Ducruet AF. The Complex Role of the Complement C3a Receptor (C3aR) in Cerebral Injury and Recovery Following Ischemic Stroke. *Cells*. 2025 Sep 15;14.0(18):1440. doi: 10.3390/cells14181440. PMID: 41002405.

14. Qiu M, Zong J, He Q, Liu Y, Wan Y, Li M, Zhou Y, Wu J, Hu B. Cell Heterogeneity Uncovered by Single-Cell RNA Sequencing Offers Potential Therapeutic Targets for Ischemic Stroke. *Aging and Disease*. 2022 Jan 1;13.0(5):1436. doi: 10.14336/ad.2022.0212. PMID: 36186129.

15. Aswendt M, Green C, Sadler R, Llovera G, Dzikowski L, Heindl S, de Agüero MG, Diedenhofen M, Vogel SN, Wieters F, Wiedermann D, Liesz A, Hoehn M. The gut microbiota modulates brain network connectivity under physiological conditions and after acute brain ischemia. *iScience*. 2021 Sep 9;24.0(10):103095. doi: 10.1016/j.isci.2021.103095. PMID: 34622150.

16. Hoehn M, Aswendt M. Functional hyperconnectivity related to brain disease: maladaptive process or element of resilience?. *Neural Regeneration Research*. 2022 Nov 18;18.0(7):1489. doi: 10.4103/1673-5374.361541. PMID: 36571347.

17. Michór P, Renardson L, Li S, Boltze J. Neurorestorative Approaches for Ischemic Stroke. *Neuroscience*. 2024 May 17;550.0:69-78. doi: 10.1016/j.neuroscience.2024.05.009. PMID: 38763225.

18. Pekna M, Siqin S, de Pablo Y, Stokowska A, Naluai ÅT, Pekny M. Astrocyte Responses to Complement Peptide C3a are Highly Context-Dependent. *Neurochemical Research*. 2022 Sep 12;48.0(4):1233-1241. doi: 10.1007/s11064-022-03743-5. PMID: 36097103.

19. Wang S, Tong S, Jin X, Li N, Dang P, Sui Y, Liu Y, Wang D. Single-cell RNA sequencing analysis of the retina under acute high intraocular pressure. *Neural Regeneration Research*. 2023 Nov 8;19.0(11):2522-2531. doi: 10.4103/1673-5374.389363. PMID: 38526288.

20. Dark HE, Shafer AT, Cordon J, An Y, Lewis A, Moghekar A, Landman BA, Resnick SM, Walker KA. Association of Plasma Biomarkers of Alzheimer Disease and Neurodegeneration With Longitudinal Intra-Network Functional Brain Connectivity. *Neurology*. 2025 Jan 31;104.0(4):e210271. doi: 10.1212/wnl.0000000000210271. PMID: 39889254.

21. Boyle BR, Berghella AP, Blanco-Suárez E. Astrocyte Regulation of Neuronal Function and Survival in Stroke Pathophysiology. *Advances in neurobiology*. 2024 Jan 1;39.0:233-267. doi: 10.1007/978-3-031-64839-7_10. PMID: 39190078.

22. He J, Yuan R, Jiang Y, Peng Y, Ye J, Wang S, Li L, Ruan Y, Li P, Yan X, He W, Li G, Chu S, Zhang Z, Chen N. Esculetin facilitates post-stroke rehabilitation by inhibiting CKLF1-mediated neutrophil infiltration. *Acta Pharmacologica Sinica*. 2024 Aug 7;46.0(1):52-65. doi: 10.1038/s41401-024-01352-2. PMID: 39112771.

23. Zhang L, Wang J, Yan Y, Xiang L, Zhai X, Cai L, Sun Z, Pi M, Xiong Q, Zhou H, Gui Y, Wang X, Shu X, Xia Y. Vimentin Fragmentation and Its Role in Amyloid-Beta Plaque Deposition in Alzheimer's Disease. *International Journal of Molecular Sciences*. 2025 Mar 21;26.0(7):2857. doi: 10.3390/ijms26072857. PMID: 40243407.

24. Ramírez-Expósito MJ, Cueto-Ureña C, Martínez-Martos JM. Neurotransmitter Systems in Alzheimer's Disease. *Current Issues in Molecular Biology*. 2026 Mar 22;48.0(3):334. doi: 10.3390/cimb48030334. PMID: 41899485.

25. Verkhratsky A, Lee CJ, Chun H, Göritz C, Harkany T, Lee J, Lee S, Lindskog M, Koh W, Mulder J, Nam M, Ottersen OP, Pekna M, Pekny M, Pękowska A, Ryu H, Sohn CH, Tretiakov EO, Untiet V, Viney TJ, Youn W, Yi C, Zorec R, Yun MR, Cheong E, Nordberg A. Curing the brain: in search for new astrocyte-specific therapies. *Experimental & Molecular Medicine*. 2026 Apr 24;58.0(4):1086-1127. doi: 10.1038/s12276-026-01712-4. PMID: 42032208.

26. Denecke KM, Reed-McBain CA, Hermes BG, Teertam SK, Farooqui MA, Virumbrales-Muñoz M, Panackal JE, Beebe DJ, Famakin BM, Ayuso JM. Microfluidic Model to Evaluate Astrocyte Activation in Penumbral Region following Ischemic Stroke. *Cells*. 2022 Jul 31;11.0(15):2356. doi: 10.3390/cells11152356. PMID: 35954200.

27. Van der Linden A, Hoehn M. Monitoring Neuronal Network Disturbances of Brain Diseases: A Preclinical MRI Approach in the Rodent Brain. *Frontiers in Cellular Neuroscience*. 2022 Jan 3;15.0:815552. doi: 10.3389/fncel.2021.815552. PMID: 35046778.

28. Kálmán M. The relative withdrawal of GFAP-An essential component of brain evolution. *Frontiers in Neuroanatomy*. 2025 Jul 17;19.0:1607603. doi: 10.3389/fnana.2025.1607603. PMID: 40747269.

29. Zhao K, Huang Z. TIMP1 promotes microglia M2 polarization through MAPK pathway to ameliorate early brain injury after ischemia. *Hereditas*. 2025 Jul 2;162.0(1):119. doi: 10.1186/s41065-025-00491-8. PMID: 40604895.

30. Yan K, Newman AG, Lange PNAMJG, Müller S, Foddis M, Koch S, Boehm-Sturm P, Mantwill M, Finke C, Deng P, Long M, Schmitz D, Tarabykin VS. Nurr1 Orchestrates Claustrum Development and Functionality. *Advanced Science*. 2025 Dec 22;13.0(13):e08999. doi: 10.1002/advs.202508999. PMID: 41431109.

31. Götz J, Wieters F, Fritz VJ, Käsgen O, Kalantari A, Fink GR, Aswendt M. Temporal and Spatial Gene Expression Profile of Stroke Recovery Genes in Mice. *Genes*. 2023 Feb 9;14.0(2):454. doi: 10.3390/genes14020454. PMID: 36833381.

32. Verkhratsky A, Butt AM. Astroglial physiology. *Neuroglia*. 2023 Jan 1;:89-197. doi: 10.1016/b978-0-12-821565-4.00009-2.

33. Kalantari A, Hambrock C, Grefkes C, Fink GR, Aswendt M. Proportional recovery in mice with cortical stroke. *Experimental Neurology*. 2025 Feb 4;386.0:115180. doi: 10.1016/j.expneurol.2025.115180. PMID: 39914643.

34. Thumu SCR, Jain M, Soman S, Das S, Verma V, Nandi A, Gutmann DH, Balaji J, Nair DK, Clement JP, Marathe SV, Ramanan N. SRF-deficient astrocytes provide neuroprotection in mouse models of excitotoxicity and neurodegeneration. *eLife*. 2024 Jan 30;13.0. doi: 10.7554/elife.95577. PMID: 38289036.

35. Gomes ALT, Teixeira-Silva B, Thomasi BBDM, Trindade P, Espírito-Santo S, Faria-Melibeu ADC, Serfaty CA, Campello-Costa P. Retinotectal plasticity induced by monocular enucleation during the critical period is dependent of A2a adenosine receptor: A possible role of astrocytes. *Experimental Neurology*. 2023 Apr 26;365.0:114427. doi: 10.1016/j.expneurol.2023.114427. PMID: 37116638.

36. Zou Y, Pei J, Wan C, Liu S, Hu B, Li Z, Tang Z. Mechanism of scutellarin inhibition of astrocyte activation to type A1 after ischemic stroke. *Journal of Stroke and Cerebrovascular Diseases*. 2024 Jan 13;33.0(3):107534. doi: 10.1016/j.jstrokecerebrovasdis.2023.107534. PMID: 38219378.

37. Zhu Z, Ma K, Yuan T, Li B, Liu L, Bai Y. Spinal cord injury repair: Mechanisms, applications, and prospects of hydrogel therapies. *APL Materials*. 2025 Jun 1;13.0(6). doi: 10.1063/5.0275183.

38. Mahani F, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic Functional Connectivity Changes in Cortical and Cortico-Striatal Strokes in Mice. *CNS Neuroscience & Therapeutics*. 2026 Aug 1;32.0(8):e71114. doi: 10.1002/cns.71114. PMID: 42661330.

39. Sitovskaya DA, Eremina AO, Соколова ТВ, Zabrodskaya YM. Changes in Immunoreactivity of Vimentin and GFAP in the Hippocampus of Patients with Drug-Resistant Epilepsy. *SciBase neurology*. 2024 Mar 28;2.0(1). doi: 10.52768/neurology/1012.

40. Ridge KM, Eriksson J, Pekny M, Goldman RD. Roles of vimentin in health and disease. *Genes & Development*. 2022 Apr 1;36.0(7-8):391-407. doi: 10.1101/gad.349358.122. PMID: 35487686.

41. Michór P, Renardson L, Boltze J. Neurorestorative Approaches for Ischemic Stroke - Challenges, Opportunities, and Recent Advances. *SSRN Electronic Journal*. 2023 Jan 1;. doi: 10.2139/ssrn.4635803.

42. Beales CT, Eibauer M, de Pablo Y, Hansen SB, Lin N, Perng M, Pekny M, Medalia O. Interdigitating coiled-coil tetramers define the helical architecture of GFAP filaments in astrocytes. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Feb 24;. doi: 10.64898/2026.02.23.707361.

43. Fu Z, Hu Y, Wang Y, Liu Z, Li M, Guo Y, Hu Z, Lai X, Hu J, Liao Y, Cai C. Human induced neural progenitor cells generated from three-dimensional aggregate-based culture significantly improve post-stroke recovery in tMCAO mice. *Stem Cell Research & Therapy*. 2025 Jun 20;16.0(1):312. doi: 10.1186/s13287-025-04433-z. PMID: 40542444.

44. Dykstra W, Matusova Z, Battaglia RA, Abaffy P, Goya-Iglesias N, Pérez-Sala D, Ahlenius H, Kubista M, Pasterkamp RJ, Li L, Chao J, Shi Y, Valihrach L, Pekny M, Hol EM. Mutations in GFAP Alter Early Lineage Commitment of Organoids. *Glia*. 2025 Jul 30;73.0(11):2167-2188. doi: 10.1002/glia.70049. PMID: 40735838.

45. Liu X, Guo Y, Popa-Wagner A, Lämmermann I, Grillari J, Hermann DM. Cellular senescence in ischemic stroke: Cell-type specificity, temporal dynamics, and response to therapeutic interventions. *Ageing Research Reviews*. 2026 Jun 14;120.0:103206. doi: 10.1016/j.arr.2026.103206. PMID: 42289192.

46. Borsa M, Lenzi I, Obrist C, Rusterholz T, Bassetti CL, Adamantidis A, Herrera CG. Optical Ministroke Reveals Dual-Role Frontal Thalamocortical Networks in Sleep Quality and Memory. *Stroke*. 2026 Jul 15;57.0(9):2884-2900. doi: 10.1161/strokeaha.126.056284. PMID: 42454410.

47. Wang BH, Wang Y, Cao W, Wang P, Yang X, Wang S, Li C, Yin Y. Multi-omics integration identifies VIM and LGALS3 as apoptosis-associated genes in astrocytes after traumatic brain injury. *Neuroscience*. 2026 Aug 19;614.0:356-372. doi: 10.1016/j.neuroscience.2026.08.027. PMID: 42617790.

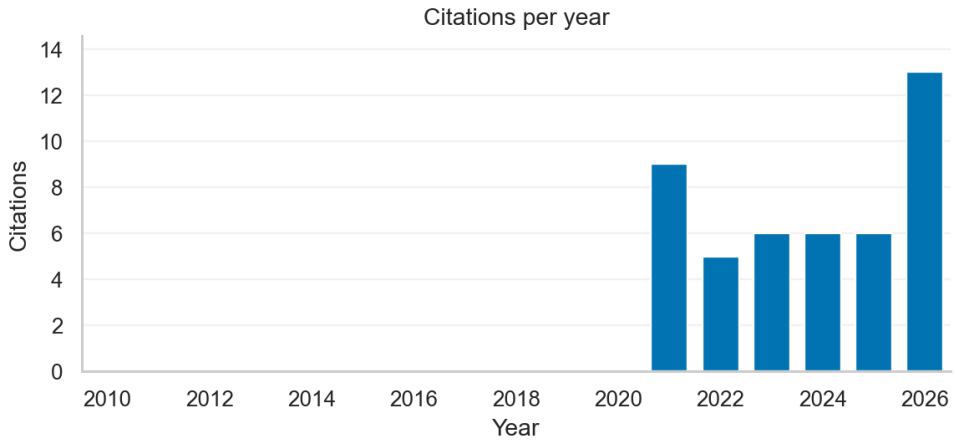
48. Jin C, Fan L, Ye Q, Pu H, Zhang W, He G, Lyu J, Wu W, Hu X, Leak RK, Yin K, Chen K, Shi Y, Chen J. A Single-cell Spatiotemporal Atlas Encoding Age-dependent Evolution of the Ischemic Penumbra. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Sep 10;. doi: 10.64898/2026.09.08.750251.

49. Shu H, Luo Q, Hu P, Xu Y, Zhu W, Yang L, Lei K, Luo M, Zhang S, Zhu X, Zhu X, Liao Q, Chen Z, Zeng Q, Zhu X, Zhu X, Zhu Z, Yan T. TRIM28 SUMOylates IRF3 in astrocytes to drive neutrophils infiltration and ischemic cerebral injury. *Cell Death and Differentiation*. 2026 Sep 15;. doi: 10.1038/s41418-026-01871-x. PMID: 42745034.

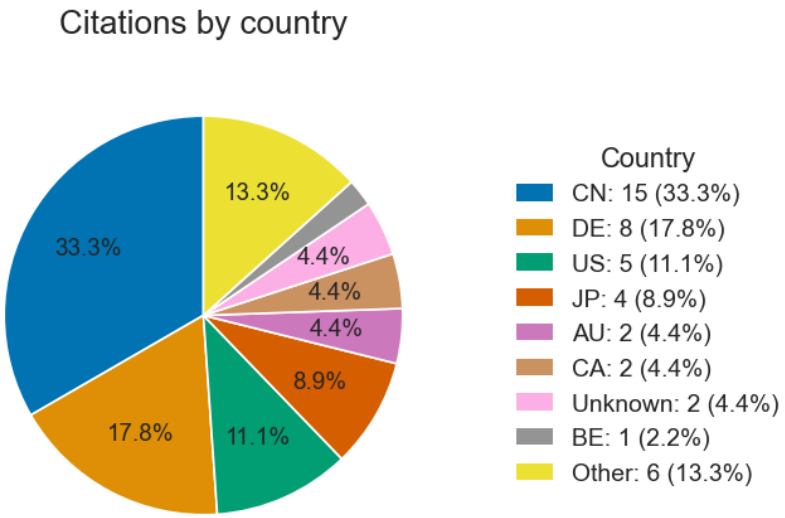
50. Poudel A, Andrews M, Srinageshwar B, MacDonald B, Malkowski C, Peruzzaro S, Resk M, Stadler J, Toth AK, Raymor G, Schurr H, Hellrung J, Kathirvelu B, Swanson DR, Sharma A, Dunbar GL, Rossignol J. Intracranial administration of surface modified PAMAM dendrimer migrates to the infarct area in the MCAo rat model of stroke. *Next Nanotechnology*. 2026 Sep 29;10.0:100817. doi: 10.1016/j.nxnano.2026.100817.

Publication

Aswendt M, Pallast N, Wieters F, Baues M, Hoehn M, Fink GR. Lesion Size- and Location-Dependent Recruitment of Contralesional Thalamus and Motor Cortex Facilitates Recovery after Stroke in Mice. *Translational Stroke Research*. 2020 Mar 12;12(1):87-97. doi: 10.1007/s12975-020-00802-3. PMID: 32166716.



Citation summary
Total citations: **45**
Average/year: **6.43**
Period: 2020-2026
SCImago cites/doc 2y (2025): **4.62**
SJR 2025: **1.39** (Q1)



Cited by

1. Stokowska A, Aswendt M, Žucha D, Lohmann S, Wieters F, Suarez JM, Atkins AL, Li Y, Miteva M, Lewin J, Wiedermann D, Diedenhofen M, Naluai ÅT, Abaffy P, Valihrach L, Kubista M, Hoehn M, Pekny M, Pekna M. Complement C3a treatment accelerates recovery after stroke via modulation of astrocyte reactivity and cortical connectivity. *Journal of Clinical Investigation*. 2023 Mar 30;133.0(10.0). doi: 10.1172/jci162253. PMID: 36995772.

2. Aswendt M, Wilhelmsson U, Wieters F, Stokowska A, Schmitt FJ, Pallast N, de Pablo Y, Mohammed L, Hoehn M, Pekna M, Pekny M. Reactive astrocytes prevent maladaptive plasticity after ischemic stroke. *Progress in Neurobiology*. 2021 Dec 15;209.0:102199. doi: 10.1016/j.pneurobio.2021.102199. PMID: 34921928.

3. Cai G, Xu J, Zhang C, Jiang J, Chen G, Chen J, Liu Q, Xu G, Lan Y. Identifying biomarkers related to motor function in chronic stroke: A fNIRS and TMS study. *CNS Neuroscience & Therapeutics*. 2024 Jul 1;30.0(7.0):e14889. doi: 10.1111/cns.14889. PMID: 39073240.

4. Sanchez-Bezanilla S, Hood RJ, Collins-Praino LE, Turner RJ, Walker FR, Nilsson M, Ong LK. More than motor impairment: A spatiotemporal analysis of cognitive impairment and associated neuropathological changes following cortical photothrombotic stroke. *Journal of Cerebral Blood Flow & Metabolism*. 2021 Mar 28;41.0(9.0):2439-2455. doi: 10.1177/0271678x211005877. PMID: 33779358.

5. Aswendt M, Green C, Sadler R, Llovera G, Dzikowski L, Heindl S, de Agüero MG, Diedenhofen M, Vogel SN, Wieters F, Wiedermann D, Liesz A, Hoehn M. The gut microbiota modulates brain network connectivity under physiological conditions and after acute brain ischemia. *iScience*. 2021 Sep 9;24.0(10.0):103095. doi: 10.1016/j.isci.2021.103095. PMID: 34622150.

6. Yu-qin M, Yu Y, Gao W, Hong Y, Shen X. Cerebral Hemodynamic Changes during Unaffected Handgrip Exercises in Stroke Patients: An fNIRS Study. *Brain Sciences*. 2023 Jan 13;13.0(1.0):141. doi: 10.3390/brainsci13010141. PMID: 36672122.

7. Weber R, Bernardoni D, Rentsch NH, Buil BA, Halliday S, Augath M, Razansky D, Tackenberg C, Rust R. A toolkit for stroke infarct volume estimation in rodents. *NeuroImage*. 2024 Jan 21;287.0:120518. doi: 10.1016/j.neuroimage.2024.120518. PMID: 38219841.

8. Li Z, Gao H, Zeng P, Jia Y, Kong X, Xu K, Bai R. Secondary Degeneration of White Matter After Focal Sensorimotor Cortical Ischemic Stroke in Rats. *Frontiers in Neuroscience*. 2021 Jan 18;14.0:611696. doi: 10.3389/fnins.2020.611696. PMID: 33536869.

9. Sato T, Nakamura Y, Takeda A, Ueno M. Lesion Area in the Cerebral Cortex Determines the Patterns of Axon Rewiring of Motor and Sensory Corticospinal Tracts After Stroke. *Frontiers in Neuroscience*. 2021 Oct 11;15.0:737034. doi: 10.3389/fnins.2021.737034. PMID: 34707476.

10. Fang S, Li L, Weng S, Guo Y, Zhong Z, Fan X, Jiang T, Wang Y. Contralesional Sensorimotor Network Participates in Motor Functional Compensation in Glioma Patients. *Frontiers in Oncology*. 2022 Apr 22;12.0:882313. doi: 10.3389/fonc.2022.882313. PMID: 35530325.

11. Pan W, Shie M, Yan W, Chang Y, Iao HM, Liu H, Tran N, Lee I, Lien H, Hsu R, Chu L, Hu S. Gut-to-brain nanoantenna-mediated in situ wireless charging activation across the brain barrier and neural regeneration. *Advanced Composites and Hybrid Materials*. 2026 Jan 21;9.0(1.0). doi: 10.1007/s42114-026-01617-w.

12. Zhang H, Zhao JY, Fan L, Gao C, Li F, Liu J, Bai C, Li X, Li B, Zhang T. Somatosensory-Thalamic Functional Dysconnectivity Associated With Poststroke Motor Function Rehabilitation: A Resting-State fMRI Study. *Brain and Behavior*. 2025 Feb 1;15.0(2.0):e70321. doi: 10.1002/brb3.70321. PMID: 39935146.

13. Tóth R, Szabó N, Törteli A, Kovács N, Horváth I, Szigeti K, Máthé D, Kincses ZT, Menyhárt Á, Farkas E. The paradoxical relationship of sensorimotor deficit and lesion volume in acute ischemic stroke. *Journal of Neuropathology & Experimental Neurology*. 2025 Apr 24;84.0(9.0):771-779. doi: 10.1093/jnen/nlaf046. PMID: 40272944.

14. Scharwächter L, Schmitt FJ, Pallast N, Fink GR, Aswendt M. Network analysis of neuroimaging in mice. *NeuroImage*. 2022 Mar 17;253.0:119110. doi: 10.1016/j.neuroimage.2022.119110. PMID: 35311664.

15. Rosenthal ZP, Raut RV, Bowen RM, Snyder AZ, Culver JP, Raichle ME, Lee J. Peripheral sensory stimulation elicits global slow waves by recruiting somatosensory cortex bilaterally. *Proceedings of the National Academy of Sciences*. 2021 Feb 17;118.0(8.0). doi: 10.1073/pnas.2021252118. PMID: 33597303.

16. Kaneko H, Namihira M, Yamamoto S, Numata N, Hyodo K. Oral administration of cyclic glycyl-proline facilitates task learning in a rat stroke model. *Behavioural Brain Research*. 2021 Sep 10;417.0:113561. doi: 10.1016/j.bbr.2021.113561. PMID: 34509530.

17. Van der Linden A, Hoehn M. Monitoring Neuronal Network Disturbances of Brain Diseases: A Preclinical MRI Approach in the Rodent Brain. *Frontiers in Cellular Neuroscience*. 2022 Jan 3;15.0:815552. doi: 10.3389/fncel.2021.815552. PMID: 35046778.

18. Kerr AL. Contralesional plasticity following constraint-induced movement therapy benefits outcome: contributions of the intact hemisphere to functional recovery. *Reviews in the Neurosciences*. 2021 Nov 10;33.0(3.0):269-283. doi: 10.1515/revneuro-2021-0085. PMID: 34761646.

19. Ostrova IV, Babkina AS, Lyubomudrov MA, Grechko AV, Golubev AM. Photochemicallly Induced Thrombosis as a Model of Ischemic Stroke. *General Reanimatology*. 2023 Jun 24;19.0(3.0):54-65. doi: 10.15360/1813-9779-2023-3-54-65.

20. Gao H, Sun M, Li M, Wang C, Yu C, Wang Y, Xu K. Force Decoding of Caudal Forelimb Area and Rostral Forelimb Area in Chronic Stroke Rats. *IEEE Transactions on Biomedical Engineering*. 2021 Mar 4;68.0(10.0):3078-3086. doi: 10.1109/tbme.2021.3063903. PMID: 33661731.

21. Götz J, Wieters F, Fritz VJ, Käsgen O, Kalantari A, Fink GR, Aswendt M. Temporal and Spatial Gene Expression Profile of Stroke Recovery Genes in Mice. *Genes*. 2023 Feb 9;14.0(2.0):454. doi: 10.3390/genes14020454. PMID: 36833381.

22. Goto T, Tsurugizawa T, Komaki Y, Takashima I, Iwaki S, Kunori N. Clemastine enhances exercise-induced motor improvement in hypoxic ischemic rats. *Brain Research*. 2024 Oct 1;1846.0:149257. doi: 10.1016/j.brainres.2024.149257. PMID: 39362477.

23. Jones TA, Nemchek V, Fracassi M. Experience-driven competition in neural reorganization after stroke. *The Journal of Physiology*. 2024 Oct 30;603.0(3.0):737-757. doi: 10.1113/jp285565. PMID: 39476290.

24. Kalantari A, Hambrock C, Grefkes C, Fink GR, Aswendt M. Proportional recovery in mice with cortical stroke. *Experimental Neurology*. 2025 Feb 4;386.0:115180. doi: 10.1016/j.expneurol.2025.115180. PMID: 39914643.

25. Stuckey SM, Hood RJ, Ong LK, Bilecki IM, Collins-Praino LE, Turner RJ. Unravelling the nexus of stroke and dementia: Deciphering the role of secondary neurodegeneration in orchestrating cognitive decline. *Neuroprotection/Neuroprotection (Chichester, England. Print)*. 2025 Feb 25;3.0(1.0):5-28. doi: 10.1002/nep3.74. PMID: 41383522.

26. Mahani FSN, Kalantari A, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic structural connectivity changes in cortical and cortico-striatal strokes in mice. *Neural Regeneration Research*. 2026 Feb 28;21.0(10.0):4945-4953. doi: 10.4103/nrr.nrr-d-25-00491. PMID: 41778731.

27. Hara S, Hori M, Kamagata K, Andica C, Inaji M, Tanaka Y, Aoki S, Nariai T, Maehara T. Increased Parenchymal Free Water May Be Decreased by Revascularization Surgery in Patients with Moyamoya Disease. *Magnetic Resonance in Medical Sciences*. 2023 Apr 17;23.0(4.0):405-416. doi: 10.2463/mrms.mp.2022-0146. PMID: 37081646.

28. Wang J, Liu Z, Ren H, Chen X, Wang J, Cai H, Wei L, Tian H, Wang J, Wang L. A novel aged mouse model of recurrent intracerebral hemorrhage in the bilateral striatum. *Neural Regeneration Research*. 2022 Jun 2;18.0(2.0):344-349. doi: 10.4103/1673-5374.346459. PMID: 35900428.

29. Mahani FSN, Kalantari A, Fink GR, Hoehn M, Aswendt M. A systematic review of the relationship between magnetic resonance imaging based resting-state and structural networks in the rodent brain. *Frontiers in Neuroscience*. 2023 Jul 24;17.0:1194630. doi: 10.3389/fnins.2023.1194630. PMID: 37554291.

30. Niu B, Wu H, Li Y, Klugah-Brown B, Hanna GP, Yao Y, Jing J, Baig TI, Xia Y, Yao D, Biswal BB. Topological functional network analysis of cortical blood flow in hyperacute ischemic rats. *Brain Structure and Function*. 2024 Dec 26;230.0(1.0):20. doi: 10.1007/s00429-024-02864-7. PMID: 39724244.

31. Guo J, Zhang H, Liu J, Wang C, Cao CY, Cheng J, Yu C, Qin W. Thalamic functional dysconnectivity in patients with left-hemisphere chronic capsular and pontine stroke. *Frontiers in Neuroscience*. 2024 Oct 15;18.0:1451307. doi: 10.3389/fnins.2024.1451307. PMID: 39474460.

32. Pini L, Salvalaggio A, Corbetta M. Considering biological limitations of lesion network mapping. *Nature Neuroscience*. 2026 May 20;29.0(9.0):2051-2053. doi: 10.1038/s41593-026-02319-8. PMID: 42162275.

33. Mahani F, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic Functional Connectivity Changes in Cortical and Cortico-Striatal Strokes in Mice. *CNS Neuroscience & Therapeutics*. 2026 Aug 1;32.0(8.0):e71114. doi: 10.1002/cns.71114. PMID: 42661330.

34. Wan H, Fu Y, Zhang S, Zhang G, Li Y, Zhu W, Wang Y. Association of Iron and Myelin Alterations in the Contralesional Dentate Nucleus and Thalamus With Functional Outcome in Acute Ischemic Stroke: A Susceptibility Source Separation Study. *Journal of Magnetic Resonance Imaging*. 2026 Jul 2;64.0(5.0):1282-1292. doi: 10.1002/jmri.70431. PMID: 42393438.

35. Jin C, Dai H, Dai D, Zhang J. A Freely Moving Photothrombotic Stroke Model Reveals Sustained Dysfunction of GABAergic Neuron in Contralesional Cortex Using Miniaturized Two-Photon Microscopy. *Translational Stroke Research*. 2025 Mar 10;16.0(5.0):1631-1643. doi: 10.1007/s12975-025-01336-2. PMID: 40063293.

36. Chary P, Rehmani S, Davidson S, Li X, Chen SD, Silasi G. Hemodynamic and electrophysiological progression of the Rose Bengal photothrombotic stroke model in mice: vasoactive properties of Rose Bengal, tissue heating, wavelength optimization, and sex differences in lesion volume. *bioRxiv (Cold Spring Harbor Laboratory)*. 2025 Sep 24;. doi: 10.1101/2025.09.22.677889.

37. Betz D, Alers VA, Kenwood MR, Zuurbier KR, Coimbra R, Rhoton P, Plautz EJ, Douglas PM, Ramirez DMO, Stowe AM, Goldberg MP. Shared Transcriptomic Signatures in Perilesional and Contralesional Cortex. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Feb 4;. doi: 10.64898/2026.02.02.703306. PMID: 41676687.

38. Chary P, Rehmani S, Davidson S, Li X, Chen SX, Silasi G. Hemodynamic and Electrophysiological Progression of the Rose Bengal Photothrombotic Stroke Model in Mice. *Journal of Neuroscience*. 2026 Feb 11;46.0(12.0):e1818252026. doi: 10.1523/jneurosci.1818-25.2026. PMID: 41672876.

39. Feng J, Li P, Fang W, Li Q, Meng Y, Ren H, Chen W, Guan C, Qiao Z, Shen Y, Wang L, Xu W. Peripheral nerve pathway reconnections promote the reorganization of central motor representations. *Science China Life Sciences*. 2026 Mar 9;69.0(7.0):2376-2389. doi: 10.1007/s11427-025-3205-6. PMID: 41840169.

40. Alasoadura M. Caractérisation de l'activité oscillatoire spontanée du cortex peri-lesionnel lors de la récupération fonctionnelle post-ischémique. 2022 Jul 7;. doi: 10.70675/b9d56b1ez4385z49dfz9234za3e7c8638338.

41. Betz D, Alers VA, Kenwood MR, Zuurbier KR, Coimbra R, Rhoton P, Plautz EJ, Douglas PM, Ramirez DMO, Stowe AM, Goldberg MP. Shared transcriptomic signatures in perilesional and contralesional cortex after ischemic stroke. *Journal of Neuroinflammation*. 2026 Jun 15;23.0(1.0). doi: 10.1186/s12974-026-03885-1. PMID: 42298604.

42. Balakrishnan R, Alam TRG, Mckeown B, Souter N, Karapanagiotidis T, Souter NE, Krieger-Redwood K, Jefferies E. Lesion-Gradient Mapping in Semantic Aphasia: Comparisons of Observed and Simulated Effects of Stroke on Connectivity Gradients. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Jun 13;. doi: 10.64898/2026.06.12.731654.

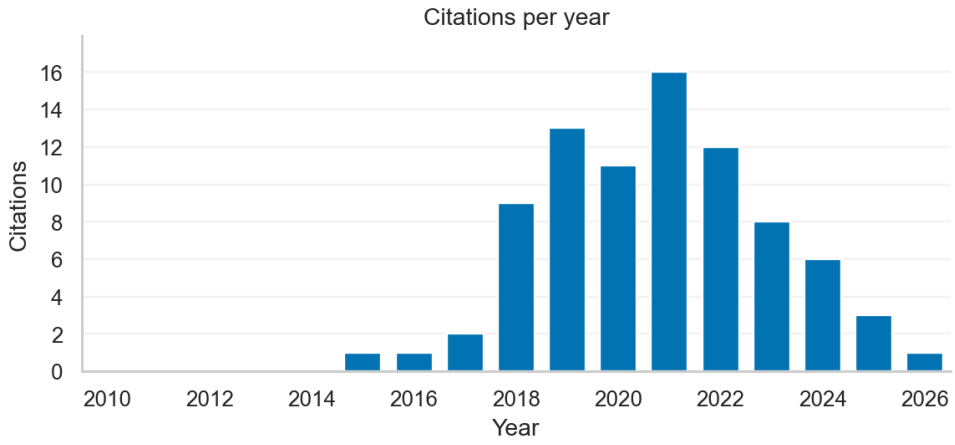
43. □□□, Wu X. From fragmented neurotechnologies to closed-loop brain health systems: integrating artificial intelligence, digital health, digital twins, and advanced materials for brain disorders. *Frontiers in Bioengineering and Biotechnology*. 2026 Aug 7;14.0:1904561. doi: 10.3389/fbioe.2026.1904561. PMID: 42630395.

44. Zhao X, Yan Y, Wang S, Zhang Y, Wang X, Chen L, Yuan M, Ju Y, Sun S. Longitudinal automated fiber quantification reveals corticospinal tract dynamics and prognostic significance after primary basal ganglia hemorrhage. *Neuroradiology*. 2026 Aug 12;. doi: 10.1007/s00234-026-04131-4. PMID: 42584657.

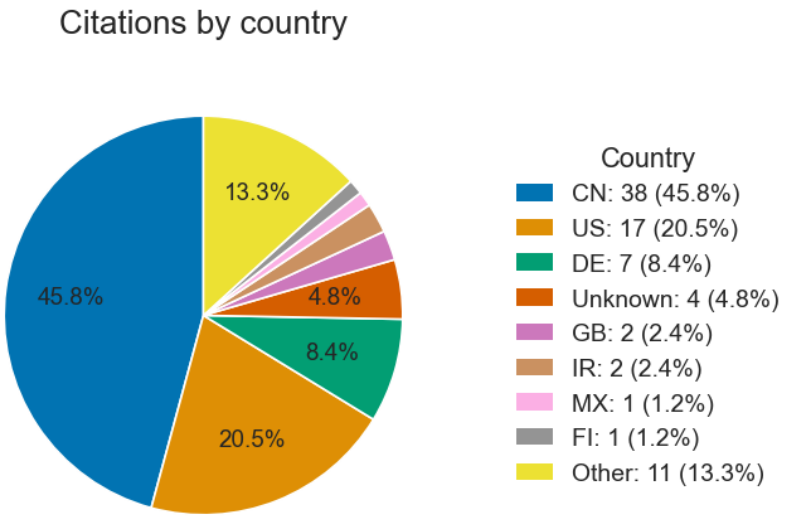
45. Shu H, Luo Q, Hu P, Xu Y, Zhu W, Yang L, Lei K, Luo M, Zhang S, Zhu X, Zhu X, Liao Q, Chen Z, Zeng Q, Zhu X, Zhu X, Zhu Z, Yan T. TRIM28 SUMOylates IRF3 in astrocytes to drive neutrophils infiltration and ischemic cerebral injury. *Cell Death and Differentiation*. 2026 Sep 15;. doi: 10.1038/s41418-026-01871-x. PMID: 42745034.

Publication

Taj SH, Kho W, Aswendt M, Collmann FM, Green C, Adamczak JM, Tennstaedt A, Hoehn M. Dynamic Modulation of Microglia/Macrophage Polarization by miR-124 after Focal Cerebral Ischemia. *Journal of Neuroimmune Pharmacology*. 2016 Aug 18;11(4):733-748. doi: 10.1007/s11481-016-9700-y. PMID: 27539642.



Citation summary
Total citations: **83**
Average/year: **7.55**
Period: 2016-2026
SCImago cites/doc 2y (2025): **6.07**
SJR 2025: **1.46** (Q1)



Cited by

1. Jurcău A, Simion A. Neuroinflammation in Cerebral Ischemia and Ischemia/Reperfusion Injuries: From Pathophysiology to Therapeutic Strategies. *International Journal of Molecular Sciences*. 2021 Dec 21;23.0(1.0):14. doi: 10.3390/ijms23010014. PMID: 35008440.

2. Jiang C, Wu W, Deng Y, Ge J. Modulators of microglia activation and polarization in ischemic stroke (Review). *Molecular Medicine Reports*. 2020 Feb 26;21.0(5.0):2006-2018. doi: 10.3892/mmr.2020.11003. PMID: 32323760.

3. Ye Y, Jin T, Zhang XD, Zeng ZR, Ye B, Wang J, Zhong Y, Xiong X, Gu L. Meisoindigo Protects Against Focal Cerebral Ischemia-Reperfusion Injury by Inhibiting NLRP3 Inflammasome Activation and Regulating Microglia/Macrophage Polarization via TLR4/NF-κB Signaling Pathway. *Frontiers in Cellular Neuroscience*. 2019 Dec 16;13.0:553. doi: 10.3389/fncel.2019.00553. PMID: 31920554.

4. Slota JA, Booth SA. MicroRNAs in Neuroinflammation: Implications in Disease Pathogenesis, Biomarker Discovery and Therapeutic Applications. *Non-Coding RNA*. 2019 Apr 24;5.0(2.0):35. doi: 10.3390/ncrna5020035. PMID: 31022830.

5. Amici SA, Dong J, Guerau-de-Arellano M. Molecular Mechanisms Modulating the Phenotype of Macrophages and Microglia. *Frontiers in Immunology*. 2017 Nov 10;8.0:1520. doi: 10.3389/fimmu.2017.01520. PMID: 29176977.

6. Lyu J, Di Xie, Bhatia TN, Leak RK, Hu X, Jiang X. Microglial/Macrophage polarization and function in brain injury and repair after stroke. *CNS Neuroscience & Therapeutics*. 2021 Mar 1;27.0(5.0):515-527. doi: 10.1111/cns.13620. PMID: 33650313.

7. Wang Y, Leak RK, Cao G. Microglia-mediated neuroinflammation and neuroplasticity after stroke. *Frontiers in Cellular Neuroscience*. 2022 Aug 16;16.0:980722. doi: 10.3389/fncel.2022.980722. PMID: 36052339.

8. Cheray M, Joseph B. Epigenetics Control Microglia Plasticity. *Frontiers in Cellular Neuroscience*. 2018 Aug 3;12.0:243. doi: 10.3389/fncel.2018.00243. PMID: 30123114.

9. Guo Y, Hong W, Wang X, Zhang P, Körner H, Tu J, Wei W. MicroRNAs in Microglia: How do MicroRNAs Affect Activation, Inflammation, Polarization of Microglia and Mediate the Interaction Between Microglia and Glioma?. *Frontiers in Molecular Neuroscience*. 2019 May 10;12.0:125. doi: 10.3389/fnmol.2019.00125. PMID: 31133802.

10. Yu A, Zhang T, Duan H, Pan Y, Zhang X, Yang G, Wang J, Deng YS, Yang Z. MiR-124 contributes to M2 polarization of microglia and confers brain inflammatory protection via the C/EBP-α pathway in intracerebral hemorrhage. *Immunology Letters*. 2016 Dec 23;182.0:1-11. doi: 10.1016/j.imlet.2016.12.003. PMID: 28025043.

11. Liu X, Feng Z, Du L, Huang Y, Ge J, Deng Y, Mei Z. The Potential Role of MicroRNA-124 in Cerebral Ischemia Injury. *International Journal of Molecular Sciences*. 2019 Dec 23;21.0(1.0):120. doi: 10.3390/ijms21010120. PMID: 31878035.

12. Yang D, Yang L, Cai J, Hu X, Li H, Zhang X, Zhang X, Chen X, Dong H, Nie H, Li Y. A sweet spot for macrophages: Focusing on polarization. *Pharmacological Research*. 2021 Mar 24;167.0:105576. doi: 10.1016/j.phrs.2021.105576. PMID: 33771700.

13. Chen Y, Sun J, Chen W, Wu G, Wang Y, Zhu K, Wang J. miR-124/VAMP3 is a novel therapeutic target for mitigation of surgical trauma-induced microglial activation. *Signal Transduction and Targeted Therapy*. 2019 Aug 15;4.0(1.0):27. doi: 10.1038/s41392-019-0061-x. PMID: 31637007.

14. Lian L, Zhang Y, Liu L, Yang L, Cai Y, Zhang J, Xu S. Neuroinflammation in Ischemic Stroke: Focus on MicroRNA-mediated Polarization of Microglia. *Frontiers in Molecular Neuroscience*. 2021 Jan 7;13.0:612439. doi: 10.3389/fnmol.2020.612439. PMID: 33488360.

15. Han D, Liu H, Gao Y. The role of peripheral monocytes and macrophages in ischemic stroke. *Neurological Sciences*. 2020 Oct 3;41.0(12.0):3589-3607. doi: 10.1007/s10072-020-04777-9. PMID: 33009963.

16. Taj SH, Le Blon D, Hoornaert C, Daans J, Quarta A, Praet J, Van der Linden A, Ponsaerts P, Hoehn M. Targeted intracerebral delivery of the anti-inflammatory cytokine IL13 promotes alternative activation of both microglia and macrophages after stroke. *Journal of Neuroinflammation*. 2018 Jun 4;15.0(1.0):174. doi: 10.1186/s12974-018-1212-7. PMID: 29866203.

17. Yang M, Wang K, Liu B, Shen Y, Liu G. Hypoxic-Ischemic Encephalopathy: Pathogenesis and Promising Therapies. *Molecular Neurobiology*. 2024 Jul 29;62.0(2.0):2105-2122. doi: 10.1007/s12035-024-04398-9. PMID: 39073530.

18. Martínez-Hernández MI, Acosta-Saavedra LC, Hernández-Kelly LC, Loaeza-Loaeza J, Ortega A. Microglial Activation in Metal Neurotoxicity: Impact in Neurodegenerative Diseases. *BioMed Research International*. 2023 Jan 1;2023.0(1.0):7389508. doi: 10.1155/2023/7389508. PMID: 36760476.

19. Yu Q, Chopp M, Chen J. Emerging role of microRNAs in ischemic stroke with comorbidities. *Experimental Neurology*. 2020 Jun 16;331.0:113382. doi: 10.1016/j.expneurol.2020.113382. PMID: 32561412.

20. Khayati S, Dehnavi S, Sadeghi M, Afshari JT, Esmaeili S, Mohammadi M. The potential role of miRNA in regulating macrophage polarization. *Heliyon*. 2023 Oct 31;9.0(11.0):e21615. doi: 10.1016/j.heliyon.2023.e21615. PMID: 38027572.

21. Xu J, Zheng Y, Wang L, Liu Y, Wang X, Li Y, Chi G. miR-124: A Promising Therapeutic Target for Central Nervous System Injuries and Diseases. *Cellular and Molecular Neurobiology*. 2021 Apr 22;42.0(7.0):2031-2053. doi: 10.1007/s10571-021-01091-6. PMID: 33886036.

22. Qiu M, Xu E, Zhan L. Epigenetic Regulations of Microglia/Macrophage Polarization in Ischemic Stroke. *Frontiers in Molecular Neuroscience*. 2021 Oct 11;14.0:697416. doi: 10.3389/fnmol.2021.697416. PMID: 34707480.

23. Quan H, Zhang R. Microglia dynamic response and phenotype heterogeneity in neural regeneration following hypoxic-ischemic brain injury. *Frontiers in Immunology*. 2023 Nov 29;14.0:1320271. doi: 10.3389/fimmu.2023.1320271. PMID: 38094292.

24. Dordoe C, Huang W, Bwalya C, Wang X, Shen B, Wang H, Wang J, Ye S, Wang P, Xiaoyan B, Li X, Lin L. The role of microglial activation on ischemic stroke: Modulation by fibroblast growth factors. *Cytokine & Growth Factor Reviews*. 2023 Jul 31;74.0:122-133. doi: 10.1016/j.cytogfr.2023.07.005. PMID: 37573252.

25. Ziegler JN, Tian C. Engineered Extracellular Vesicles: Emerging Therapeutic Strategies for Translational Applications. *International Journal of Molecular Sciences*. 2023 Oct 15;24.0(20.0):15206. doi: 10.3390/ijms242015206. PMID: 37894887.

26. Singh S, Chen CC, Kim S, Singh A, Singh G. Role of Extracellular vesicle microRNAs and RNA binding proteins on glioblastoma dynamics and therapeutics development. *Extracellular Vesicle*. 2024 Sep 10;4.0:100049. doi: 10.1016/j.vesic.2024.100049.

27. Dong X, Luo S, Hu D, Cao R, Wang Q, Meng Z, Feng Z, Zhou W, Song W. Gallic acid inhibits neuroinflammation and reduces neonatal hypoxic-ischemic brain damages. *Frontiers in Pediatrics*. 2022 Dec 22;10.0:973256. doi: 10.3389/fped.2022.973256. PMID: 36619526.

28. Zeng H, Liu X, Liu P, Jia S, Wei G, Chen G, Zhao L. Exercise's protective role in chronic obstructive pulmonary disease via modulation of M1 macrophage phenotype through the miR-124-3p/ERN1 axis. *Science Progress*. 2025 Jul 1;108.0(3.0):368504251360892. doi: 10.1177/00368504251360892. PMID: 40671533.

29. Fang M, Zhong L, Jin X, Cui R, Yang W, Gao S, Lv J, Li B, Liu T. Effect of Inflammation on the Process of Stroke Rehabilitation and Poststroke Depression. *Frontiers in Psychiatry*. 2019 Apr 11;10.0:184. doi: 10.3389/fpsyt.2019.00184. PMID: 31031649.

30. Gandy KAO, Zhang J, Nagarkatti PS, Nagarkatti M. Resveratrol (3, 5, 4′-Trihydroxy-trans-Stilbene) Attenuates a Mouse Model of Multiple Sclerosis by Altering the miR-124/Sphingosine Kinase 1 Axis in Encephalitogenic T Cells in the Brain. *Journal of Neuroimmune Pharmacology*. 2019 Apr 2;14.0(3.0):462-477. doi: 10.1007/s11481-019-09842-5. PMID:

30941623.

31. Lyu J, Jiang X, Leak RK, Shi Y, Hu X, Chen J. Microglial Responses to Brain Injury and Disease: Functional Diversity and New Opportunities. *Translational Stroke Research*. 2020 Oct 31;12.0(3.0):474-495. doi: 10.1007/s12975-020-00857-2. PMID: 33128703.

32. Shen PF, Chen Y, Luo S, Fan Z, Wang J, Chang J, Deng J. Applications of Biomaterials for Immunosuppression in Tissue Repair and Regeneration. *Acta Biomaterialia*. 2020 Jan 1;126.0:31-44. doi: 10.1016/j.actbio.2021.03.019. PMID: 33722787.

33. Collmann FM, Pijnenburg R, Taj SH, Minassian A, Folz-Donahue K, Kukat C, Aswendt M, Hoehn M. Individual in vivo Profiles of Microglia Polarization After Stroke, Represented by the Genes iNOS and Ym1. *Frontiers in Immunology*. 2019 Jun 4;10.0:1236. doi: 10.3389/fimmu.2019.01236. PMID: 31214190.

34. Tsuyama J, Nakamura A, Ooboshi H, Yoshimura A, Shichita T. Pivotal role of innate myeloid cells in cerebral post-ischemic sterile inflammation. *Seminars in Immunopathology*. 2018 Sep 11;40.0(6.0):523-538. doi: 10.1007/s00281-018-0707-8. PMID: 30206661.

35. Zheng Q, Zhao Y, Su Y, Cao B, Yang J, Xing Q. Serum Extracellular Vesicle-Derived miR-124-3p as a Diagnostic and Predictive Marker for Early-Stage Acute Ischemic Stroke. *Frontiers in Molecular Biosciences*. 2021 Jul 1;8.0:685088. doi: 10.3389/fmolb.2021.685088. PMID: 34277703.

36. Saraiva C, Talhada D, Rai A, Ferreira R, Ferreira L, Bernardino LI, Ruscher K. MicroRNA-124-loaded nanoparticles increase survival and neuronal differentiation of neural stem cells in vitro but do not contribute to stroke outcome in vivo. *PLoS ONE*. 2018 Mar 1;13.0(3.0):e0193609. doi: 10.1371/journal.pone.0193609. PMID: 29494665.

37. Zhao YS, Yang J, Li C, Zhou G, Wan H, Wan H, Ding Z, Wan H, Wan H, Zhou H. Role of the neurovascular unit in the process of cerebral ischemic injury. *Pharmacological Research*. 2020 Jul 30;160.0:105103. doi: 10.1016/j.phrs.2020.105103. PMID: 32739425.

38. Mo Y, Xu W, Fu K, Chen H, Wen J, Huang Q, Guo F, Mo L, Yan J. The dual function of microglial polarization and its treatment targets in ischemic stroke. *Frontiers in Neurology*. 2022 Sep 23;13.0:921705. doi: 10.3389/fneur.2022.921705. PMID: 36212660.

39. Blaschke SJ, Demir S, König A, Abraham J, Vay SU, Rabenstein M, Olschewski DN, Hoffmann C, Hoffmann M, Hersch N, Merkel R, Hoffmann B, Schroeter M, Fink GR, Rueger MA. Substrate Elasticity Exerts Functional Effects on Primary Microglia. *Frontiers in Cellular Neuroscience*. 2020 Nov 5;14.0:590500. doi: 10.3389/fncel.2020.590500. PMID: 33250714.

40. Cohen D, Mashiach R, Houben L, Gálisová A, Addadi Y, Kain D, Lubart A, Blinder P, Allouche-Arnon H, Bar-Shir A. Glyconanofluorides as Immunotracers with a Tunable Core Composition for Sensitive Hotspot Magnetic Resonance Imaging of Inflammatory Activity. *ACS Nano*. 2021 Apr 19;15.0(4.0):7563-7574. doi: 10.1021/acsnano.1c01040. PMID: 33872494.

41. Lee CY, Chooi WH, Ng S, Chew SY. Modulating neuroinflammation through molecular, cellular and biomaterial-based approaches to treat spinal cord injury. *Bioengineering & Translational Medicine*. 2022 Aug 31;8.0(2.0):e10389. doi: 10.1002/btm2.10389. PMID: 36925680.

42. Yi S, Shi W, Wang H, Ma C, Zhang X, Wang S, Cong B, Li Y. Endoplasmic Reticulum Stress PERK-ATF4-CHOP Pathway Is Associated with Hypothalamic Neuronal Injury in Different Durations of Stress in Rats. *Frontiers in Neuroscience*. 2017 Mar 24;11.0:152. doi: 10.3389/fnins.2017.00152. PMID: 28392758.

43. Kim T, Valera E, Desplats P. Alterations in Striatal microRNA-mRNA Networks Contribute to Neuroinflammation in Multiple System Atrophy. *Molecular Neurobiology*. 2019 Apr 9;56.0(10.0):7003-7021. doi: 10.1007/s12035-019-1577-3. PMID: 30968343.

44. Wang S, Shi W, Zhang G, Zhang X, Ma C, Zhao K, Cong B, Li Y. Endoplasmic Reticulum Stress-Mediated Basolateral Amygdala GABAergic Neuron Injury Is Associated With Stress-Induced Mental Disorders in Rats. *Frontiers in Cellular Neuroscience*. 2019 Nov 14;13.0:511. doi: 10.3389/fncel.2019.00511. PMID: 31798418.

45. Amato S, Arnold A. Modeling Microglia Activation and Inflammation-Based Neuroprotectant Strategies During Ischemic Stroke. *Bulletin of Mathematical Biology*. 2021 May 12;83.0(6.0):72. doi: 10.1007/s11538-021-00905-4. PMID: 33982158.

46. Kashif H, Shah D, Sukumari-Ramesh S. Dysregulation of microRNA and Intracerebral Hemorrhage: Roles in Neuroinflammation. *International Journal of Molecular Sciences*. 2021 Jul 29;22.0(15.0):8115. doi: 10.3390/ijms22158115. PMID: 34360881.

47. Torrente D, Su EJ, Schielke GP, Warnock M, Mann KM, Lawrence DA. Opposing effects of β-2 and β-1 adrenergic receptor signaling on neuroinflammation and dopaminergic neuron survival in α-synuclein-mediated neurotoxicity. *Journal of Neuroinflammation*. 2023 Mar 2;20.0(1.0):56. doi: 10.1186/s12974-023-02748-3. PMID: 36864439.

48. Gao Y, Fang C, Wang J, Ye Y, Li Y, Xu Q, Kang X, Gu L. Neuroinflammatory Biomarkers in the Brain, Cerebrospinal Fluid, and Blood After Ischemic Stroke. *Molecular Neurobiology*. 2023 May 31;60.0(9.0):5117-5136. doi: 10.1007/s12035-023-03399-4. PMID: 37258724.

49. Chavda VK, Madhwani K. Coding and non-coding nucleotides': The future of stroke gene therapeutics. *Genomics*. 2021 Mar 5;113.0(3.0):1291-1307. doi: 10.1016/j.ygeno.2021.03.003. PMID: 33677059.

50. Fullerton JL, Cosgrove CC, Rooney RA, Work LM. Extracellular vesicles and their microRNA cargo in ischaemic stroke. *The Journal of Physiology*. 2022 Apr 14;601.0(22.0):4907-4921. doi: 10.1113/jp282050. PMID: 35421904.

51. Wang S, Liu X, Shi W, Qian Q, Zhang G, Li Y, Cong B, Zuo M. Mechanism of Chronic Stress-Induced Glutamatergic Neuronal Damage in the Basolateral Amygdaloid Nucleus. *Analytical Cellular Pathology*. 2021 Nov 23;2021.0:1-14. doi: 10.1155/2021/8388527. PMID: 34858775.

52. Breßer M, Siemens KD, Schneider LH, Lunnebach JE, Leven P, Glowka TR, Oberländer K, De Domenico E, Schultze JL, Schmidt J, Kalff JC, Schneider A, Wehner S, Schneider R. Macrophage-induced enteric neurodegeneration leads to motility impairment during gut inflammation. *EMBO Molecular Medicine*. 2025 Jan 6;17.0(2.0):301-335. doi: 10.1038/s44321-024-00189-w. PMID: 39762650.

53. McGurran H, Kumbol VW, Krüger C, Wallach T, Lehnardt S. miR-154-5p Is a Novel Endogenous Ligand for TLR7 Inducing Microglial Activation and Neuronal Injury. *Cells*. 2024 Feb 26;13.0(5.0):407. doi: 10.3390/cells13050407. PMID: 38474371.

54. Shi Y, Yang Y, Liu J, Zheng J. Avicularin Treatment Ameliorates Ischemic Stroke Damage by Regulating Microglia Polarization and its Exosomes via the NLRP3 Pathway. *Journal of Integrative Neuroscience*. 2024 Oct 30;23.0(11.0):196. doi: 10.31083/j.jin2311196. PMID: 39613475.

55. Yang Y, Liu R, Sun Y, Wu B, He B, Jia Y, Yan T. Schisandrin B restores M1/M2 balance through miR-124 in lipopolysaccharide-induced BV2 cells. *Journal of Pharmacy and Pharmacology*. 2024 Jul 18;76.0(10.0):1352-1361. doi: 10.1093/jpp/rgae079. PMID: 39024474.

56. Torrente D, Su EJ, Citalán-Madrid AF, Schielke GP, Magaoay D, Warnock M, Stevenson TK, Mann KM, Lesept F, Delétage N, Blanc M, Norris EH, Vivien D, Lawrence DA. The interaction of tPA with NMDAR1 drives neuroinflammation and neurodegeneration in α-synuclein-mediated neurotoxicity. *Journal of Neuroinflammation*. 2025 Jan 14;22.0(1.0):8. doi: 10.1186/s12974-025-03336-3. PMID: 39810216.

57. Kołosowska N, Gotkiewicz M, Dhungana H, Giudice L, Giugno R, Box D, Huuskonen MT, Korhonen P, Scoyni F, Kanninen KM, Ylä-Herttuala S, Turunen TA, Turunen MP, Koıstınaho J, Malm T. Intracerebral overexpression of miR-669c is protective in mouse ischemic stroke model by targeting MyD88 and inducing alternative microglial/macrophage activation. *Journal of Neuroinflammation*. 2020 Jun 19;17.0(1.0):194. doi: 10.1186/s12974-020-01870-w. PMID: 32560730.

58. Wen M, Ye J, Han Y, Huang L, Yang H, Jiang W, Chen S, Zhong W, Zeng H, Li D. Hypertonic saline regulates microglial M2 polarization via miR-200b/KLF4 in cerebral edema treatment. *Biochemical and Biophysical Research Communications*. 2018 Mar 27;499.0(2.0):345-353. doi: 10.1016/j.bbrc.2018.03.161. PMID: 29577903.

59. Zhong W, Huang Q, Zeng L, Hu Z, Tang X. Caveolin-1 and MLRs: A potential target for neuronal growth and neuroplasticity after ischemic stroke. *International Journal of Medical Sciences*. 2019 Jan 1;16.0(11.0):1492-1503. doi: 10.7150/ijms.35158. PMID: 31673241.

60. Wu C, Sinha B, Li Y, Benowitz LI, Chen Q, Zhang Z, Patel NJ, Aziz-Sultan MA, Chiocca EA, Wang X. Monogenic, Polygenic, and MicroRNA Markers for Ischemic Stroke. *Molecular Neurobiology*. 2018 Jun 8;56.0(2.0):1330-1343. doi: 10.1007/s12035-018-1055-3. PMID: 29948938.

61. Yu W, Lv Z, Zhang L, Gao Z, Chen X, Yang X, Zhong M. RETRACTED: Astragaloside IV reduces the hypoxia-induced injury in PC-12 cells by inhibiting expression of miR-124. *Biomedicine & Pharmacotherapy*. 2018 Jul 12;106.0:419-425. doi: 10.1016/j.biopha.2018.06.127. PMID: 29990829.

62. Ehtesham N, Mosallaei M, Karimzadeh MR, Kazerouni HM, Sharifi M. microRNAs: key modulators of disease-modifying therapies in multiple sclerosis. *International Reviews of Immunology*. 2020 Jun 18;39.0(6.0):264-279. doi: 10.1080/08830185.2020.1779712. PMID: 32552273.

63. Pei H, Peng Q, Guo S, Gu Y, Sun T, D X, Jiang Y, Xie J, Zhang L, Zhu Z. MiR-367 alleviates inflammatory injury of microglia by promoting M2 polarization via targeting CEBPA. *In Vitro Cellular & Developmental Biology - Animal*. 2020 Nov 4;56.0(10.0):878-887. doi: 10.1007/s11626-020-00519-5. PMID: 33150481.

64. Kim K, Hong EP, Lee Y, McLean ZL, Elezi E, Lee R, Kwak S, McAllister B, Massey TH, Lobanov SV, Holmans P, Orth M, Ciosi M, Monckton DG, Long JD, Lucente DE, Wheeler VC, MacDonald ME, Gusella JF, Lee J. Posttranscriptional regulation of FAN1 by miR-124-3p at rs3512 underlies onset-delaying genetic modification in Huntington's disease. *Proceedings of the National Academy of Sciences*. 2024 Apr 12;121.0(16.0):e2322924121. doi: 10.1073/pnas.2322924121. PMID: 38607933.

65. Dukhinova MS, Kopeikina E, Ponomarev ED. Usage of Multiparameter Flow Cytometry to Study Microglia and Macrophage Heterogeneity in the Central Nervous System During Neuroinflammation and Neurodegeneration. *Methods in molecular biology*. 2018 Jan 1;1745.0:167-177. doi: 10.1007/978-1-4939-7680-5_10. PMID: 29476469.

66. Deng X, Zeng Y, Ding D. MiR-30c-5p-Targeted Regulation of GNAI2 Improves Neural Function Injury and Inflammation in Cerebral Ischemia-Reperfusion Injury. *Applied Biochemistry and Biotechnology*. 2023 Dec 28;196.0(8.0):5235-5248. doi: 10.1007/s12010-023-04802-5. PMID: 38153649.

67. Wei Z, Luo L, Hu S, Tian R, Liu Z. KDM2B overexpression prevents myocardial ischemia-reperfusion injury in rats through regulating inflammatory response via the TLR4/NF-κB p65 axis. *Experimental and Therapeutic Medicine*. 2021 Dec 17;23.0(2.0):154. doi: 10.3892/etm.2021.11077. PMID: 35069835.

68. Green C, Minassian A, Vogel SN, Diedenhofen M, Wiedermann D, Hoehn M. Persistent Quantitative Vitality of Stem Cell Graft Is Necessary for Stabilization of Functional Brain Networks After Stroke. *Frontiers in Neurology*. 2019 Apr 5;10.0:335. doi: 10.3389/fneur.2019.00335. PMID: 31024429.

69. Sun A, Blecharz-Lang KG, Małecki A, Meybohm P, Nowacka-Chmielewska MM, Burek M. Role of microRNAs in the regulation of blood-brain barrier function in ischemic stroke and under hypoxic conditions in vitro. *Frontiers in Drug Delivery*. 2022 Nov 1;2.0. doi: 10.3389/fddev.2022.1027098.

70. Berdasco C, Vega MD, Rosato-Siri MV, Goldstein J. Environmental Cues Modulate Microglial Cell Behavior Upon Shiga Toxin 2 From Enterohemorrhagic Escherichia coli Exposure. *Frontiers in Cellular and Infection Microbiology*. 2020 Jan 8;9.0:442. doi: 10.3389/fcimb.2019.00442. PMID: 31970091.

71. Huang J, Chen D, Lin X, Lin X, Yang C, Lin X, Lin X. miRNA -124 alleviated memory impairment induced by d -galactose rapidly in male rats via microglia polarization. *Hippocampus*. 2022 Dec 21;33.0(2.0):96-111. doi: 10.1002/hipo.23491. PMID: 36541921.

72. MacKenzie SM, van Kralingen JC, Martin H, Davies E. MicroRNAs in Aldosterone Production and Action. *IntechOpen eBooks*. 2019 Sep 25;. doi: 10.5772/intechopen.87226.

73. Chao Y, Wang F, Wang Y, Han B. Correlation analysis of miRNA-124, miRNA-210 with brain injury and inflammatory response in patients with craniocerebral injury. *PubMed*. 2022 Jan 1;14.0(1.0):285-294. PMID: 35173845.

74. Zhou H, Yang X, Yu J, Xu J, Zhang R, Zhang T, Wang X, Ma J. Reference gene identification for normalisation of RT-qPCR analysis in plasma samples of the rat middle cerebral artery occlusion model. *Veterinary Medicine and Science*. 2022 Jul 27;8.0(5.0):2076-2085. doi: 10.1002/vms3.879. PMID: 35894780.

75. Collmann FM, Pijnenburg R, Schneider G, Schäfer C, Folz-Donahue K, Kukat C, Hoehn M. Imaging Reporter Strategy to Monitor Gene Activation of Microglia Polarisation States under Stimulation. *Journal of Neuroimmune Pharmacology*. 2018 May 22;13.0(3.0):371-382. doi: 10.1007/s11481-018-9789-2. PMID: 29790106.

76. Spellicy SE, Scheulin KM, Baker EW, Jurgielewicz BJ, Kinder HA, Waters ES, Grimes JA, Stice SL, West FD. Semi-Automated Cell and Tissue Analyses Reveal Regionally Specific Morphological Alterations of Immune and Neural Cells in a Porcine Middle Cerebral Artery Occlusion Model of Stroke. *Frontiers in Cellular Neuroscience*. 2021 Jan 22;14.0:600441. doi: 10.3389/fncel.2020.600441. PMID: 33551749.

77. García-Bonilla L, Iadecola C, Anrather J. Inflammation and Immune Response. *Stroke*. 2021 Apr 24;:117-128.e5. doi: 10.1016/b978-0-323-69424-7.00010-7.

78. Anrather J, Iadecola C, Hallenbeck JM. Inflammation and Immune Response. *Stroke*. 2015 Aug 27;:129-140.e5. doi: 10.1016/b978-0-323-29544-4.00010-4.

79. Oh SS, Lee YB, Jeon JS, An S, Choi J. Preliminary Study on Safety Assessment of 10 Hz Transcranial Alternating Current Stimulation in Rat Brain. *Applied Sciences*. 2022 May 24;12.0(11.0):5299. doi: 10.3390/app12115299.

80. Wang J, Chen X, Sun L, Chen X, Li H, Xiong B, Wang H. [Long noncoding RNA ZEB1-AS1 aggravates cerebral ischemia/reperfusion injury in rats through the HMGB1/TLR-4 signaling axis]. *PubMed*. 2022 Aug 20;42.0(8.0):1134-1142. doi: 10.12122/j.issn.1673-4254.2022.08.04. PMID: 36073211.

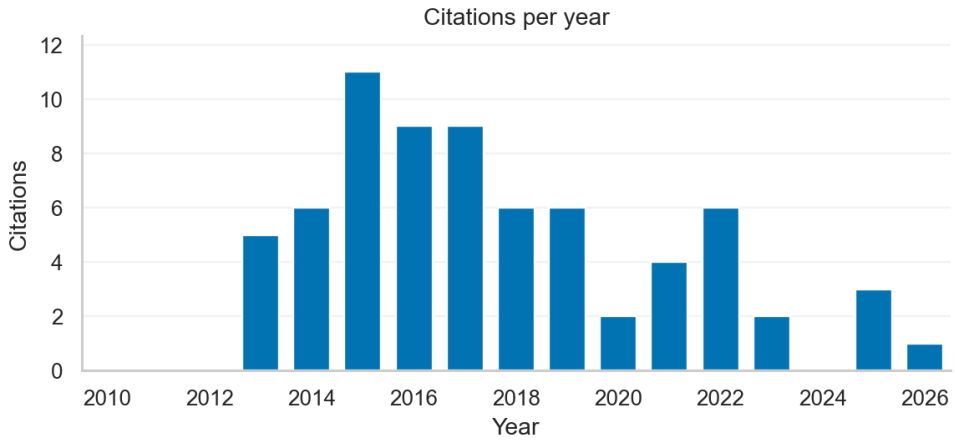
81. Torrente D, Su EJ, Schielke GP, Warnock M, Stevenson TK, Mann KM, Vivien D, Lawrence DA. Tissue Plasminogen Activator Interaction with NMDAR1 Promotes Dopaminergic Neuron Degeneration in a Model of A-Synuclein-Mediated Neurotoxicity. *SSRN Electronic Journal*. 2022 Jan 1;. doi: 10.2139/ssrn.4059525.

82. Zhang H, Xu S, Chi F, Li G, Zhou Y, Li C, Wang A. Remifentanil Protects PC-12 Cells From OGD Damage by Up-Regulating miR-124. *Research Square*. 2021 Jan 14;. doi: 10.21203/rs.3.rs-143379/v1.

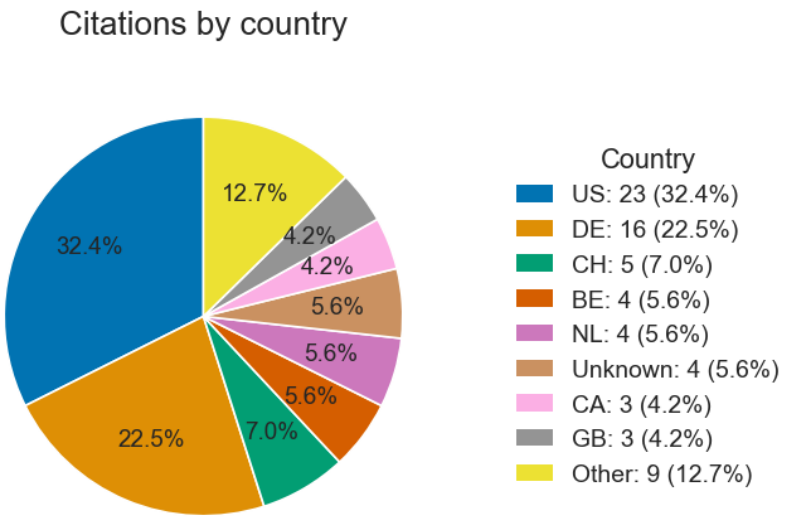
83. Cong J, Cong J, Wang L, Wang Y, Xin H, Xin H, Cong L, Cong L, Zhang G, Zhu W, Cong B, Shi W. Cardiac and brain pathological alterations with potential biomarker screening in stress-related fatalities. *Journal of Forensic and Legal Medicine*. 2026 Apr 16;120.0:103132. doi: 10.1016/j.jflm.2026.103132. PMID: 42001635.

Publication

Aswendt M, Adamczak JM, Couillard-Després S, Hoehn M. Boosting Bioluminescence Neuroimaging: An Optimized Protocol for Brain Studies. *PLoS ONE*. 2013 Feb 6;8(2):e55662. doi: 10.1371/journal.pone.0055662. PMID: 23405190.



Citation summary
Total citations: **71**
Average/year: **5.07**
Period: 2013-2026
SCImago cites/doc 2y (2025): **2.95**
SJR 2025: **0.726** (Q1)



Cited by

1. Bala S, Csák T, Momen-Heravi F, Lippai D, Kodys K, Catalano DE, Satishchandran A, Ambros V, Szabó G. Biodistribution and function of extracellular miRNA-155 in mice. *Scientific Reports*. 2015 May 29;5.0(1.0):10721. doi: 10.1038/srep10721. PMID: 26024046.

2. Hall MP, Woodroofe CC, Wood MG, Que I, Root MV', Ridwan YR, Shi C, Kirkland TA, Encell LP, Wood KV, Löwik CW, Mezzanotte L. Click beetle luciferase mutant and near infrared naphthyl-luciferins for improved bioluminescence imaging. *Nature Communications*. 2018 Jan 3;9.0(1.0):132. doi: 10.1038/s41467-017-02542-9. PMID: 29317625.

3. Tung JK, Berglund K, Gutekunst C, Hochgeschwender U, Gross RE. Bioluminescence imaging in live cells and animals. *Neurophotonics*. 2016 Apr 5;3.0(2.0):1-6. doi: 10.1117/1.nph.3.2.025001. PMID: 27226972.

4. Yang M, Jin L, Wu Z, Xie Y, Zhang P, Wang Q, Yan S, Chen B, Liang H, Naman CB, Zhang J, He S, Yan X, Zhao L, Cui W. PLGA-PEG Nanoparticles Facilitate In Vivo Anti-Alzheimer's Effects of Fucoxanthin, a Marine Carotenoid Derived from Edible Brown Algae. *Journal of Agricultural and Food Chemistry*. 2021 Aug 18;69.0(34.0):9764-9777. doi: 10.1021/acs.jafc.1c00569. PMID: 34404210.

5. Su Y, Walker JR, Hall MP, Klein MA, Wu XY, Encell LP, Casey KM, Liu LX, Hong G, Lin MZ, Kirkland TA. An optimized bioluminescent substrate for non-invasive imaging in the brain. *Nature Chemical Biology*. 2023 Feb 9;19.0(6.0):731-739. doi: 10.1038/s41589-023-01265-x. PMID: 36759751.

6. Rust R, Weber R, Generali M, Kehl D, Bodenmann C, Uhr D, Wanner D, Zürcher KJ, Saito H, Hoerstrup SP, Nitsch RM, Tackenberg C. Xeno-free induced pluripotent stem cell-derived neural progenitor cells for in vivo applications. *Journal of Translational Medicine*. 2022 Sep 16;20.0(1.0):421. doi: 10.1186/s12967-022-03610-5. PMID: 36114512.

7. De Niz M, Helm S, Horstmann S, Annoura T, Del Portillo HA, Khan SM, Heussler VT. In Vivo and In Vitro Characterization of a Plasmodium Liver Stage-Specific Promoter. *PLoS ONE*. 2015 Apr 15;10.0(4.0):e0123473. doi: 10.1371/journal.pone.0123473. PMID: 25874388.

8. Myburgh E, Coles JA, Ritchie R, Kennedy PGE, McLatchie AP, Rodgers J, Taylor MC, Barrett MP, Brewer JM, Mottram JC. In Vivo Imaging of Trypanosome-Brain Interactions and Development of a Rapid Screening Test for Drugs against CNS Stage Trypanosomiasis. *PLoS neglected tropical diseases*. 2013 Aug 22;7.0(8.0):e2384. doi: 10.1371/journal.pntd.0002384. PMID: 23991236.

9. Wu W, Su J, Tang C, Bai H, Ma Z, Zhang T, Yuan Z, Li Z, Zhou W, Zhang H, Liu Z, Wang Y, Zhou Y, Du L, Gu L, Li M. cybLuc: An Effective Aminoluciferin Derivative for Deep Bioluminescence Imaging. *Analytical Chemistry*. 2017 Apr 5;89.0(9.0):4808-4816. doi: 10.1021/acs.analchem.6b03510. PMID: 28378575.

10. Madala HR, Punganuru SR, Ali-Osman F, Zhang R, Srivenugopal KS. Brain- and brain tumor-penetrating disulfiram nanoparticles: Sequence of cytotoxic events and efficacy in human glioma cell lines and intracranial xenografts. *Oncotarget*. 2017 Dec 15;9.0(3.0):3459-3482. doi: 10.18632/oncotarget.23320. PMID: 29423059.

11. Boehm-Sturm P, Aswendt M, Minassian A, Michalk S, Mengler L, Adamczak JM, Mezzanotte L, Löwik CW, Hoehn M. A multi-modality platform to image stem cell graft survival in the naïve and stroke-damaged mouse brain. *Biomaterials*. 2013 Dec 16;35.0(7.0):2218-2226. doi: 10.1016/j.biomaterials.2013.11.085. PMID: 24355489.

12. Bakhsheshian J, Wei B, Chang K, Shukla S, Ambudkar SV, Simpson RJ, Gottesman MM, Hall MD. Bioluminescent imaging of drug efflux at the blood-brain barrier mediated by the transporter ABCG2. *Proceedings of the National Academy of Sciences*. 2013 Dec 2;110.0(51.0):20801-20806. doi: 10.1073/pnas.1312159110. PMID: 24297888.

13. Tennstaedt A, Aswendt M, Adamczak JM, Collienne U, Selt M, Schneider G, Henn N, Schaefer C, Lagouge M, Wiedermann D, Kloppenburg P, Hoehn M. Human neural stem cell intracerebral grafts show spontaneous early neuronal differentiation after several weeks. *Biomaterials*. 2015 Jan 12;44.0:143-154. doi: 10.1016/j.biomaterials.2014.12.038. PMID: 25617134.

14. Mannheim JG, Kara F, Doorduin J, Fuchs K, Reischl G, Liang S, Verhoye M, Gremse F, Mezzanotte L, Huisman MC. Standardization of Small Animal Imaging-Current Status and Future Prospects. *Molecular Imaging and Biology*. 2017 Oct 2;20.0(5.0):716-731. doi: 10.1007/s11307-017-1126-2. PMID: 28971332.

15. Green C, Minassian A, Vogel SN, Diedenhofen M, Beyrau A, Wiedermann D, Hoehn M. Sensorimotor Functional and Structural Networks after Intracerebral Stem Cell Grafts in the Ischemic Mouse Brain. *Journal of Neuroscience*. 2018 Jan 10;38.0(7.0):1648-1661. doi: 10.1523/jneurosci.2715-17.2018. PMID: 29321138.

16. Keuters MH, Aswendt M, Tennstaedt A, Wiedermann D, Pikhovych A, Rotthues S, Fink GR, Schroeter M, Hoehn M, Rueger MA. Transcranial direct current stimulation promotes the mobility of engrafted NSCs in the rat brain. *NMR in Biomedicine*. 2014 Dec 17;28.0(2.0):231-239. doi: 10.1002/nbm.3244. PMID: 25521600.

17. Coleman S, McGregor A. A bright Future for Bioluminescent Imaging in Viral Research. *Future Virology*. 2015 Feb 1;10.0(2.0):169-183. doi: 10.2217/fvl.14.96. PMID: 26413138.

18. Aswendt M, Vogel S, Schäfer C, Jathoul AP, Pulé M, Hoehn M. Quantitative in vivo dual-color bioluminescence imaging in the mouse brain. *Neurophotonics*. 2019 May 7;6.0(2.0):1-11. doi: 10.1117/1.nph.6.2.025006. PMID: 31093514.

19. Adamczak JM, Aswendt M, Kreutzer C, Rotheneichner P, Riou A, Selt M, Beyrau A, Uhlenküken U, Diedenhofen M, Nelles M, Aigner L, Couillard-Després S, Hoehn M. Neurogenesis upregulation on the healthy hemisphere after stroke enhances compensation for age-dependent decrease of basal neurogenesis. *Neurobiology of Disease*. 2016 Dec 20;99.0:47-57. doi: 10.1016/j.nbd.2016.12.015. PMID: 28007584.

20. Amadeo F, Plagge A, Chacko AR, Wilm B, Hanson V, Liptrott NJ, Murray PA, Taylor A. Firefly luciferase offers superior performance to AkaLuc for tracking the fate of administered cell therapies. *European Journal of Nuclear Medicine and Molecular Imaging*. 2021 Jul 27;49.0(3.0):796-808. doi: 10.1007/s00259-021-05439-4. PMID: 34313817.

21. Aswendt M, Adamczak JM, Tennstaedt A. A review of novel optical imaging strategies of the stroke pathology and stem cell therapy in stroke. *Frontiers in Cellular Neuroscience*. 2014 Aug 14;8.0:226. doi: 10.3389/fncel.2014.00226. PMID: 25177269.

22. Yevtodiyenko A, Bazhin AA, Khodakivskyi PV, Godinat A, Budin G, Maric T, Pietramaggiori G, Scherer SS, Kunchulia M, Eppeldauer GP, Polyakov SV, Francis KP, Bryan JN, Goun EA. Portable bioluminescent platform for in vivo monitoring of biological processes in non-transgenic animals. *Nature Communications*. 2021 May 11;12.0(1.0):2680. doi: 10.1038/s41467-021-22892-9. PMID: 33976191.

23. Sivanesan E, Sanchez KR, Zhang C, He S, Linderoth B, Stephens K, Raja SN, Guan Y. Spinal Cord Stimulation Increases Chemoefficacy and Prevents Paclitaxel-Induced Pain via CX3CL1. *Neuromodulation Technology at the Neural Interface*. 2023 Apr 11;26.0(5.0):938-949. doi: 10.1016/j.neurom.2023.03.006. PMID: 37045646.

24. Stout D, Berr SS, LeBlanc AK, Kalen JD, Osborne D, Price JC, Schiffer WK, Kuntner C, Wall JS. Guidance for Methods Descriptions Used in Preclinical Imaging Papers. *Molecular Imaging*. 2013 Oct 1;12.0(7.0):1-15. doi: 10.2310/7290.2013.00055. PMID: 23920252.

25. Vanherp L, Ristani A, Poelmans J, Hillen A, Lagrou K, Janbon G, Brock M, Himmelreich U, Velde GV. Sensitive bioluminescence imaging of fungal dissemination to the brain in mouse models of cryptococcosis. *Disease Models & Mechanisms*. 2019 May 17;12.0(6.0). doi: 10.1242/dmm.039123. PMID: 31101657.

26. Naumova AV, Balu N, Yarnykh VL, Reinecke H, Murry CE, Yuan C. Magnetic Resonance Imaging Tracking of Graft Survival in the Infarcted Heart. *Journal of Cardiovascular Pharmacology and Therapeutics*. 2014 Mar 30;19.0(4.0):358-367. doi: 10.1177/1074248414525999. PMID: 24685664.

27. Rotschafer JH, Hu S, Little MR, Erickson M, Low WC, Cheeran MC. Modulation of neural stem/progenitor cell proliferation during experimental Herpes Simplex encephalitis is mediated by differential FGF-2 expression in the adult

brain. *Neurobiology of Disease*. 2013 Jun 5;58.0:144-155. doi: 10.1016/j.nbd.2013.05.018. PMID: 23748078.

28. Branchini BR, Southworth TL, Fontaine DM, Kohrt D, Welcome FS, Florentine CM, Henricks ER, DeBartolo DB, Michelini E, Cevenini L, Roda A, Grossel MJ. Red-emitting chimeric firefly luciferase for in vivo imaging in low ATP cellular environments. *Analytical Biochemistry*. 2017 Jul 4;534.0:36-39. doi: 10.1016/j.ab.2017.07.001. PMID: 28687486.

29. Chehade M, Srivastava AK, Bulte JWM. Co-Registration of Bioluminescence Tomography, Computed Tomography, and Magnetic Resonance Imaging for Multimodal In Vivo Stem Cell Tracking. *Tomography*. 2016 Jun 1;2.0(2.0):158-165. doi: 10.18383/j.tom.2016.00160. PMID: 27478872.

30. Reséndiz-Sharpe A, da Silva RP, Geib E, Vanderbeke L, Seldeslachts L, Hupko C, Brock M, Lagrou K, Velde GV. Longitudinal multimodal imaging-compatible mouse model of triazole-sensitive and -resistant invasive pulmonary aspergillosis. *Disease Models & Mechanisms*. 2022 Mar 1;15.0(3.0). doi: 10.1242/dmm.049165. PMID: 35352801.

31. Van Reet N, Van de Vyver H, Pyana PP, Van der Linden AM, Büscher P. A Panel of Trypanosoma brucei Strains Tagged with Blue and Red-Shifted Luciferases for Bioluminescent Imaging in Murine Infection Models. *PLoS neglected tropical diseases*. 2014 Aug 21;8.0(8.0):e3054. doi: 10.1371/journal.pntd.0003054. PMID: 25144573.

32. Tennstaedt A, Mastropietro A, Nelles M, Beyrau A, Hoehn M. In Vivo Fate Imaging of Intracerebral Stem Cell Grafts in Mouse Brain. *PLoS ONE*. 2015 Dec 7;10.0(12.0):e0144262. doi: 10.1371/journal.pone.0144262. PMID: 26641453.

33. Ayzenberg I, Schlevogt S, Metzdorf J, Stahlke S, Pedreiturria X, Hunfeld A, Couillard-Després S, Kleiter I. Analysis of Neurogenesis during Experimental Autoimmune Encephalomyelitis Reveals Pitfalls of Bioluminescence Imaging. *PLoS ONE*. 2015 Mar 17;10.0(3.0):e0118550. doi: 10.1371/journal.pone.0118550. PMID: 25780928.

34. Bernau K, Lewis CM, Petelinsek AM, Benink HA, Zimprich C, Meyerand ME, Suzuki M, Svendsen CN. In vivo tracking of human neural progenitor cells in the rat brain using bioluminescence imaging. *Journal of Neuroscience Methods*. 2014 Mar 24;228.0:67-78. doi: 10.1016/j.jneumeth.2014.03.005. PMID: 24675049.

35. Patterson AP, Booth SA, Saba R. The Emerging Use ofIn VivoOptical Imaging in the Study of Neurodegenerative Diseases. *BioMed Research International*. 2014 Jan 1;2014.0:1-14. doi: 10.1155/2014/401306. PMID: 25147799.

36. Vogel SN, Aswendt M, Nelles M, Henn N, Schneider G, Hoehn M. Initial graft size and not the innate immune response limit survival of engrafted neural stem cells. *Journal of Tissue Engineering and Regenerative Medicine*. 2017 Jun 9;12.0(3.0):784-793. doi: 10.1002/term.2497. PMID: 28599089.

37. Weber R, Bodenmann C, Uhr D, Zürcher KJ, Wanner D, Generali M, Nitsch RM, Rust R, Tackenberg C. Intracerebral Transplantation and In Vivo Bioluminescence Tracking of Human Neural Progenitor Cells in the Mouse Brain. *Journal of Visualized Experiments*. 2022 Jan 27;(179.0). doi: 10.3791/63102. PMID: 35156654.

38. Weber R, Bodenmann C, Uhr D, Zürcher KJ, Wanner D, Generali M, Nitsch RM, Rust R, Tackenberg C. Intracerebral Transplantation and In Vivo Bioluminescence Tracking of Human Neural Progenitor Cells in the Mouse Brain. *Journal of Visualized Experiments*. 2022 Jan 27;(179.0). doi: 10.3791/63102-v.

39. Fricke IB, Schelhaas S, Zinnhardt B, Viel T, Hermann S, Couillard-Després S, Jacobs AH. In vivo bioluminescence imaging of neurogenesis - the role of the blood brain barrier in an experimental model of Parkinson's disease. *European Journal of Neuroscience*. 2017 Feb 13;45.0(7.0):975-986. doi: 10.1111/ejn.13540. PMID: 28194885.

40. Aswendt M, Henn N, Michalk S, Schneider G, Steiner M, Bissa U, Dose C, Hoehn M. Novel Bimodal Iron Oxide Particles for Efficient Tracking of Human Neural Stem Cells in Vivo. *Nanomedicine*. 2015 Aug 1;10.0(16.0):2499-2512. doi: 10.2217/nnm.15.94. PMID: 26296195.

41. Vogel SN, Schäfer C, Heß S, Folz-Donahue K, Nelles M, Minassian A, Schwarz MK, Kukat C, Ehrlich M, Zaehres H, Kloppenburg P, Hoehn M, Aswendt M. The in vivo timeline of differentiation of engrafted human neural progenitor cells. *Stem Cell Research*. 2019 Mar 25;37.0:101429. doi: 10.1016/j.scr.2019.101429. PMID: 30933718.

42. Vanherp L, Poelmans J, Hillen A, Janbon G, Brock M, Lagrou K, Velde GV, Himmelreich U. The Added Value of Longitudinal Imaging for Preclinical In Vivo Efficacy Testing of Therapeutic Compounds against Cerebral Cryptococcosis. *Antimicrobial Agents and Chemotherapy*. 2020 Apr 11;64.0(7.0). doi: 10.1128/aac.00070-20. PMID: 32284382.

43. Mezzanotte L, Iljas JD, Que I, Chan AB, Kaijzel EL, Hoeben RC, Löwik CW. Optimized Longitudinal Monitoring of Stem Cell Grafts in Mouse Brain Using a Novel Bioluminescent/Near Infrared Fluorescent Fusion Reporter. *Cell Transplantation*. 2017 Dec 1;26.0(12.0):1878-1889. doi: 10.1177/0963689717739718. PMID: 29390874.

44. Cao JH, Hu Y, Shazeeb MS, Pedraza CE, Pande N, Weinstock D, Polites GH, Zhang W, Chandross KJ, Ying X. In Vivo Optical Imaging of Myelination Events in a Myelin Basic Protein Promoter-Driven Luciferase Transgenic Mouse Model. *ASN NEURO*. 2018 Jan 1;10.0(1.0):1759091418777329. doi: 10.1177/1759091418777329. PMID: 29806482.

45. Maysinger D, Zhang I. Nutritional and Nanotechnological Modulators of Microglia. *Frontiers in Immunology*. 2016 Jul 15;7.0:270. doi: 10.3389/fimmu.2016.00270. PMID: 27471505.

46. Selt M, Tennstaedt A, Beyrau A, Nelles M, Schneider G, Löwik CW, Hoehn M. In Vivo Non-Invasive Tracking of Macrophage Recruitment to Experimental Stroke. *PLoS ONE*. 2016 Jun 24;11.0(6.0):e0156626. doi: 10.1371/journal.pone.0156626. PMID: 27341631.

47. Uyar O, Plante P, Piret J, Venable M, Carbonneau J, Corbeil J, Boivin G. A novel bioluminescent herpes simplex virus 1 for in vivo monitoring of herpes simplex encephalitis. *Scientific Reports*. 2021 Sep 21;11.0(1.0):18688. doi: 10.1038/s41598-021-98047-z. PMID: 34548521.

48. Green C, Minassian A, Vogel SN, Diedenhofen M, Wiedermann D, Hoehn M. Persistent Quantitative Vitality of Stem Cell Graft Is Necessary for Stabilization of Functional Brain Networks After Stroke. *Frontiers in Neurology*. 2019 Apr 5;10.0:335. doi: 10.3389/fneur.2019.00335. PMID: 31024429.

49. Madsen SD, Giler MK, Bunnell BA, O'Connor KC. Illuminating the Regenerative Properties of Stem Cells In Vivo with Bioluminescence Imaging. *Biotechnology Journal*. 2020 Oct 22;16.0(3.0):e2000248. doi: 10.1002/biot.202000248. PMID:

33089922.

50. Bakhsheshian J, Wei B, Hall MD, Simpson RJ, Gottesman MM. In Vivo Bioluminescent Imaging of ATP-Binding Cassette Transporter-Mediated Efflux at the Blood-Brain Barrier. *Methods in molecular biology*. 2016 Jan 1;1461.0:227-239. doi: 10.1007/978-1-4939-3813-1_19. PMID: 27424909.

51. Boitet M, Eun H, Lee T, Kim J, Grailhe R. Non-invasive In Vivo Brain Astrogenesis and Astrogliosis Quantification Using a Far-red E2-Crimson Transgenic Reporter Mouse. *Molecular Neurobiology*. 2022 Aug 24;59.0(11.0):6740-6753. doi: 10.1007/s12035-022-02997-y. PMID: 36001234.

52. Esimbekova EN, Kratasyuk VA, Rozanova NA, Lonshakova-Mukina VI, Torgashina IG, Komleva YK, Saridis MR, Korsakova SA, Yurchenko SO, Salmina AB. Current and prospective trends in the application of bioluminescent analysis in experimental brain studies. *Talanta*. 2025 Jun 24;296.0:128510. doi: 10.1016/j.talanta.2025.128510. PMID: 40582108.

53. Martin EC, Aarons LJ, Yates JW. Pharmacodynamic modelling of resistance to epidermal growth factor receptor inhibition in brain metastasis mouse models. *Cancer Chemotherapy and Pharmacology*. 2018 Jul 27;82.0(4.0):669-675. doi: 10.1007/s00280-018-3630-8. PMID: 30054711.

54. Aswendt M, Vogel S, Hoehn M. In vivo bioluminescence imaging to elucidate stem cell graft differentiation. *Neural Regeneration Research*. 2019 Sep 16;15.0(1.0):61. doi: 10.4103/1673-5374.264449. PMID: 31535647.

55. Schomann T, Mezzanotte L, de Groot JC, Löwik CW, Frijns JHM, Huisman MA. Imaging Bioluminescent Exogenous Stem Cells in the Intact Guinea Pig Cochlea. *The Anatomical Record*. 2019 Jan 12;303.0(3.0):427-440. doi: 10.1002/ar.24068. PMID: 30635981.

56. Zare-Shahabadi A, Ataei A, Rezaei N. Proteins brighten the brain. *Life Sciences*. 2016 Oct 22;167.0:1-5. doi: 10.1016/j.lfs.2016.10.024. PMID: 27780734.

57. Stout D, David J. Optical Bioluminescence Protocol for Imaging Mice. *Methods in molecular biology*. 2018 Jan 1;1790.0:29-40. doi: 10.1007/978-1-4939-7860-1_3. PMID: 29858781.

58. Mastropietro A, Tennstaedt A, Beyrau A, Henn N, Hoehn M, Baselli G. Proof of concept of an automatic tool for bioluminescence imaging data analysis. *PubMed*. 2015 Aug 1;2015.0:6269-6272. doi: 10.1109/embc.2015.7319825. PMID: 26737725.

59. Rose LC, Bulte JWM. Cell Tracking and Transplant Imaging. *Springer eBooks*. 2017 Jan 1;:593-603. doi: 10.1007/978-3-319-42202-2_23.

60. Aswendt M, Collmann FM, Hoehn M. Neurobiological insights from bioluminescence imaging. *Oncotarget*. 2017 Aug 13;8.0(41.0):69198-69199. doi: 10.18632/oncotarget.20302. PMID: 29050195.

61. Mofford DM, Miller SC. Luciferins Behave Like Drugs. *ACS Chemical Neuroscience*. 2015 Jul 30;6.0(8.0):1273-1275. doi: 10.1021/acschemneuro.5b00195. PMID: 26225810.

62. Saito-Moriya R, Iwano S, Kitada N, Yamasaki N, Kamiya G, Ogo K, Sugiyama T, Chuluun-Erdene A, Isagawa T, Obata R, Ijuin R, Abe H, Kashiwakura Y, Mabuchi Y, Takahashi M, Yamamoto Y, Sawaki D, Sato T, Sato S, Nishikii H, Yoshimura K, Hioki H, Ohmori T, Nakashiba T, Hirano T, Aoyama H, Takeda N, Miyawaki A, Maki S, Kuchimaru T. A bright synthetic near-infrared luciferin enhances the capabilities of deep-tissue bioluminescence imaging using firefly luciferases. *bioRxiv (Cold Spring Harbor Laboratory)*. 2025 Jan 22;. doi: 10.1101/2025.01.20.633992.

63. Rotschafer JH. Immune Modulation of Adult Neurogenesis during Experimental Herpes Simplex Encephalitis. *University of Minnesota Digital Conservancy (University of Minnesota)*. 2015 May 1;.

64. Mofford DM. Pushing The Boundaries of Bioluminescence Using Synthetic Luciferins: A Dissertation. *eScholarship@UMassChan*. 2015 Jan 1;. doi: 10.13028/m2fk5x.

65. Tandle AT, Shankavaram UT, Schlaff CD, Camphausen KA, Krauze AV. Preclinical Models of Glioblastoma in Radiobiology: Evolving Protocols and Research Methods. *Current clinical pathology*. 2016 Jan 1;:255-274. doi: 10.1007/978-3-319-28305-0_16.

66. Brown AK. Investigating the Effect of Energy Substrates and LPS-activation on the In Vitro Energy Metabolism of BV-2, RAW264.7 and VM-M3 Cells. 2016 Jan 1;.

67. Mistretta A, Sorrentino F, Caltagirone S, Inclimona V. [Electron microscopy observations of the small coronary vessels after prolonged lipid load]. *PubMed*. 1967 Jan 1;2.0:119-20. PMID: 5589542.

68. Shah N. Investigational Chemotherapy and Novel Pharmacokinetics for the Treatment of Cancers of the CNS. *The Research Repository @ WVU (West Virginia University)*. 2018 Jan 1;. doi: 10.33915/etd.7250.

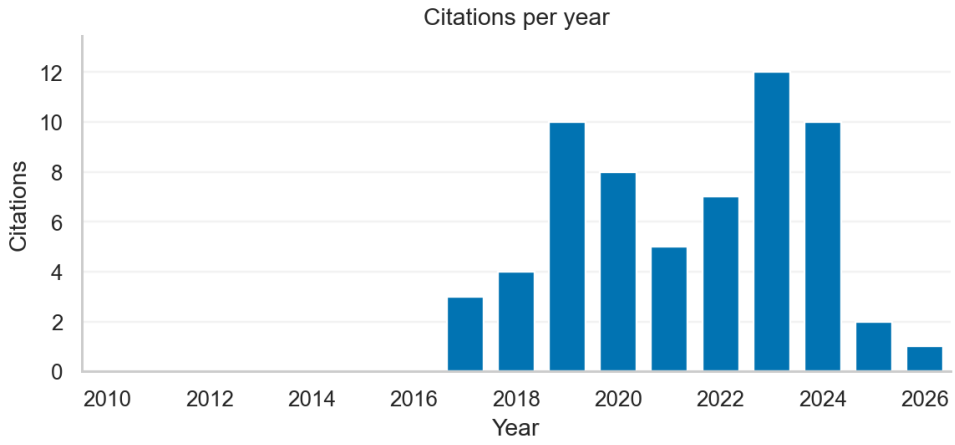
69. Curia G, Estrada-Camarena EM, Manjarrez E, Mizuno H. Editorial: In vivo investigations on neurological disorders: From traditional approaches to forefront technologies. *Frontiers in Neuroscience*. 2022 Oct 18;16.0:1052089. doi: 10.3389/fnins.2022.1052089. PMID: 36330344.

70. Nakanishi T, Matsumaru D, Ishida K. Advances in Bioluminescence Imaging for Assessment of Developmental Neurotoxicity. *BPB Reports*. 2025 Jan 1;8.0(6.0):176-181. doi: 10.1248/bpbreports.8.6_176.

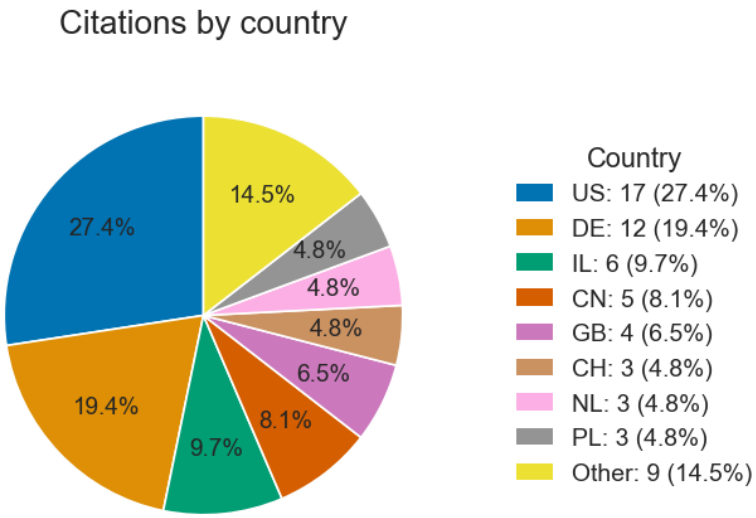
71. Bhattarai A, Moret R, Zhang X, Maresh GA, Yip H, Haupt C, Graham R, Latsis M, Matrana MR, Rose K, Bardot SF, Li L. Optimizing Sequential Targeted Therapies in Advanced Renal Cell Carcinoma Using Patient-Derived Orthotopic Xenograft Mouse Avatars. *Cancers*. 2026 May 16;18.0(10.0):1615. doi: 10.3390/cancers18101615. PMID: 42192974.

Publication

Leuze CW, Aswendt M, Ferenczi E, Liu CW, Hsueh B, Goubran M, Tian Q, Steinberg GK, Zeineh M, Deisseroth KA, McNab JA. The separate effects of lipids and proteins on brain MRI contrast revealed through tissue clearing. *NeuroImage*. 2017 Apr 12;156:412-422. doi: 10.1016/j.neuroimage.2017.04.021. PMID: 28411157.



Citation summary
Total citations: **62**
Average/year: **6.20**
Period: 2017-2026
SCImago cites/doc 2y (2025): **5.31**
SJR 2025: **1.93** (Q1)



Cited by

1. Weiskopf N, Edwards L, Helms G, Mohammadi S, Kirilina EP. Quantitative magnetic resonance imaging of brain anatomy and in vivo histology. *Nature Reviews Physics*. 2021 Jun 28;3.0(8.0):570-588. doi: 10.1038/s42254-021-00326-1.

2. Edwards L, Kirilina EP, Mohammadi S, Weiskopf N. Microstructural imaging of human neocortex in vivo. *NeuroImage*. 2018 Mar 26;182.0:184-206. doi: 10.1016/j.neuroimage.2018.02.055. PMID: 29588229.

3. Roebroeck AF, Miller KL, Aggarwal M. Ex vivo diffusion MRI of the human brain: Technical challenges and recent advances. *NMR in Biomedicine*. 2018 Jun 4;32.0(4.0):e3941. doi: 10.1002/nbm.3941. PMID: 29863793.

4. Gradinaru V, Treweek JB, Overton K, Deisseroth KA. Hydrogel-Tissue Chemistry: Principles and Applications. *Annual Review of Biophysics*. 2018 May 20;47.0(1.0):355-376. doi: 10.1146/annurev-biophys-070317-032905. PMID: 29792820.

5. Yendiki A, Aggarwal M, Axer M, Howard AF, van Walsum AVC, Haber SN. Post mortem mapping of connectional anatomy for the validation of diffusion MRI. *NeuroImage*. 2022 Mar 25;256.0:119146. doi: 10.1016/j.neuroimage.2022.119146. PMID: 35346838.

6. Morawski M, Kirilina EP, Scherf N, Jäger C, Reimann K, Trampel R, Gavriilidis F, Geyer S, Biedermann B, Arendt T, Weiskopf N. Developing 3D microscopy with CLARITY on human brain tissue: Towards a tool for informing and validating MRI-based histology. *NeuroImage*. 2017 Nov 28;182.0:417-428. doi: 10.1016/j.neuroimage.2017.11.060. PMID: 29196268.

7. Weinberger DR, Radulescu E. Structural Magnetic Resonance Imaging All Over Again. *JAMA Psychiatry*. 2020 Jul 22;78.0(1.0):11. doi: 10.1001/jamapsychiatry.2020.1941. PMID: 32697315.

8. Goubran M, Leuze CW, Hsueh B, Aswendt M, Li Y, Tian Q, Cheng M, Crow A, Steinberg GK, McNab JA, Deisseroth KA, Zeineh M. Multimodal image registration and connectivity analysis for integration of connectomic data from microscopy to MRI. *Nature Communications*. 2019 Dec 3;10.0(1.0):5504. doi: 10.1038/s41467-019-13374-0. PMID: 31796741.

9. Matryba P, Kaczmarek L, Gołąb J. Advances in Ex Situ Tissue Optical Clearing. *Laser & Photonics Review*. 2019 Jul 15;13.0(8.0). doi: 10.1002/lpor.201800292.

10. Filo S, Shtangel O, Salamon N, Kol A, Weisinger B, Shifman S, Mezer A. Disentangling molecular alterations from water-content changes in the aging human brain using quantitative MRI. *Nature Communications*. 2019 Jul 30;10.0(1.0):3403. doi: 10.1038/s41467-019-11319-1. PMID: 31363094.

11. Johnson JTE, Irfanoglu MO, Manninen E, Ross TJ, Yang Y, Laun FB, Martin J, Topgaard D, Benjamini D. In vivo disentanglement of diffusion frequency-dependence, tensor shape, and relaxation using multidimensional MRI. *Human Brain Mapping*. 2024 May 1;45.0(7.0):e26697. doi: 10.1002/hbm.26697. PMID: 37987005.

12. Tisdall L, MacNiven KH, Leong JK, Frey R, Rieskamp J, Hertwig R, Knutson BD, Mata R. Structural projections to the nucleus accumbens link to impulsive components of human risk preference. *Imaging Neuroscience*. 2024 Jan 1;2.0:1-15. doi: 10.1162/imag_a_00344. PMID: 40800310.

13. Williamson NH, Ravin R, Benjamini D, Merkle H, Falgairolle M, O'Donovan MJ, Blivis D, Ide D, Cai TX, Ghorashi NS, Bai R, Basser PJ. Magnetic resonance measurements of cellular and sub-cellular membrane structures in live and fixed neural tissue. *eLife*. 2019 Dec 12;8.0. doi: 10.7554/elife.51101. PMID: 31829935.

14. Turner R. Myelin and Modeling: Bootstrapping Cortical Microcircuits. *Frontiers in Neural Circuits*. 2019 May 8;13.0:34. doi: 10.3389/fncir.2019.00034. PMID: 31133821.

15. Kwon D, Pfefferbaum A, Sullivan EV, Pohl KM. Regional growth trajectories of cortical myelination in adolescents and young adults: longitudinal validation and functional correlates. *Brain Imaging and Behavior*. 2018 Nov 8;14.0(1.0):242-266. doi: 10.1007/s11682-018-9980-3. PMID: 30406353.

16. Leuze CW, Goubran M, Baraković M, Aswendt M, Tian Q, Hsueh B, Crow A, Weber EMM, Steinberg GK, Zeineh M, Plowey ED, Daducci A, Innocenti G, Thiran J, Deisseroth KA, McNab JA. Comparison of diffusion MRI and CLARITY fiber orientation estimates in both gray and white matter regions of human and primate brain. *NeuroImage*. 2020 Dec 29;228.0:117692. doi: 10.1016/j.neuroimage.2020.117692. PMID: 33385546.

17. MacNiven KH, Leong JK, Knutson BD. Medial forebrain bundle structure is linked to human impulsivity. *Science Advances*. 2020 Sep 16;6.0(38.0). doi: 10.1126/sciadv.aba4788. PMID: 32938676.

18. Bogusiewicz J, Burlikowska K, Łuczykowski K, Jaroch K, Birski M, Furtak J, Harat M, Pawliszyn JB, Bojko B. New chemical biopsy tool for spatially resolved profiling of human brain tissue in vivo. *Scientific Reports*. 2021 Sep 30;11.0(1.0):19522. doi: 10.1038/s41598-021-98973-y. PMID: 34593948.

19. Kenkhuis B, Jonkman LE, Bulk M, Buijs M, Boon BD, Bouwman FH, Geurts JJ, van de Berg WDJ, van der Weerd L. 7T MRI allows detection of disturbed cortical lamination of the medial temporal lobe in patients with Alzheimer's disease. *NeuroImage Clinical*. 2019 Jan 1;21.0:101665. doi: 10.1016/j.nicl.2019.101665. PMID: 30642758.

20. Paquola C, Hong S. The Potential of Myelin-Sensitive Imaging: Redefining Spatiotemporal Patterns of Myeloarchitecture. *Biological Psychiatry*. 2022 Sep 9;93.0(5.0):442-454. doi: 10.1016/j.biopsych.2022.08.031. PMID: 36481065.

21. Gülban ÖF, Bollmann S, Huber L, Wagstyl K, Goebel R, Poser BA, Kay K, Ivanov D. Mesoscopic in vivo human T 2 * dataset acquired using quantitative MRI at 7 Tesla. *NeuroImage*. 2022 Nov 11;264.0:119733. doi: 10.1016/j.neuroimage.2022.119733. PMID: 36375782.

22. Panchuelo RMS, Mougin OE, Turner R, Francis S. Quantitative T1 mapping using multi-slice multi-shot inversion recovery EPI. *NeuroImage*. 2021 Mar 26;234.0:117976. doi: 10.1016/j.neuroimage.2021.117976. PMID: 33781969.

23. Tisdall L, MacNiven KH, Padula CB, Leong JK, Knutson BD. Brain tract structure predicts relapse to stimulant drug use. *Proceedings of the National Academy of Sciences*. 2022 Jun 21;119.0(26.0):e2116703119. doi: 10.1073/pnas.2116703119. PMID: 35727973.

24. Haenelt D, Trampel R, Nasr S, Polimeni JR, Tootell RBH, Sereno MI, Pine K, Edwards L, Helbling S, Weiskopf N. High-resolution quantitative and functional MRI indicate lower myelination of thin and thick stripes in human secondary visual cortex. *eLife*. 2023 Feb 9;12.0. doi: 10.7554/elife.78756. PMID: 36888685.

25. Kauppinen RA, Thotland J, Pisharady PK, Lenglet C, Garwood M. White matter microstructure and longitudinal relaxation time anisotropy in human brain at 3 and 7 T. *NMR in Biomedicine*. 2022 Aug 22;36.0(1.0):e4815. doi: 10.1002/nbm.4815. PMID: 35994269.

26. Lee K, Lai HM, Sørensen MH, Hui ESK, Wan-San V, Cho WCS, Ho JYS, Chang RC. Optimised tissue clearing minimises distortion and destruction during tissue delipidation. *Neuropathology and Applied Neurobiology*. 2020 Oct 26;47.0(3.0):441-453. doi: 10.1111/nan.12673. PMID: 33107057.

27. Perens J, Salinas CG, Roostalu U, Skytte JL, Gundlach C, Hecksher-Sørensen J, Dahl AB, Dyrby TB. Multimodal 3D Mouse Brain Atlas Framework with the Skull-Derived Coordinate System. *Neuroinformatics*. 2023 Feb 21;21.0(2.0):269-286. doi: 10.1007/s12021-023-09623-9. PMID: 36809643.

28. Leong JK, Ho TC, Colich NL, Sisk LM, Knutson BD, Gotlib IH. White-matter tract connecting anterior insula to nucleus accumbens predicts greater future motivation in adolescents. *Developmental Cognitive Neuroscience*. 2020 Nov 10;47.0:100881. doi: 10.1016/j.dcn.2020.100881. PMID: 33373886.

29. Williamson NH, Witherspoon VJ, Cai TX, Ravin R, Horkay F, Basser PJ. Low-field, high-gradient NMR shows diffusion contrast consistent with localization or motional averaging of water near surfaces. *Magnetic Resonance Letters*. 2023 Apr 8;3.0(2.0):90-107. doi: 10.1016/j.mrl.2023.03.009. PMID: 40918000.

30. Menzel M, Gräßel D, Rajković I, Zeineh M, Georgiadis M. Using light and X-ray scattering to untangle complex neuronal orientations and validate diffusion MRI. *eLife*. 2023 May 11;12.0. doi: 10.7554/elife.84024. PMID: 37166005.

31. Yablonskiy DA, Sukstanskii AL. Quantum dipole interactions and transient hydrogen bond orientation order in cells, cellular membranes and myelin sheath: Implications for MRI signal relaxation, anisotropy, and T1 magnetic field dependence. *Magnetic Resonance in Medicine*. 2024 Jan 19;91.0(6.0):2597-2611. doi: 10.1002/mrm.29996. PMID: 38241135.

32. Shanshool AS, Ziaee S, Ansari MA, Tuchin VV. Advances in the transport of laser radiation to the brain with optical clearing: From simulation to reality. *Progress in Quantum Electronics*. 2024 Feb 24;94.0:100506. doi: 10.1016/j.pquantelec.2024.100506.

33. Lee S, Kim KW, Initiative ADN. Associations between texture of T1-weighted magnetic resonance imaging and radiographic pathologies in Alzheimer's disease. *European Journal of Neurology*. 2020 Oct 24;28.0(3.0):735-744. doi: 10.1111/ene.14609. PMID: 33098172.

34. Schyboll F, Jaekel U, Petruccione F, Neeb H. Dipolar induced spin-lattice relaxation in the myelin sheath: A molecular dynamics study. *Scientific Reports*. 2019 Oct 15;9.0(1.0):14813. doi: 10.1038/s41598-019-51003-4. PMID: 31616004.

35. Autio JA, Uematsu A, Ikeda T, Ose T, Hou Y, Magrou L, Kimura I, Ohno M, Murata K, Coalson T, Kennedy H, Glasser MF, Van Essen DC, Hayashi T. Charting cortical-layer specific area boundaries using Gibbs' ringing attenuated T1w/T2w-FLAIR myelin MRI. *Proceedings on CD-ROM - International Society for Magnetic Resonance in Medicine. Scientific Meeting and Exhibition/Proceedings of the International Society for Magnetic Resonance in Medicine, Scientific Meeting and Exhibition*. 2025 Sep 16;. doi: 10.58530/2025/4869. PMID: 39386722.

36. Willett MR, Codd SL, Seymour JD, Kirkland CM. Relaxation-weighted MRI analysis of biofilm EPS: Differentiating biopolymers, cells, and water. *Biofilm*. 2024 Oct 25;8.0:100235. doi: 10.1016/j.bioflm.2024.100235. PMID: 39610831.

37. Donadieu M, Lee NJ, Gaitán MI, Ha S, Luciano NJ, Roy S, Ineichen BV, Leibovitch EC, Yen CC, Pham DL, Silva AC, Johnson M, Jacobson S, Sati P, Reich DS. In vivo MRI is sensitive to remyelination in a nonhuman primate model of multiple sclerosis. *eLife*. 2023 Apr 21;12.0. doi: 10.7554/elife.73786. PMID: 37083540.

38. Radunsky D, Solomon C, Stern N, Blumenfeld-Katzir T, Filo S, Mezer A, Karsa A, Shmueli K, Soustelle L, Duhamel G, Girard OM, Kepler G, Shrot S, Hoffmann C, Ben-Eliezer N. A comprehensive protocol for quantitative magnetic resonance imaging of the brain at 3 Tesla. *PLoS ONE*. 2024 May 31;19.0(5.0):e0297244. doi: 10.1371/journal.pone.0297244. PMID: 38820354.

39. Di Zheng, Li Y, Kashif MF, Piscopo L, De Vittorio M, Pisanello F. Nanomaterial-Integrated Fiber Neural Probes for Deep Brain Monitoring and Modulation: Challenges and Opportunities. *Advanced Functional Materials*. 2025 Sep 13;36.0(11.0). doi: 10.1002/adfm.202517054.

40. Baek K, Jung S, Lee J, Min E, Jung W, Cho H. Quantitative assessment of regional variation in tissue clearing efficiency using optical coherence tomography (OCT) and magnetic resonance imaging (MRI): A feasibility study. *Scientific Reports*. 2019 Feb 27;9.0(1.0):2923. doi: 10.1038/s41598-019-39634-z. PMID: 30814611.

41. Shtangel O, Mezer A. A phantom system for assessing the effects of membrane lipids on water proton relaxation. *NMR in Biomedicine*. 2020 Jan 3;33.0(4.0):e4209. doi: 10.1002/nbm.4209. PMID: 31899589.

42. Thankarajan E, Oz S, Saady A, Kulbitski K, Kompanets M, Eisen MS, Berlin S. SNAP-Tag-Targeted MRI-Fluorescent Multimodal Probes. *ChemBioChem*. 2023 Apr 24;24.0(16.0):e202300172. doi: 10.1002/cbic.202300172. PMID: 37092744.

43. Mahani FSN, Kalantari A, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic structural connectivity changes in cortical and cortico-striatal strokes in mice. *Neural Regeneration Research*. 2026 Feb 28;21.0(10.0):4945-4953. doi: 10.4103/nrr.nrr-d-25-00491. PMID: 41778731.

44. Mahani FSN, Kalantari A, Fink GR, Hoehn M, Aswendt M. A systematic review of the relationship between magnetic resonance imaging based resting-state and structural networks in the rodent brain. *Frontiers in Neuroscience*. 2023 Jul 24;17.0:1194630. doi: 10.3389/fnins.2023.1194630. PMID: 37554291.

45. Schulz J, Brandl F, Grothe MJ, Kirschner M, Kaiser S, Schmidt A, Borgwardt S, Priller J, Sorg C, Avram M. Basal-Forebrain Cholinergic Nuclei Alterations are Associated With Medication and Cognitive Deficits Across the Schizophrenia Spectrum. *Schizophrenia Bulletin*. 2023 Aug 22;49.0(6.0):1530-1541. doi: 10.1093/schbul/sbad118. PMID: 37606273.

46. Aumiller M, Arazar A, Sroka RC, Dietrich O, Rühm A. Investigations on correlations between changes of optical tissue properties and NMR relaxation times. *Photodiagnosis and Photodynamic Therapy*. 2024 Jan 10;45.0:103968. doi: 10.1016/j.pdpdt.2024.103968. PMID: 38215958.

47. Duan J, Gong X, Womer FY, Sun K, Tang L, Liu J, Zheng J, Zhu Y, Tang Y, Zhang X, Wang F. Neurodevelopmental trajectories, polygenic risk, and lipometabolism in vulnerability and resilience to schizophrenia. *BMC Psychiatry*. 2023 Mar 9;23.0(1.0):153. doi: 10.1186/s12888-023-04597-z. PMID: 36894907.

48. Kruggel F, Solodkin A. Analyzing the cortical fine structure as revealed by ex-vivo anatomical MRI. *The Journal of Comparative Neurology*. 2023 Jul 31;531.0(18.0):2146-2161. doi: 10.1002/cne.25532. PMID: 37522626.

49. Tyson AL, Akhtar AM, Andreae LC. Optimisation and validation of hydrogel-based brain tissue clearing shows uniform expansion across anatomical regions and spatial scales. *Scientific Reports*. 2019 Aug 19;9.0(1.0):12084. doi: 10.1038/s41598-019-48460-2. PMID: 31427619.

50. Liu J, Liu Z, Wei Y, Zhang Y, Womer FY, Duan J, Wei S, Wu F, Kong L, Jiang X, Zhang L, Tang Y, Zhang X, Wang F. Combinatorial panel with endophenotypes from multilevel information of diffusion tensor imaging and lipid profile as predictors for depression. *Australian & New Zealand Journal of Psychiatry*. 2022 May 27;56.0(9.0):1187-1198. doi: 10.1177/00048674211031477. PMID: 35632993.

51. Oz S, Saar G, Olszakier S, Heinrich R, Kompanets MO, Berlin S. Revealing the MRI-Contrast in Optically Cleared Brains. *Advanced Science*. 2024 Apr 22;11.0(22.0):e2400316. doi: 10.1002/advs.202400316. PMID: 38647385.

52. MacKay AL, Laule C. Relaxometry: Applications in the Brain. *Advances in magnetic resonance technology and applications*. 2020 Jan 1;:149-184. doi: 10.1016/b978-0-12-817057-1.00009-3.

53. Williamson NH, Ravin R, Benjamini D, Merkle H, Falgairolle M, O'Donovan MJ, Blivis D, Ide D, Cai TX, Ghorashi NS, Bai R, Basser PJ. Magnetic resonance measurements of cellular and sub-cellular membrane structures in live and fixed neural tissue. *bioRxiv (Cold Spring Harbor Laboratory)*. 2019 Jul 7;. doi: 10.1101/694661.

54. Magdoom KN, Avram AV, Witzel T, Huang SY, Basser PJ. Water Diffusion in the Live Human Brain is Gaussian at the Mesoscale. *bioRxiv (Cold Spring Harbor Laboratory)*. 2024 Apr 14;. doi: 10.1101/2024.04.10.588939. PMID: 38645264.

55. Oz S, Saar G, Olszakier S, Heinrich R, Kompanets M, Berlin S. Tissue expansion of ‘cleared’ whole brains reduces contrast in ex vivo MRI. *Research Square*. 2021 Sep 14;. doi: 10.21203/rs.3.rs-877945/v1.

56. Oz S, Saar G, Olszakier S, Heinrich R, Kompanets MO, Berlin S. Datasets assessing lipid-content in optically cleared brains. *Data in Brief*. 2023 Nov 30;52.0:109795. doi: 10.1016/j.dib.2023.109795. PMID: 38146303.

57. Wu Q. Bring CLARITY to Temporal Lobe Epilepsy: 3D Visualization of p-Tau(Ser262) and 14-3-3 Zeta. *International Journal of Biomedical Science and Engineering*. 2017 Jan 1;5.0(6.0):63. doi: 10.11648/j.ijbse.20170506.11.

58. Song H. Application of CLARITY to Investigate the 3D Architecture of 14-3-3 Zeta Protein in AD. *International Journal of Biomedical Science and Engineering*. 2017 Jan 1;5.0(6.0):68. doi: 10.11648/j.ijbse.20170506.12.

59. Bojko B, Bogusiewicz J, Burlikowska K, Łuczykowski K, Jaroch K, Birski M, Furtak J, Harat M, Pawliszyn JB. SPME Probes: A Novel Chemical Biopsy Tool for Spatially Resolved Profiling of Human Brain Tissue in vivo. *Research Square*. 2021 Jul 7;. doi: 10.21203/rs.3.rs-668958/v1.

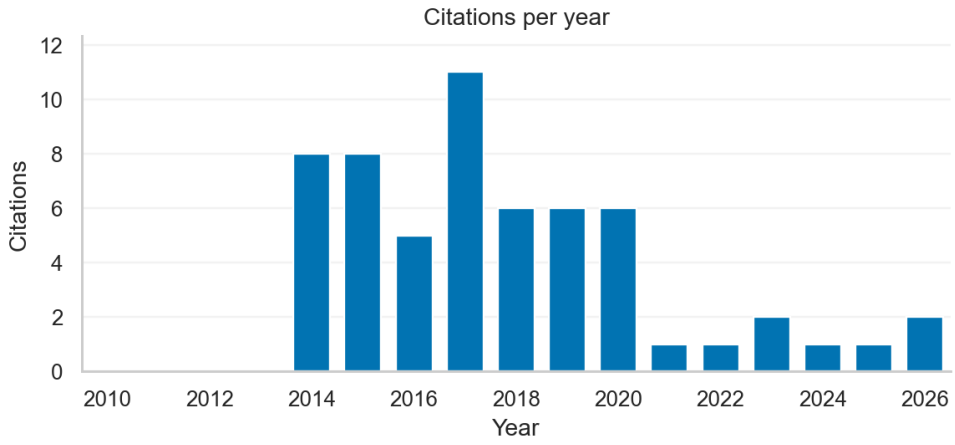
60. Tisdall L, MacNiven KH, Leong JK, Frey R, Rieskamp J, Hertwig R, Knutson BD, Mata R. Structure of connections to the nucleus accumbens link to specific but not general measures of human risk preference. *MPG.PuRe (Max Planck Society)*. 2023 Oct 31;. doi: 10.31219/osf.io/ab4vr.

61. Helbling S, Meyer SS, Weiskopf N. Myelin-informed forward models for M/EEG source reconstruction. *bioRxiv (Cold Spring Harbor Laboratory)*. 2024 Jul 2;. doi: 10.1101/2024.06.30.601378.

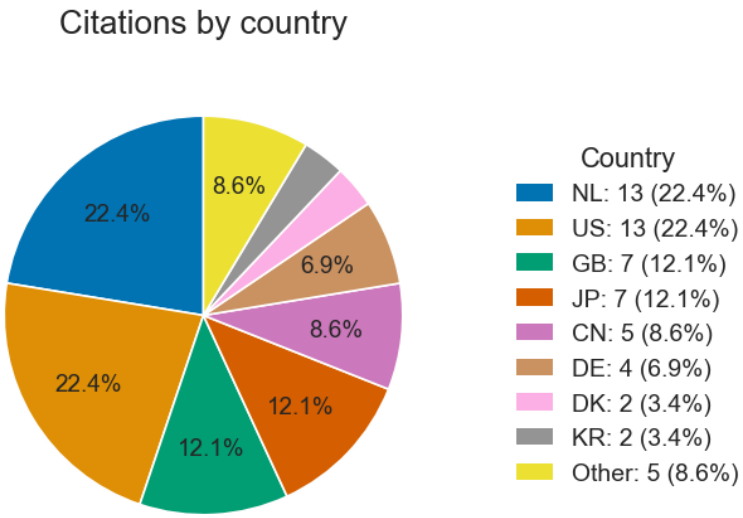
62. Leong JK, Padula CB, Knutson BD, MacNiven KH, Tisdall L. Brain tract structure predicts relapse to stimulant drug use. *OSF Preprints (OSF Preprints)*. 2022 Jun 20;.

Publication

Mezzanotte L, Aswendt M, Tennstaedt A, Hoeben RC, Hoehn M, Löwik CW. Evaluating reporter genes of different luciferases for optimized in vivo bioluminescence imaging of transplanted neural stem cells in the brain. *Contrast Media & Molecular Imaging*. 2013 Nov 1;8(6):505-513. doi: 10.1002/cmmi.1549. PMID: 24375906.



Citation summary
Total citations: **58**
Average/year: **4.14**
Period: 2013-2026



Cited by

1. Iwano S, Sugiyama M, Hama H, Watakabe A, Hasegawa N, Kuchimaru T, Tanaka KZ, Takahashi M, Ishida Y, Hata J, Shimozono S, Namiki K, Fukano T, Kiyama M, Okano H, Kizaka-Kondoh S, McHugh TJ, Yamamori T, Hioki H, Maki S, Miyawaki A. Single-cell bioluminescence imaging of deep tissue in freely moving animals. *Science*. 2018 Feb 22;359.0(6378.0):935-939. doi: 10.1126/science.aaq1067. PMID: 29472486.

2. Chu J, Oh Y, Sens A, Ataie N, Dana H, Macklin JJ, Laviv T, Welf ES, Dean KM, Zhang F, Kim BB, Tang CT, Hu M, Baird MA, Davidson MW, Kay MA, Fiolka RP, Yasuda R, Kim DS, Ng HL, Lin MZ. A bright cyan-excitable orange fluorescent protein facilitates dual-emission microscopy and enhances bioluminescence imaging in vivo. *Nature Biotechnology*. 2016 May 30;34.0(7.0):760-767. doi: 10.1038/nbt.3550. PMID: 27240196.

3. Kim JE, Kalimuthu S, Ahn B. In Vivo Cell Tracking with Bioluminescence Imaging. *Nuclear Medicine and Molecular Imaging*. 2014 Nov 26;49.0(1.0):3-10. doi: 10.1007/s13139-014-0309-x. PMID: 25774232.

4. Hall MP, Woodroofe CC, Wood MG, Que I, Root MV', Ridwan YR, Shi C, Kirkland TA, Encell LP, Wood KV, Löwik CW, Mezzanotte L. Click beetle luciferase mutant and near infrared naphthyl-luciferins for improved bioluminescence imaging. *Nature Communications*. 2018 Jan 3;9.0(1.0):132. doi: 10.1038/s41467-017-02542-9. PMID: 29317625.

5. Adams ST, Miller SC. Beyond D-luciferin: expanding the scope of bioluminescence imaging in vivo. *Current Opinion in Chemical Biology*. 2014 Aug 1;21.0:112-120. doi: 10.1016/j.cbpa.2014.07.003. PMID: 25078002.

6. Schaub FX, Reza MS, Flaveny CA, Li W, Musicant AM, Hoxha S, Guo M, Cleveland JL, Amelio AL. Fluorophore-NanoLuc BRET Reporters Enable Sensitive In Vivo Optical Imaging and Flow Cytometry for Monitoring Tumorigenesis. *Cancer Research*. 2015 Sep 30;75.0(23.0):5023-5033. doi: 10.1158/0008-5472.can-14-3538. PMID: 26424696.

7. Zhang B, Yan W, Zhu Y, Yang W, Le W, Chen B, Zhu R, Cheng L. Nanomaterials in Neural-Stem-Cell-Mediated Regenerative Medicine: Imaging and Treatment of Neurological Diseases. *Advanced Materials*. 2018 Mar 15;30.0(17.0):e1705694. doi: 10.1002/adma.201705694. PMID: 29543350.

8. Tung JK, Berglund K, Gutekunst C, Hochgeschwender U, Gross RE. Bioluminescence imaging in live cells and animals. *Neurophotonics*. 2016 Apr 5;3.0(2.0):1-6. doi: 10.1117/1.nph.3.2.025001. PMID: 27226972.

9. Serkova NJ, Glunde K, Haney CR, Farhoud M, De Lille A, Redente EF, Simberg D, Westerly D, Griffin LR, Mason RP. Preclinical Applications of Multi-Platform Imaging in Animal Models of Cancer. *Cancer Research*. 2020 Dec 1;81.0(5.0):1189-1200. doi: 10.1158/0008-5472.can-20-0373. PMID: 33262127.

10. De Niz M, Helm S, Horstmann S, Annoura T, Del Portillo HA, Khan SM, Heussler VT. In Vivo and In Vitro Characterization of a Plasmodium Liver Stage-Specific Promoter. *PLoS ONE*. 2015 Apr 15;10.0(4.0):e0123473. doi: 10.1371/journal.pone.0123473. PMID: 25874388.

11. Suzuki K, Nagai T. Recent progress in expanding the chemiluminescent toolbox for bioimaging. *Current Opinion in Biotechnology*. 2017 May 5;48.0:135-141. doi: 10.1016/j.copbio.2017.04.001. PMID: 28482221.

12. Serganova IS, Blasberg RG. Molecular Imaging with Reporter Genes: Has Its Promise Been Delivered?. *Journal of Nuclear Medicine*. 2019 Dec 1;60.0(12.0):1665-1681. doi: 10.2967/jnumed.118.220004. PMID: 31792128.

13. Tennstaedt A, Aswendt M, Adamczak JM, Collienne U, Selt M, Schneider G, Henn N, Schaefer C, Lagouge M, Wiedermann D, Kloppenburg P, Hoehn M. Human neural stem cell intracerebral grafts show spontaneous early neuronal differentiation after several weeks. *Biomaterials*. 2015 Jan 12;44.0:143-154. doi: 10.1016/j.biomaterials.2014.12.038. PMID: 25617134.

14. Coutant EP, Janin YL. Synthetic Routes to Coelenterazine and Other Imidazo[1,2- a]pyrazin-3-one Luciferins: Essential Tools for Bioluminescence-Based Investigations. *Chemistry - A European Journal*. 2015 Sep 25;21.0(48.0):17158-17171. doi: 10.1002/chem.201501531. PMID: 26403487.

15. Li R, Liang J, He Y, Qin J, He H, Lee S, Pang Z, Wang J. Sustained Release of Immunosuppressant by Nanoparticle-anchoring Hydrogel Scaffold Improved the Survival of Transplanted Stem Cells and Tissue Regeneration. *Theranostics*. 2017 Dec 27;8.0(4.0):878-893. doi: 10.7150/thno.22072. PMID: 29463988.

16. Kleinovink JW, Mezzanotte L, Zambito G, Fransen MF, Cruz LJ, Verbeek JS, Chan AB, OSSENDORP FA, Löwik CW. A Dual-Color Bioluminescence Reporter Mouse for Simultaneous in vivo Imaging of T Cell Localization and Function. *Frontiers in Immunology*. 2019 Jan 8;9.0:3097. doi: 10.3389/fimmu.2018.03097. PMID: 30671062.

17. Jurgielewicz P, Harmsen S, Wei EY, Bachmann MH, Ting R, Aras Ö. New imaging probes to track cell fate: reporter genes in stem cell research. *Cellular and Molecular Life Sciences*. 2017 Jul 3;74.0(24.0):4455-4469. doi: 10.1007/s00018-017-2584-z. PMID: 28674728.

18. Comenge J, Sharkey JW, Fragueiro O, Wilm B, Brust M, Murray PA, Lévy R, Plagge A. Multimodal cell tracking from systemic administration to tumour growth by combining gold nanorods and reporter genes. *eLife*. 2018 Jun 26;7.0. doi: 10.7554/elife.33140. PMID: 29949503.

19. Aswendt M, Vogel S, Schäfer C, Jathoul AP, Pulé M, Hoehn M. Quantitative in vivo dual-color bioluminescence imaging in the mouse brain. *Neurophotonics*. 2019 May 7;6.0(2.0):1-11. doi: 10.1117/1.nph.6.2.025006. PMID: 31093514.

20. Aswendt M, Adamczak JM, Tennstaedt A. A review of novel optical imaging strategies of the stroke pathology and stem cell therapy in stroke. *Frontiers in Cellular Neuroscience*. 2014 Aug 14;8.0:226. doi: 10.3389/fncel.2014.00226. PMID: 25177269.

21. Kim SB, Paulmurugan R. Bioluminescent Imaging Systems for Assay Developments. *Analytical Sciences*. 2020 Sep 17;37.0(2.0):233-247. doi: 10.2116/analsci.20r003. PMID: 32963202.

22. Yuan M, Ma X, Jiang T, Zhang C, Chen H, Gao Y, Yang X, Du L, Li M. A novel coelenterate luciferin-based luminescent probe for selective and sensitive detection of thiophenols. *Organic & Biomolecular Chemistry*. 2016 Jan 1;14.0(43.0):10267-10274. doi: 10.1039/c6ob02038k. PMID: 27747358.

23. Bajgain P, Chavez AGT, Balasubramanian K, Fleckenstein L, Lulla P, Heslop HE, Vera JF, Leen AM. Secreted Fas Decoys Enhance the Antitumor Activity of Engineered and Bystander T Cells in Fas Ligand-Expressing Solid Tumors. *Cancer Immunology Research*. 2022 Sep 19;10.0(11.0):1370-1385. doi: 10.1158/2326-6066.cir-22-0115. PMID: 36122411.

24. Branchini BR, Southworth TL, Fontaine DM, Kohrt D, Welcome FS, Florentine CM, Henricks ER, DeBartolo DB, Michelini E, Cevenini L, Roda A, Grossel MJ. Red-emitting chimeric firefly luciferase for in vivo imaging in low ATP cellular environments. *Analytical Biochemistry*. 2017 Jul 4;534.0:36-39. doi: 10.1016/j.ab.2017.07.001. PMID: 28687486.

25. Nomura N, Nishihara R, Nakajima T, Kim SB, Iwasawa N, Hiruta Y, Nishiyama S, Sato M, Citterio D, Suzuki KAO. Biothiol-Activatable Bioluminescent Coelenterazine Derivative for Molecular Imaging in Vitro and in Vivo. *Analytical Chemistry*. 2019 Jul 11;91.0(15.0):9546-9553. doi: 10.1021/acs.analchem.9b00694. PMID: 31291724.

26. Zambito G, Gaspar N, Ridwan YR, Hall MP, Shi C, Kirkland TA, Encell LP, Löwik CW, Mezzanotte L. Evaluating Brightness and Spectral Properties of Click Beetle and Firefly Luciferases Using Luciferin Analogues: Identification of Preferred Pairings of Luciferase and Substrate for In Vivo Bioluminescence Imaging. *Molecular Imaging and Biology*. 2020 Sep 14;22.0(6.0):1523-1531. doi: 10.1007/s11307-020-01523-7. PMID: 32926287.

27. Ashraf S, Taylor A, Sharkey JW, Barrow M, Murray PA, Wilm B, Poptani H, Rosseinsky MJ, Adams DJ, Lévy R. In vivo fate of free and encapsulated iron oxide nanoparticles after injection of labelled stem cells. *Nanoscale Advances*. 2018 Sep 25;1.0(1.0):367-377. doi: 10.1039/c8na00098k. PMID: 36132463.

28. Birkner E, Berglund K, Klein ME, Augustine GJ, Hochgeschwender U. Non-invasive activation of optogenetic actuators. *Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE*. 2014 Mar 5;8928.0. doi: 10.1117/12.2044157. PMID: 27965518.

29. Yuan M, Ma X, Jiang T, Gao Y, Cui Y, Zhang C, Yang X, Huang Y, Du L, Yampolsky IV, Li M. Prolonged bioluminescence imaging in living cells and mice using novel pro-substrates for Renilla luciferase. *Organic & Biomolecular Chemistry*. 2017 Jan 1;15.0(48.0):10238-10244. doi: 10.1039/c7ob01656e. PMID: 29177293.

30. Yasunaga M, Nakajima Y, Ohmiya Y. Dual-color bioluminescence imaging assay using green- and red-emitting beetle luciferases at subcellular resolution. *Analytical and Bioanalytical Chemistry*. 2014 Jul 11;406.0(23.0):5735-5742. doi: 10.1007/s00216-014-7981-7. PMID: 25015042.

31. Dimond A, Van de Pette M, Fisher AG. Illuminating Epigenetics and Inheritance in the Immune System with Bioluminescence. *Trends in Immunology*. 2020 Oct 6;41.0(11.0):994-1005. doi: 10.1016/j.it.2020.09.001. PMID: 33036908.

32. Nakashiba T, Ogoh K, Iwano S, Sugiyama T, Mizuno-Iijima S, Nakashima K, Mizuno S, Sugiyama F, Yoshiki A, Miyawaki A, Abe K. Development of two mouse strains conditionally expressing bright luciferases with distinct emission spectra as new tools for in vivo imaging. *Lab Animal*. 2023 Sep 7;52.0(10.0):247-257. doi: 10.1038/s41684-023-01238-6. PMID: 37679611.

33. Kim SB, Furuta T. Bioluminescence from the bright and dark sides. *Frontiers in Chemical Biology*. 2024 Aug 19;3.0. doi: 10.3389/fchbi.2024.1459397.

34. Wang C, Chen Q, Yang Z, Zhang C, Song M, Xing X, Ma W, Chen C, Zhang L, Gong M, Zhang K, Yuan Z, Liu Z, Zhang J. A Triple-Fusion Protein Probe for Multimodal Imaging of Neural Lineage Cells In Vitro and After Transplantation in Brain. *Advanced Functional Materials*. 2025 May 22;35.0(44.0). doi: 10.1002/adfm.202422870.

35. Cevenini L, Calabretta MM, Calabria D, Roda A, Michelini E. Luciferase Genes as Reporter Reactions: How to Use Them in Molecular Biology?. *Advances in biochemical engineering, biotechnology*. 2015 Jan 1;154.0:3-17. doi: 10.1007/10_2015_325. PMID: 25898810.

36. van Zaane F, Gamm UA, van Driel PBAA, Snoeks TJA, de Bruijn HS, Heuvel AVDPD, Sterenborg HJCM, Löwik CW, Amelink A, Robinson DJ. Intrinsic photosensitizer fluorescence measured using multi-diameter single-fiber spectroscopy in vivo. *Journal of Biomedical Optics*. 2014 Jan 29;19.0(1.0):1-10. doi: 10.1117/1.jbo.19.1.015010. PMID: 24477382.

37. Mezzanotte L, Iljas JD, Que I, Chan AB, Kaijzel EL, Hoeben RC, Löwik CW. Optimized Longitudinal Monitoring of Stem Cell Grafts in Mouse Brain Using a Novel Bioluminescent/Near Infrared Fluorescent Fusion Reporter. *Cell Transplantation*. 2017 Dec 1;26.0(12.0):1878-1889. doi: 10.1177/0963689717739718. PMID: 29390874.

38. Mata Á, Azevedo HS, Botto L, Gavara N, Su L. New Bioengineering Breakthroughs and Enabling Tools in Regenerative Medicine. *Current Stem Cell Reports*. 2017 May 3;3.0(2.0):83-97. doi: 10.1007/s40778-017-0081-9. PMID: 28596936.

39. Green C, Minassian A, Vogel SN, Diedenhofen M, Wiedermann D, Hoehn M. Persistent Quantitative Vitality of Stem Cell Graft Is Necessary for Stabilization of Functional Brain Networks After Stroke. *Frontiers in Neurology*. 2019 Apr 5;10.0:335. doi: 10.3389/fneur.2019.00335. PMID: 31024429.

40. Madsen SD, Giler MK, Bunnell BA, O'Connor KC. Illuminating the Regenerative Properties of Stem Cells In Vivo with Bioluminescence Imaging. *Biotechnology Journal*. 2020 Oct 22;16.0(3.0):e2000248. doi: 10.1002/biot.202000248. PMID: 33089922.

41. van Zaane F, van Driel PBAA, Gamm UA, Snoeks TJA, de Bruijn HS, Heuvel AVDPD, Löwik CW, Sterenborg HJCM, Amelink A, Robinson DJ. Microscopic analysis of the localization of two chlorin-based photosensitizers in OSC19 tumors in the mouse oral cavity. *Lasers in Surgery and Medicine*. 2014 Jan 16;46.0(3.0):224-234. doi: 10.1002/lsm.22220. PMID: 24436109.

42. Aelvoet S, Pascual-Brazo J, Libbrecht S, Reumers V, Gijsbers R, Haute CVD, Baekelandt V. Long-Term Fate Mapping Using Conditional Lentiviral Vectors Reveals a Continuous Contribution of Radial Glia-Like Cells to Adult Hippocampal Neurogenesis in Mice. *PLoS ONE*. 2015 Nov 23;10.0(11.0):e0143772. doi: 10.1371/journal.pone.0143772. PMID: 26600383.

43. Falk S, Al-Dihaissy T, Mezzanotte L, Heegaard A. Effect of sex in the MRMT-1 model of cancer-induced bone pain. *F1000Research*. 2015 Sep 18;4.0:445. doi: 10.12688/f1000research.6827.2.

44. Schomann T, Mezzanotte L, de Groot JC, Rivolta MN, Hendriks SH, Frijns JHM, Huisman MA. Neuronal differentiation of hair-follicle-bulge-derived stem cells co-cultured with mouse cochlear modiolus explants. *PLoS ONE*. 2017 Oct 30;12.0(10.0):e0187183. doi: 10.1371/journal.pone.0187183. PMID: 29084289.

45. Gleneadie HJ, Dimond A, Fisher AG. Harnessing bioluminescence for drug discovery and epigenetic research. *Frontiers in Drug Discovery*. 2023 Sep 5;3.0. doi: 10.3389/fddsv.2023.1249507.

46. Falk S, Al-Dihaissy T, Mezzanotte L, Heegaard A. Sex-difference affects disease progression in the MRMT-1 model of cancer-induced bone pain. *F1000Research*. 2015 Jul 31;4.0:445. doi: 10.12688/f1000research.6827.1. PMID: 26834983.

47. Zambito G, Mezzanotte L. Near-infrared bioluminescence imaging of two cell populations in living mice. *STAR Protocols*. 2021 Jul 7;2.0(3.0):100662. doi: 10.1016/j.xpro.2021.100662. PMID: 34286293.

48. Schomann T, Mezzanotte L, Lourens IM, de Groot JC, Frijns JHM, Huisman MA. Lentiviral transduction and subsequent loading with nanoparticles do not affect cell viability and proliferation in hair-follicle-bulge-derived stem cellsin vitro. *Contrast Media & Molecular Imaging*. 2016 Nov 1;11.0(6.0):550-560. doi: 10.1002/cmmi.1717. PMID: 27976505.

49. Schomann T, Mezzanotte L, de Groot JC, Löwik CW, Frijns JHM, Huisman MA. Imaging Bioluminescent Exogenous Stem Cells in the Intact Guinea Pig Cochlea. *The Anatomical Record*. 2019 Jan 12;303.0(3.0):427-440. doi: 10.1002/ar.24068. PMID: 30635981.

50. Schomann T, Iljas JD, Que I, Li Y, Suidgeest E, Cruz LJ, Frijns JHM, Chan AB, Löwik CM, Huisman MA, Mezzanotte L. Multimodal imaging of hair follicle bulge-derived stem cells in a mouse model of traumatic brain injury. *Cell and Tissue Research*. 2020 Feb 8;381.0(1.0):55-69. doi: 10.1007/s00441-020-03173-1. PMID: 32036485.

51. Heo H, Jo J, Jung JI, Han Y, Lee S, Kim S, Kwon S, Kim K, Hwang BJ, Kee Y, Lee BD, Kang D, Her S. Improved dynamic monitoring of transcriptional activity during longitudinal analysis in the mouse brain. *Biology Open*. 2018 Jan 1;8.0(1.0). doi: 10.1242/bio.037168. PMID: 30341106.

52. Rose LC, Bulte JWM. Cell Tracking and Transplant Imaging. *Springer eBooks*. 2017 Jan 1;:593-603. doi: 10.1007/978-3-319-42202-2_23.

53. Aswendt M, Collmann FM, Hoehn M. Neurobiological insights from bioluminescence imaging. *Oncotarget*. 2017 Aug 13;8.0(41.0):69198-69199. doi: 10.18632/oncotarget.20302. PMID: 29050195.

54. Lyons SK. Bioluminescence Imaging. *Springer eBooks*. 2014 Jan 1;:1-6. doi: 10.1007/978-3-642-27841-9_640-3.

55. Kersemans V. Imaging of Cell Trafficking and Cell Tissue Homing. *Imaging in medical diagnosis and therapy*. 2016 Feb 4;:509-525. doi: 10.1201/b19052-37.

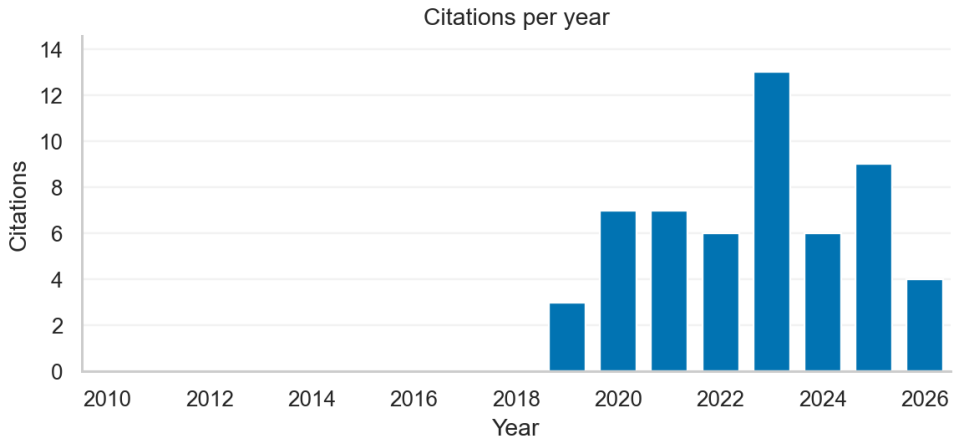
56. Lyons SK. Bioluminescence Imaging. *Springer eBooks*. 2017 Jan 1;:502-507. doi: 10.1007/978-3-662-46875-3_640.

57. Cambiaso E, Motais B, Tieland RG, Pleijte C, Cuenca M, Peperzak V. 3D bioluminescence imaging assays for measuring in vitro CAR T cell cytotoxicity. *Methods in cell biology*. 2026 Jan 1;209.0:123-139. doi: 10.1016/bs.mcb.2026.04.006. PMID: 42442857.

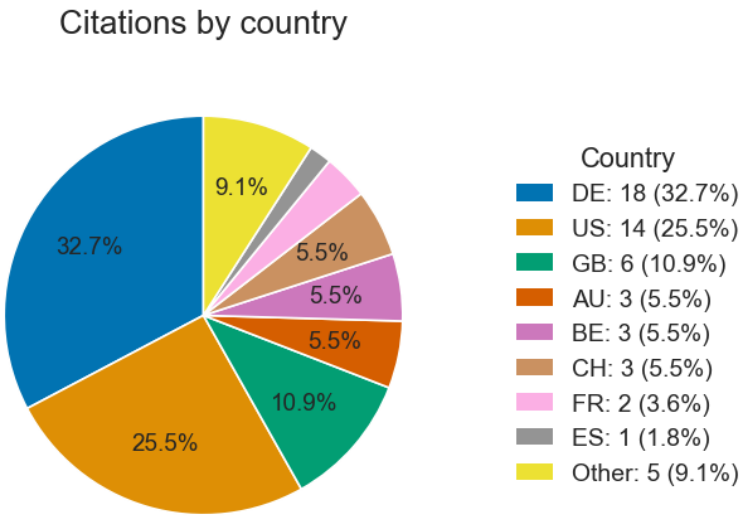
58. Tieland RG, Pleijte C, Rockx-Brouwer D, Motais B, Cambiaso E, Cuenca M, Peperzak V. Measuring tumor cell killing using in vivo bioluminescence imaging. *Methods in cell biology*. 2026 Jan 1;. doi: 10.1016/bs.mcb.2026.05.024.

Publication

Pallast N, Diedenhofen M, Blaschke SJ, Wieters F, Wiedermann D, Hoehn M, Fink GR, Aswendt M. Processing Pipeline for Atlas-Based Imaging Data Analysis of Structural and Functional Mouse Brain MRI (AIDAmri). *Frontiers in Neuroinformatics*. 2019 Jun 4;13:42. doi: 10.3389/fninf.2019.00042. PMID: 31231202.



Citation summary
Total citations: **55**
Average/year: **6.88**
Period: 2019-2026
SCImago cites/doc 2y (2025): **2.8**
SJR 2025: **0.783** (Q2)



Cited by

1. Wang Q, Ding S, Li Y, Royall JJ, Feng DH, Lesnar P, Graddis N, Naeemi M, Facer BA, Ho A, Dolbeare TA, Blanchard B, Dee N, Wakeman W, Hirokawa KE, Szafer A, Sunkin SM, Oh SW, Bernard A, Phillips JW, Hawrylycz M, Koch C, Zeng H, Harris JA, Ng L. The Allen Mouse Brain Common Coordinate Framework: A 3D Reference Atlas. *Cell*. 2020 May 1;181.0(4.0):936-953.e20. doi: 10.1016/j.cell.2020.04.007. PMID: 32386544.

2. Chon U, Vanselow DJ, Cheng KC, Kim Y. Enhanced and unified anatomical labeling for a common mouse brain atlas. *Nature Communications*. 2019 Nov 7;10.0(1.0):5067. doi: 10.1038/s41467-019-13057-w. PMID: 31699990.

3. Brunner C, Grillet M, Urban A, Roska B, Montaldo G, Macé É. Whole-brain functional ultrasound imaging in awake head-fixed mice. *Nature Protocols*. 2021 Jun 4;16.0(7.0):3547-3571. doi: 10.1038/s41596-021-00548-8. PMID: 34089019.

4. Renton A, Dao TT, Johnstone T, Civier O, Sullivan RP, White D, Lyons P, Slade B, Abbott DF, Amos TJ, Bollmann S, Bollmann S, Botting A, Campbell MEJ, Chang J, Close T, Dörig M, Eckstein K, Egan GF, Evas S, Flandin G, Garner K, Garrido MI, Ghosh S, Grignard M, Halchenko YO, Hannan AJ, Heinsfeld AS, Huber L, Hughes ME, Kaczmarzyk J, Kasper L, Kuhlmann L, Lou K, Mantilla-Ramos Y, Mattingley JB, Meier ML, Morris J, Narayanan A, Pestilli F, Puce A, Ribeiro FL, Rogasch NC, Rorden C, Schira MM, Shaw TB, Sowman PF, Spitz G, Stewart AW, Ye X, Zhu JD, Narayanan A, Rorden C, Rorden C. Neurodesk: an accessible, flexible and portable data analysis environment for reproducible neuroimaging. *Nature Methods*. 2024 Jan 8;21.0(5.0):804-808. doi: 10.1038/s41592-023-02145-x. PMID: 38191935.

5. Carey H, Pegios M, Martin L, Saleeba C, Turner AJ, Everett NA, Bjerke IE, Puchades MA, Bjaalie JG, McMullan S. DeepSlice: rapid fully automatic registration of mouse brain imaging to a volumetric atlas. *Nature Communications*. 2023 Sep 21;14.0(1.0):5884. doi: 10.1038/s41467-023-41645-4. PMID: 37735467.

6. Aswendt M, Wilhelmsson U, Wieters F, Stokowska A, Schmitt FJ, Pallast N, de Pablo Y, Mohammed L, Hoehn M, Pekna M, Pekny M. Reactive astrocytes prevent maladaptive plasticity after ischemic stroke. *Progress in Neurobiology*. 2021 Dec 15;209.0:102199. doi: 10.1016/j.pneurobio.2021.102199. PMID: 34921928.

7. Aswendt M, Pallast N, Wieters F, Baues M, Hoehn M, Fink GR. Lesion Size- and Location-Dependent Recruitment of Contralesional Thalamus and Motor Cortex Facilitates Recovery after Stroke in Mice. *Translational Stroke Research*. 2020 Mar 12;12.0(1.0):87-97. doi: 10.1007/s12975-020-00802-3. PMID: 32166716.

8. Blaschke SJ, Hensel L, Minassian A, Vlachakis S, Tscherpel C, Vay SU, Rabenstein M, Schroeter M, Fink GR, Hoehn M, Grefkes C, Rueger MA. Translating Functional Connectivity After Stroke: Functional Magnetic Resonance Imaging Detects Comparable Network Changes in Mice and Humans. *Stroke*. 2021 Jul 20;52.0(9.0):2948-2960. doi: 10.1161/strokeaha.120.032511. PMID: 34281374.

9. Aswendt M, Green C, Sadler R, Llovera G, Dzikowski L, Heindl S, de Agüero MG, Diedenhofen M, Vogel SN, Wieters F, Wiedermann D, Liesz A, Hoehn M. The gut microbiota modulates brain network connectivity under physiological conditions and after acute brain ischemia. *iScience*. 2021 Sep 9;24.0(10.0):103095. doi: 10.1016/j.isci.2021.103095. PMID: 34622150.

10. Krämer J, Brück W, Zipp F, Cerina M, Groppa S, Meuth SG. Imaging in mice and men: Pathophysiological insights into multiple sclerosis from conventional and advanced MRI techniques. *Progress in Neurobiology*. 2019 Jul 30;182.0:101663. doi: 10.1016/j.pneurobio.2019.101663. PMID: 31374243.

11. Pallast N, Wieters F, Nill M, Fink GR, Aswendt M. Graph theoretical quantification of white matter reorganization after cortical stroke in mice. *NeuroImage*. 2020 May 4;217.0:116873. doi: 10.1016/j.neuroimage.2020.116873. PMID: 32380139.

12. Yamasaki E, Thakore P, Ali S, Solano AS, Wang X, Gao XP, Labelle-Dumais C, Chaumeil MM, Gould DB, Earley S. Impaired intracellular Ca 2+ signaling contributes to age-related cerebral small vessel disease in Col4a1 mutant mice. *Science Signaling*. 2023 Nov 14;16.0(811.0):eadi3966. doi: 10.1126/scisignal.adi3966. PMID: 37963192.

13. Pallast N, Wieters F, Fink GR, Aswendt M. Atlas-based imaging data analysis tool for quantitative mouse brain histology (AIDAhisto). *Journal of Neuroscience Methods*. 2019 Aug 12;326.0:108394. doi: 10.1016/j.jneumeth.2019.108394. PMID: 31415844.

14. Blaschke SJ, Vlachakis S, Pallast N, Walter HL, Volz LJ, Wiedermann D, Fink GR, Hoehn M, Aswendt M, Schroeter M, Rueger MA. Transcranial Direct Current Stimulation Reverses Stroke-Induced Network Alterations in Mice. *Stroke*. 2023 Jun 28;54.0(8.0):2145-2155. doi: 10.1161/strokeaha.123.042808. PMID: 37377015.

15. Uselman TW, Medina CS, Gray HB, Jacobs RE, Bearer EL. Longitudinal manganese-enhanced magnetic resonance imaging of neural projections and activity. *NMR in Biomedicine*. 2022 Mar 6;35.0(6.0):e4675. doi: 10.1002/nbm.4675. PMID: 35253280.

16. Lee JJ, Prag HA, Chary K, Abe J, Uno S, Sorby-Adams A, Yu CS, Sauchanka O, Mottahedin A, Kaggie JD, Gallagher FA, Murphy MP, Krieg T. Local arterial administration of acidified malonate as an adjunct therapy to mechanical thrombectomy in ischaemic stroke. *Cardiovascular Research*. 2025 Jun 27;121.0(9.0):1407-1418. doi: 10.1093/cvr/cvaf118. PMID: 40577592.

17. Tustison NJ, Chen M, Kronman FN, Duda J, Gamlin CR, Tustison MG, Kunst M, Dalley RA, Sorensen SA, Wang Q, Ng L, Kim Y, Gee JC. The ANTsX ecosystem for mapping the mouse brain. *Nature Communications*. 2025 Nov 22;16.0(1.0):11548. doi: 10.1038/s41467-025-66741-5. PMID: 38746199.

18. Scharwächter L, Schmitt FJ, Pallast N, Fink GR, Aswendt M. Network analysis of neuroimaging in mice. *NeuroImage*. 2022 Mar 17;253.0:119110. doi: 10.1016/j.neuroimage.2022.119110. PMID: 35311664.

19. Célestine M, Nadkarni NA, Garin CM, Bougacha S, Dhénain M. Sammba-MRI: A Library for Processing SmAll-MaMmal BrAin MRI Data in Python. *Frontiers in Neuroinformatics*. 2020 May 28;14.0:24. doi: 10.3389/fninf.2020.00024. PMID: 32547380.

20. Piekarski DJ, Zahr NM, Zhao Q, Sullivan EV, Pfefferbaum A. Alcohol's effects on the mouse brain are modulated by age and sex. *Addiction Biology*. 2022 Jul 12;27.0(5.0):e13209. doi: 10.1111/adb.13209. PMID: 36001428.

21. Gurdon B, Kaczorowski CC. Pursuit of precision medicine: Systems biology approaches in Alzheimer's disease mouse models. *Neurobiology of Disease*. 2021 Nov 9;161.0:105558. doi: 10.1016/j.nbd.2021.105558. PMID: 34767943.

22. Takata N, Sato N, Komaki Y, Okano H, Tanaka KF. Flexible annotation atlas of the mouse brain: combining and dividing brain structures of the Allen Brain Atlas while maintaining anatomical hierarchy. *Scientific Reports*. 2021 Mar 18;11.0(1.0):6234. doi: 10.1038/s41598-021-85807-0. PMID: 33737651.

23. Young DM, Duhn C, Gilson MC, Nojima M, Yuruk D, Kumar A, Yu W, Sanders S. Whole-Brain Image Analysis and Anatomical Atlas 3D Generation Using MagellanMapper. *Current Protocols in Neuroscience*. 2020 Sep 27;94.0(1.0):e104. doi: 10.1002/cpns.104. PMID: 32981139.

24. Zahr NM, Zhao Q, Goodcase R, Pfefferbaum A. Systemic Administration of the TLR7/8 Agonist Resiquimod (R848) to Mice Is Associated with Transient, In Vivo-Detectable Brain Swelling. *Biology*. 2022 Feb 10;11.0(2.0):274. doi: 10.3390/biology11020274. PMID: 35205140.

25. Xu N, Zhang LY, Larson S, Li Z, Anumba N, Daley L, Pan W, Chuang K, Keilholz SD. Rodent Whole-Brain fMRI Data Preprocessing Toolbox. *Aperture Neuro*. 2023 Aug 9;3.0. doi: 10.52294/001c.85075. PMID: 37654427.

26. Scarpelli M, Healey DR, Mehta SV, Kodibagkar VD, Quarles CC. A practical method for multimodal registration and assessment of whole-brain disease burden using PET, MRI, and optical imaging. *Scientific Reports*. 2020 Oct 14;10.0(1.0):17324. doi: 10.1038/s41598-020-74459-1. PMID: 33057180.

27. Heller C, Kimmig AS, Kubicki MR, Derntl B, Kikinis Z. Imaging the human brain on oral contraceptives: A review of structural imaging methods and implications for future research goals. *Frontiers in Neuroendocrinology*. 2022 Aug

20;67.0:101031. doi: 10.1016/j.yfrne.2022.101031. PMID: 35998859.

28. Ostrova IV, Babkina AS, Lyubomudrov MA, Grechko AV, Golubev AM. Photochemicallly Induced Thrombosis as a Model of Ischemic Stroke. *General Reanimatology*. 2023 Jun 24;19.0(3.0):54-65. doi: 10.15360/1813-9779-2023-3-54-65.

29. Götz J, Wieters F, Fritz VJ, Käsgen O, Kalantari A, Fink GR, Aswendt M. Temporal and Spatial Gene Expression Profile of Stroke Recovery Genes in Mice. *Genes*. 2023 Feb 9;14.0(2.0):454. doi: 10.3390/genes14020454. PMID: 36833381.

30. Aljuraysi S, Platt M, Pulix M, Poptani H, Plagge A. Microcephaly with a disproportionate hippocampal reduction, stem cell loss and neuronal lipid droplet symptoms in Trappc9 KO mice. *Neurobiology of Disease*. 2024 Feb 6;192.0:106431. doi: 10.1016/j.nbd.2024.106431. PMID: 38331351.

31. Llambrich S, Tielemans B, Saliën E, Atzori M, Wouters K, Van Bulck V, Platt M, Vanherp L, Fernandez NG, de la Fuente LG, Poptani H, Verlinden L, Himmelreich U, Croitor A, Attanasio C, Callaerts-Vegh Z, Gsell W, Martínez-Abadías N, Velde GV. Pleiotropic effects of trisomy and pharmacologic modulation on structural, functional, molecular, and genetic systems in a Down syndrome mouse model. *eLife*. 2024 Mar 18;12.0. doi: 10.7554/elife.89763.3.

32. Kalantari A, Hambrock C, Grefkes C, Fink GR, Aswendt M. Proportional recovery in mice with cortical stroke. *Experimental Neurology*. 2025 Feb 4;386.0:115180. doi: 10.1016/j.expneurol.2025.115180. PMID: 39914643.

33. Mahani FSN, Kalantari A, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic structural connectivity changes in cortical and cortico-striatal strokes in mice. *Neural Regeneration Research*. 2026 Feb 28;21.0(10.0):4945-4953. doi: 10.4103/nrr.nrr-d-25-00491. PMID: 41778731.

34. Aswendt M, Wieters F. Structural integrity and remodeling underlying functional recovery after stroke. *Neural Regeneration Research*. 2020 Dec 17;16.0(7.0):1423. doi: 10.4103/1673-5374.301004. PMID: 33318437.

35. Kamouh ME, Spajer M, Singhabahu R, Marolany AJ, Sailor KA, Mouton L, Doukhi D, Koundal S, Metcalfe T, Langui D, Grine A, Boyé K, Leser FS, Ninnemann J, Gottschalk J, Sadeghi C, Han X, Boisserand LSB, Akbar D, Van Wassenhove J, Ruze A, Nasry A, Bourrienne M, Buscher E, Hong Y, Mazighi M, Lledo P, Benveniste H, Santin MD, Eichmann A, Lehéricy S, Thomas J, Lenck S. Cerebral venous blood flow regulates intracerebral pressure and brain clearance via meningeal lymphatic vessels. *Nature Neuroscience*. 2026 Jul 22;29.0(9.0):2190-2203. doi: 10.1038/s41593-026-02358-1. PMID: 42487032.

36. Llambrich S, Tielemans B, Saliën E, Atzori M, Wouters K, Van Bulck V, Platt M, Vanherp L, Fernandez NG, de la Fuente LG, Poptani H, Verlinden L, Himmelreich U, Croitor A, Attanasio C, Callaerts-Vegh Z, Gsell W, Martínez-Abadías N, Velde GV. Pleiotropic effects of trisomy and pharmacologic modulation on structural, functional, molecular, and genetic systems in a Down syndrome mouse model. *eLife*. 2023 Oct 6;12.0. doi: 10.7554/elife.89763. PMID: 38497812.

37. Kalantari A, Shahbazi M, Schneider M, Raikes AC, Frazão VV, Bhattrai A, Carnevale L, Diao Y, Franx BAA, Gammaraccio F, Goncalves L, Lee SD, van Leeuwen EM, Michalek A, Mueller SG, Rivera-Olvera A, Padró D, Selim MK, van der Toorn A, Varriano F, Vrooman RM, Wenk P, Albers HE, Boehm-Sturm P, Budinger E, Canals S, De Santis S, Brinton RD, Dijkhuizen RM, Eixarch E, Forloni G, Grandjean J, Hekmatyar K, Jacobs RE, Jelescu I, Kurniawan ND, Lembo G, Longo DL, Maria NSS, Micotti E, Muñoz-Moreno E, Ramos-Cabrer P, Reichardt W, Sòria G, Ielacqua GD, Aswendt M. Automated quality control of small animal MR neuroimaging data. *Imaging Neuroscience*. 2024 Jan 1;2.0:1-23. doi: 10.1162/imag_a_00317. PMID: 40212822.

38. Uselman TW, Jacobs RE, Bearer EL. Reconfiguration of brain-wide neural activity after early life adversity. *Proceedings of the National Academy of Sciences*. 2025 Oct 17;122.0(42.0):e2506140122. doi: 10.1073/pnas.2506140122. PMID: 38328213.

39. Doelemeyer A, Vaishampayan S, Zurbruegg S, Morvan F, Locatelli G, Shimshek DR, Beckmann N. Deep learning-driven MRI for accurate brain volumetry in murine models of neurodegenerative diseases. *Frontiers in Neuroscience*. 2025 Nov 3;19.0:1632169. doi: 10.3389/fnins.2025.1632169. PMID: 41255818.

40. Gao X, Wang X, Labelle-Dumais C, Gould DB, Chaumeil MM. Multimodal neuroimaging of Col4a1-mutant mouse models of Gould syndrome. *Frontiers in Neuroscience*. 2025 Nov 12;19.0:1639871. doi: 10.3389/fnins.2025.1639871. PMID: 41311701.

41. Mahani F, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic Functional Connectivity Changes in Cortical and Cortico-Striatal Strokes in Mice. *CNS Neuroscience & Therapeutics*. 2026 Aug 1;32.0(8.0):e71114. doi: 10.1002/cns.71114. PMID: 42661330.

42. Im S, Suh J, Shim J, Baek H. Deterministic Tractography Analysis of Rat Brain Using SIGMA Atlas in 9.4T MRI. *Brain Sciences*. 2021 Dec 18;11.0(12.0):1656. doi: 10.3390/brainsci11121656. PMID: 34942958.

43. Renton A, Dao TT, Abbott DF, Amos TJ, Bollmann S, Bollmann S, Botting A, Campbell MEJ, Chang J, Civier O, Close T, Eckstein K, Egan GF, Evas S, Flandin G, Garner K, Garrido MI, Ghosh S, Grignard M, Hannan AJ, Huber L, Hughes ME, Johnstone T, Kaczmarzyk J, Kasper L, Kuhlmann L, Lou K, Lyons P, Mantilla-Ramos Y, Mattingley JB, Morris J, Narayanan A, Pestilli F, Puce A, Ribeiro FL, Rogasch NC, Rorden C, Schira MM, Shaw TB, Slade BM, Spitz G, Stewart AW, Sullivan RP, White D, Ye X, Zhu JD, Narayanan A, Rorden C, Rorden C. Neurodesk: An accessible, flexible, and portable data analysis environment for reproducible neuroimaging. *bioRxiv (Cold Spring Harbor Laboratory)*. 2022 Dec 23;. doi: 10.1101/2022.12.23.521691.

44. Kalantari A, Szczepanik M, Heunis S, Mönch C, Hanke M, Wächtler T, Aswendt M. How to establish and maintain a multimodal animal research dataset using DataLad. *Scientific Data*. 2023 Jun 5;10.0(1.0):357. doi: 10.1038/s41597-023-02242-8. PMID: 37277500.

45. Rorden C, Renton A, Dao TT, Johnstone T, Civier O, Sullivan R, White D, Lyons P, Slade B, Abbott DF, Amos TJ, Bollmann S, Botting A, Campbell MEJ, Chang J, Close T, Eckstein K, Egan GF, Evas S, Flandin G, Garner K, Garrido MI, Ghosh S, Grignard M, Hannan AJ, Huber L, Kaczmarzyk J, Kasper L, Kuhlmann L, Lou K, Mantilla-Ramos Y, Mattingley JB, Morris J, Narayanan A, Pestilli F, Puce A, Ribeiro FL, Rogasch NC, Rorden C, Schira MM, Shaw TB, Sowman PF, Spitz G, Stewart AW, Ye X, Zhu JD, Hughes ME, Narayanan A. Neurodesk: An accessible, flexible, and portable data analysis

environment for reproducible neuroimaging. *Research Square*. 2023 Mar 13;. doi: 10.21203/rs.3.rs-2649734/v1. PMID: 36993557.

46. Llambrich S, Tielemans B, Saliën E, Atzori M, Wouters K, Van Bulck V, Platt M, Vanherp L, Fernandez NG, de la Fuente LG, Poptani H, Verlinden L, Himmelreich U, Croitor A, Attanasio C, Callaerts-Vegh Z, Gsell W, Martínez-Abadías N, Velde GV. Pleiotropic effects of trisomy and pharmacologic modulation on structural, functional, molecular, and genetic systems in a Down syndrome mouse model. *bioRxiv (Cold Spring Harbor Laboratory)*. 2023 Aug 1;. doi: 10.1101/2023.07.31.551282.

47. Aljuraysi S, Platt M, Pulix M, Poptani H, Plagge A. Microcephaly with a disproportionate hippocampal reduction, stem cell loss and neuronal lipid droplet symptoms in Trappc9 KO mice. *bioRxiv (Cold Spring Harbor Laboratory)*. 2023 Nov 20;. doi: 10.1101/2023.11.20.567859.

48. Wendlandt TA, Wenk P, Henschke JU, Michalek A, Noesselt T, Pakan JM, Budinger E. Neural networks representing temporal expectation in mice. *bioRxiv (Cold Spring Harbor Laboratory)*. 2024 Apr 26;. doi: 10.1101/2024.04.25.589683.

49. Llambrich S, Tielemans B, Saliën E, Atzori M, Wouters K, Van Bulck V, Platt M, Vanherp L, Fernandez NG, de la Fuente LG, Poptani H, Verlinden L, Himmelreich U, Croitor A, Attanasio C, Callaerts-Vegh Z, Gsell W, Martínez-Abadías N, Velde GV. Pleiotropic effects of trisomy and pharmacologic modulation on structural, functional, molecular, and genetic systems in a Down syndrome mouse model. *eLife*. 2023 Oct 6;. doi: 10.7554/elife.89763.1.

50. Marques FC, Wehrle E, Müller R. Correlative Multimodal Imaging for Multiscale Analysis of Complex Biological Systems: Validation Methods and Performance Analysis. *Reference series in biomedical engineering*. 2023 Jan 1;:1-36. doi: 10.1007/978-3-030-85569-7_17-1.

51. Llambrich S, Tielemans B, Saliën E, Atzori M, Wouters K, Van Bulck V, Platt M, Vanherp L, Fernandez NG, de la Fuente LG, Poptani H, Verlinden L, Himmelreich U, Croitor A, Attanasio C, Callaerts-Vegh Z, Gsell W, Martínez-Abadías N, Velde GV. Pleiotropic effects of trisomy and pharmacologic modulation on structural, functional, molecular, and genetic systems in a Down syndrome mouse model. *eLife*. 2024 Feb 15;. doi: 10.7554/elife.89763.2.

52. Marques FC, Wehrle E, Müller R. Correlative Multimodal Imaging for Multiscale Analysis of Complex Biological Systems: Validation Methods and Performance Analysis. *Reference series in biomedical engineering*. 2025 Jan 1;:387-422. doi: 10.1007/978-3-031-91544-4_17.

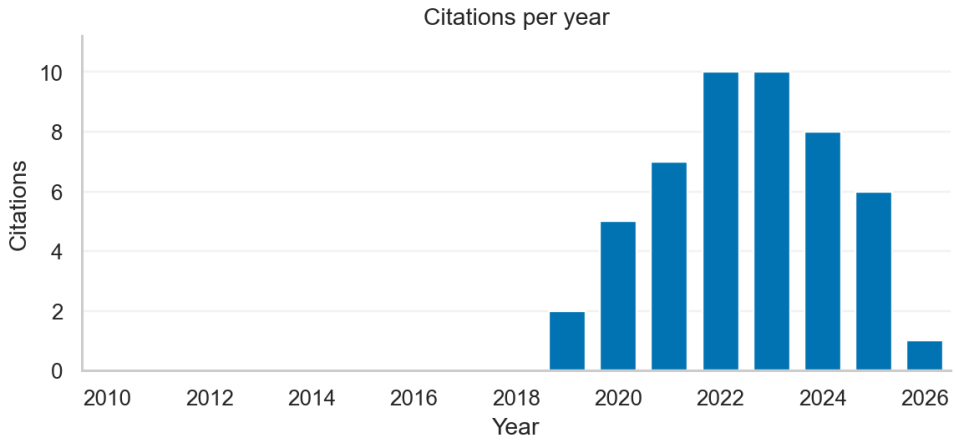
53. González-Alday R, Ferreiro A, Arias-Ramos N, Lizarbe B, López-Larrubia P. Resomapper: A Versatile Pipeline for Multiparametric MRI Processing. A Demonstrative Application in Mouse Brain Imaging. *bioRxiv (Cold Spring Harbor Laboratory)*. 2025 Aug 24;. doi: 10.1101/2025.08.20.671222.

54. Blaschke SJ, Backes H, Vlachakis S, Rautenberg N, Demir S, Wiedermann D, Aswendt M, Fink GR, Schroeter M, Rueger MA. Subacute cathodal transcranial direct current stimulation rescues secondary thalamic neurodegeneration after cortical stroke in mice. *Experimental Neurology*. 2025 Dec 23;398.0:115604. doi: 10.1016/j.expneurol.2025.115604. PMID: 41448520.

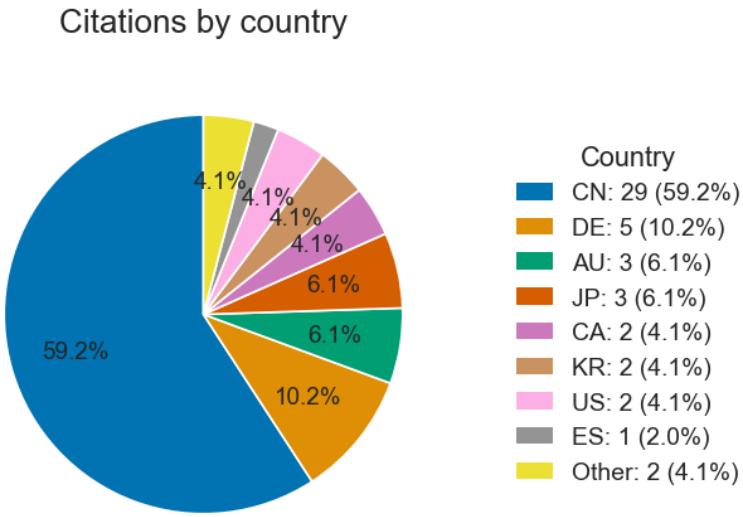
55. Shilikbay T, Nawaz A, Sun M, Doon M, Olmo I, Cumbie L, Benson J, Ibrahim BA, Tsai N, Llano DA, Goense JBM, Gritton HJ, Ceman S. Reduced MOV10 reveals novel functional cortical connections in an increased fear response. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 May 19;. doi: 10.64898/2026.05.18.725995. PMID: 42239320.

Publication

Collmann FM, Pijnenburg R, Taj SH, Minassian A, Folz-Donahue K, Kukat C, Aswendt M, Hoehn M. Individual in vivo Profiles of Microglia Polarization After Stroke, Represented by the Genes iNOS and Ym1. *Frontiers in Immunology*. 2019 Jun 4;10:1236. doi: 10.3389/fimmu.2019.01236. PMID: 31214190.



Citation summary
Total citations: **49**
Average/year: **6.12**
Period: 2019-2026
SCImago cites/doc 2y (2025): **6.38**
SJR 2025: **1.96** (Q1)



Cited by

1. Jiang C, Wu W, Deng Y, Ge J. Modulators of microglia activation and polarization in ischemic stroke (Review). *Molecular Medicine Reports*. 2020 Feb 26;21(5):2006-2018. doi: 10.3892/mmr.2020.11003. PMID: 32323760.

2. Sen MK, Mahns DA, Coorssen JR, Shortland PJ. The roles of microglia and astrocytes in phagocytosis and myelination: Insights from the cuprizone model of multiple sclerosis. *Glia*. 2022 Feb 2;70(7):1215-1250. doi: 10.1002/glia.24148. PMID: 35107839.

3. Planas AM. Role of microglia in stroke. *Glia*. 2024 Jan 4;72(6):1016-1053. doi: 10.1002/glia.24501. PMID: 38173414.

4. Dong R, Huang R, Wang J, Liu H, Xu Z. Effects of Microglial Activation and Polarization on Brain Injury After Stroke. *Frontiers in Neurology*. 2021 Jul 1;12:620948. doi: 10.3389/fneur.2021.620948. PMID: 34276530.

5. Hardeland R. Melatonin and Microglia. *International Journal of Molecular Sciences*. 2021 Aug 2;22(15):8296. doi: 10.3390/ijms22158296. PMID: 34361062.

6. Shui X, Chen J, Fu Z, Zhu H, Tao H, Li Z. Microglia in Ischemic Stroke: Pathogenesis Insights and Therapeutic Challenges. *Journal of Inflammation Research*. 2024 May 1;Volume 17:3335-3352. doi: 10.2147/jir.s461795. PMID: 38800598.

7. Wang S, Fu X, Duan R, Wei B, Cao H, E Y, Chen S, Zhang Y, Jiang T. The Alzheimer's disease-associated gene TREML2 modulates inflammation by regulating microglia polarization and NLRP3 inflammasome activation. *Neural Regeneration Research*. 2022 Jun 2;18(2):434-438. doi: 10.4103/1673-5374.346468. PMID: 35900442.

8. Meleady L, Towriss M, Kim JS, Bacarac V, Dang V, Rowland ME, Ciernia AV. Histone deacetylase 3 regulates microglial function through histone deacetylation. *Epigenetics*. 2023 Jul 28;18(1):2241008. doi: 10.1080/15592294.2023.2241008. PMID: 37506371.

9. Kang Q, Li L, Pang Y, Zhu W, Meng L. An update on Ym1 and its immunoregulatory role in diseases. *Frontiers in Immunology*. 2022 Jul 28;13:891220. doi: 10.3389/fimmu.2022.891220. PMID: 35967383.

10. Li T, Xu T, Zhao J, Gao H, Xie W. Depletion of iNOS-positive inflammatory cells decelerates neuronal degeneration and alleviates cerebral ischemic damage by suppressing the inflammatory response. *Free Radical Biology and Medicine*. 2022 Feb 9;181:209-220. doi: 10.1016/j.freeradbiomed.2022.02.008. PMID: 35150825.

11. Guo M, Cao Q, Xia S, Cao X, Chen J, Qian Y, Bao X, Xu Y. A newly-synthesized compound CP-07 alleviates microglia-mediated neuroinflammation and ischemic brain injury via inhibiting STAT3 phosphorylation. *Journal of Translational Internal Medicine*. 2023 Jun 1;11(2):156-168. doi: 10.2478/jtim-2023-0090. PMID: 37408577.

12. Lu W, Wang Y, Wen J. The Roles of RhoA/ROCK/NF-κB Pathway in Microglia Polarization Following Ischemic Stroke. *Journal of Neuroimmune Pharmacology*. 2024 May 16;19(1):19. doi: 10.1007/s11481-024-10118-w. PMID: 38753217.

13. Wang H, Yiqiang Z, Cai J, Guo J. Immune-mediated blood-brain barrier disruption after ischemic stroke: mechanisms and therapeutic targets. *Frontiers in Immunology*. 2025 Dec 10;16:1716041. doi: 10.3389/fimmu.2025.1716041. PMID: 41451205.

14. Xiao R, Huang X, Gao S, Duan J, Zhang Y, Zhang M, Zhang M. Microglia in retinal diseases: From pathogenesis towards therapeutic strategies. *Biochemical Pharmacology*. 2024 Sep 20;230(Pt 1):116550. doi: 10.1016/j.bcp.2024.116550. PMID: 39307318.

15. Ulbrich F, Goebel U, Scheid S, Spassov SG, Schallner N, Wollborn J, Buerkle H. Argon reduces microglial activation and inflammatory cytokine expression in retinal ischemia/reperfusion injury. *Neural Regeneration Research*. 2020 Aug 10;16(1):192. doi: 10.4103/1673-5374.290098. PMID: 32788476.

16. Wieters F, Lucas CW, Gruhn M, Büschges A, Fink GR, Aswendt M. Introduction to spasticity and related mouse models. *Experimental Neurology*. 2020 Sep 29;335:113491. doi: 10.1016/j.expneurol.2020.113491. PMID: 33007294.

17. Qiu M, Zong J, He Q, Liu Y, Wan Y, Li M, Zhou Y, Wu J, Hu B. Cell Heterogeneity Uncovered by Single-Cell RNA Sequencing Offers Potential Therapeutic Targets for Ischemic Stroke. *Aging and Disease*. 2022 Jan 1;13(5):1436. doi: 10.14336/ad.2022.0212. PMID: 36186129.

18. Mo Y, Xu W, Fu K, Chen H, Wen J, Huang Q, Guo F, Mo L, Yan J. The dual function of microglial polarization and its treatment targets in ischemic stroke. *Frontiers in Neurology*. 2022 Sep 23;13:921705. doi: 10.3389/fneur.2022.921705. PMID: 36212660.

19. Aswendt M, Vogel S, Schäfer C, Jathoul AP, Pulé M, Hoehn M. Quantitative in vivo dual-color bioluminescence imaging in the mouse brain. *Neurophotonics*. 2019 May 7;6(02):1-11. doi: 10.1117/1.nph.6.2.025006. PMID: 31093514.

20. Gamdzyk M, Doycheva DM, Kang R, Tang HC, Travis ZD, Tang J, Zhang JH. GW0742 activates miR-17-5p and inhibits TXNIP/NLRP3-mediated inflammation after hypoxic-ischaemic injury in rats and in PC12 cells. *Journal of Cellular and Molecular Medicine*. 2020 Oct 9;24(21):12318-12330. doi: 10.1111/jcmm.15698. PMID: 33034416.

21. Shin HJ, Lee KY, Kang JW, Choi SG, Kim DW, Yi YY. Perampanel Reduces Brain Damage via Induction of M2 Microglia in a Neonatal Rat Stroke Model. *International Journal of Nanomedicine*. 2022 Jun 1;Volume 17:2791-2804. doi: 10.2147/ijn.s361377. PMID: 35782016.

22. Saeed KM, Jo MH, Park JS, Alam SI, Khan I, Ahmad R, Khan A, Ullah R, Kim MO. 17β-Estradiol Abrogates Oxidative Stress and Neuroinflammation after Cortical Stab Wound Injury. *Antioxidants*. 2021 Oct 25;10(11):1682. doi: 10.3390/antiox10111682. PMID: 34829553.

23. Lin X, Zhan J, Jiang J, Ren Y. Upregulation of Neuronal Cylindromatosis Expression is Essential for Electroacupuncture-Mediated Alleviation of Neuroinflammatory Injury by Regulating Microglial Polarization in Rats Subjected to Focal Cerebral Ischemia/Reperfusion. *Journal of Inflammation Research*. 2021 May 1;Volume 14:2061-2078. doi: 10.2147/jir.s307841. PMID: 34045881.

24. Sun X, Yao Z, Chen L, Huang J, Dong S. Metabolic reprogramming regulates microglial polarization and its role in cerebral ischemia reperfusion. *Fundamental and Clinical Pharmacology*. 2023 Jun 20;37(6):1065-1078. doi: 10.1111/fcp.12928. PMID: 37339781.

25. Ma Z, Wu Y, Xu J, Cao H, Du M, Jiang H, Qiu F. Sodium Tanshinone IIA Sulfonate Ameliorates Oxygen-glucose Deprivation/Reoxygenation-induced Neuronal Injury via Protection of Mitochondria and Promotion of Autophagy. *Neurochemical Research*. 2023 Jul 12;48(11):3378-3390. doi: 10.1007/s11064-023-03985-x. PMID: 37436612.

26. Au NPB, Wu T, Kumar G, Jin Y, Li YYT, Chan SL, Lai JHC, Chan KWY, Yu K, Wang X, Ma CHE. Low-dose ionizing radiation promotes motor recovery and brain rewiring by resolving inflammatory response after brain injury and stroke. *Brain Behavior and Immunity*. 2023 Sep 27;115:43-63. doi: 10.1016/j.bbi.2023.09.015. PMID: 37774892.

27. Zhou C, Li J, Zheng C, Zhou X, Chen C, Qiu S, Liu W, Hua L. Neuroprotective effects of Jie-du-huo-xue decoction on microglia pyroptosis after cerebral ischemia and reperfusion--From the perspective of glial-vascular unit. *Journal of Ethnopharmacology*. 2023 Aug 1;318(Pt B):116990. doi: 10.1016/j.jep.2023.116990. PMID: 37536647.

28. Hu X, Fukui Y, Feng T, Bian Z, Yu H, Morihara R, Hu X, Bian Y, Sun H, Takemoto M, Nakano Y, Yunoki T, Abe K, Yamashita T. Neuroprotective effects of carnosine in a mice stroke model concerning oxidative stress and inflammatory response. *Journal of the Neurological Sciences*. 2023 Mar 3;447:120608. doi: 10.1016/j.jns.2023.120608. PMID: 36906993.

29. Fu T, Wang G, Li Y, Xu Z, Wang C, Zhang R, Ma Q, Ma Y, Guo Y, Guo Y, Dai X, Guo Y, Guo Y. Mobilizing endogenous neuroprotection: the mechanism of the protective effect of acupuncture on the brain after stroke. *Frontiers in Neuroscience*. 2024 Apr 25;18:1181670. doi: 10.3389/fnins.2024.1181670. PMID: 38737099.

30. Yang L, Li Q, Wang K, Liu H, Yang X, Xu Y, Chen Y, Teng J, Gong Z, Jia Y. Blocking cerebral lymphatic system reduces central and peripheral inflammatory response in ischemic stroke. *Brain Research*. 2024 Feb 23;1831:148825. doi: 10.1016/j.brainres.2024.148825. PMID: 38403041.

31. Guo Y, Shang Y, Yao J, Bi X. Role and Mechanism of Microglia in White Matter Injury Recovery in Ischemic Stroke. *Immunity Inflammation and Disease*. 2025 Aug 1;13(8):e70226. doi: 10.1002/iid3.70226. PMID: 40864831.

32. Lai C, He W, Yang H, Lai J, Huang S. Electroacupuncture improved depressive behaviors and synaptic plasticity of post-stroke depressed mice via inhibiting the JNK signaling pathway. *Neurological Research*. 2025 Jun 25;48(1):12-27. doi: 10.1080/01616412.2025.2520017. PMID: 40566816.

33. An Z, He Q, Jiang L, Wang Y, Zhang Y, Sun Y, Wang M, Yang S, Huang L, Li H, Hao Y, Liang X, Wang S. A One-Stone-Two-Birds Strategy of Targeting Microbubbles with “Dual” Anti-Inflammatory and Blood-Brain Barrier “Switch” Function for Ischemic Stroke Treatment. *ACS Biomaterials Science & Engineering*. 2024 Feb 29;10(3):1774-1787. doi: 10.1021/acsbiomaterials.3c01561. PMID: 38420991.

34. Chen J, Wang H, Su Y, Ho Y, Liang C, Lee Y, Chu T, Lin Y, Wu C. HDAC1 dysregulation promotes pro-inflammatory microglial activation and aggravates post-stroke neuroinflammation. *Annals of Medicine*. 2025 Dec 12;57(1):2597624. doi: 10.1080/07853890.2025.2597624. PMID: 41388741.

35. Al-Khishman NU, Qi Q, Roseborough AD, Levit A, Allman BL, Anazodo U, Fox MS, Whitehead SN, Thiessen JD. TSPO PET detects acute neuroinflammation but not diffuse chronically activated MHCII microglia in the rat. *EJNMMI Research*. 2020 Sep 29;10(1):113. doi: 10.1186/s13550-020-00699-x. PMID: 32990808.

36. Pan L, Sze YH, Yang M, Tang J, Zhao S, Yi I, To CH, Lam C, Chen DF, Cho K, Wai C. Baicalein-A Potent Pro-Homeostatic Regulator of Microglia in Retinal Ischemic Injury. *Frontiers in Immunology*. 2022 Feb 21;13:837497. doi: 10.3389/fimmu.2022.837497. PMID: 35265083.

37. Liao L, Qian Z, Li X, Yang D, Lei B, Li H, Hong X. Disrupting RhoA activity by blocking Arhgef3 expression mitigates microglia-induced neuroinflammation post spinal cord contusion. *Journal of Neuroimmunology*. 2021 Aug 8;359:577688. doi: 10.1016/j.jneuroim.2021.577688. PMID: 34390950.

38. He X, Gou X, Fan D, Yang J, Yang J, Fu X, Luo Y, Yang T. Repurposing TAK875 as a novel STAT3 inhibitor for treating inflammatory bowel disease. *Biochemical Pharmacology*. 2023 Dec 2;219:115957. doi: 10.1016/j.bcp.2023.115957. PMID: 38049007.

39. Jin N, Sha W, Gao L. Shaping the Microglia in Retinal Degenerative Diseases Using Stem Cell Therapy: Practice and Prospects. *Frontiers in Cell and Developmental Biology*. 2021 Dec 13;9:741368. doi: 10.3389/fcell.2021.741368. PMID: 34966736.

40. Wang Y, Wang Y, Chen W, Zhou J, Wang Y, Wang Y, Wang H, Wang Y, Wang Y. Nitrate Metabolism and Ischemic Cerebrovascular Disease: A Narrative Review. *Frontiers in Neurology*. 2022 Mar 3;13:735181. doi: 10.3389/fneur.2022.735181. PMID: 35309590.

41. Wang X, Rong C, Leng W, Niu P, He Z, Wang G, Qi X, Zhao D, Li J. Effect and mechanism of Dichloroacetate in the treatment of stroke and the resolution strategy for side effect. *European journal of medical research*. 2025 Mar 3;30(1):148. doi: 10.1186/s40001-025-02399-5. PMID: 40025562.

42. Jiang Q, Zhang Z, Sun Y, Zhang W, Tao S, Zhao L, Chen L. miR-24 protects against ischemia-induced brain damage in rats via regulating microglia polarization by targeting Clcn3. *Neuroscience Letters*. 2021 May 29;759:135998. doi: 10.1016/j.neulet.2021.135998. PMID: 34062195.

43. Fu X, Chen S, Lian H, Deng Y, Duan R, Zhang Y, Jiang T. The TREM2 H157Y Variant Influences Microglial Phagocytosis, Polarization, and Inflammatory Cytokine Release. *Brain Sciences*. 2023 Apr 9;13(4):642. doi: 10.3390/brainsci13040642. PMID: 37190607.

44. Shimada A, Murata M, Aoyagi S, Asano H, Obara A, Hasegawa-Ishii S. Delayed microglial activation associated with the resolution of neuroinflammation in a mouse model of sublethal endotoxemia-induced systemic inflammation. *Toxicology Reports*. 2022 Jan 1;9:1380-1390. doi: 10.1016/j.toxrep.2022.06.015. PMID: 36518431.

45. Okawa K, Tabata E, Kida Y, Uno K, Suzuki H, Kamaya M, Bauer PO, Oyama F. Irreversible evolutionary loss of chitin-degrading ability in the chitinase-like protein Y m1 under positive selection in rodents. *Protein Science*. 2023 Mar 8;32(4):e4620. doi: 10.1002/pro.4620. PMID: 36883357.

46. Niu X, Zhang Z, Zhou Q, Wuethrich A, Lobb RJ, Trau M. Analysis of secreted small extracellular vesicles from activated human microglial cell lines reveals distinct pro- and anti-inflammatory proteomic profiles. *PROTEOMICS*. 2024 Feb 11;24(11):e2300094. doi: 10.1002/pmic.202300094. PMID: 38343172.

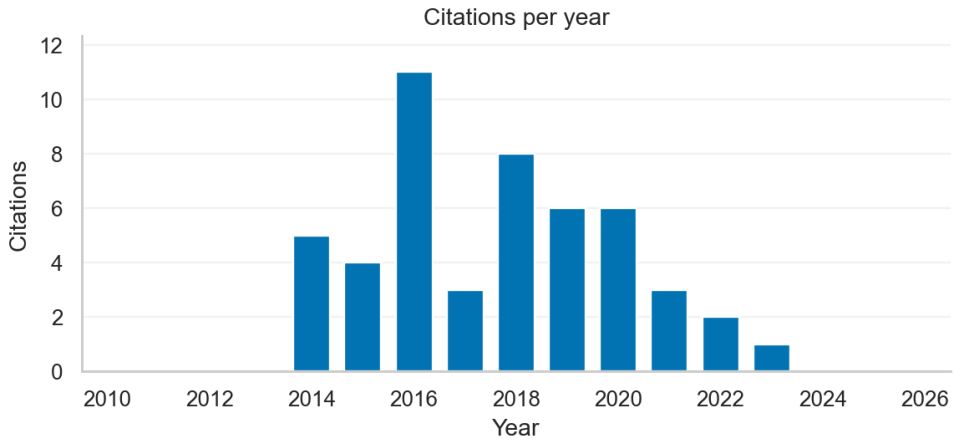
47. Meng D, Su Q, Su Q, Jia K, Jin M, Liu W, Yang S, Su Z. ETV5 improves ischemic stroke by targeting HACE1/Nrf2 axis to inhibit microglia M1 polarization and neuron pyroptosis. *Biochemical Pharmacology*. 2025 Jun 23;240:117057. doi: 10.1016/j.bcp.2025.117057. PMID: 40562126.

48. Aswendt M, Vogel S, Hoehn M. In vivo bioluminescence imaging to elucidate stem cell graft differentiation. *Neural Regeneration Research*. 2019 Sep 16;15(1):61. doi: 10.4103/1673-5374.264449. PMID: 31535647.

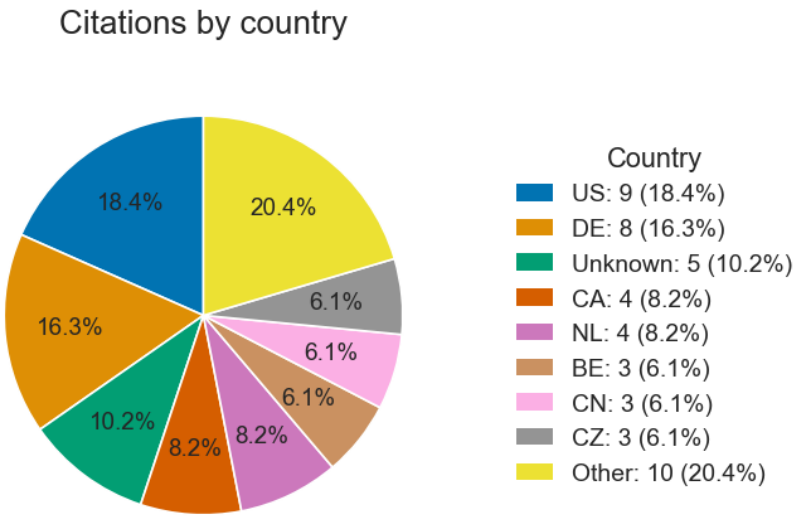
49. Mahmoudi N, Harvey A, Moriarty N, Mahmoudi M, Tong W, Goris T, Reynolds N, Ahmed NY, Dehorter N, Lisowski L, Parish CL, Williams RJ, Nisbet DR. Spatially Precise Neuron Formation via Hydrogel-Mediated Modulation of the Host Astrocyte Response. *ACS Nano*. 2026 Aug 12;20(33):23200-23222. doi: 10.1021/acsnano.6c03424. PMID: 42674445.

Publication

Boehm-Sturm P, Aswendt M, Minassian A, Michalk S, Mengler L, Adamczak JM, Mezzanotte L, Löwik CW, Hoehn M. A multi-modality platform to image stem cell graft survival in the naïve and stroke-damaged mouse brain. *Biomaterials*. 2013 Dec 16;35(7):2218-2226. doi: 10.1016/j.biomaterials.2013.11.085. PMID: 24355489.



Citation summary
Total citations: **49**
Average/year: **3.50**
Period: 2013-2026
SCImago cites/doc 2y (2025): **14.1**
SJR 2025: **2.85** (Q1)



Cited by

1. Hall MP, Woodroofe CC, Wood MG, Que I, Root MV', Ridwan YR, Shi C, Kirkland TA, Encell LP, Wood KV, Löwik CW, Mezzanotte L. Click beetle luciferase mutant and near infrared naphthyl-luciferins for improved bioluminescence imaging. *Nature Communications*. 2018 Jan 3;9(1.0):132. doi: 10.1038/s41467-017-02542-9. PMID: 29317625.

2. Krekorian M, Fruhwirth GO, Srinivas M, Figdor CG, Heskamp S, Witney TH, Aarntzen EH. Imaging of T-cells and their responses during anti-cancer immunotherapy. *Theranostics*. 2019 Jan 1;9(25.0):7924-7947. doi: 10.7150/thno.37924. PMID: 31656546.

3. Gaudet JM, Ribot EJ, Chen Y, Gilbert KM, Foster PJ. Tracking the Fate of Stem Cell Implants with Fluorine-19 MRI. *PLoS ONE*. 2015 Mar 13;10(3.0):e0118544. doi: 10.1371/journal.pone.0118544. PMID: 25767871.

4. Jirák D, Gálisová A, Kolouchová K, Babuka D, Hrubý M. Fluorine polymer probes for magnetic resonance imaging: quo vadis?. *Magnetic Resonance Materials in Physics Biology and Medicine*. 2018 Nov 29;32(1.0):173-185. doi: 10.1007/s10334-018-0724-6. PMID: 30498886.

5. Wu L, Liu F, Liu S, Xu X, Liu Z, Sun X. Perfluorocarbons-Based 19F Magnetic Resonance Imaging in Biomedicine. *International Journal of Nanomedicine*. 2020 Oct 1;Volume 15:7377-7395. doi: 10.2147/ijn.s255084. PMID: 33061385.

6. Staal AHJ, Becker K, Tagit O, van Riessen NK, Koshkina O, Veltien AA, Bouvain P, Cortenbach KR, Scheenen TWJ, Flögel U, Temme S, Srinivas M. In vivo clearance of 19F MRI imaging nanocarriers is strongly influenced by nanoparticle ultrastructure. *Biomaterials*. 2020 Aug 12;261:120307. doi: 10.1016/j.biomaterials.2020.120307. PMID: 32927288.

7. Kakaei N, Amirian R, Azadi M, Mohammadi G, Izadi Z. Perfluorocarbons: A perspective of theranostic applications and challenges. *Frontiers in Bioengineering and Biotechnology*. 2023 Aug 3;11:1115254. doi: 10.3389/fbioe.2023.1115254.

PMID: 37600314.

8. Gervois P, Wolfs E, Ratajczak J, Dillen Y, Vangansewinkel T, Hilkens P, Bronckaers A, Lambrichts I, Struys T. Stem Cell-Based Therapies for Ischemic Stroke: Preclinical Results and the Potential of Imaging-Assisted Evaluation of Donor Cell Fate and Mechanisms of Brain Regeneration. *Medicinal Research Reviews*. 2016 Jul 21;36(6.0):1080-1126. doi: 10.1002/med.21400. PMID: 27439773.

9. Muhammad G, Xu J, Bulte JWM, Jabłońska A, Walczak P, Janowski M. Transplanted adipose-derived stem cells can be short-lived yet accelerate healing of acid-burn skin wounds: a multimodal imaging study. *Scientific Reports*. 2017 Jun 29;7(1.0):4644. doi: 10.1038/s41598-017-04484-0. PMID: 28680144.

10. Moonshi SS, Zhang C, Peng H, Puttick SG, Rose S, Fisk NM, Bhakoo KK, Stringer BW, Qiao GG, Gurr PA, Whittaker AK. A unique 19 F MRI agent for the tracking of non phagocytic cells in vivo. *Nanoscale*. 2018 Jan 1;10(17.0):8226-8239. doi: 10.1039/c8nr00703a. PMID: 29682654.

11. Tennstaedt A, Aswendt M, Adamczak JM, Collienne U, Selt M, Schneider G, Henn N, Schaefer C, Lagouge M, Wiedermann D, Kloppenburg P, Hoehn M. Human neural stem cell intracerebral grafts show spontaneous early neuronal differentiation after several weeks. *Biomaterials*. 2015 Jan 12;44:143-154. doi: 10.1016/j.biomaterials.2014.12.038. PMID: 25617134.

12. Fernández-Susavila H, Bugallo-Casal A, Castillo J, Campos F. Adult Stem Cells and Induced Pluripotent Stem Cells for Stroke Treatment. *Frontiers in Neurology*. 2019 Aug 28;10:908. doi: 10.3389/fneur.2019.00908. PMID: 31555195.

13. Qian C, Li P, Jiao Y, Yao H, Chen Y, Yang JZ, Ding J, Yang X, Teng G. Precise Characterization of the Penumbra Revealed by MRI: A Modified Photothrombotic Stroke Model Study. *PLoS ONE*. 2016 Apr 19;11(4.0):e0153756. doi: 10.1371/journal.pone.0153756. PMID: 27093556.

14. Bose RJ, Mattrey RF. Accomplishments and challenges in stem cell imaging in vivo. *Drug Discovery Today*. 2018 Oct 17;24(2.0):492-504. doi: 10.1016/j.drudis.2018.10.007. PMID: 30342245.

15. Gálisová A, Herynek V, Swider E, Sticová E, Pátíková A, Košinová L, Kříž J, Hájek M, Srinivas M, Jirák D. A Trimodal Imaging Platform for Tracking Viable Transplanted Pancreatic Islets In Vivo: F-19 MR, Fluorescence, and Bioluminescence Imaging. *Molecular Imaging and Biology*. 2018 Aug 30;21(3.0):454-464. doi: 10.1007/s11307-018-1270-3. PMID: 30167995.

16. Aswendt M, Adamczak JM, Tennstaedt A. A review of novel optical imaging strategies of the stroke pathology and stem cell therapy in stroke. *Frontiers in Cellular Neuroscience*. 2014 Aug 14;8:226. doi: 10.3389/fncel.2014.00226. PMID: 25177269.

17. Makela AV, Murrell DH, Parkins KM, Kara J, Gaudet JM, Foster PJ. Cellular Imaging With MRI. *Topics in Magnetic Resonance Imaging*. 2016 Sep 30;25(5.0):177-186. doi: 10.1097/rmr.0000000000000101. PMID: 27748707.

18. Kolouchová K, Jirák D, Groborz O, Sedláček O, Ziółkowska N, Vít M, Sticová E, Gálisová A, Švec P, Trousil J, Hájek M, Hrubý M. Implant-forming polymeric 19F MRI-tracer with tunable dissolution. *Journal of Controlled Release*. 2020 Jul 27;327:50-60. doi: 10.1016/j.jconrel.2020.07.026. PMID: 32730953.

19. Hu G, Tang J, Bai X, Xu S, Wang L. Superfluorinated copper sulfide nanoprobes for simultaneous 19F magnetic resonance imaging and photothermal ablation. *Nano Research*. 2016 Apr 8;9(6.0):1630-1638. doi: 10.1007/s12274-016-1057-2.

20. Hébert JM, Vijg J. Cell Replacement to Reverse Brain Aging: Challenges, Pitfalls, and Opportunities. *Trends in Neurosciences*. 2018 Mar 18;41(5.0):267-279. doi: 10.1016/j.tins.2018.02.008. PMID: 29548515.

21. Pongrac IM, Radmilović MD, Ahmed LB, Mlinarić H, Regul J, Škokić S, Babič M, Horák D, Hoehn M, Gajović S. D-mannose-Coating of Maghemite Nanoparticles Improved Labeling of Neural Stem Cells and Allowed Their Visualization by ex vivo MRI after Transplantation in the Mouse Brain. *Cell Transplantation*. 2019 May 1;28(5.0):553-567. doi: 10.1177/0963689719834304. PMID: 31293167.

22. Modo M. 19F Magnetic Resonance Imaging and Spectroscopy in Neuroscience. *Neuroscience*. 2021 Mar 23;474:37-50. doi: 10.1016/j.neuroscience.2021.03.016. PMID: 33766776.

23. Lee JM, Ryu JH, Kim EA, Jo S, Kim B, Lee H, Im G. Adhesive barrier/directional controlled release for cartilage repair by endogenous progenitor cell recruitment. *Biomaterials*. 2014 Nov 22;39:173-181. doi: 10.1016/j.biomaterials.2014.11.006. PMID: 25468369.

24. Gaudet JM, Hamilton AM, Chen Y, Fox MS, Foster PJ. Application of dual19F and iron cellular MRI agents to track the infiltration of immune cells to the site of a rejected stem cell transplant. *Magnetic Resonance in Medicine*. 2016 Sep 9;78(2.0):713-720. doi: 10.1002/mrm.26400. PMID: 27610596.

25. Tennstaedt A, Mastropietro A, Nelles M, Beyrau A, Hoehn M. In Vivo Fate Imaging of Intracerebral Stem Cell Grafts in Mouse Brain. *PLoS ONE*. 2015 Dec 7;10(12.0):e0144262. doi: 10.1371/journal.pone.0144262. PMID: 26641453.

26. Praet J, Santermans E, Daans J, Le Blon D, Hoornaert C, Goossens H, Hens N, Van der Linden A, Berneman ZN, Ponsaerts P. Early Inflammatory Responses following Cell Grafting in the CNS Trigger Activation of the Subventricular Zone: A Proposed Model of Sequential Cellular Events. *Cell Transplantation*. 2014 Jul 15;24(8.0):1481-1492. doi: 10.3727/096368914x682800. PMID: 25197881.

27. Xiao Z, Puré E. Imaging of T-cell Responses in the Context of Cancer Immunotherapy. *Cancer Immunology Research*. 2021 May 1;9(5.0):490-502. doi: 10.1158/2326-6066.cir-20-0678. PMID: 33941536.

28. Moonshi SS, Wu Y, Ta HT. Visualizing stem cells in vivo using magnetic resonance imaging. *Wiley Interdisciplinary Reviews Nanomedicine and Nanobiotechnology*. 2021 Oct 15;14(2.0):e1760. doi: 10.1002/wnan.1760. PMID: 34651465.

29. Vogel SN, Aswendt M, Nelles M, Henn N, Schneider G, Hoehn M. Initial graft size and not the innate immune response limit survival of engrafted neural stem cells. *Journal of Tissue Engineering and Regenerative Medicine*. 2017 Jun

9;12(3.0):784-793. doi: 10.1002/term.2497. PMID: 28599089.

30. Jabłońska A, Drela K, Stanaszek L, Janowski M, Zalewska T, Łukomska B. Short-Lived Human Umbilical Cord-Blood-Derived Neural Stem Cells Influence the Endogenous Secretome and Increase the Number of Endogenous Neural Progenitors in a Rat Model of Lacunar Stroke. *Molecular Neurobiology*. 2015 Nov 25;53(9.0):6413-6425. doi: 10.1007/s12035-015-9530-6. PMID: 26607630.

31. Vogel SN, Schäfer C, Heß S, Folz-Donahue K, Nelles M, Minassian A, Schwarz MK, Kukat C, Ehrlich M, Zaehres H, Kloppenburg P, Hoehn M, Aswendt M. The in vivo timeline of differentiation of engrafted human neural progenitor cells. *Stem Cell Research*. 2019 Mar 25;37:101429. doi: 10.1016/j.scr.2019.101429. PMID: 30933718.

32. Shin SH, Park SH, Kang SH, Kim SW, Kim MS, Kim D. Fluorine-19 Magnetic Resonance Imaging and Positron Emission Tomography of Tumor-Associated Macrophages and Tumor Metabolism. *Contrast Media & Molecular Imaging*. 2017 Jan 1;2017:1-8. doi: 10.1155/2017/4896310. PMID: 29362559.

33. Selt M, Tennstaedt A, Beyrau A, Nelles M, Schneider G, Löwik CW, Hoehn M. In Vivo Non-Invasive Tracking of Macrophage Recruitment to Experimental Stroke. *PLoS ONE*. 2016 Jun 24;11(6.0):e0156626. doi: 10.1371/journal.pone.0156626. PMID: 27341631.

34. Martín A, Ramos-Cabrer P, Planas AM. Noninvasive Brain Imaging in Small Animal Stroke Models: MRI, PET, and SPECT. *Neuromethods*. 2016 Jan 1;:147-186. doi: 10.1007/978-1-4939-5620-3_11.

35. Green C, Minassian A, Vogel SN, Diedenhofen M, Wiedermann D, Hoehn M. Persistent Quantitative Vitality of Stem Cell Graft Is Necessary for Stabilization of Functional Brain Networks After Stroke. *Frontiers in Neurology*. 2019 Apr 5;10:335. doi: 10.3389/fneur.2019.00335. PMID: 31024429.

36. Madsen SD, Giler MK, Bunnell BA, O'Connor KC. Illuminating the Regenerative Properties of Stem Cells In Vivo with Bioluminescence Imaging. *Biotechnology Journal*. 2020 Oct 22;16(3.0):e2000248. doi: 10.1002/biot.202000248. PMID: 33089922.

37. Liang S, Louchami K, Holvoet B, Verbeke R, Deroose CM, Manshian BB, Soenen SJ, Lentacker I, Himmelreich U. Tri-modal In vivo Imaging of Pancreatic Islets Transplanted Subcutaneously in Mice. *Molecular Imaging and Biology*. 2018 Apr 18;20(6.0):940-951. doi: 10.1007/s11307-018-1192-0. PMID: 29671177.

38. Duffy BA, Weitz AJ, Lee JH. In vivo imaging of transplanted stem cells in the central nervous system. *Current Opinion in Genetics & Development*. 2014 Oct 1;28:83-88. doi: 10.1016/j.gde.2014.09.007. PMID: 25461455.

39. Khalil AA, Mueller SG, Foddis M, Mosch L, Lips J, Przesdzing I, Temme S, Flögel U, Dirnagl U, Boehm-Sturm P. Longitudinal 19F magnetic resonance imaging of brain oxygenation in a mouse model of vascular cognitive impairment using a cryogenic radiofrequency coil. *Magnetic Resonance Materials in Physics Biology and Medicine*. 2018 Nov 12;32(1.0):105-114. doi: 10.1007/s10334-018-0712-x. PMID: 30421249.

40. Temme S, Grapentin C, Güden-Silber T, Flögel U. Chapter 4 Active Targeting of Perfluorocarbon Nanoemulsions. *Pan Stanford Publishing eBooks*. 2016 Oct 21;:103-140. doi: 10.1201/9781315364605-5.

41. Swider E, Srinivas M. Spinning to a Different Beat: 19F Agents for "Hot-Spot" Cellular MR Imaging. *Springer eBooks*. 2016 Nov 26;:153-170. doi: 10.1007/978-3-319-42169-8_7.

42. Schomann T, Iljas JD, Que I, Li Y, Suidgeest E, Cruz LJ, Frijns JHM, Chan AB, Löwik CM, Huisman MA, Mezzanotte L. Multimodal imaging of hair follicle bulge-derived stem cells in a mouse model of traumatic brain injury. *Cell and Tissue Research*. 2020 Feb 8;381(1.0):55-69. doi: 10.1007/s00441-020-03173-1. PMID: 32036485.

43. Modo M. Gaining Mechanistic Insights into Cell Therapy Using Magnetic Resonance Imaging. *Current Stem Cell Reports*. 2016 Jul 7;2(3.0):221-227. doi: 10.1007/s40778-016-0059-z.

44. Shaikh S. Molecular Imaging and Stem Cell Imaging. *Springer eBooks*. 2022 Jan 1;:237-256. doi: 10.1007/978-981-16-9535-3_20.

45. Mahoney R. Characterization and in vivo Tracking of transplanted Oligodendrocyte Progenitor Cells in the Injured Rat Spinal Cord. *University of Minnesota Digital Conservancy (University of Minnesota)*. 2014 Dec 1;.

46. Aswendt M, Boehm-Sturm P, Hoehn M. Chapter 10 Neural Stem Cells. *Pan Stanford Publishing eBooks*. 2016 Oct 21;:283-310. doi: 10.1201/9781315364605-11.

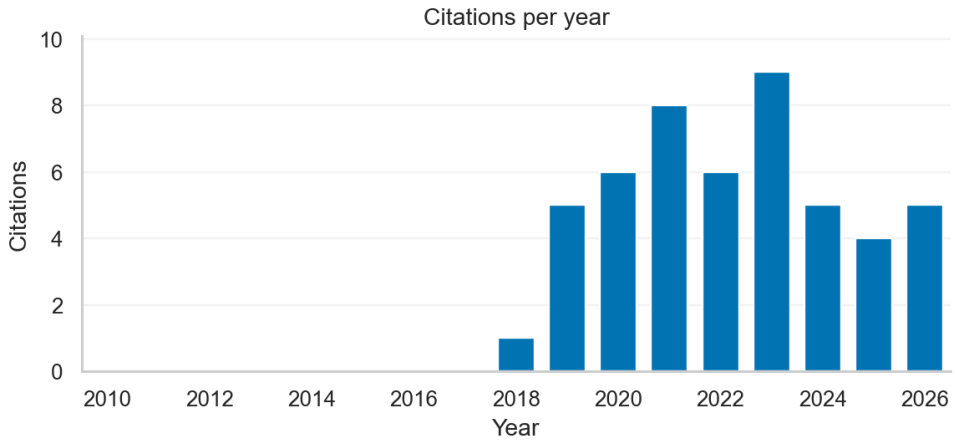
47. Weir N. Developing a neural implant to enable controlled alterations to brain architecture. *Nottingham Trent University's Institutional Repository (Nottingham Trent Repository)*. 2019 Dec 1;.

48. Ganesh MG. Grafting purified cortical pericytes into mouse motor cortex. *Otago University Research Archive (University of Otago)*. 2020 Jan 1;.

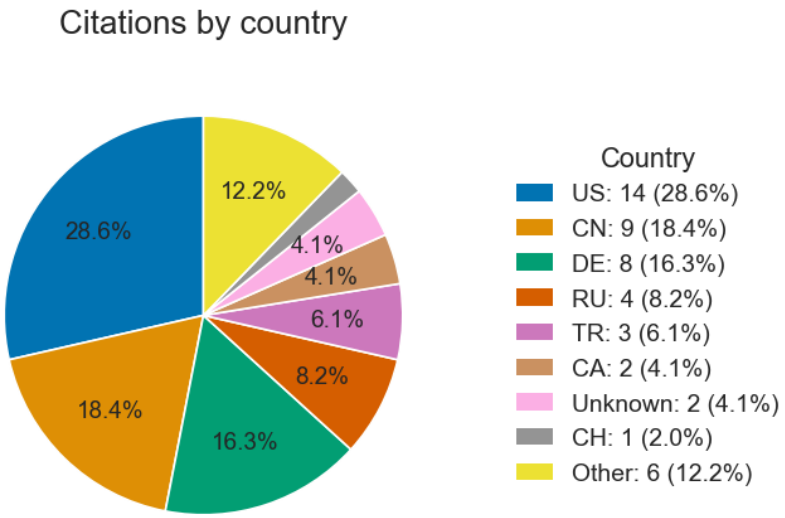
49. Akbar MA, Martin AR, Pfyffer D, Cadotte DW, Kurpad SN, Freund P, Fehlings MG. Advanced imaging for spinal cord injury. *Elsevier eBooks*. 2022 Jan 1;:105-124. doi: 10.1016/b978-0-12-819835-3.00028-9.

Publication

Ito M, Aswendt M, Lee AG, Ishizaka S, Cao Z, Wang EH, Levy SL, Smerin D, McNab JA, Zeineh M, Leuze CW, Goubran M, Cheng M, Steinberg GK. RNA-Sequencing Analysis Revealed a Distinct Motor Cortex Transcriptome in Spontaneously Recovered Mice After Stroke. *Stroke*. 2018 Sep 1;49(9):2191-2199. doi: 10.1161/strokeaha.118.021508. PMID: 30354987.



Citation summary
Total citations: **49**
Average/year: **5.44**
Period: 2018-2026
SCImago cites/doc 2y (2025): **6.75**
SJR 2025: **2.86** (Q1)



Cited by

1. Li J, Pan L, Pembroke WG, Rexach JE, Godoy MI, Condro MC, Alvarado AG, Harteni M, Chen Y, Stiles L, Chen AY, Wanner IB, Yang X, Goldman SA, Geschwind DH, Kornblum HI, Zhang Y. Conservation and divergence of vulnerability and responses to stressors between human and mouse astrocytes. *Nature Communications*. 2021 Jun 25;12.0(1.0):3958. doi: 10.1038/s41467-021-24232-3. PMID: 34172753.

2. Xing Y, Bai Y. A Review of Exercise-Induced Neuroplasticity in Ischemic Stroke: Pathology and Mechanisms. *Molecular Neurobiology*. 2020 Jul 20;57.0(10.0):4218-4231. doi: 10.1007/s12035-020-02021-1. PMID: 32691303.

3. Goubran M, Leuze CW, Hsueh B, Aswendt M, Li Y, Tian Q, Cheng M, Crow A, Steinberg GK, McNab JA, Deisseroth KA, Zeineh M. Multimodal image registration and connectivity analysis for integration of connectomic data from microscopy to MRI. *Nature Communications*. 2019 Dec 3;10.0(1.0):5504. doi: 10.1038/s41467-019-13374-0. PMID: 31796741.

4. Schädlich IS, Winzer R, Stabernack J, Tolosa E, Magnus TU, Rissiek B. The role of the ATP-adenosine axis in ischemic stroke. *Seminars in Immunopathology*. 2023 Mar 14;45.0(3.0):347-365. doi: 10.1007/s00281-023-00987-3. PMID: 36917241.

5. Cao Z, Harvey SS, Chiang T, Foltz AG, Lee AG, Cheng M, Steinberg GK. Unique Subtype of Microglia in Degenerative Thalamus After Cortical Stroke. *Stroke*. 2021 Jan 8;52.0(2.0):687-698. doi: 10.1161/strokeaha.120.032402. PMID: 33412903.

6. Rust R, Nih LR, Liberale L, Yin HR, Amki ME, Ong LK, Zloković BV. Brain repair mechanisms after cell therapy for stroke. *Brain*. 2024 Jun 22;147.0(10.0):3286-3305. doi: 10.1093/brain/awae204. PMID: 38916992.

7. Aswendt M, Wilhelmsson U, Wieters F, Stokowska A, Schmitt FJ, Pallast N, de Pablo Y, Mohammed L, Hoehn M, Pekna M, Pekny M. Reactive astrocytes prevent maladaptive plasticity after ischemic stroke. *Progress in Neurobiology*. 2021 Dec 15;209.0:102199. doi: 10.1016/j.pneurobio.2021.102199. PMID: 34921928.

8. Aswendt M, Pallast N, Wieters F, Baues M, Hoehn M, Fink GR. Lesion Size- and Location-Dependent Recruitment of Contralesional Thalamus and Motor Cortex Facilitates Recovery after Stroke in Mice. *Translational Stroke Research*. 2020 Mar 12;12.0(1.0):87-97. doi: 10.1007/s12975-020-00802-3. PMID: 32166716.

9. Luo F, Wang J, Zhang Z, You Z, Bedolla A, Okwubido-Williams FV, Huang LF, Silver J, Luo Y. Inhibition of CSPG receptor PTPσ promotes migration of newly born neuroblasts, axonal sprouting, and recovery from stroke. *Cell Reports*. 2022 Jul 1;40.0(4.0):111137. doi: 10.1016/j.celrep.2022.111137. PMID: 35905716.

10. Zhang Y, Shen L, Xie J, Li L, Xi W, Li B, Bai Y, Yao H, Zhang S, Zhang S. Pushen capsule treatment promotes functional recovery after ischemic stroke. *Phytomedicine*. 2023 Jan 11;111.0:154664. doi: 10.1016/j.phymed.2023.154664. PMID: 36682301.

11. Gong L, Liang J, Xie L, Zhang Z, Mei Z, Zhang W. Metabolic Reprogramming in Gliocyte Post-cerebral Ischemia/Reperfusion: From Pathophysiology to Therapeutic Potential. *Current Neuropharmacology*. 2024 Feb 6;22.0(10.0):1672-1696. doi: 10.2174/1570159x22666240131121032. PMID: 38362904.

12. Gong Y, Liu Z, Zhou P, Li J, Miao Y. Biomimetic nanocarriers harnessing microbial metabolites usher the path for brain disease therapy. *Nano TransMed*. 2023 Dec 1;2.0(4.0):100020. doi: 10.1016/j.ntm.2023.100020.

13. Pallast N, Diedenhofen M, Blaschke SJ, Wieters F, Wiedermann D, Hoehn M, Fink GR, Aswendt M. Processing Pipeline for Atlas-Based Imaging Data Analysis of Structural and Functional Mouse Brain MRI (AIDAmri). *Frontiers in Neuroinformatics*. 2019 Jun 4;13.0:42. doi: 10.3389/fninf.2019.00042. PMID: 31231202.

14. Aswendt M, Green C, Sadler R, Llovera G, Dzikowski L, Heindl S, de Agüero MG, Diedenhofen M, Vogel SN, Wieters F, Wiedermann D, Liesz A, Hoehn M. The gut microbiota modulates brain network connectivity under physiological conditions and after acute brain ischemia. *iScience*. 2021 Sep 9;24.0(10.0):103095. doi: 10.1016/j.isci.2021.103095. PMID: 34622150.

15. Dergunova LV, Filippenkov IB, Limborska SA, Myasoedov NF. Neuroprotective Peptides and New Strategies for Ischemic Stroke Drug Discoveries. *Genes*. 2023 Apr 22;14.0(5.0):953. doi: 10.3390/genes14050953. PMID: 37239313.

16. Xiao C, Yang Y, Li W, Liu M, Zhang S, Wang Y, Du G. Dynamic Alterations of Brain Injury, Functional Recovery, and Metabolites Profile after Cerebral Ischemia/Reperfusion in Rats Contributes to Potential Biomarkers. *Journal of Molecular Neuroscience*. 2020 Jan 6;70.0(5.0):667-676. doi: 10.1007/s12031-019-01474-x. PMID: 31907865.

17. Choi I, Yun JH, Kim J, Kim HY, Choi D, Lee J. Sequential Transcriptome Changes in the Penumbra after Ischemic Stroke. *International Journal of Molecular Sciences*. 2019 Dec 16;20.0(24.0):6349. doi: 10.3390/ijms20246349. PMID: 31888302.

18. Beker MÇ, Çağlayan AB, Altunay S, Özbay E, Ateş N, Keleştemur T, Çağlayan B, Kılıç Ü, Doeppner TR, Hermann DM, Kılıç E. Phosphodiesterase 10A Is a Critical Target for Neuroprotection in a Mouse Model of Ischemic Stroke. *Molecular Neurobiology*. 2021 Nov 4;59.0(1.0):574-589. doi: 10.1007/s12035-021-02621-5. PMID: 34735672.

19. Pallast N, Wieters F, Fink GR, Aswendt M. Atlas-based imaging data analysis tool for quantitative mouse brain histology (AIDAhisto). *Journal of Neuroscience Methods*. 2019 Aug 12;326.0:108394. doi: 10.1016/j.jneumeth.2019.108394. PMID: 31415844.

20. Hsu CG, Fazal F, Rahman A, Berk BC, Chen Y. Phosphodiesterase 10A Is a Key Mediator of Lung Inflammation. *The Journal of Immunology*. 2021 Jun 1;206.0(12.0):3010-3020. doi: 10.4049/jimmunol.2001026. PMID: 34117108.

21. Filippenkov IB, Remizova JA, Denisova AE, Stavchansky VV, Golovina KD, Gubsky LV, Limborska SA, Dergunova LV. Differential gene expression in the contralateral hemisphere of the rat brain after focal ischemia. *Scientific Reports*. 2023 Jan 11;13.0(1.0):573. doi: 10.1038/s41598-023-27663-8. PMID: 36631528.

22. Sun X, Yao Z, Chen L, Huang J, Dong S. Metabolic reprogramming regulates microglial polarization and its role in cerebral ischemia reperfusion. *Fundamental and Clinical Pharmacology*. 2023 Jun 20;37.0(6.0):1065-1078. doi: 10.1111/fcp.12928. PMID: 37339781.

23. Yamaguchi S, Yoshida M, Horie N, Satoh K, Fukuda Y, Ishizaka S, Ogawa K, Morofuji Y, Hiu T, Izumo T, Kawakami S, Nishida N, Matsuo T. Stem Cell Therapy for Acute/Subacute Ischemic Stroke with a Focus on Intraarterial Stem Cell Transplantation: From Basic Research to Clinical Trials. *Bioengineering*. 2022 Dec 27;10.0(1.0):33. doi: 10.3390/bioengineering10010033. PMID: 36671605.

24. Adhikari Y, Ma C, Chai Z, Jin X. Preventing development of post-stroke hyperexcitability by optogenetic or pharmacological stimulation of cortical excitatory activity. *Neurobiology of Disease*. 2023 Jul 17;184.0:106233. doi: 10.1016/j.nbd.2023.106233. PMID: 37468047.

25. Lyu T, Qiu XY, Wang Y, Wang Y, Zhang L, Dai Y, Wang X, Zhao S, Xiang M, Cui L, Cheng S, Liu Y, Gu H, Jiang Y, Meng X, Wang Y, Wang Y, Zhao X, Wang X, Li Q, Wang M, Jiang Y, Xu Z, Huang X, Li H, Wang Y, Wang Y, Li Z. DNMT3A dysfunction promotes neuroinflammation and exacerbates acute ischemic stroke. *MedComm*. 2024 Jul 1;5.0(7.0):e652. doi: 10.1002/mco2.652. PMID: 39006763.

26. Beker MÇ, Altıntaş MÖ, Dogan E, Bayraktaroğlu Ç, Balaban B, Ozpinar A, Şengün N, Altunay S, Kılıç E. Inhibition of phosphodiesterase 10A mitigates neuronal injury by modulating apoptotic pathways in cold-induced traumatic brain injury. *Molecular and Cellular Neuroscience*. 2024 Oct 20;131.0:103977. doi: 10.1016/j.mcn.2024.103977. PMID: 39437931.

27. Berk BC, Chávez CL, Hsu CG. PDE10A Inhibition Reduces NLRP3 Activation and Pyroptosis in Sepsis and Nerve Injury. *International Journal of Molecular Sciences*. 2025 May 8;26.0(10.0):4498. doi: 10.3390/ijms26104498. PMID: 40429643.

28. Love CJ, Selim MH, Spector M, Lo EH. Biomaterials for Stroke Therapy. *Stroke*. 2019 Jun 10;50.0(8.0):2278-2284. doi: 10.1161/strokeaha.118.023721. PMID: 31177979.

29. Коростовцева ЛС. Ischemic Stroke and Sleep: The Linking Genetic Factors. *Cardiology and Therapy*. 2021 Jun 30;10.0(2.0):349-375. doi: 10.1007/s40119-021-00231-9. PMID: 34191267.

30. Király B, Hangya B. Navigating the Statistical Minefield of Model Selection and Clustering in Neuroscience. *eNeuro*. 2022 Jul 1;9.0(4.0):ENEURO.0066-22.2022. doi: 10.1523/eneuro.0066-22.2022. PMID: 35835556.

31. Kerr AL. Contralesional plasticity following constraint-induced movement therapy benefits outcome: contributions of the intact hemisphere to functional recovery. *Reviews in the Neurosciences*. 2021 Nov 10;33.0(3.0):269-283. doi: 10.1515/revneuro-2021-0085. PMID: 34761646.

32. Livingston JM, Lee TT, Enbar T, Daniele E, Phillips C, Krassikova A, Bang KA, Kortebi I, Donville B, Ibragimov OS, Sachewsky N, Casasbuenas DL, Olfat A, Morshead CM. Ectopic Expression of Neurod1 Is Sufficient for Functional Recovery following a Sensory-Motor Cortical Stroke. *Biomedicines*. 2024 Mar 15;12.0(3.0):663. doi: 10.3390/biomedicines12030663. PMID: 38540276.

33. Chen Y, Wang K, Cai J, Li Y, Yu H, Wu Q, Meng W, Wang H, Yin C, Wu J, Huang M, Li R, Guan D. Detecting Key Functional Components Group and Speculating the Potential Mechanism of Xiao-Xu-Ming Decoction in Treating Stroke. *Frontiers in Cell and Developmental Biology*. 2022 May 12;10.0:753425. doi: 10.3389/fcell.2022.753425. PMID: 35646921.

34. Beker MÇ, Pence ME, Yagmur S, Çağlayan B, Çağlayan A, Kılıç Ü, Yelkenci HE, Altıntaş MÖ, Çağlayan AB, Doeppner TR, Hermann DM, Kılıç E. Phosphodiesterase 10A deactivation induces long-term neurological recovery, Peri-infarct remodeling and pyramidal tract plasticity after transient focal cerebral ischemia in mice. *Experimental Neurology*. 2022 Sep 5;358.0:114221. doi: 10.1016/j.expneurol.2022.114221. PMID: 36075453.

35. Götz J, Wieters F, Fritz VJ, Käsgen O, Kalantari A, Fink GR, Aswendt M. Temporal and Spatial Gene Expression Profile of Stroke Recovery Genes in Mice. *Genes*. 2023 Feb 9;14.0(2.0):454. doi: 10.3390/genes14020454. PMID: 36833381.

36. Bernal-Vicente BN, Ponce I, Gutiérrez MS, Ríos-Castro E, Tovar-y-Romo LB. Neuroprotective Proteins in Hypoxia-stressed Astrocyte-Derived Extracellular Vesicles. *Current Neuropharmacology*. 2025 Jun 24;23.0(14.0):1962-1978. doi: 10.2174/011570159x359837250611052037. PMID: 40551681.

37. Filippenkov IB, Shpetko YY, Stavchansky VV, Denisova AE, Yuzhakov VV, Fomina NK, Gubsky LV, Limborska SA, Dergunova LV. Differentially Expressed Genes in Rat Brain Regions with Different Degrees of Ischemic Damage. *International Journal of Molecular Sciences*. 2025 Mar 6;26.0(5.0):2347. doi: 10.3390/ijms26052347. PMID: 40076966.

38. Aswendt M, Wieters F. Structural integrity and remodeling underlying functional recovery after stroke. *Neural Regeneration Research*. 2020 Dec 17;16.0(7.0):1423. doi: 10.4103/1673-5374.301004. PMID: 33318437.

39. Thapa K, Khan H, Grewal AK, Kanojia N, Kaur G, Sood P, Singh TG. Insights into the Therapeutic Potential of Phytoconstituents as an Epigenetic Modulators in Diabetic Nephropathy. *Current Pharmacology Reports*. 2023 Oct 26;9.0(6.0):404-426. doi: 10.1007/s40495-023-00345-9.

40. Li J, Pan L, Godoy MI, Pembroke WG, Rexach JE, Condro MC, Alvarado AG, Harteni M, Chen Y, Stiles L, Chen AY, Wanner IB, Yang X, Goldman SA, Geschwind DH, Kornblum HI, Zhang Y. Conservation and divergence of vulnerability and responses to stressors between human and mouse astrocytes. *bioRxiv (Cold Spring Harbor Laboratory)*. 2020 Apr 18;. doi: 10.1101/2020.04.17.044222.

41. Livingston JM, Lee T, Daniele E, Phillips C, Krassikova A, Enbar T, Kortebi I, Bang KA, Donville B, Ibragimov OS, Sachewsky N, Morshead CM, Faiz M. Direct reprogramming of astrocytes to neurons leads to functional recovery after stroke. *bioRxiv (Cold Spring Harbor Laboratory)*. 2020 Feb 3;. doi: 10.1101/2020.02.02.929091.

42. Chen H, Frank JA, Tan C, Lee AG, Kopchock R, Chiang T, Kim AY, Galvan MD, Fraser JF, Dornbos D, Nour HA, Millson N, Tomlinson S, McCullough LD, Pennypacker KR, Cheng M, Bliss TM, Steinberg GK. Luminal Vascular Dysfunction Drives Rapid Blood Brain Barrier Injury in Hyperglycemic Stroke: Key Roles for Luminal Glycocalyx and Complement. *bioRxiv (Cold Spring Harbor Laboratory)*. 2025 Jun 30;. doi: 10.1101/2025.06.26.661858. PMID: 40667062.

43. Talan J. At the Bench-Stroke Recovery. *Neurology Today*. 2018 Oct 4;18.0(19.0). doi: 10.1097/01.nt.0000547369.27412.7b.

44. Higo N. Motor Cortex Plasticity During Functional Recovery Following Brain Damage. *Journal of Robotics and Mechatronics*. 2022 Aug 19;34.0(4.0):700-709. doi: 10.20965/jrm.2022.p0700.

45. Betz D, Alers VA, Kenwood MR, Zuurbier KR, Coimbra R, Rhoton P, Plautz EJ, Douglas PM, Ramirez DMO, Stowe AM, Goldberg MP. Shared Transcriptomic Signatures in Perilesional and Contralesional Cortex. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Feb 4;. doi: 10.64898/2026.02.02.703306. PMID: 41676687.

46. Dong X, Zhang J, Xu G, Zhu J, Wan H, He Y. Dopamine receptors: Potential therapeutic targets for ischemia-reperfusion injury: A review. *International Journal of Biological Macromolecules*. 2026 Mar 10;354.0:151328. doi: 10.1016/j.ijbiomac.2026.151328. PMID: 41819328.

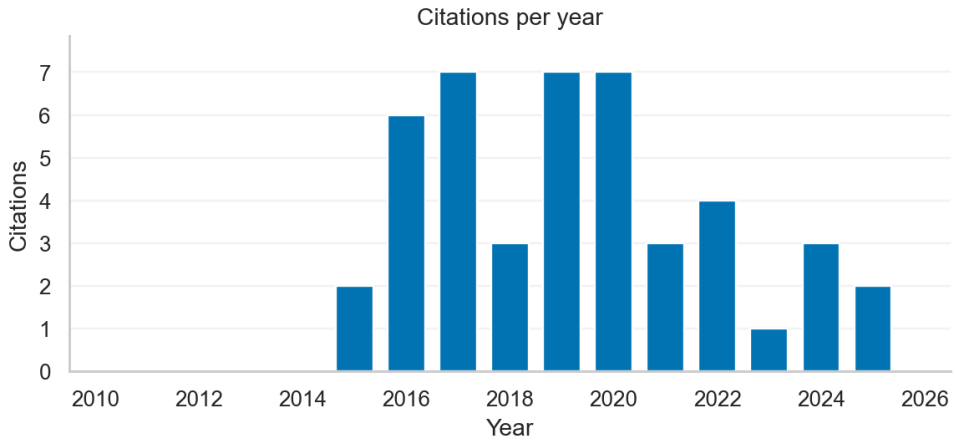
47. Tampé J, Monni E, Shen Y, Palma-Tortosa S, Brogårdh E, Lindgren AG, Kokaia Z. Time-dependent changes in monocyte subsets and gene expression patterns are associated with long-term recovery in patients with ischemic stroke. *Research Square*. 2026 Mar 16;. doi: 10.21203/rs.3.rs-8980124/v1. PMID: 41890845.

48. Betz D, Alers VA, Kenwood MR, Zuurbier KR, Coimbra R, Rhoton P, Plautz EJ, Douglas PM, Ramirez DMO, Stowe AM, Goldberg MP. Shared transcriptomic signatures in perilesional and contralesional cortex after ischemic stroke. *Journal of Neuroinflammation*. 2026 Jun 15;23.0(1.0). doi: 10.1186/s12974-026-03885-1. PMID: 42298604.

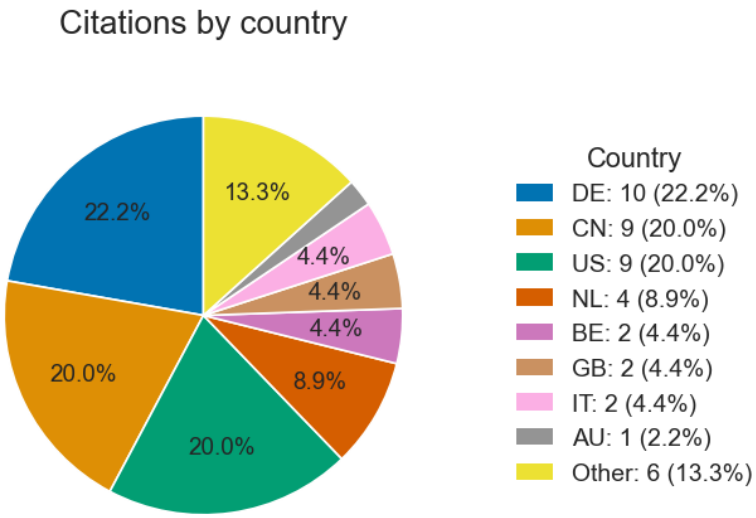
49. Vitrac C, Branscheidt M, Mahmoud WJ, Luft AR. Impaired peri-infarct long-term potentiation in anaesthetized rats suggests alternative mechanisms of post-stroke recovery. *The Journal of Physiology*. 2026 Aug 27;604.0(18.0):7819-7844. doi: 10.1113/jp291416. PMID: 42661355.

Publication

Tennstaedt A, Aswendt M, Adamczak JM, Collienne U, Selt M, Schneider G, Henn N, Schaefer C, Lagouge M, Wiedermann D, Kloppenburg P, Hoehn M. Human neural stem cell intracerebral grafts show spontaneous early neuronal differentiation after several weeks. *Biomaterials*. 2015 Jan 12;44:143-154. doi: 10.1016/j.biomaterials.2014.12.038. PMID: 25617134.



Citation summary
Total citations: **45**
Average/year: **3.75**
Period: 2015-2026
SCImago cites/doc 2y (2025): **14.1**
SJR 2025: **2.85** (Q1)



Cited by

1. Arulmoli J, Wright HJ, Phan DTT, Sheth U, Que RA, Botten GA, Keating M, Botvinick EL, Pathak MM, Zarembinski TI, Yanni DS, Razorenova OV, Hughes CC, Flanagan LA. Combination scaffolds of salmon fibrin, hyaluronic acid, and laminin for human neural stem cell and vascular tissue engineering. *Acta Biomaterialia*. 2016 Jul 28;43.0:122-138. doi: 10.1016/j.actbio.2016.07.043. PMID: 27475528.

2. Huang D, Cao Y, Yang X, Liu Y, Zhang Y, Li C, Chen G, Wang Q. A Nanoformulation-Mediated Multifunctional Stem Cell Therapy with Improved Beta-Amyloid Clearance and Neural Regeneration for Alzheimer's Disease. *Advanced Materials*. 2021 Feb 24;33.0(13.0):e2006357. doi: 10.1002/adma.202006357. PMID: 33624894.

3. Medda X, Mertens L, Versweyveld S, Diels A, Barnham L, Bretteville A, Buist A, Verheyen A, Royaux IE, Ebneth A, Cabrera-Socorro A. Development of a Scalable, High-Throughput-Compatible Assay to Detect Tau Aggregates Using iPSC-Derived Cortical Neurons Maintained in a Three-Dimensional Culture Format. *SLAS DISCOVERY*. 2016 Mar 17;21.0(8.0):804-815. doi: 10.1177/1087057116638029. PMID: 26984927.

4. Scarfe L, Brillant N, Kumar JD, Ali N, Alrumayh A, Amali MO, Barbellion S, Jones V, Niemeijer M, Potdevin S, Roussignol G, Vaganov A, Barbaric I, Barrow M, Burton NC, Connell JJ, Dazzi F, Edsbagge J, French NS, Holder JC, Hutchinson C, Jones D, Kalber TL, Lovatt CA, Lythgoe MF, Patel S, Patrick PS, Piner J, Reinhardt J, Ricci E, Sidaway JE, Stacey GN, Lewis PJS, Sullivan GJ, Taylor A, Wilm B, Poptani H, Murray PA, Goldring CE, Park BK. Preclinical imaging methods for assessing the safety and efficacy of regenerative medicine therapies. *npj Regenerative Medicine*. 2017 Oct 11;2.0(1.0):28. doi: 10.1038/s41536-017-0029-9. PMID: 29302362.

5. Boese AC, Eckert A, Hamblin MH, Lee J. Human neural stem cells improve early stage stroke outcome in delayed tissue plasminogen activator-treated aged stroke brains. *Experimental Neurology*. 2020 Mar 5;329.0:113275. doi:

10.1016/j.expneurol.2020.113275. PMID: 32147438.

6. Yang C, Tian R, Liu T, Liu G. MRI Reporter Genes for Noninvasive Molecular Imaging. *Molecules*. 2016 May 18;21.0(5.0):580. doi: 10.3390/molecules21050580. PMID: 27213309.

7. Green C, Minassian A, Vogel SN, Diedenhofen M, Beyrau A, Wiedermann D, Hoehn M. Sensorimotor Functional and Structural Networks after Intracerebral Stem Cell Grafts in the Ischemic Mouse Brain. *Journal of Neuroscience*. 2018 Jan 10;38.0(7.0):1648-1661. doi: 10.1523/jneurosci.2715-17.2018. PMID: 29321138.

8. Aswendt M, Vogel S, Schäfer C, Jathoul AP, Pulé M, Hoehn M. Quantitative in vivo dual-color bioluminescence imaging in the mouse brain. *Neurophotonics*. 2019 May 7;6.0(2.0):1-11. doi: 10.1117/1.nph.6.2.025006. PMID: 31093514.

9. Spurlock MS, Ahmed AI, Rivera KN, Yokobori S, Lee SW, Sam PN, Shear DA, Hefferan MP, Hazel TG, Johe KK, Gajavelli S, Tortella FC, Bullock RM. Amelioration of Penetrating Ballistic-Like Brain Injury Induced Cognitive Deficits after Neuronal Differentiation of Transplanted Human Neural Stem Cells. *Journal of Neurotrauma*. 2017 Mar 1;34.0(11.0):1981-1995. doi: 10.1089/neu.2016.4602. PMID: 28249550.

10. Crane AT, Voth JP, Shen FX, Low WC. Concise Review: Human-Animal Neurological Chimeras: Humanized Animals or Human Cells in an Animal?. *Stem Cells*. 2019 Jan 10;37.0(4.0):444-452. doi: 10.1002/stem.2971. PMID: 30629789.

11. Hamblin MH, Murad R, Yin J, Vallim G, Lee J. Modulation of gene expression on a transcriptome-wide level following human neural stem cell transplantation in aged mouse stroke brains. *Experimental Neurology*. 2021 Nov 6;347.0:113913. doi: 10.1016/j.expneurol.2021.113913. PMID: 34752785.

12. Pongrac IM, Radmilović MD, Ahmed LB, Mlinarić H, Regul J, Škokić S, Babič M, Horák D, Hoehn M, Gajović S. D-mannose-Coating of Maghemite Nanoparticles Improved Labeling of Neural Stem Cells and Allowed Their Visualization by ex vivo MRI after Transplantation in the Mouse Brain. *Cell Transplantation*. 2019 May 1;28.0(5.0):553-567. doi: 10.1177/0963689719834304. PMID: 31293167.

13. McMillan N, McMillan A, Kiliaan PC, Yahya T, Thakkar R, Weiner HL, Smirnakis SM, Yavagal DR, Izzy S. Stem cell-mediated recovery in stroke: partnering with the immune system. *Nature reviews. Neuroscience*. 2025 Nov 13;27.0(1.0):23-43. doi: 10.1038/s41583-025-00985-4. PMID: 41233589.

14. Han C, Chen S, Shi Z, Guan X, Zou W, Liu J. An innovative treatment for hypoxic-ischemic encephalopathy: Silk fibroin nanomaterials improve neural stem cell axon formation and facilitate cognitive improvement. *Neural Regeneration Research*. 2025 Jun 19;21.0(7.0):3017-3025. doi: 10.4103/nrr.nrr-d-24-01178. PMID: 40537000.

15. Ortuño-Lizarán I, Vilariño-Feltrer G, Martínez-Ramos C, Pradas MM, Vallés-Lluch A. Influence of synthesis parameters on hyaluronic acid hydrogels intended as nerve conduits. *Biofabrication*. 2016 Oct 24;8.0(4.0):045011. doi: 10.1088/1758-5090/8/4/045011. PMID: 27775923.

16. Tennstaedt A, Mastropietro A, Nelles M, Beyrau A, Hoehn M. In Vivo Fate Imaging of Intracerebral Stem Cell Grafts in Mouse Brain. *PLoS ONE*. 2015 Dec 7;10.0(12.0):e0144262. doi: 10.1371/journal.pone.0144262. PMID: 26641453.

17. Kassi A, Mahavadi AK, Clavijo A, Caliz D, Lee SW, Ahmed AI, Yokobori S, Hu Z, Spurlock MS, Wasserman JM, Rivera KN, Nodal S, Powell HR, Di L, Torres R, Leung LY, Rubiano AM, Bullock RM, Gajavelli S. Enduring Neuroprotective Effect of Subacute Neural Stem Cell Transplantation After Penetrating TBI. *Frontiers in Neurology*. 2019 Jan 17;9.0:1097. doi: 10.3389/fneur.2018.01097. PMID: 30719019.

18. Minassian A, Green C, Diedenhofen M, Vogel SN, Heß S, Stoeber M, Radmilović MD, Wiedermann D, Kloppenburg P, Hoehn M. Human Neural Stem Cell Induced Functional Network Stabilization After Cortical Stroke: A Longitudinal Resting-State fMRI Study in Mice. *Frontiers in Cellular Neuroscience*. 2020 Apr 7;14.0:86. doi: 10.3389/fncel.2020.00086. PMID: 32317940.

19. Vogel SN, Aswendt M, Nelles M, Henn N, Schneider G, Hoehn M. Initial graft size and not the innate immune response limit survival of engrafted neural stem cells. *Journal of Tissue Engineering and Regenerative Medicine*. 2017 Jun 9;12.0(3.0):784-793. doi: 10.1002/term.2497. PMID: 28599089.

20. Van der Linden A, Hoehn M. Monitoring Neuronal Network Disturbances of Brain Diseases: A Preclinical MRI Approach in the Rodent Brain. *Frontiers in Cellular Neuroscience*. 2022 Jan 3;15.0:815552. doi: 10.3389/fncel.2021.815552. PMID: 35046778.

21. Liu T, Li Z, Li X, Zhao R, Wei X, Wang Z, Xin X. In vivo visualization of murine melanoma cells B16-derived exosomes through magnetic resonance imaging. *Biochimica et Biophysica Acta (BBA) - General Subjects*. 2021 Nov 22;1866.0(2.0):130062. doi: 10.1016/j.bbagen.2021.130062. PMID: 34822924.

22. Fan H, Duan H, Hao P, Gao Y, Zhao W, Hao F, Li X, Yang Z. Cellular regeneration treatments for traumatic brain injury. *Medicine in Novel Technology and Devices*. 2022 Oct 19;16.0:100182. doi: 10.1016/j.medntd.2022.100182.

23. Lin G, He X, Liang F, Guo Y, Sunnassee G, Chen J, Cao X, Chen Y, Pan G, Pei Z, Tan SHD. Transplanted human neural precursor cells integrate into the host neural circuit and ameliorate neurological deficits in a mouse model of traumatic brain injury. *Neuroscience Letters*. 2018 Mar 1;674.0:11-17. doi: 10.1016/j.neulet.2018.02.064. PMID: 29501684.

24. Aswendt M, Henn N, Michalk S, Schneider G, Steiner M, Bissa U, Dose C, Hoehn M. Novel Bimodal Iron Oxide Particles for Efficient Tracking of Human Neural Stem Cells in Vivo. *Nanomedicine*. 2015 Aug 1;10.0(16.0):2499-2512. doi: 10.2217/nnm.15.94. PMID: 26296195.

25. Vogel SN, Schäfer C, Heß S, Folz-Donahue K, Nelles M, Minassian A, Schwarz MK, Kukat C, Ehrlich M, Zaehres H, Kloppenburg P, Hoehn M, Aswendt M. The in vivo timeline of differentiation of engrafted human neural progenitor cells. *Stem Cell Research*. 2019 Mar 25;37.0:101429. doi: 10.1016/j.scr.2019.101429. PMID: 30933718.

26. Mezzanotte L, Iljas JD, Que I, Chan AB, Kaijzel EL, Hoeben RC, Löwik CW. Optimized Longitudinal Monitoring of Stem Cell Grafts in Mouse Brain Using a Novel Bioluminescent/Near Infrared Fluorescent Fusion Reporter. *Cell Transplantation*. 2017 Dec 1;26.0(12.0):1878-1889. doi: 10.1177/0963689717739718. PMID: 29390874.

27. Pottmeier P, Doszyn O, Peuckert C, JAZIN EE. Increased Expression of Y-Encoded Demethylases During Differentiation of Human Male Neural Stem Cells. *Stem Cells and Development*. 2020 Oct 12;29.0(23.0):1497-1509. doi: 10.1089/scd.2020.0138. PMID: 33040644.

28. Cai J, He X, Zhou Y, Mu T, Liao Y, Jiang L, Qin Y. Magnetic resonance imaging focused on the ferritin heavy chain 1 reporter gene detects neuronal differentiation in stem cells. *Neural Regeneration Research*. 2022 Oct 14;18.0(7.0):1563. doi: 10.4103/1673-5374.358608. PMID: 36571363.

29. Green C, Minassian A, Vogel SN, Diedenhofen M, Wiedermann D, Hoehn M. Persistent Quantitative Vitality of Stem Cell Graft Is Necessary for Stabilization of Functional Brain Networks After Stroke. *Frontiers in Neurology*. 2019 Apr 5;10.0:335. doi: 10.3389/fneur.2019.00335. PMID: 31024429.

30. Mousavinejad M, Skidmore S, Barone FG, Tyers P, Pisupati V, Poptani H, Plagge A, Barker R, Murray PA, Taylor A, Hill CJ. Assessing Human Embryonic Stem Cell-Derived Dopaminergic Neuron Progenitor Transplants Using Non-invasive Imaging Techniques. *Molecular Imaging and Biology*. 2020 May 6;22.0(5.0):1244-1254. doi: 10.1007/s11307-020-01499-4. PMID: 32378000.

31. Madsen SD, Giler MK, Bunnell BA, O'Connor KC. Illuminating the Regenerative Properties of Stem Cells In Vivo with Bioluminescence Imaging. *Biotechnology Journal*. 2020 Oct 22;16.0(3.0):e2000248. doi: 10.1002/biot.202000248. PMID: 33089922.

32. Chen J, Li D, Li H, Zhu K, Shi L, Xu Y. Cell membrane-targeting NIR fluorescent probes with large Stokes shifts for ultralong-term transplanted neural stem cell tracking. *Frontiers in Bioengineering and Biotechnology*. 2023 Feb 9;11.0:1139668. doi: 10.3389/fbioe.2023.1139668. PMID: 36845195.

33. Tao Z, Gao P, Yan Y, Li H, Song J, Yang J. Osthole Enhances the Therapeutic Efficiency of Stem Cell Transplantation in Neuroendoscopy Caused Traumatic Brain Injury. *Biological and Pharmaceutical Bulletin*. 2017 Jan 1;40.0(7.0):1043-1054. doi: 10.1248/bpb.b17-00072. PMID: 28674247.

34. Schomann T, Mezzanotte L, de Groot JC, Rivolta MN, Hendriks SH, Frijns JHM, Huisman MA. Neuronal differentiation of hair-follicle-bulge-derived stem cells co-cultured with mouse cochlear modiolus explants. *PLoS ONE*. 2017 Oct 30;12.0(10.0):e0187183. doi: 10.1371/journal.pone.0187183. PMID: 29084289.

35. Shokrollahi N, Shahbazzadeh D, Bagheri KP, Habibi-Anbouhi M, Jahanian-Najafabadi A, Behdani M. A Model to Study the Phenotypic Changes of Insect Cell Transfection by Copepod Super Green Fluorescent Protein (cop-GFP) in Baculovirus Expression System. *PubMed*. 2016 Jul 1;20.0(3.0):182-6. doi: 10.7508/ibj.2016.03.008. PMID: 26518237.

36. Collmann FM, Pijnenburg R, Schneider G, Schäfer C, Folz-Donahue K, Kukat C, Hoehn M. Imaging Reporter Strategy to Monitor Gene Activation of Microglia Polarisation States under Stimulation. *Journal of Neuroimmune Pharmacology*. 2018 May 22;13.0(3.0):371-382. doi: 10.1007/s11481-018-9789-2. PMID: 29790106.

37. Gyimesi M, Oikari LE, Yu C, Sutherland HGE, Nyholt DR, Griffiths LR, van Wijnen AJ, Okolicsanyi RK, Haupt LM. CpG methylation changes in human mesenchymal and neural stem cells in response to in vitro niche modifications. *Biochimie*. 2024 Apr 17;223.0:147-157. doi: 10.1016/j.biochi.2024.04.007. PMID: 38640996.

38. Aswendt M, Vogel S, Hoehn M. In vivo bioluminescence imaging to elucidate stem cell graft differentiation. *Neural Regeneration Research*. 2019 Sep 16;15.0(1.0):61. doi: 10.4103/1673-5374.264449. PMID: 31535647.

39. Schomann T, Iljas JD, Que I, Li Y, Suidgeest E, Cruz LJ, Frijns JHM, Chan AB, Löwik CM, Huisman MA, Mezzanotte L. Multimodal imaging of hair follicle bulge-derived stem cells in a mouse model of traumatic brain injury. *Cell and Tissue Research*. 2020 Feb 8;381.0(1.0):55-69. doi: 10.1007/s00441-020-03173-1. PMID: 32036485.

40. O'Donnell JC, Swanson RL, Wofford KL, Grovola MR, Purvis EM, Petrov D, Cullen DK. Emerging Approaches for Regenerative Rehabilitation Following Traumatic Brain Injury. *Physiology in health and disease*. 2022 Jan 1;:409-459. doi: 10.1007/978-3-030-95884-8_13.

41. Aswendt M, Collmann FM, Hoehn M. Neurobiological insights from bioluminescence imaging. *Oncotarget*. 2017 Aug 13;8.0(41.0):69198-69199. doi: 10.18632/oncotarget.20302. PMID: 29050195.

42. Merighi F, De Vincentiis S, Onorati M, Raffa V. Long-term Mouse Spinal Cord Organotypic Slice Culture as a Platform for Validating Cell Transplantation in Spinal Cord Injury. *Journal of Visualized Experiments*. 2024 Apr 12;(206.0). doi: 10.3791/66704. PMID: 38682940.

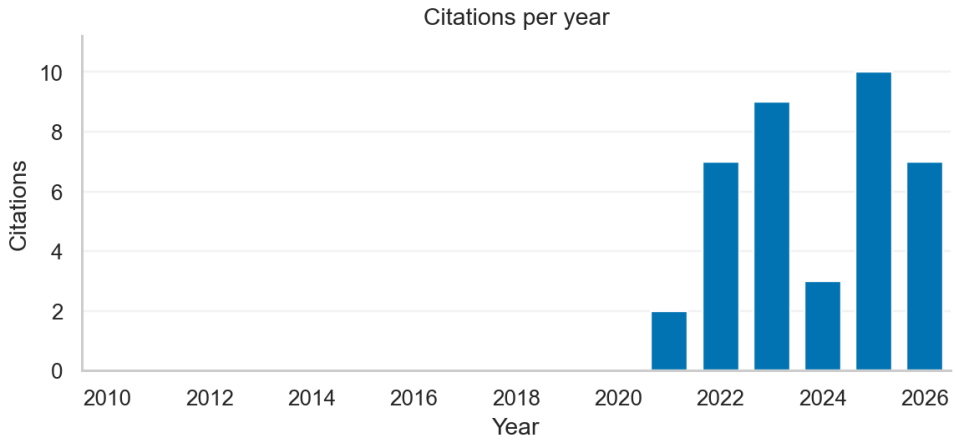
43. Lois T. One Step at a Time, Stem Cell Therapy for Traumatic Brain Injury Needs Two More Breakthroughs. 2016 Jan 1;.

44. Merighi F, De Vincentiis S, Onorati M, Raffa V. JoVE Video Dataset. 2024 Apr 12;. doi: 10.3791/66704-v.

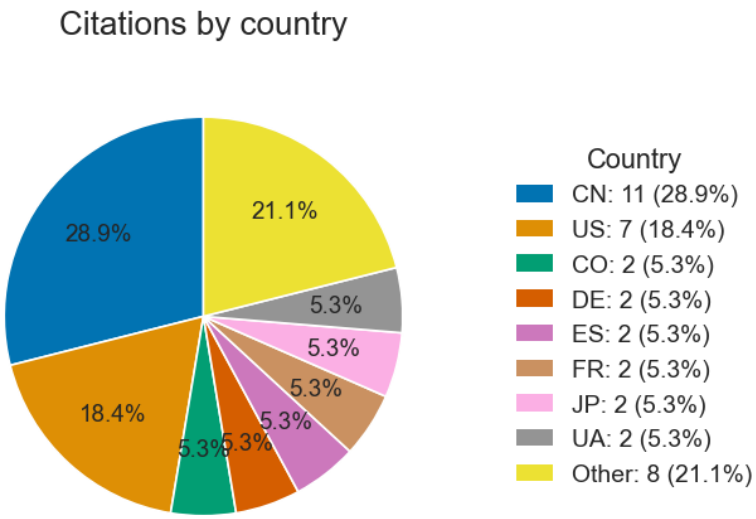
45. Le Friec A. Evolution of corticospinal tract integrity in stroked marmoset monkeys : Towards a bioimplant and stem cell therapeutic strategy. *Thèses en ligne de l'Université Toulouse III (Université Toulouse III)*. 2020 Jan 20;. doi: 10.70675/2412aa7cz73eez4ce3za0f4z1529f930f5f4.

Publication

Wieters F, Lucas CW, Gruhn M, Büschges A, Fink GR, Aswendt M. Introduction to spasticity and related mouse models. *Experimental Neurology*. 2020 Sep 29;335:113491. doi: 10.1016/j.expneurol.2020.113491. PMID: 33007294.



Citation summary
Total citations: **38**
Average/year: **5.43**
Period: 2020-2026
SCImago cites/doc 2y (2025): **5.03**
SJR 2025: **1.39** (Q1)



Cited by

1. Bhagwani A, Chopra M, Kumar H. Spinal Cord Injury Provoked Neuropathic Pain and Spasticity, and Their GABAergic Connection. *Neurospine*. 2022 Sep 29;19.0(3):646-668. doi: 10.14245/ns.2244368.184. PMID: 36203291.

2. Wang Z, Liu Q, Li D, Wei L, Chu E, Bao Y, Liu Z, Jin L, Chen X, Zhao K. Clinical efficacy observation of repetitive magnetic stimulation for treating upper limb spasticity after stroke. *Scientific Reports*. 2025 May 21;15.0(1):17722. doi: 10.1038/s41598-025-02443-8. PMID: 40399427.

3. Iwasawa E, Brown FN, Shula C, Kahn F, Lee SH, Berta T, Ladle DR, Campbell K, Mangano FT, Goto J. The Anti-Inflammatory Agent Bindarit Attenuates the Impairment of Neural Development through Suppression of Microglial Activation in a Neonatal Hydrocephalus Mouse Model. *Journal of Neuroscience*. 2022 Jan 6;42.0(9):1820-1844. doi: 10.1523/jneurosci.1160-21.2021. PMID: 34992132.

4. Iwasawa E, Brown FN, Shula C, Kahn F, Lee SH, Berta T, Ladle DR, Campbell K, Mangano FT, Goto J. The Anti-Inflammatory Agent Bindarit Attenuates the Impairment of Neural Development through Suppression of Microglial Activation in a Neonatal Hydrocephalus Mouse Model. *Journal of Bioresource Management*. 2023 Oct 24;.

5. Hörner M, Groh J, Klein D, Ilg W, Schöls LJ, Santos SD, Bergmann A, Klebe S, Cauhapé M, Branchu J, Hachimi KHE, Stévanin G, Darios FD, Martini R. CNS-associated T-lymphocytes in a mouse model of Hereditary Spastic Paraplegia type 11 (SPG11) are therapeutic targets for established immunomodulators. *Experimental Neurology*. 2022 May 20;355.0:114119. doi: 10.1016/j.expneurol.2022.114119. PMID: 35605667.

6. Blaschke SJ, Vlachakis S, Pallast N, Walter HL, Volz LJ, Wiedermann D, Fink GR, Hoehn M, Aswendt M, Schroeter M, Rueger MA. Transcranial Direct Current Stimulation Reverses Stroke-Induced Network Alterations in Mice. *Stroke*. 2023 Jun 28;54.0(8):2145-2155. doi: 10.1161/strokeaha.123.042808. PMID: 37377015.

7. Yokota K, Kawano O, Sakai H, Morishita Y, Masuda M, Hayashi T, Kubota K, Ideta R, Ariji Y, Koga R, Murai S, Ifuku R, Uemura M, Katoh H, Nakashima Y, Maeda T. Predicting the Progression of Spasticity in the Early Phase of Spinal Cord Injury: A Prospective Cohort Study. *Journal of Neurotrauma*. 2023 Sep 29;41.0(9-10):1122-1132. doi: 10.1089/neu.2023.0191. PMID: 37772699.

8. Zhang Z, Ma L, Mu J, Mu J, Chen M, Qian X, Zhang Q, Ma L, Ma L. Acupuncture waggle needling alleviates spastic movement disorder in post-stroke rats via GLT-1 activation to modulate Glu/GABA-Gln cycle. *Physiology & Behavior*. 2025 Aug 7;301.0:115059. doi: 10.1016/j.physbeh.2025.115059. PMID: 40783057.

9. He JN, Ma L, Ma L, Zhuang Y, Wen J, Ma L, Ma L, Xiu J, Chen M. Acupuncture Modulates NMDAR-PP1/Calpain1-KCC2 Pathway to Ameliorate Spinal Hyperexcitability and Spastic Hemiplegia Induced by Ischemic Stroke. *Journal of Integrative Neuroscience*. 2025 Dec 26;24.0(12):46980. doi: 10.31083/jin46980. PMID: 41503987.

10. Benson CA, Olson K, Patwa S, Kauer SD, King JF, Waxman SG, Tan AM. Conditional Astrocyte Rac1KO Attenuates Hyperreflexia after Spinal Cord Injury. *Journal of Neuroscience*. 2023 Nov 14;44.0(1):e1670222023. doi: 10.1523/jneurosci.1670-22.2023. PMID: 37963762.

11. Zhang Z, Mu J, Liangxiao MA, Wenyan YU, Tianyi S, Xu Q, Tian Y, Yidan Z, Wang J. Waggle needling Yanglingquan (GB34) enhances cerebral blood flow and ameliorates condition of spastic paresis in rats with middle cerebral artery occlusion induced by surgical nylon suture. *PubMed*. 2023 Jun 1;43.0(3):514-522. doi: 10.19852/j.cnki.jtcm.20230328.001. PMID: 37147753.

12. Goltash S, Khodr R, Bui TV, Laliberté AM. An optogenetic mouse model of hindlimb spasticity after spinal cord injury. *Experimental Neurology*. 2025 Jan 23;386.0:115157. doi: 10.1016/j.expneurol.2025.115157. PMID: 39863244.

13. Othman AM, Abdallah OY, Elnaggar YS. Topical hyalubilosomes of dantrolene sodium as muscle targeted nanocarrier for muscle spasms: fabrication, ex-vivo permeation and behavioral animal model. *Pharmaceutical Development and Technology*. 2025 May 15;30.0(5):590-603. doi: 10.1080/10837450.2025.2504999. PMID: 40371687.

14. Yu J, Li Y, Yang L, Li Y, Zhang S, Yang S. The highest region of muscle spindle abundance should be the optimal target of botulinum toxin A injection to block muscle spasms in rats. *Frontiers in Neurology*. 2023 Feb 23;14.0:1061849. doi: 10.3389/fneur.2023.1061849. PMID: 36908586.

15. Mei J, Xue Y, Li J, Zhang L, Zhang J, Wang Y, Su K, Gao J, Guo J, Li R. Effects of Functional Acupuncture on Upper Limb Spasticity After Ischemic Stroke: A Protocol for a Randomized Controlled Parallel Clinical Trial. *Frontiers in Neurology*. 2022 May 18;13.0:835408. doi: 10.3389/fneur.2022.835408. PMID: 35665035.

16. Wang S, Qiu L, Zhou Q, Chen C, Wu J. Serotonin syndrome caused by escitalopram in Parkinson's disease psychosis: a case report. *BMC Geriatrics*. 2024 Sep 18;24.0(1):769. doi: 10.1186/s12877-024-05371-w. PMID: 39294572.

17. Barbay T, Pecchi É, Ramírez-Franco J, Ivanov A, Brocard F, Rouach N, Bos R. Functional contribution of astrocytic Kir4.1 channels to spasticity after spinal cord injury. *Brain*. 2025 May 6;148.0(10):3576-3591. doi: 10.1093/brain/awaf147. PMID: 40333399.

18. Yan J, Luo Q, Chen Z, Wang Z, Guo X, Xie Q, Oetomo D, Tan Y, Niu CM. Spike-Based Neuromorphic Model of Spasticity for Generation of Affected Neural Activity. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. 2025 Jan 1;33.0:1360-1371. doi: 10.1109/tnsre.2025.3557044. PMID: 40173059.

19. de Lima MSN, Paz CCDSC, Ribeiro TG, Fachin-Martins E. Assessment of Passive Upper Limb Stiffness and Its Function in Post-Stroke Individuals Wearing an Inertial Sensor during the Pendulum Test. *Sensors*. 2023 Mar 27;23.0(7):3487. doi: 10.3390/s23073487. PMID: 37050547.

20. Jia J, Gong S, Zhang A, Jiang L, Yao Y. Stiffening of the gluteal muscle increased the intramuscular stress: An in-silico implication of deep tissue injury. *Heliyon*. 2023 Feb 1;9.0(2):e13459. doi: 10.1016/j.heliyon.2023.e13459. PMID: 36816309.

21. Пизова НВ, Pizov NA, Mareev M. Post-stroke pain and rehabilitation (case report). *Russian Medical Inquiry*. 2023 Jan 1;7.0(10):672-679. doi: 10.32364/2587-6821-2023-7-10-10.

22. Hanasaki T, Hanaki K, Sako Y, Uchiyama Y, Lee S. Effect of continuous Ia fibre activity suppression on hyperreflexia-related spasticity and maladaptive synaptic connections in the spinal cord after injury. *Scientific Reports*. 2025 Jul 3;15.0(1):23764. doi: 10.1038/s41598-025-09397-x. PMID: 40610652.

23. Charalampopoulou A, Taga A, Rust K, Luciani EG, Marshall KL, MONTGOMERY E, MANSINGHKA A, Singh R, Zhao Y, O'Keefe CM, Wang T, Venkatesan A, Habela CW, Maragakis NJ. Development of a human iPSC-derived corticospinal tract-on-a-chip. *Cell Reports Methods*. 2026 May 19;6.0(7):101457. doi: 10.1016/j.crmeth.2026.101457. PMID: 42161268.

24. Chen L, Li W, Cheng S, Liang S, Huang M, Lei T, Hu X, Liang Z, Zheng H. Correlation of N30 somatosensory evoked potentials with spasticity and neurological function after stroke: A cross-sectional study. *Journal of Rehabilitation Medicine*. 2021 Jan 1;53.0(9):jrm00223. doi: 10.2340/16501977-2867. PMID: 34435643.

25. García MF, Nossa S, Jimenez-Ramirez JD, Romero-Barreto C, Mendoza-Pulido C. Terminal nerve entry points' locations to muscles of the thigh for selective peripheral neurectomy in the adult population: a cadaveric study. *Surgical and Radiologic Anatomy*. 2022 Dec 16;45.0(1):3-9. doi: 10.1007/s00276-022-03047-x. PMID: 36522468.

26. Shunenkov DA, Loginov AA, Bosenko SA, Saveliev OG, Kovaleva NY, Vorobiev AV, Lebedev AS, Kanarskii MM. Present problem of neurororehabilitology: methods for quantitative assessment of pathological increased muscle tone. *Medical and Social Expert Evaluation and Rehabilitation*. 2021 Dec 1;24.0(3):23-34. doi: 10.17816/mser83089.

27. García MF, Nossa S, Jimenez-Ramirez JD, Romero-Barreto C, Mendoza-Pulido C. Terminal Nerve Entry Points Locations to Muscles of the Thigh for Selective Peripheral Neurectomy in the Adult Population: A Cadaveric Study. *Research Square*. 2022 Oct 17;. doi: 10.21203/rs.3.rs-2133370/v1.

28. Ivanchenko O, Melnikova O, Lukina HM, Mikaelyan GR. ROSPECTS FOR THE USE OF H-REFLEXOMETRY IN THE STUDY OF HUMAN MOTOR FUNCTION. *Здобутки клінічної і експериментальної медицини*. 2024 Mar 28;(1):6-16. doi: 10.11603/1811-2471.2024.v.i1.14522.

29. Marand LA, Azizi MR, Dehkordi SN. The relationship between trunk function and spasticity in people with multiple sclerosis. *Journal of Bodywork and Movement Therapies*. 2024 Dec 12;42.0:162-166. doi: 10.1016/j.jbmt.2024.12.003. PMID: 40325663.

30. Charalampopoulou A, Taga A, Rust K, Luciani EG, Marshall KL, Montgomery E, MANSINGHKA A, Singh R, Zhao Y, O'Keefe CM, Wang T, Venkatesan A, Habela CW, Maragakis NJ. Development of a Human iPSC-Derived "Corticospinal Tract-on-a-Chip" for Neurodegenerative Disease Research. *bioRxiv (Cold Spring Harbor Laboratory)*. 2025 Sep 25;. doi: 10.1101/2025.09.24.678092. PMID: 41040384.

31. Sahaidak V, Rybachuk OA, Melikov ZK, Medvediev VV. Effect of implantation of a PHPMA hydrogel containing human mesenchymal stromal cells of different origins on motor function recovery in rats with spinal cord injury. *Cell and Organ Transplantology*. 2025 Nov 30;13.0(2). doi: 10.22494/cot.v13-2.186.

32. Kauer SD, Effraim PR, Bangalore L, Waxman SG, Tan AM. Dendritic spine dysgenesis in spinal cord injury: A structural contributor to pain and spasticity. *Experimental Neurology*. 2026 Feb 5;399.0:115679. doi: 10.1016/j.expneurol.2026.115679. PMID: 41654287.

33. Kaul S, Liu L, Candelario-Jalil E, Vahdat S. Ischemic injury to primary motor and premotor cortices is linked to hyperreflexia in mice. *Neurobiology of Disease*. 2026 Mar 14;222.0:107354. doi: 10.1016/j.nbd.2026.107354. PMID: 41839378.

34. Crugeiras-Sampedro M, Zas-Veiga L, Piñeiro-Ramil M, Pazos-Pérez A, López-López V, Jorge-Mora AA, Alonso-Pérez A, Gómez R. Baclofen Promotes Osteochondrogenic Commitment of Mesenchymal Stem Cells: Implications for Heterotopic Ossification Risk. *International Journal of Molecular Sciences*. 2026 Mar 19;27.0(6):2783. doi: 10.3390/ijms27062783. PMID: 41898644.

35. Canon L. Mechanistic and structural basis for tuning of myosin activity. 2022 Oct 19;. doi: 10.70675/94badd12z7c22z450bz8651zbe319fc38481.

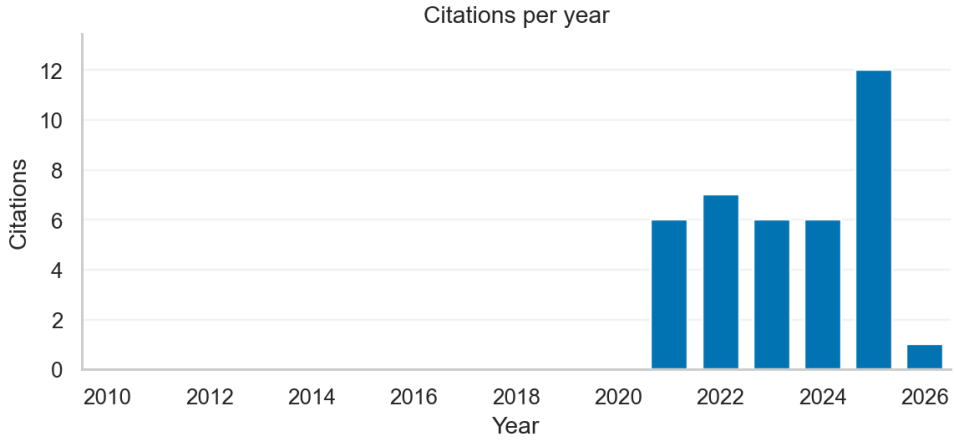
36. Sahaidak V, Medvediev VV. pHPMA-hydrogel does not improve intensive autogenic recovery of motor function after lateral spinal cord fragment excision in adult rats. *Ukrainian Scientific Medical Youth Journal*. 2026 Mar 31;160.0(1):50-61. doi: 10.32345/usmyj.1(160).2026.50-61.

37. González-Mingorance L, Andreu-Caravaca L, Rubio-Arias JÁ. Effects of physical exercise on spasticity in people with multiple sclerosis: A systematic review. *Physiotherapy*. 2026 Apr 29;133.0:102334. doi: 10.1016/j.physio.2026.102334. PMID: 42424757.

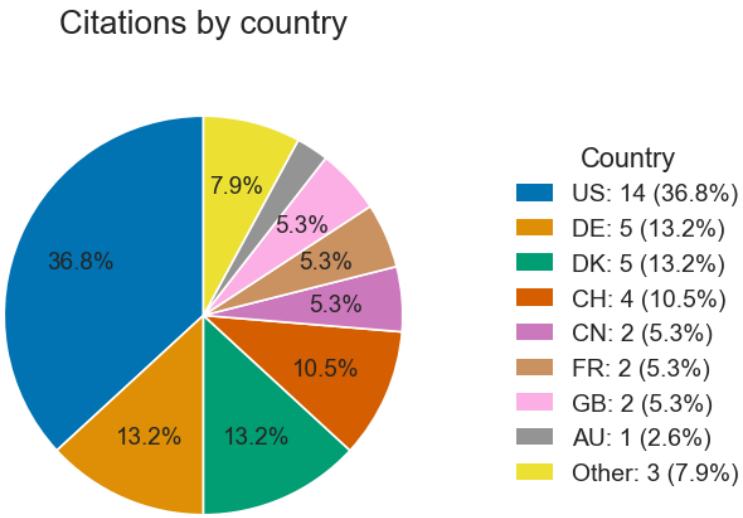
38. Sun T, Ma L, Ma L, Ma L, Ma L, Zhang Q, Qian X, Xiu J, Wang X. Spastic Hypertonia Following Middle Cerebral Artery Occlusion in Rats: Assessment of Behavioral, Electrophysiological, and Morphological Parameters in Dynamics. *Bulletin of Experimental Biology and Medicine*. 2026 Apr 1;180.0(6):688-695. doi: 10.1007/s10517-026-06700-2. PMID: 42525344.

Publication

Leuze CW, Goubran M, Baraković M, Aswendt M, Tian Q, Hsueh B, Crow A, Weber EMM, Steinberg GK, Zeineh M, Plowey ED, Daducci A, Innocenti G, Thiran J, Deisseroth KA, McNab JA. Comparison of diffusion MRI and CLARITY fiber orientation estimates in both gray and white matter regions of human and primate brain. *NeuroImage*. 2020 Dec 29;228:117692. doi: 10.1016/j.neuroimage.2020.117692. PMID: 33385546.



Citation summary
Total citations: **38**
Average/year: **5.43**
Period: 2020-2026
SCImago cites/doc 2y (2025): **5.31**
SJR 2025: **1.93** (Q1)



Cited by

1. Yendiki A, Aggarwal M, Axer M, Howard AF, van Walsum AVC, Haber SN. Post mortem mapping of connectional anatomy for the validation of diffusion MRI. *NeuroImage*. 2022 Mar 25;256.0:119146. doi: 10.1016/j.neuroimage.2022.119146. PMID: 35346838.

2. Grisot G, Haber SN, Yendiki A. Diffusion MRI and anatomic tracing in the same brain reveal common failure modes of tractography. *NeuroImage*. 2021 Jun 22;239.0:118300. doi: 10.1016/j.neuroimage.2021.118300. PMID: 34171498.

3. Yang JY, Yeh C, Poupon C, Calamante F. Diffusion MRI tractography for neurosurgery: the basics, current state, technical reliability and challenges. *Physics in Medicine and Biology*. 2021 Jun 22;66.0(15.0):15TR01. doi: 10.1088/1361-6560/ac0d90. PMID: 34157706.

4. Georgiadis M, Schroeter A, Gao Z, Guizar-Sicairos M, Liebi M, Leuze CW, McNab JA, Balolia A, Veraart J, Ades-Aron B, Kim S, Shepherd TM, Lee CH, Walczak P, Chodankar S, DiGiacomo PS, Dávid G, Augath MA, Zerbi V, Sommer S, Rajković I, Weiss TM, Bunk O, Yang L, Zhang J, Novikov DS, Zeineh M, Fieremans E, Rudin M. Nanostructure-specific X-ray tomography reveals myelin levels, integrity and axon orientations in mouse and human nervous tissue. *Nature Communications*. 2021 May 19;12.0(1.0):2941. doi: 10.1038/s41467-021-22719-7. PMID: 34011929.

5. Barbosa DAN, Gattas S, Salgado JS, Kuijper FM, Wang AR, Huang Y, Kakusa BW, Leuze CW, Luczak A, Rapp PE, Malenka RC, Hermes D, Miller KJ, Heifets BD, Bohon C, McNab JA, Halpern CH. An orexigenic subnetwork within the human hippocampus. *Nature*. 2023 Aug 30;621.0(7978.0):381-388. doi: 10.1038/s41586-023-06459-w. PMID: 37648849.

6. Tisdall L, MacNiven KH, Leong JK, Frey R, Rieskamp J, Hertwig R, Knutson BD, Mata R. Structural projections to the nucleus accumbens link to impulsive components of human risk preference. *Imaging Neuroscience*. 2024 Jan 1;2.0:1-15.

doi: 10.1162/imag_a_00344. PMID: 40800310.

7. Tisdall L, MacNiven KH, Padula CB, Leong JK, Knutson BD. Brain tract structure predicts relapse to stimulant drug use. *Proceedings of the National Academy of Sciences*. 2022 Jun 21;119.0(26.0):e2116703119. doi: 10.1073/pnas.2116703119. PMID: 35727973.

8. Amunts K, Lippert T. Brain research challenges supercomputing. *Science*. 2021 Nov 25;374.0(6571.0):1054-1055. doi: 10.1126/science.abl8519. PMID: 34822267.

9. Blanke N, Chang S, Novoseltseva A, Wang H, Boas DA, Bigio IJ. Multiscale label-free imaging of myelin in human brain tissue with polarization-sensitive optical coherence tomography and birefringence microscopy. *Biomedical Optics Express*. 2023 Sep 7;14.0(11.0):5946. doi: 10.1364/boe.499354. PMID: 38021128.

10. Menzel M, Gräßel D, Rajković I, Zeineh M, Georgiadis M. Using light and X-ray scattering to untangle complex neuronal orientations and validate diffusion MRI. *eLife*. 2023 May 11;12.0. doi: 10.7554/elife.84024. PMID: 37166005.

11. Schilling KG, Howard AF, Grussu F, Ianuș A, Hansen B, Barrett RLC, Aggarwal M, Michielse S, Nasrallah FA, Syeda WT, Wang N, Veraart J, Roebroeck AF, Bagdasarian FA, Eichner C, Sepehrband F, Zimmermann J, Soustelle L, Bowman C, Tendler BC, Hertanu A, Jeurissen B, Verhoye M, Frydman L, van de Looij Y, Hike D, Dunn JF, Miller KL, Landman BA, Shemesh N, Anderson AW, McKinnon ET, Farquharson S, Dell'Acqua F, Pierpaoli CM, Drobnjak I, Leemans A, Harkins KD, Descoteaux M, Xu D, Huang H, Santin MD, Grant SC, Obenaus A, Kim GS, Wu DJ, Le Bihan D, Blackband SJ, Ciobanu L, Fieremans E, Bai R, Leergaard TB, Zhang J, Dyrby TB, Johnson GA, Cohen-Adad J, Budde MD, Jelescu I. Considerations and recommendations from the ISMRM Diffusion Study Group for preclinical diffusion MRI : Part 3-Ex vivo imaging: Data processing, comparisons with microscopy, and tractography. *Magnetic Resonance in Medicine*. 2025 Feb 26;93.0(6.0):2561-2582. doi: 10.1002/mrm.30424. PMID: 40008460.

12. Liu CJ, Ammon W, Jones RJ, Nolan JC, Gong D, Maffei C, Blanke N, Edlow BL, Augustinack JC, Magnain CV, Yendiki A, Villiger M, Fischl BR, Wang H. Three-dimensional fiber orientation mapping of ex vivo human brain at micrometer resolution. *npj Imaging*. 2025 Apr 8;3.0(1.0):13. doi: 10.1038/s44303-025-00074-2. PMID: 40213097.

13. Chourrout M, Rositi H, Ong E, Hubert V, Paccalet A, Foucault L, Autret A, C BF, Olivier C, Bolbos RI, Peyrin F, Da Silva CC, Meyronet D, Raineteau O, Elleaume H, Brun E, Chauveau F, Wiart M. Brain virtual histology with X-ray phase-contrast tomography Part I: whole-brain myelin mapping in white-matter injury models. *Biomedical Optics Express*. 2021 Nov 5;13.0(3.0):1620. doi: 10.1364/boe.438832. PMID: 35415001.

14. Rusch H, Brammerloh M, Stieler J, Sonntag M, Mohammadi S, Weiskopf N, Arendt T, Kirilina EP, Morawski M. Finding the best clearing approach - Towards 3D wide-scale multimodal imaging of aged human brain tissue. *NeuroImage*. 2021 Dec 17;247.0:118832. doi: 10.1016/j.neuroimage.2021.118832. PMID: 34929383.

15. Caspers S, Axer M, Gräßel D, Amunts K. Additional fiber orientations in the sagittal stratum-noise or anatomical fine structure?. *Brain Structure and Function*. 2022 Feb 3;227.0(4.0):1331-1345. doi: 10.1007/s00429-021-02439-w. PMID: 35113243.

16. Baxi M, Karayumak SC, Papadimitriou G, Makris N, van der Kouwe AJW, Jenkins BG, Moore TL, Rosene DL, Kubicki MR, Rathi Y. Investigating the contribution of cytoarchitecture to diffusion MRI measures in gray matter using histology. *Frontiers in Neuroimaging*. 2022 Sep 13;1.0:947526. doi: 10.3389/fnimg.2022.947526. PMID: 37555179.

17. Kolluru C, Joseph NM, Seckler JM, Fereidouni F, Levenson RM, Shoffstall AJ, Jenkins MW, Wilson DL. NerveTracker: a Python-based software toolkit for visualizing and tracking groups of nerve fibers in serial block-face microscopy with ultraviolet surface excitation images. *Journal of Biomedical Optics*. 2024 Jun 18;29.0(7.0):1-16. doi: 10.1117/1.jbo.29.7.076501. PMID: 38912214.

18. Deng Y, Xu J, Yu T, Zhu DY. Advances in tissue optical clearing for 3D imaging in large animal. *Frontiers of Optoelectronics*. 2025 Aug 18;18.0(1.0):18. doi: 10.1007/s12200-025-00162-6. PMID: 40824479.

19. Kjer HM, Andersson M, He Y, Pacureanu A, Daducci A, Pizzolato M, Salditt T, Robisch A, Eckermann M, Töpperwien M, Dahl AB, Elkjaer ML, Illés Z, Ptito M, Dahl VA, Dyrby TB. Bridging the 3D geometrical organisation of white matter pathways across anatomical length scales and species. *eLife*. 2025 Feb 28;13.0. doi: 10.7554/elife.94917.3.

20. Zabri SH, Ahmad AH, Zakaria R, Abdullah AN, Mukhtar SF, Abdullah JM. Probabilistic Tractography Between Nucleus Accumbens and Other Reward-related Brain Areas in Malay Female Adolescents. *Malaysian Journal of Medicine and Health Sciences*. 2023 Mar 13;19.0(2.0):61-68. doi: 10.47836/mjmhs.19.2.11.

21. Dyrby TB, Sarubbo S, Rheault F, Fieremans E, Anderson AW, Palombo M, Neher P, Rockland KS, Schilling KG. Tractography validation Part 2: The use of anatomical model systems and measures for validation. *Elsevier eBooks*. 2025 Jan 1;:511-542. doi: 10.1016/b978-0-12-818894-1.00020-3.

22. Gray B, Smith AW, MacKenzie-Graham AJ, Shattuck DW, Tward DJ. Validation of structure tensor analysis for orientation estimation in brain tissue microscopy. *Journal of Neuroscience Methods*. 2025 Jul 28;423.0:110539. doi: 10.1016/j.jneumeth.2025.110539. PMID: 39868114.

23. Stones R, Descoteaux M, Dell'Acqua F. From diffusion models to fiber orientations. *Elsevier eBooks*. 2025 Jan 1;:221-238. doi: 10.1016/b978-0-12-818894-1.00035-5.

24. Song M, Yang Z, Jiang T. Multimodal Brain Imaging Fusion for the White-Matter Fiber Architecture in the Human Brain. *Neuroscience Bulletin*. 2022 Jan 31;38.0(5.0):561-564. doi: 10.1007/s12264-022-00822-z. PMID: 35099675.

25. Kjer HM, Andersson M, He Y, Pacureanu A, Daducci A, Pizzolato M, Salditt T, Robisch A, Eckermann M, Töpperwien M, Dahl AB, Elkjaer ML, Illés Z, Ptito M, Dahl VA, Dyrby TB. Bridging the 3D geometrical organisation of white matter pathways across anatomical length scales and species. *eLife*. 2024 Apr 10;13.0. doi: 10.7554/elife.94917. PMID: 40019134.

26. Kjer HM, Andersson M, He Y, Pacureanu A, Daducci A, Pizzolato M, Salditt T, Robisch A, Eckermann M, Töpperwien M, Dahl AB, Elkjaer ML, Illés Z, Ptito M, Dahl VA, Dyrby TB. Bridging the 3D geometrical organisation of white matter

pathways across anatomical length scales and species. *eLife*. 2024 Apr 10;. doi: 10.7554/elife.94917.1.

27. Liu CJ, Ammon W, Jones RJ, Nolan JC, Gong D, Maffei C, Edlow BL, Augustinack JC, Magnain CV, Yendiki A, Villiger M, Fischl BR, Wang H. Quantitative imaging of three-dimensional fiber orientation in the human brain via two illumination angles using polarization-sensitive optical coherence tomography. *bioRxiv (Cold Spring Harbor Laboratory)*. 2023 Oct 23;. doi: 10.1101/2023.10.20.563298. PMID: 37961162.

28. Kjer HM, Andersson M, He Y, Pacureanu A, Daducci A, Pizzolato M, Salditt T, Robisch A, Eckermann M, Töpperwien M, Dahl AB, Elkjaer ML, Illés Z, Ptito M, Dahl VA, Dyrby TB. Bridging the 3D geometrical organisation of white matter pathways across anatomical length scales and species. *eLife*. 2024 Dec 20;. doi: 10.7554/elife.94917.2.

29. Legeay S. Mesoscopic mapping of the human structural connectome using high-performance global tractography. 2025 Jan 31;. doi: 10.70675/f036d118z8937z4350zb52cz2c55acdaa577.

30. Schmitz D, Benning K, Schubert N, Minnerop M, Amunts K, Axer M. Fast data-driven computation and intuitive visualization of fiber orientation uncertainty in 3D-polarized light imaging. *Frontiers in Physics*. 2022 Sep 26;10.0. doi: 10.3389/fphy.2022.958364.

31. Tisdall L, MacNiven KH, Leong JK, Frey R, Rieskamp J, Hertwig R, Knutson BD, Mata R. Structure of connections to the nucleus accumbens link to specific but not general measures of human risk preference. *MPG.PuRe (Max Planck Society)*. 2023 Oct 31;. doi: 10.31219/osf.io/ab4vr.

32. Wang H, Ammon W, Jones RJ, Nolan JC, Gong D, Maffei C, Edlow BL, Augustinack JC, Magnain CV, Yendiki A, Villiger M, Fischl BR, Liu CJ. Three-dimensional fiber orientation mapping of the human brain at micrometer resolution. *Research Square*. 2024 Aug 7;. doi: 10.21203/rs.3.rs-4725871/v1. PMID: 39149445.

33. Aggarwal M. Preclinical and ex vivo tractography: Techniques and applications at high field. *Elsevier eBooks*. 2025 Jan 1;:653-667. doi: 10.1016/b978-0-12-818894-1.00025-2.

34. Schilling KG, Rheault F, Dyrby TB. Tractography validation part 3: Lessons learned through validation studies. *Elsevier eBooks*. 2025 Jan 1;:543-564. doi: 10.1016/b978-0-12-818894-1.00004-5.

35. Chau GR, Weber EMM, Batra A, Ni L, Zeineh M, Chaudhari AS, Adeli E, Knowles JK, McNab JA. Non-parametric prediction of brain MRI microstructure using transfer learning. *Imaging Neuroscience*. 2025 Jan 1;3.0. doi: 10.1162/imag_a_00548. PMID: 40800932.

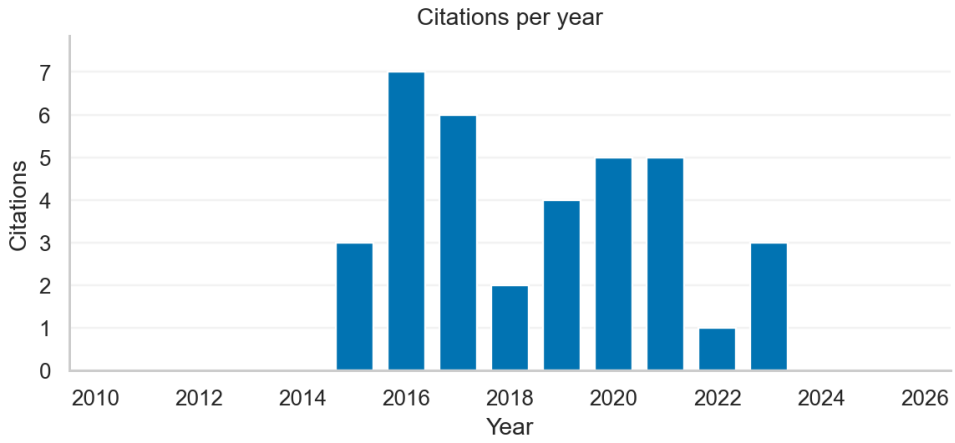
36. Trisal A, Singh AK. CLARITY in Alzheimer's Research: Merging Tissue Transparency with Next-Gen Neurotechnologies. *Journal of Neuroscience Methods*. 2025 Aug 13;423.0:110554. doi: 10.1016/j.jneumeth.2025.110554. PMID: 40816537.

37. Wanjau E, Chourrout M, Maffei C, Balbastre Y, Keenlyside A, Brunet J, Sharma A, Huang SY, Tafforeau P, Fischl BR, Yendiki A, Lee PD, Walsh C. Bridging the microstructural gap in human connectomics using hierarchical phase-contrast tomography as a reference for diffusion MRI in the human brain. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Apr 6;. doi: 10.64898/2026.04.02.715729. PMID: 41993538.

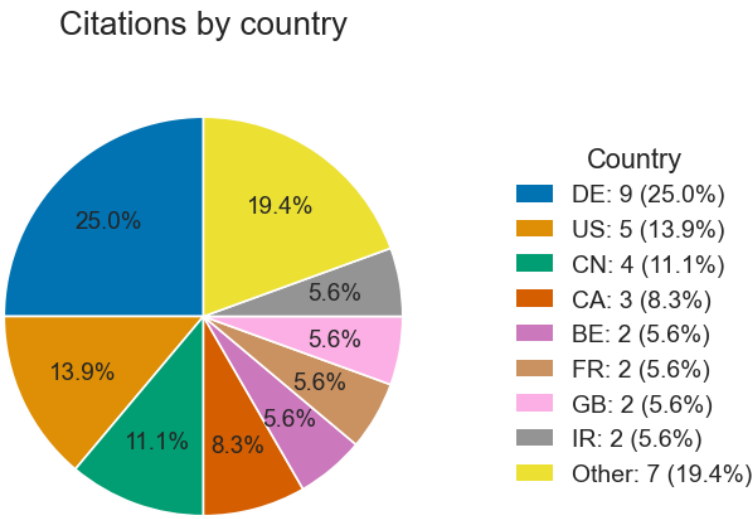
38. Leong JK, Padula CB, Knutson BD, MacNiven KH, Tisdall L. Brain tract structure predicts relapse to stimulant drug use. *OSF Preprints (OSF Preprints)*. 2022 Jun 20;.

Publication

Keuters MH, Aswendt M, Tennstaedt A, Wiedermann D, Pikhovych A, Rotthues S, Fink GR, Schroeter M, Hoehn M, Rueger MA. Transcranial direct current stimulation promotes the mobility of engrafted NSCs in the rat brain. *NMR in Biomedicine*. 2014 Dec 17;28(2):231-239. doi: 10.1002/nbm.3244. PMID: 25521600.



Citation summary
Total citations: **36**
Average/year: **2.77**
Period: 2014-2026
SCImago cites/doc 2y (2025): **3.13**
SJR 2025: **0.822** (Q1)



Cited by

1. Antal A, Alekseichuk I, Bikson M, Brockmöller J, Brunoni AR, Chen RE, Cohen LG, Dowthwaite G, Ellrich J, Flöel A, Fregni F, George MS, Hamilton RH, Haueisen J, Herrmann CS, Hummel FC, Lefaucheur JP, Liebetanz D, Loo CKA, McCaig C, Miniussi C, Miranda PC, Moliadze V, Nitsche MA, Nowak R, Padberg F, Pascual-Leone Á, Poppendieck W, Priori A, Rossi S, Rossini PM, Rothwell JC, Rueger MA, Ruffini G, Schellhorn K, Siebner HR, Ugawa Y, Wexler A, Ziemann U, Hallett M, Paulus W. Low intensity transcranial electric stimulation: Safety, ethical, legal regulatory and application guidelines. *Clinical Neurophysiology*. 2017 Jun 20;128(9.0):1774-1809. doi: 10.1016/j.clinph.2017.06.001. PMID: 28709880.

2. Cirillo G, Di Pino G, Capone F, Ranieri F, Florio L, Todisco V, Tedeschi G, Funke K, Di Lazzaro V. Neurobiological after-effects of non-invasive brain stimulation. *Brain stimulation*. 2016 Nov 17;10(1.0):1-18. doi: 10.1016/j.brs.2016.11.009. PMID: 27931886.

3. Morya E, Monte-Silva KK, Bikson M, Esmaeilpour Z, Biazoli CE, Fonseca ARO, Bocci T, Farzan F, Chatterjee R, Hausdorff JM, Machado DGDS, Brunoni AR, Mezger E, Moscaleski LA, Pegado R, Sato JR, Caetano MS, Sá KN, Tanaka C, Li LM, Baptista AF, Okano AH. Beyond the target area: an integrative view of tDCS-induced motor cortex modulation in patients and athletes. *Journal of NeuroEngineering and Rehabilitation*. 2019 Nov 15;16(1.0):141. doi: 10.1186/s12984-019-0581-1. PMID: 31730494.

4. Lefaucheur J. A comprehensive database of published tDCS clinical trials (2005-2016). *Neurophysiologie Clinique*. 2016 Nov 17;46(6.0):319-398. doi: 10.1016/j.neucli.2016.10.002. PMID: 27865707.

5. Ahuja CS, Mothe AJ, Khazaei M, Badhiwala JH, Gilbert EAB, van der Kooy D, Morshead CM, Tator CH, Fehlings MG. The leading edge: Emerging neuroprotective and neuroregenerative cell-based therapies for spinal cord injury. *Stem Cells Translational Medicine*. 2020 Jul 21;9(12.0):1509-1530. doi: 10.1002/sctm.19-0135. PMID: 32691994.

6. Chalah MA, Riachi NJ, Ahdab R, Créange A, Lefaucheur J, Ayache SS. Fatigue in Multiple Sclerosis: Neural Correlates and the Role of Non-Invasive Brain Stimulation. *Frontiers in Cellular Neuroscience*. 2015 Nov 30;9:460. doi: 10.3389/fncel.2015.00460. PMID: 26648845.

7. Hosseini SM, Borys B, Karimi-Abdolrezaee S. Neural stem cell therapies for spinal cord injury repair: an update on recent preclinical and clinical advances. *Brain*. 2023 Nov 17;147(3.0):766-793. doi: 10.1093/brain/awad392. PMID: 37975820.

8. Braun R, Klein RM, Walter HL, Ohren M, Freudenmacher L, Getachew K, Ladwig A, Lülling J, Neumaier B, Endepols H, Graf R, Hoehn M, Fink GR, Schroeter M, Rueger MA. Transcranial direct current stimulation accelerates recovery of function, induces neurogenesis and recruits oligodendrocyte precursors in a rat model of stroke. *Experimental Neurology*. 2016 Feb 27;279:127-136. doi: 10.1016/j.expneurol.2016.02.018. PMID: 26923911.

9. Dillen Y, Kemps H, Gervois P, Wolfs E, Bronckaers A. Adult Neurogenesis in the Subventricular Zone and Its Regulation After Ischemic Stroke: Implications for Therapeutic Approaches. *Translational Stroke Research*. 2019 Jul 15;11(1.0):60-79. doi: 10.1007/s12975-019-00717-8. PMID: 31309427.

10. Bierman RD, Safarians G, Huang J, Rajput B, Chen JY, Wang ZZ, Seidlits SK. Engineering Tissues of the Central Nervous System: Interfacing Conductive Biomaterials with Neural Stem/Progenitor Cells. *Advanced Healthcare Materials*. 2021 Nov 22;11(7.0):e2101577. doi: 10.1002/adhm.202101577. PMID: 34808031.

11. Pikhovych A, Stolberg NP, Flitsch LJ, Walter HL, Graf R, Fink GR, Schroeter M, Rueger MA. Transcranial Direct Current Stimulation Modulates Neurogenesis and Microglia Activation in the Mouse Brain. *Stem Cells International*. 2016 Jan 1;2016(1.0):2715196. doi: 10.1155/2016/2715196. PMID: 27403166.

12. Shen C, Cheng WN, Yu P, Wang L, Zhou L, Zeng L, Yang Q. Resveratrol pretreatment attenuates injury and promotes proliferation of neural stem cells following oxygen-glucose deprivation/reoxygenation by upregulating the expression of Nrf2, HO-1 and NQO1 in vitro. *Molecular Medicine Reports*. 2016 Aug 24;14(4.0):3646-3654. doi: 10.3892/mmr.2016.5670. PMID: 27573874.

13. Huang YH, Li Y, Chen J, Zhou H, Tan SHD. Electrical Stimulation Elicits Neural Stem Cells Activation: New Perspectives in CNS Repair. *Frontiers in Human Neuroscience*. 2015 Oct 19;9:586. doi: 10.3389/fnhum.2015.00586. PMID: 26539102.

14. Rabenstein M, Unverricht-Yeboah M, Keuters MH, Pikhovych A, Hucklenbroich J, Vay SU, Blaschke SJ, Ladwig A, Walter HL, Beiderbeck M, Fink GR, Schroeter M, Kriehuber R, Rueger MA. Transcranial Current Stimulation Alters the Expression of Immune-Mediating Genes. *Frontiers in Cellular Neuroscience*. 2019 Oct 25;13:461. doi: 10.3389/fncel.2019.00461. PMID: 31708742.

15. Yuan T, Dong YW, Zhang L, Qi J, Yao C, Wang Y, Chai R, Liu Y, So K. Neuromodulation-Based Stem Cell Therapy in Brain Repair: Recent Advances and Future Perspectives. *Neuroscience Bulletin*. 2021 Apr 19;37(5.0):735-745. doi: 10.1007/s12264-021-00667-y. PMID: 33871821.

16. Mojaverrostami S, Khadivi F, Zarini D, Mohammadi A. Combination effects of mesenchymal stem cells transplantation and anodal transcranial direct current stimulation on a cuprizone-induced mouse model of multiple sclerosis. *Journal of Molecular Histology*. 2022 Aug 10;53(5.0):817-831. doi: 10.1007/s10735-022-10092-8. PMID: 35947228.

17. Sánchez-León CA, Ammann C, Medina JF, Márquez-Ruiz J. Using Animal Models to Improve the Design and Application of Transcranial Electrical Stimulation in Humans. *Current Behavioral Neuroscience Reports*. 2018 Apr 25;5(2.0):125-135. doi: 10.1007/s40473-018-0149-6. PMID: 30013890.

18. Sprugnoli G, Rossi S, Rotenberg A, Pascual-Leone Á, Fakhri GE, Golby AJ, Santarnecchi E. Personalised, image-guided, noninvasive brain stimulation in gliomas: Rationale, challenges and opportunities. *EBioMedicine*. 2021 Aug 1;70:103514. doi: 10.1016/j.ebiom.2021.103514. PMID: 34391090.

19. Iwasa SN, Babona-Pilipos R, Morshead CM. Environmental Factors That Influence Stem Cell Migration: An “Electric Field”. *Stem Cells International*. 2017 Jan 1;2017:1-9. doi: 10.1155/2017/4276927. PMID: 28588621.

20. Broeder S, Heremans E, Pereira MP, Nackaerts E, Meesen R, Verheyden G, Nieuwboer A. Does transcranial direct current stimulation during writing alleviate upper limb freezing in people with Parkinson’s disease? A pilot study. *Human Movement Science*. 2018 Feb 28;65:142-153. doi: 10.1016/j.humov.2018.02.012. PMID: 29501349.

21. Winkler C, Reis J, Hoffmann N, Gellner A, Münkel C, Curado MR, Furlanetti LL, García J, Döbrössy MD, Fritsch B. Anodal Transcranial Direct Current Stimulation Enhances Survival and Integration of Dopaminergic Cell Transplants in a Rat Parkinson Model. *eNeuro*. 2017 Sep 1;4(5.0):ENEURO.0063-17.2017. doi: 10.1523/eneuro.0063-17.2017. PMID: 28966974.

22. Xia Y, Khalid W, Yin Z, Huang G, Bikson M, Fu BM. Modulation of solute diffusivity in brain tissue as a novel mechanism of transcranial direct current stimulation (tDCS). *Scientific Reports*. 2020 Oct 28;10(1.0):18488. doi: 10.1038/s41598-020-75460-4. PMID: 33116214.

23. Pikhovych A, Walter HL, Mahabir E, Fink GR, Graf R, Schroeter M, Rueger MA. Transcranial direct current stimulation in the male mouse to promote recovery after stroke. *Laboratory Animals*. 2015 Oct 6;50(3.0):212-216. doi: 10.1177/0023677215610708. PMID: 26442519.

24. Thibaut A, Shie VL, Ryan CM, ZAFONTE RD, Ohrtman EA, Schneider JC, Fregni F. A review of burn symptoms and potential novel neural targets for non-invasive brain stimulation for treatment of burn sequelae. *Burns*. 2020 Jun 20;47(3.0):525-537. doi: 10.1016/j.burns.2020.06.005. PMID: 33293156.

25. Kubelt C, Molkewehrum H, Lucius R, Synowitz M, Held-Feindt J, Helmers A. Influence of Simulated Deep Brain Stimulation on the Expression of Inflammatory Mediators by Human Central Nervous System Cells In Vitro. *NeuroMolecular Medicine*. 2021 Jul 3;24(2.0):169-182. doi: 10.1007/s12017-021-08674-y. PMID: 34216357.

26. Tahtis V, Kaski DN. Parkinson's disease treatments: focus on transcranial direct current stimulation (tDCS). *Journal of Parkinsonism and Restless Legs Syndrome*. 2017 Oct 1;Volume 7:55-70. doi: 10.2147/jprls.s128146.

27. Latchoumane CV, Barany DA, Karumbaiah L, Singh T. Neurostimulation and Reach-to-Grasp Function Recovery Following Acquired Brain Injury: Insight From Pre-clinical Rodent Models and Human Applications. *Frontiers in Neurology*. 2020 Jul 21;11:835. doi: 10.3389/fneur.2020.00835. PMID: 32849253.

28. Schönfeld L, Jahanshahi A, Lemmens E, Bauwens M, Hescham S, Schipper S, Lagière M, Hendrix S, Temel Y. Motor cortex stimulation does not lead to functional recovery after experimental cortical injury in rats. *Restorative Neurology and Neuroscience*. 2017 May 9;35(3.0):295-305. doi: 10.3233/rnn-160703. PMID: 28506001.

29. Finnegan J, Ye H. Cell Therapy for Spinal Cord Injury Informed by Electromagnetic Waves. *Regenerative Medicine*. 2016 Sep 6;11(7.0):675-691. doi: 10.2217/rme-2016-0019. PMID: 27599240.

30. Wandrey JD, Kastelik J, Fritzsche T, Denke C, Schäfer M, Tafelski S. Supplementing transcranial direct current stimulation to local infiltration series for refractory neuropathic craniocephalic pain: A randomized controlled pilot trial. *Frontiers in Neurology*. 2023 Mar 1;14:1069434. doi: 10.3389/fneur.2023.1069434. PMID: 36937523.

31. Zarifkar AH, Zarifkar AH, Zarifkar A, Zarifkar A, Safaei S. Different paradigms of transcranial electrical stimulation induce structural changes in the CA1 region of the hippocampus in a rat model of Alzheimer’s disease. *Neuroscience Letters*. 2023 Nov 22;818:137570. doi: 10.1016/j.neulet.2023.137570. PMID: 38000774.

32. Yiu HHP, Chari DM. How Can Nanoparticles Help Neural Cell Transplantation Therapy?. *Nanomedicine*. 2020 Aug 13;15(22.0):2103-2106. doi: 10.2217/nnm-2020-0279. PMID: 32787617.

33. Schönfeldt L. Neurostimulation to treat brain injury?. *Research Publications (Maastricht University)*. 2016 Jan 1;. doi: 10.26481/dis.20160610ls.

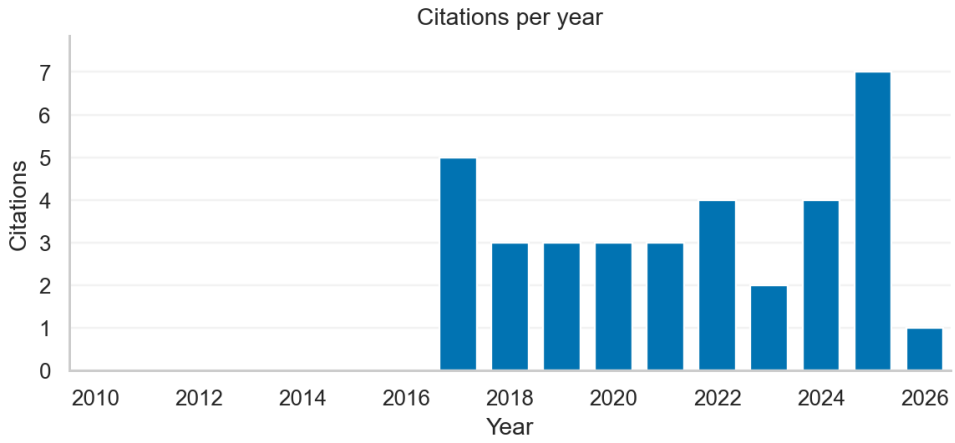
34. Rabenstein M, Rueger MA. Mobilization of Endogenous Neural Stem Cells to Promote Regeneration After Stroke. *Springer series in translational stroke research*. 2017 Nov 5;:101-112. doi: 10.1007/978-3-319-66679-2_5.

35. □ □. Advances in Research of Repetitive Transcranial Magnetic Stimulation for Post-Stroke Depression. *Asian Case Reports in Vascular Medicine*. 2019 Jan 1;07(3.0):50-56. doi: 10.12677/acrvm.2019.73009.

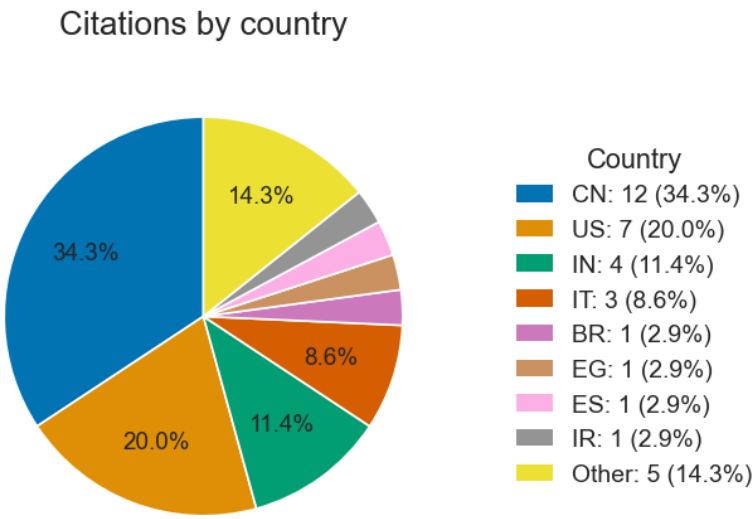
36. Song H, Yang R, Lai R, Guo J, Li K, Ni B, Liu Y. Application of a simple device for “attracting” and “trapping” glioma cells in situ. *bioRxiv (Cold Spring Harbor Laboratory)*. 2021 Dec 23;. doi: 10.1101/2021.12.21.473650.

Publication

Cruz LJ, Que I, Aswendt M, Chan AB, Hoehn M, Löwik CW. Targeted nanoparticles for the non-invasive detection of traumatic brain injury by optical imaging and fluorine magnetic resonance imaging. *Nano Research*. 2016 Apr 14;9(5):1276-1289. doi: 10.1007/s12274-016-1023-z.



Citation summary
Total citations: **35**
Average/year: **3.18**
Period: 2016-2026
SCImago cites/doc 2y (2025): **9.09**
SJR 2025: **1.98** (Q1)



Cited by

1. Hong G, Antaris AL, Dai H. Near-infrared fluorophores for biomedical imaging. *Nature Biomedical Engineering*. 2017 Jan 10;1.0(1.0). doi: 10.1038/s41551-016-0010.

2. Farzin A, Etesami SA, Quint JP, Memić A, Tamayol A. Magnetic Nanoparticles in Cancer Therapy and Diagnosis. *Advanced Healthcare Materials*. 2020 Mar 20;9.0(9.0):e1901058. doi: 10.1002/adhm.201901058. PMID: 32196144.

3. Yang Q, Hu Z, Zhu S, Ma R, Ma H, Ma Z, Wan H, Zhu T, Jiang Z, Liu W, Jiao L, Sun H, Liang Y, Dai H. Donor Engineering for NIR-II Molecular Fluorophores with Enhanced Fluorescent Performance. *Journal of the American Chemical Society*. 2018 Jan 16;140.0(5.0):1715-1724. doi: 10.1021/jacs.7b10334. PMID: 29337545.

4. Wang X, Ding Q, Groleau RR, Wu L, Mao Y, Che F, Kotova OV, Scanlan EM, Lewis SE, Li P, Tang B, James TD, Gunnlaugsson T. Fluorescent Probes for Disease Diagnosis. *Chemical Reviews*. 2024 May 17;124.0(11.0):7106-7164. doi: 10.1021/acs.chemrev.3c00776. PMID: 38760012.

5. Zhang C, Yan KQ, Fu C, Peng H, Hawker CJ, Whittaker AK. Biological Utility of Fluorinated Compounds: from Materials Design to Molecular Imaging, Therapeutics and Environmental Remediation. *Chemical Reviews*. 2021 Oct 5;122.0(1.0):167-208. doi: 10.1021/acs.chemrev.1c00632. PMID: 34609131.

6. Bharadwaj VN, Nguyen DT, Kodibagkar VD, Stabenfeldt SE. Nanoparticle-Based Therapeutics for Brain Injury. *Advanced Healthcare Materials*. 2017 Oct 16;7.0(1.0). doi: 10.1002/adhm.201700668. PMID: 29034608.

7. Wang H, Mu X, Jiang Y, Liang Y, Zhang X, Ming D. Brain imaging with near-infrared fluorophores. *Coordination Chemistry Reviews*. 2018 Nov 20;380.0:550-571. doi: 10.1016/j.ccr.2018.11.003.

8. Kumar A, Tan A, Wong JKL, Spagnoli JC, Lam JM, Blevins B, Natasha G, Thorne LW, Ashkan K, Xie J, Liu H. Nanotechnology for Neuroscience: Promising Approaches for Diagnostics, Therapeutics and Brain Activity Mapping.

Advanced Functional Materials. 2017 Aug 14;27.0(39.0). doi: 10.1002/adfm.201700489. PMID: 30853878.

9. Sim TM, Dinesh T, Dheen ST, Bay B, Srinivasan DK. Nanoparticle-Based Technology Approaches to the Management of Neurological Disorders. *International Journal of Molecular Sciences*. 2020 Aug 23;21.0(17.0):6070. doi: 10.3390/ijms21176070. PMID: 32842530.

10. Da Silva CG, Peters GJ, OSSENDORP FA, Cruz LJ. The potential of multi-compound nanoparticles to bypass drug resistance in cancer. *Cancer Chemotherapy and Pharmacology*. 2017 Sep 8;80.0(5.0):881-894. doi: 10.1007/s00280-017-3427-1. PMID: 28887666.

11. Silva DF, Melo ALP, Uchôa AFC, Pereira GMA, Alves ÁEF, Vasconcellos MC, Júnior FHX, Passos MF. Biomedical Approach of Nanotechnology and Biological Risks: A Mini-Review. *International Journal of Molecular Sciences*. 2023 Nov 24;24.0(23.0):16719. doi: 10.3390/ijms242316719. PMID: 38069043.

12. Joseph JM, Gigliobianco MR, Firouzabadi BM, Censi R, Di Martino P. Nanotechnology as a Versatile Tool for 19F-MRI Agent's Formulation: A Glimpse into the Use of Perfluorinated and Fluorinated Compounds in Nanoparticles. *Pharmaceutics*. 2022 Feb 9;14.0(2.0):382. doi: 10.3390/pharmaceutics14020382. PMID: 35214114.

13. Ye J, Wang D, Wang Z, Cao M, Shen Y, Yu B, Cong H. Recent advances in NIR small molecule activable probes for imaging and therapy. *Coordination Chemistry Reviews*. 2025 Jul 1;543.0:216904. doi: 10.1016/j.ccr.2025.216904.

14. Chi W, He Y, Chen S, Guo L, Yuan Y, Li R, Liu R, Zhou D, Du J, Xu T, Yu Y. Advances in research on biomaterials and stem cell/exosome-based strategies in the treatment of traumatic brain injury. *Acta Pharmaceutica Sinica B*. 2025 May 21;15.0(7.0):3511-3544. doi: 10.1016/j.apsb.2025.05.010. PMID: 40698148.

15. Zhang P, Ran Y, Han L, Yao L, Tian W, Sun X, Jiao M, Jing L, Luo X. Nanomaterial technologies for precision diagnosis and treatment of brain hemorrhage. *Biomaterials*. 2025 Mar 19;321.0:123269. doi: 10.1016/j.biomaterials.2025.123269. PMID: 40174300.

16. Ping Z, Liu T, Xu HJ, Meng Y, Li W, Xu X, Zhang L. Construction of highly stable selenium nanoparticles embedded in hollow nanofibers of polysaccharide and their antitumor activities. *Nano Research*. 2017 May 26;10.0(11.0):3775-3789. doi: 10.1007/s12274-017-1590-7.

17. Bony BA, Kievit FM. A Role for Nanoparticles in Treating Traumatic Brain Injury. *Pharmaceutics*. 2019 Sep 13;11.0(9.0):473. doi: 10.3390/pharmaceutics11090473. PMID: 31540234.

18. Bharadwaj VN, Rowe RK, Harrison JL, Wu C, Anderson TR, Lifshitz J, Adelson PD, Kodibagkar VD, Stabenfeldt SE. Blood-brainbarrier disruption dictates nanoparticle accumulation following experimental brain injury. *Nanomedicine Nanotechnology Biology and Medicine*. 2018 Jun 19;14.0(7.0):2155-2166. doi: 10.1016/j.nano.2018.06.004. PMID: 29933022.

19. Zhang D, Gao MQ, Jin Q, Ni Y, Zhang J. Updated developments on molecular imaging and therapeutic strategies directed against necrosis. *Acta Pharmaceutica Sinica B*. 2019 Feb 13;9.0(3.0):455-468. doi: 10.1016/j.apsb.2019.02.002. PMID: 31193829.

20. Zhang D, Jin Q, Jiang C, Gao MQ, Ni Y, Zhang J. Imaging Cell Death: Focus on Early Evaluation of Tumor Response to Therapy. *Bioconjugate Chemistry*. 2020 Mar 9;31.0(4.0):1025-1051. doi: 10.1021/acs.bioconjchem.0c00119. PMID: 32150392.

21. Roffo F, Ponsiglione AM, Netti PA, Torino E. coupled Hydrodynamic Flow Focusing (cHFF) to Engineer Lipid-Polymer Nanoparticles (LiPoNs) for Multimodal Imaging and Theranostic Applications. *Biomedicines*. 2022 Feb 14;10.0(2.0):438. doi: 10.3390/biomedicines10020438. PMID: 35203647.

22. Chen Z, Yu Y, Deng LJ, Wen L, Han Y, Zhao X, Huo L, Ru J, Gong Q, Wang Y, Zhang H, Song G, Liang C. Trianthracene Derivative Based Nanoparticles for in vivo Afterglow Imaging of Traumatic Brain Injury. *Chinese Journal of Chemistry*. 2025 Apr 11;43.0(14.0):1619-1626. doi: 10.1002/cjoc.70001.

23. Bakhshandeh B, Jahanafrooz Z, Haghighi DM, Rahmani-Kukia N, Abbasi A, Pourramezanali A. Recent Trends in the Application of Materials for Cancer Therapy and Diagnosis. *Springer eBooks*. 2023 Jan 1;:305-364. doi: 10.1007/978-981-99-2119-5_12.

24. Thangaleela S, Wang C, Revathi S, Sivamaruthi BS, Chaiyasut C. Nanoparticle-Based Imaging Techniques in Neurological Disorders. *Nanotechnology in the life sciences*. 2025 Jan 1;:43-107. doi: 10.1007/978-3-031-80185-3_3.

25. Deng LJ, Xiang J, Zeng X, Li Z, Chen Z, Zhang C, He H, Ye T, Gong Q, Song G, Liang CF. A Ratiometric Afterglow Luminescence Probe for in Vivo Imaging of Neuroinflammation in Traumatic Brain Injury. *Analytical Chemistry*. 2026 Mar 4;98.0(10.0):7877-7887. doi: 10.1021/acs.analchem.6c00591. PMID: 41779627.

26. Iqbal J, Courville E, Kazim SF, Kogan M, Schmidt MH, Bowers CA. Role of nanotechnology in neurosurgery: A review of recent advances and their applications. *World Neurosurgery X*. 2024 Feb 24;22.0:100298. doi: 10.1016/j.wnsx.2024.100298. PMID: 38455250.

27. Gaglio SC, De Rosa C, Piccinelli F, Romeo A, Perduca M. Complexes of rare earth ions embedded in poly(lactic-co-glycolic acid) (PLGA) nanoparticles: Characterization and spectroscopic study. *Optical Materials*. 2019 May 31;94.0:249-256. doi: 10.1016/j.optmat.2019.05.034.

28. Ovejero JG, Wang E, Veintemillas-Verdaguer S, Morales MP, Sorolla A. Nanoparticles for Neural Applications. *Springer eBooks*. 2021 Aug 3;:149-184. doi: 10.1007/978-3-030-81400-7_7.

29. Khanam N, Nath D. Traumatic brain injury: Future application of nanomedicine. *GSC Advanced Research and Reviews*. 2021 Feb 25;6.0(2.0):020-027. doi: 10.30574/gscarr.2021.6.2.0019.

30. Abdel-Mottaleb MM, Pellequer Y, Abd-Allah H, El-Gogary RI, Nasr M. A futuristic insight into functionalized magnetic nanosystems for diagnostic tools and devices: From lab to commercial applications. *Elsevier eBooks*. 2024 Jan 1;:745-766. doi: 10.1016/b978-0-443-19012-4.00016-3.

31. Kumar R. Treatment of Traumatic Brain Injury: Nanotherapeutics. *Journal of Clinical Haematology*. 2024 Jan 1;5.0(1.0):1-3. doi: 10.33696/haematology.5.056.

32. Miller HA, Kievit FM. Theranostic Nanomaterials for Brain Injury. *Springer eBooks*. 2022 Jan 1;:307-350. doi: 10.1007/978-3-031-11409-0_8.

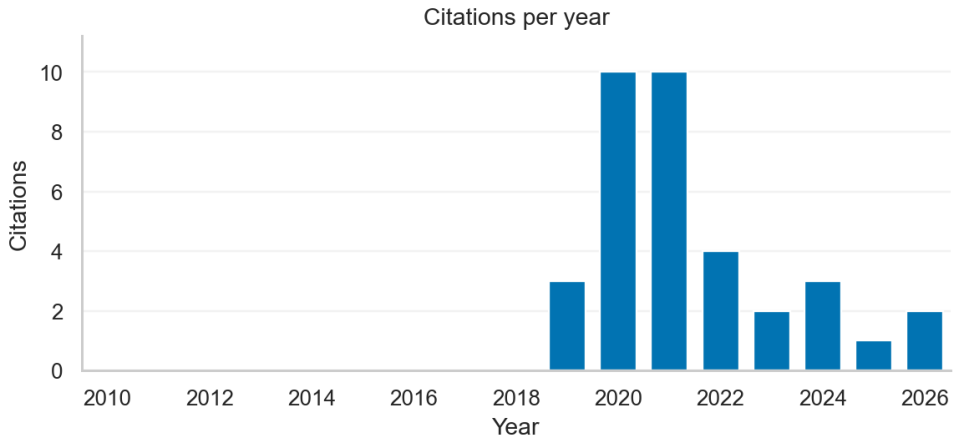
33. Hatipoglu MK, Süt PA. Synthesis and Biological Use of Nanomaterials. *Topics in applied physics*. 2022 Jan 1;:793-858. doi: 10.1007/978-3-030-93460-6_29.

34. Singh MP, Waseem A, Kumar V, Prakash R, Siddiqui AJ, Prakash A, Raza SS. A Novel Drug Delivery System Utilizing Nanoparticles to Treat Neurological Disorders. *Smart nanomaterials technology*. 2025 Jan 1;:413-430. doi: 10.1007/978-981-96-2112-5_16.

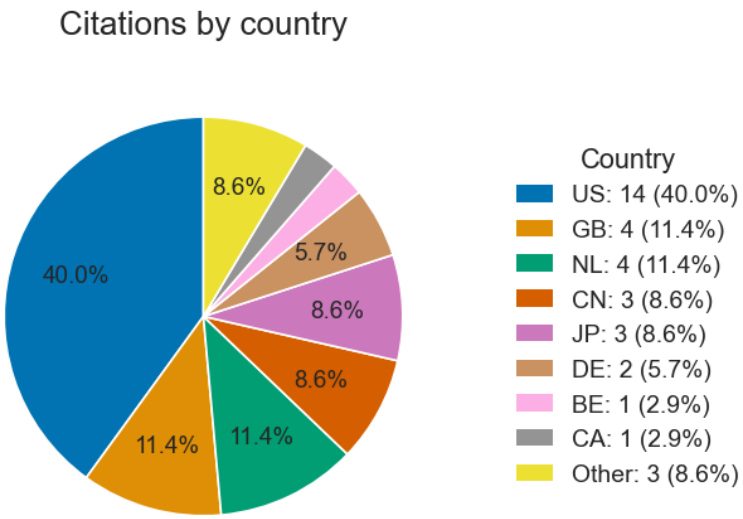
35. Mukherjee S, Madamsetty VS, Reddy R, Pradhan L, Yenurkar D, Manna S, Banerjee D. Imaging, Diagnostics, and Theranostics Applications of Nanocrystals. *Synthesis lectures on biomedical engineering*. 2025 Oct 22;:113-137. doi: 10.1007/978-3-032-04133-3_6.

Publication

Aswendt M, Vogel S, Schäfer C, Jathoul AP, Pulé M, Hoehn M. Quantitative in vivo dual-color bioluminescence imaging in the mouse brain. *Neurophotonics*. 2019 May 7;6(02):1-11. doi: 10.1117/1.nph.6.2.025006. PMID: 31093514.



Citation summary
Total citations: **35**
Average/year: **4.38**
Period: 2019-2026
SCImago cites/doc 2y (2025): **3.47**
SJR 2025: **1.1** (Q1)



Cited by

1. Liu S, Su Y, Lin MZ, Ronald JA. Brightening up Biology: Advances in Luciferase Systems for in Vivo Imaging. *ACS Chemical Biology*. 2021 Nov 15;16.0(12.0):2707-2718. doi: 10.1021/acschembio.1c00549. PMID: 34780699.

2. Ashmore-Harris C, Iafrate M, Saleem A, Fruhwirth GO. Non-invasive Reporter Gene Imaging of Cell Therapies, including T Cells and Stem Cells. *Molecular Therapy*. 2020 Mar 20;28.0(6.0):1392-1416. doi: 10.1016/j.ymthe.2020.03.016. PMID: 32243834.

3. Rai H, Gupta S, Kumar S, Yang J, Singh SK, Ran C, Modi GP. Near-Infrared Fluorescent Probes as Imaging and Theranostic Modalities for Amyloid-Beta and Tau Aggregates in Alzheimer's Disease. *Journal of Medicinal Chemistry*. 2022 Jun 27;65.0(13.0):8550-8595. doi: 10.1021/acs.jmedchem.1c01619. PMID: 35759679.

4. Zambito G, Chawda C, Mezzanotte L. Emerging tools for bioluminescence imaging. *Current Opinion in Chemical Biology*. 2021 Mar 23;63.0:86-94. doi: 10.1016/j.cbpa.2021.02.005. PMID: 33770744.

5. Collmann FM, Pijnenburg R, Taj SH, Minassian A, Folz-Donahue K, Kukat C, Aswendt M, Hoehn M. Individual in vivo Profiles of Microglia Polarization After Stroke, Represented by the Genes iNOS and Ym1. *Frontiers in Immunology*. 2019 Jun 4;10.0:1236. doi: 10.3389/fimmu.2019.01236. PMID: 31214190.

6. Endo M, Ozawa T. Advanced Bioluminescence System for In Vivo Imaging with Brighter and Red-Shifted Light Emission. *International Journal of Molecular Sciences*. 2020 Sep 7;21.0(18.0):6538. doi: 10.3390/ijms21186538. PMID: 32906768.

7. Zambito G, Hall MP, Wood MG, Gaspar N, Ridwan YR, Stellari FF, Shi C, Kirkland TA, Encell LP, Löwik CM, Mezzanotte L. Red-shifted click beetle luciferase mutant expands the multicolor bioluminescent palette for deep tissue imaging. *iScience*. 2020 Dec 26;24.0(1.0):101986. doi: 10.1016/j.isci.2020.101986. PMID: 33490896.

8. Iafrate M, Fruhwirth GO. How Non-invasive in vivo Cell Tracking Supports the Development and Translation of Cancer Immunotherapies. *Frontiers in Physiology*. 2020 Apr 3;11.0:154. doi: 10.3389/fphys.2020.00154. PMID: 32327996.

9. Yao ZW, Caldwell DR, Love AC, Kolbaba-Kartchner B, Mills JH, Schnermann MJ, Prescher JA. Coumarin luciferins and mutant luciferases for robust multi-component bioluminescence imaging. *Chemical Science*. 2021 Jan 1;12.0(35.0):11684-11691. doi: 10.1039/d1sc03114g. PMID: 34659703.

10. Jiang Y, Shi X, Tang CC, Wang F. Beyond luciferase-luciferin system: Modification, improved imaging and biomedical application. *Coordination Chemistry Reviews*. 2023 Jan 31;481.0:215045. doi: 10.1016/j.ccr.2023.215045.

11. Hesemans E, Buttiens K, Manshian BB, Soenen SJ. The Role of Optical Imaging in Translational Nanomedicine. *Journal of Functional Biomaterials*. 2022 Aug 31;13.0(3.0):137. doi: 10.3390/jfb13030137. PMID: 36135572.

12. Brennan CK, Yao ZW, Ionkina AA, Rathbun CM, Sathishkumar B, Prescher JA. Multiplexed bioluminescence imaging with a substrate unmixing platform. *Cell chemical biology*. 2022 Oct 24;29.0(11.0):1649-1660.e4. doi: 10.1016/j.chembiol.2022.10.004. PMID: 36283402.

13. Bettini A, Patrick PS, Day RM, Stuckey DJ. CT-Visible Microspheres Enable Whole-Body In Vivo Tracking of Injectable Tissue Engineering Scaffolds. *Advanced Healthcare Materials*. 2024 Apr 28;13.0(17.0):e2303588. doi: 10.1002/adhm.202303588. PMID: 38678393.

14. Pyon WS, Gray DT, Barnes CA. An Alternative to Dye-Based Approaches to Remove Background Autofluorescence From Primate Brain Tissue. *Frontiers in Neuroanatomy*. 2019 Jul 18;13.0:73. doi: 10.3389/fnana.2019.00073. PMID: 31379520.

15. Saito-Moriya R, Nakayama J, Kamiya G, Kitada N, Obata R, Maki S, Aoyama H. How to Select Firefly Luciferin Analogues for In Vivo Imaging. *International Journal of Molecular Sciences*. 2021 Feb 12;22.0(4.0):1848. doi: 10.3390/ijms22041848. PMID: 33673331.

16. Zambito G, Gaspar N, Ridwan YR, Hall MP, Shi C, Kirkland TA, Encell LP, Löwik CW, Mezzanotte L. Evaluating Brightness and Spectral Properties of Click Beetle and Firefly Luciferases Using Luciferin Analogues: Identification of Preferred Pairings of Luciferase and Substrate for In Vivo Bioluminescence Imaging. *Molecular Imaging and Biology*. 2020 Sep 14;22.0(6.0):1523-1531. doi: 10.1007/s11307-020-01523-7. PMID: 32926287.

17. Raikwar SP, Thangavel R, Ahmed ME, Selvakumar GP, Kempuraj D, Wu K, Khan O, Bazley K, Bussinger B, Kukulka K, Zaheer SA, Iyer SS, Govindarajan R, Burton C, James D, Zaheer A. Real-Time Noninvasive Bioluminescence, Ultrasound and Photoacoustic Imaging in NFκB-RE-Luc Transgenic Mice Reveal Glia Maturation Factor-Mediated Immediate and Sustained Spatio-Temporal Activation of NFκB Signaling Post-Traumatic Brain Injury in a Gender-Specific Manner. *Cellular and Molecular Neurobiology*. 2020 Aug 12;41.0(8.0):1687-1706. doi: 10.1007/s10571-020-00937-9. PMID: 32785863.

18. Rathbun CM, Ionkina AA, Yao ZW, Jones KA, Porterfield W, Prescher JA. Rapid Multicomponent Bioluminescence Imaging via Substrate Unmixing. *ACS Chemical Biology*. 2021 Mar 17;16.0(4.0):682-690. doi: 10.1021/acschembio.0c00959. PMID: 33729750.

19. Dimond A, Van de Pette M, Fisher AG. Illuminating Epigenetics and Inheritance in the Immune System with Bioluminescence. *Trends in Immunology*. 2020 Oct 6;41.0(11.0):994-1005. doi: 10.1016/j.it.2020.09.001. PMID: 33036908.

20. Kim I, Park J, Kim B, Hwang K, Song B. Recent advances in stem cell therapy for neurodegenerative disease: Three dimensional tracing and its emerging use. *World Journal of Stem Cells*. 2021 Sep 18;13.0(9.0):1215-1230. doi: 10.4252/wjsc.v13.i9.1215. PMID: 34630859.

21. Williams SJ, Hwang CS, Prescher JA. Orthogonal Bioluminescent Probes from Disubstituted Luciferins. *Biochemistry*. 2021 Feb 18;60.0(8.0):563-572. doi: 10.1021/acs.biochem.0c00894. PMID: 33599497.

22. Townsend KM, Prescher JA. Recent advances in bioluminescent probes for neurobiology. *Neurophotonics*. 2024 Feb 22;11.0(2.0):1-17. doi: 10.1117/1.nph.11.2.024204. PMID: 38390217.

23. Silvagnoli AD, Taylor KA, Slaviero AN, Petersen ED. Method for optimizing imaging parameters to record neuronal and cellular activity at depth with bioluminescence. *Neurophotonics*. 2024 Mar 28;11.0(2.0):1-17. doi: 10.1117/1.nph.11.2.024206. PMID: 38550774.

24. Xia Y, Bao H, Huang J, Li X, Yu C, Zhang Z, Wang H. Near-infrared-persistent luminescence/bioluminescence imaging tracking of transplanted mesenchymal stem cells in pulmonary fibrosis. *Biomaterials Science*. 2020 Jan 1;8.0(11.0):3095-3105. doi: 10.1039/d0bm00063a. PMID: 32347847.

25. Navarro MX, Brennan CK, Love AC, Prescher JA. Caged luciferins enable rapid multicomponent bioluminescence imaging. *Photochemistry and Photobiology*. 2023 May 31;100.0(1.0):67-74. doi: 10.1111/php.13814. PMID: 37259257.

26. Madsen SD, Giler MK, Bunnell BA, O'Connor KC. Illuminating the Regenerative Properties of Stem Cells In Vivo with Bioluminescence Imaging. *Biotechnology Journal*. 2020 Oct 22;16.0(3.0):e2000248. doi: 10.1002/biot.202000248. PMID: 33089922.

27. Li X, Yu C, Bao H, Chen Z, Liu X, Huang J, Zhang Z. CT/bioluminescence dual-modal imaging tracking of stem cells labeled with Au@PEI@PEG nanotracers and RfLuc in nintedanib-assisted pulmonary fibrosis therapy. *Nanomedicine Nanotechnology Biology and Medicine*. 2022 Jan 13;41.0:102517. doi: 10.1016/j.nano.2022.102517. PMID: 35032629.

28. Harward SC, Southwell DG. Interneuron transplantation: a prospective surgical therapy for medically refractory epilepsy. *Neurosurgical FOCUS*. 2020 Apr 1;48.0(4.0):E18. doi: 10.3171/2020.2.focus19955. PMID: 32234982.

29. Yawata S, Noda K, Shimomura A, Kuroda A. Mutant firefly luciferase enzymes resistant to the inhibition by sodium chloride. *Biotechnology Letters*. 2021 May 4;43.0(8.0):1585-1594. doi: 10.1007/s10529-021-03109-x. PMID: 33945054.

30. Zambito G, Mezzanotte L. Near-infrared bioluminescence imaging of two cell populations in living mice. *STAR Protocols*. 2021 Jul 7;2.0(3.0):100662. doi: 10.1016/j.xpro.2021.100662. PMID: 34286293.

31. Esimbekova EN, Kratasyuk VA, Rozanova NA, Lonshakova-Mukina VI, Torgashina IG, Komleva YK, Saridis MR, Korsakova SA, Yurchenko SO, Salmina AB. Current and prospective trends in the application of bioluminescent analysis in experimental brain studies. *Talanta*. 2025 Jun 24;296.0:128510. doi: 10.1016/j.talanta.2025.128510. PMID: 40582108.

32. Aswendt M, Vogel S, Hoehn M. In vivo bioluminescence imaging to elucidate stem cell graft differentiation. *Neural Regeneration Research*. 2019 Sep 16;15.0(1.0):61. doi: 10.4103/1673-5374.264449. PMID: 31535647.

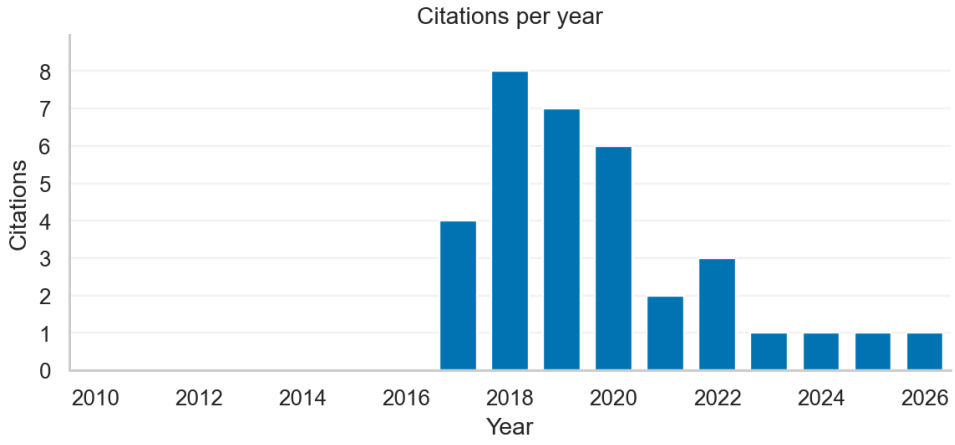
33. Yao ZW, Caldwell D, Love AC, Kolbaba-Kartchner B, Mills J, Schnermann MJ, Prescher JA. Coumarin Luciferins and Mutant Luciferases for Robust Multicomponent Bioluminescence Imaging. *ChemRxiv*. 2021 May 10;. doi: 10.26434/chemrxiv.14544630.

34. Taylor KA, Covell J, Pinderi Z, Galvin H, Onge BS, Petersen ED. Neurotransmitter Sensor-Driven Neuromodulation. *Springer eBooks*. 2026 Jan 1;:103-113. doi: 10.1007/978-3-032-18121-3_7.

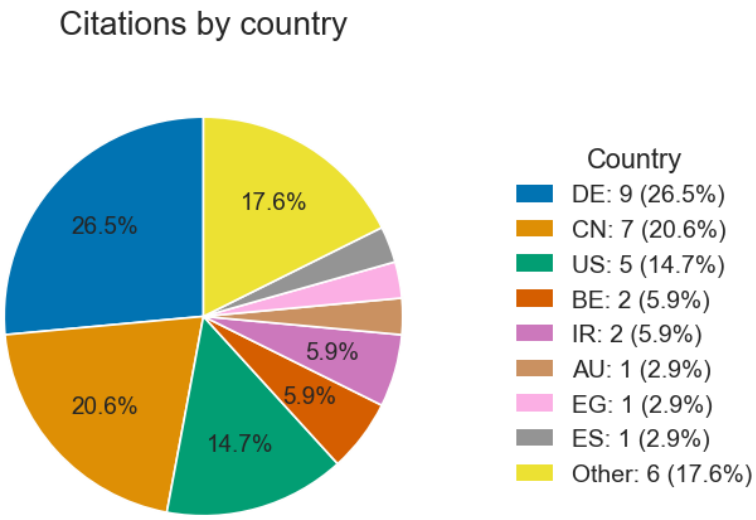
35. Chatterton M, Dubey O, VanDusen A, Robinette E, Abel JD, Stewart AN, Rossignol J, Bakke J, Petersen ED. Bioluminescent Kinase Sensors (BlinKS) as Optogenetic Integrators. *Springer eBooks*. 2026 Jan 1;:143-155. doi: 10.1007/978-3-032-18121-3_10.

Publication

Adamczak JM, Aswendt M, Kreutzer C, Rotheneichner P, Riou A, Selt M, Beyrau A, Uhlenküken U, Diedenhofen M, Nelles M, Aigner L, Couillard-Després S, Hoehn M. Neurogenesis upregulation on the healthy hemisphere after stroke enhances compensation for age-dependent decrease of basal neurogenesis. *Neurobiology of Disease*. 2016 Dec 20;99:47-57. doi: 10.1016/j.nbd.2016.12.015. PMID: 28007584.



Citation summary
Total citations: **34**
Average/year: **3.09**
Period: 2016-2026
SCImago cites/doc 2y (2025): **6.07**
SJR 2025: **2.16** (Q1)



Cited by

1. Candelario-Jalil E, Paul S. Impact of aging and comorbidities on ischemic stroke outcomes in preclinical animal models: A translational perspective. *Experimental Neurology*. 2020 Oct 6;335.0:113494. doi: 10.1016/j.expneurol.2020.113494. PMID: 33035516.

2. Xu J, Zheng Y, Wang L, Liu Y, Wang X, Li Y, Chi G. miR-124: A Promising Therapeutic Target for Central Nervous System Injuries and Diseases. *Cellular and Molecular Neurobiology*. 2021 Apr 22;42.0(7.0):2031-2053. doi: 10.1007/s10571-021-01091-6. PMID: 33886036.

3. Anckaerts C, Blockx I, Summer P, Michael J, Hamaide J, Kreutzer C, Boutin H, Couillard-Després S, Verhoye M, Van der Linden A. Early functional connectivity deficits and progressive microstructural alterations in the TgF344-AD rat model of Alzheimer's Disease: A longitudinal MRI study. *Neurobiology of Disease*. 2018 Nov 13;124.0:93-107. doi: 10.1016/j.nbd.2018.11.010. PMID: 30445024.

4. Ni G, Liang C, Wang J, Duan C, Wang P, Wang Y. Astragaloside IV improves neurobehavior and promotes hippocampal neurogenesis in MCAO rats though BDNF-TrkB signaling pathway. *Biomedicine & Pharmacotherapy*. 2020 Jul 16;130.0:110353. doi: 10.1016/j.biopha.2020.110353. PMID: 32682983.

5. Collmann FM, Pijnenburg R, Taj SH, Minassian A, Folz-Donahue K, Kukat C, Aswendt M, Hoehn M. Individual in vivo Profiles of Microglia Polarization After Stroke, Represented by the Genes iNOS and Ym1. *Frontiers in Immunology*. 2019 Jun 4;10.0:1236. doi: 10.3389/fimmu.2019.01236. PMID: 31214190.

6. Green C, Minassian A, Vogel SN, Diedenhofen M, Beyrau A, Wiedermann D, Hoehn M. Sensorimotor Functional and Structural Networks after Intracerebral Stem Cell Grafts in the Ischemic Mouse Brain. *Journal of Neuroscience*. 2018 Jan 10;38.0(7.0):1648-1661. doi: 10.1523/jneurosci.2715-17.2018. PMID: 29321138.

7. Aswendt M, Vogel S, Schäfer C, Jathoul AP, Pulé M, Hoehn M. Quantitative in vivo dual-color bioluminescence imaging in the mouse brain. *Neurophotonics*. 2019 May 7;6.0(2.0):1-11. doi: 10.1117/1.nph.6.2.025006. PMID: 31093514.

8. Paniagua-Torija B, Norenberg MD, Arévalo-Martín Á, Carballosa-Gautam MM, Campos-Martín Y, Molina-Holgado E, García-Ovejero D. Cells in the adult human spinal cord ependymal region do not proliferate after injury. *The Journal of Pathology*. 2018 Aug 8;246.0(4.0):415-421. doi: 10.1002/path.5151. PMID: 30091291.

9. Trout AL, Kahle M, Roberts J, Marcelo AJ, de Hoog L, Boychuk JA, Grupke SL, Berretta A, Gowing EK, Boychuk CR, Gorman AA, Edwards DN, Rutkai I, Biose IJ, Ishibashi-Ueda H, Ihara M, Smith BN, Clarkson AN, Bix GJ. Perlecan Domain-V Enhances Neurogenic Brain Repair After Stroke in Mice. *Translational Stroke Research*. 2020 Apr 7;12.0(1.0):72-86. doi: 10.1007/s12975-020-00800-5. PMID: 32253702.

10. Oh B, Santhanam S, Azadian MM, Swaminathan V, Lee AG, McConnell KW, Levinson A, Song S, Patel JJ, Gardner EE, George PM. Electrical modulation of transplanted stem cells improves functional recovery in a rodent model of stroke. *Nature Communications*. 2022 Mar 15;13.0(1.0):1366. doi: 10.1038/s41467-022-29017-w. PMID: 35292643.

11. Yang XY, Geng K, Zhang Y, Zhang J, Yang K, Shao J, Xia W. Sirt3 deficiency impairs neurovascular recovery in ischemic stroke. *CNS Neuroscience & Therapeutics*. 2018 May 18;24.0(9.0):775-783. doi: 10.1111/cns.12853. PMID: 29777578.

12. Rao J, Li H, Zhang H, Xiang X, Ding X, Li L, Geng F, Qi H. Periplaneta Americana (L.) extract activates the ERK/CREB/BDNF pathway to promote post-stroke neuroregeneration and recovery of neurological functions in rats. *Journal of Ethnopharmacology*. 2023 Nov 11;321.0:117400. doi: 10.1016/j.jep.2023.117400. PMID: 37952730.

13. Rogall R, Rabenstein M, Vay SU, Bach A, Pikhovych A, Bärmann J, Hoehn M, Couillard-Després S, Fink GR, Schroeter M, Rueger MA. Bioluminescence imaging visualizes osteopontin-induced neurogenesis and neuroblast migration in the mouse brain after stroke. *Stem Cell Research & Therapy*. 2018 Jul 4;9.0(1.0):182. doi: 10.1186/s13287-018-0927-9. PMID: 29973246.

14. Damavandi AR, Mirmosayyeb O, Ebrahimi N, Zalpoor H, Khalilian P, Yahiazadeh S, Eskandari N, Rahdar A, Kumar PS, Pandey S. Advances in nanotechnology versus stem cell therapy for the theranostics of multiple sclerosis disease. *Applied Nanoscience*. 2022 Dec 5;13.0(6.0):4043-4073. doi: 10.1007/s13204-022-02698-x.

15. Ding Y, Jie K, Liu X, Shao B. Astragaloside IV plays a neuroprotective role by promoting PPARγ in cerebral ischemia-reperfusion rats. *Behavioural Brain Research*. 2024 Sep 27;476.0:115267. doi: 10.1016/j.bbr.2024.115267. PMID: 39341463.

16. Popa-Wagner A, Glavan D, Olaru A, Olaru D, Mărgăritescu OC, Tica OS, Surugiu R, Sandu RE. Present Status and Future Challenges of New Therapeutic Targets in Preclinical Models of Stroke in Aged Animals with/without Comorbidities. *International Journal of Molecular Sciences*. 2018 Jan 25;19.0(2.0):356. doi: 10.3390/ijms19020356. PMID: 29370078.

17. Zuo Y, Wang J, Enkhjargal B, Doycheva DM, Yan X, Zhang JH, Liu F. Neurogenesis changes and the fate of progenitor cells after subarachnoid hemorrhage in rats. *Experimental Neurology*. 2018 Oct 23;311.0:274-284. doi: 10.1016/j.expneurol.2018.10.011. PMID: 30359565.

18. Williamson MR, Jones TA, Drew MR. Functions of subventricular zone neural precursor cells in stroke recovery. *Behavioural Brain Research*. 2019 Sep 4;376.0:112209. doi: 10.1016/j.bbr.2019.112209. PMID: 31493429.

19. Glover JC, Aswendt M, Boulland J, Lojk J, Stamenković S, Anđjus PR, Fiori F, Hoehn M, Mitrečić D, Pavlin M, Cavalli S, Frati C, Quaini F, (BIONECA) OBOTECA1. In vivo Cell Tracking Using Non-invasive Imaging of Iron Oxide-Based Particles with Particular Relevance for Stem Cell-Based Treatments of Neurological and Cardiac Disease. *Molecular Imaging and Biology*. 2019 Dec 4;22.0(6.0):1469-1488. doi: 10.1007/s11307-019-01440-4. PMID: 31802361.

20. Bălşeanu TA, Grigore MM, Pinosanu LR, SLEVIN MA, Hermann DM, Glăvan D, Popa-Wagner A. Electric Stimulation of Neurogenesis Improves Behavioral Recovery After Focal Ischemia in Aged Rats. *Frontiers in Neuroscience*. 2020 Jul 9;14.0:732. doi: 10.3389/fnins.2020.00732. PMID: 32742258.

21. Sun Y, Sun X, Qu H, Zhao S, Xiao T, Zhao C. Neuroplasticity and behavioral effects of fluoxetine after experimental stroke. *Restorative Neurology and Neuroscience*. 2017 Aug 24;35.0(5.0):457-468. doi: 10.3233/rnn-170725. PMID: 28854520.

22. Vogel SN, Schäfer C, Heß S, Folz-Donahue K, Nelles M, Minassian A, Schwarz MK, Kukat C, Ehrlich M, Zaehres H, Kloppenburg P, Hoehn M, Aswendt M. The in vivo timeline of differentiation of engrafted human neural progenitor cells. *Stem Cell Research*. 2019 Mar 25;37.0:101429. doi: 10.1016/j.scr.2019.101429. PMID: 30933718.

23. Vanhunsel S, Beckers AM, Moons L. Designing neuroreparative strategies using aged regenerating animal models. *Ageing Research Reviews*. 2020 May 31;62.0:101086. doi: 10.1016/j.arr.2020.101086. PMID: 32492480.

24. Green C, Minassian A, Vogel SN, Diedenhofen M, Wiedermann D, Hoehn M. Persistent Quantitative Vitality of Stem Cell Graft Is Necessary for Stabilization of Functional Brain Networks After Stroke. *Frontiers in Neurology*. 2019 Apr 5;10.0:335. doi: 10.3389/fneur.2019.00335. PMID: 31024429.

25. Khodanovich MY, Евгеньевич AA, Anan'ina TV, Kudabaeva M, Pishchelko AO, Павловна KE, Nemirovich-Danchenko NM, Светлик MB, Tumentceva YA, Haute CVD, Gijsbers R, Daniëls V, Thiry I, Pershina AG, Shadrina MM, Naumova AV. Tissue-Specific Ferritin- and GFP-Based Genetic Vectors Visualize Neurons by MRI in the Intact and Post-Ischemic Rat Brain. *International Journal of Molecular Sciences*. 2020 Nov 25;21.0(23.0):8951. doi: 10.3390/ijms21238951. PMID: 33255702.

26. Zaghary WA, Elansary MM, Shouman DN, Abdelrahim AA, Abu-Zied KM, Sakr TM. Can nanotechnology overcome challenges facing stem cell therapy? A review. *Journal of Drug Delivery Science and Technology*. 2021 Sep 29;66.0:102883. doi: 10.1016/j.jddst.2021.102883.

27. Shabani Z, Zangbar HS, Nasrolahi A. Cerebral dopamine neurotrophic factor increases proliferation, Migration and differentiation of subventricular zone neuroblasts in photothrombotic stroke model of mouse. *Journal of Stroke and Cerebrovascular Diseases*. 2022 Sep 15;31.0(11.0):106725. doi: 10.1016/j.jstrokecerebrovasdis.2022.106725. PMID: 36116218.

28. Vargas-Saturno L, Ayala-Grosso CA. Adaptive neurogenesis in the cerebral cortex and contralateral subventricular zone induced by unilateral cortical devascularization: Possible modulation by dopamine neurotransmission. *European Journal of Neuroscience*. 2018 Nov 7;48.0(12.0):3514-3533. doi: 10.1111/ejn.14260. PMID: 30402991.

29. Aswendt M, Vogel S, Hoehn M. In vivo bioluminescence imaging to elucidate stem cell graft differentiation. *Neural Regeneration Research*. 2019 Sep 16;15.0(1.0):61. doi: 10.4103/1673-5374.264449. PMID: 31535647.

30. Alawieh AM, Langley F, Tomlinson S. The Inflammatory Response and Its Effect on Rehabilitation-Induced Repair Processes After Stroke. *Springer series in translational stroke research*. 2017 Nov 6;:509-520. doi: 10.1007/978-3-319-66679-2_25.

31. Aswendt M, Collmann FM, Hoehn M. Neurobiological insights from bioluminescence imaging. *Oncotarget*. 2017 Aug 13;8.0(41.0):69198-69199. doi: 10.18632/oncotarget.20302. PMID: 29050195.

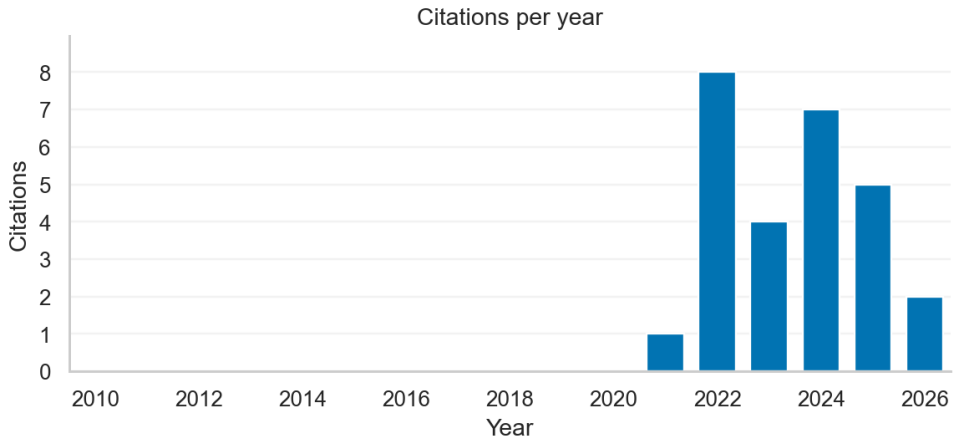
32. Rabenstein M, Rueger MA. Mobilization of Endogenous Neural Stem Cells to Promote Regeneration After Stroke. *Springer series in translational stroke research*. 2017 Nov 5;:101-112. doi: 10.1007/978-3-319-66679-2_5.

33. Nakanishi T, Matsumaru D, Ishida K. Advances in Bioluminescence Imaging for Assessment of Developmental Neurotoxicity. *BPB Reports*. 2025 Jan 1;8.0(6.0):176-181. doi: 10.1248/bpbreports.8.6_176.

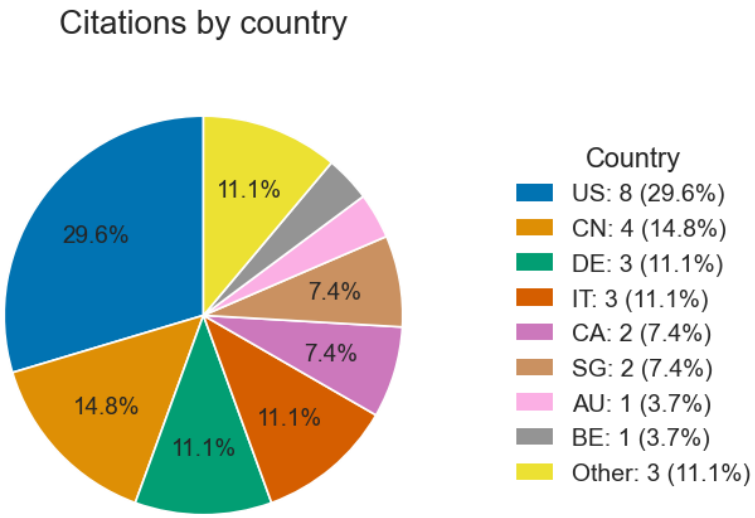
34. Hartung T, Freyer D, Zemella A, Radbruch H, Weiner JM, El-Din JJ, Meisel A, Priller J. Non-hematopoietic erythropoietin splice variant is produced in the diseased human brain and confers neuroprotection. *Frontiers in Cellular Neuroscience*. 2026 Jan 12;19.0:1677505. doi: 10.3389/fncel.2025.1677505. PMID: 41602576.

Publication

Aswendt M, Green C, Sadler R, Llovera G, Dzikowski L, Heindl S, de Agüero MG, Diedenhofen M, Vogel SN, Wieters F, Wiedermann D, Liesz A, Hoehn M. The gut microbiota modulates brain network connectivity under physiological conditions and after acute brain ischemia. *iScience*. 2021 Sep 9;24(10):103095. doi: 10.1016/j.isci.2021.103095. PMID: 34622150.



Citation summary
Total citations: **27**
Average/year: **4.50**
Period: 2021-2026
SCImago cites/doc 2y (2025): **4.5**
SJR 2025: **1.36** (Q1)



Cited by

1. Paeslack N, Mimmler M, Becker S, Gao Z, Khuu MP, Mann A, Malinarich F, Regen T, Reinhardt C. Microbiota-derived tryptophan metabolites in vascular inflammation and cardiovascular disease. *Amino Acids*. 2022 Apr 22;54.0(10.0):1339-1356. doi: 10.1007/s00726-022-03161-5. PMID: 35451695.

2. Carloni S, Rescigno M. Unveiling the gut-brain axis: structural and functional analogies between the gut and the choroid plexus vascular and immune barriers. *Seminars in Immunopathology*. 2022 Jul 21;44.0(6.0):869-882. doi: 10.1007/s00281-022-00955-3. PMID: 35861857.

3. Darch H, McCafferty CP. Gut microbiome effects on neuronal excitability & activity: Implications for epilepsy. *Neurobiology of Disease*. 2022 Jan 13;165.0:105629. doi: 10.1016/j.nbd.2022.105629. PMID: 35033659.

4. Pini L, Salvalaggio A, Wennberg AMV, Dimakou A, Matteoli M, Corbetta M. The pollutome-connectome axis: a putative mechanism to explain pollution effects on neurodegeneration. *Ageing Research Reviews*. 2023 Jan 28;86.0:101867. doi: 10.1016/j.arr.2023.101867. PMID: 36720351.

5. Drobyshevsky A, Synowiec S, Goussakov IV, Fabres RB, Lu JN, Caplan MS. Intestinal microbiota modulates neuroinflammatory response and brain injury after neonatal hypoxia-ischemia. *Gut Microbes*. 2024 Mar 27;16.0(1.0):2333808. doi: 10.1080/19490976.2024.2333808. PMID: 38533575.

6. Sinha N, Joshi RB, Sandhu MRS, Netoff TI, Zaveri HP, Lehnertz K. Perspectives on Understanding Aberrant Brain Networks in Epilepsy. *Frontiers in Network Physiology*. 2022 Apr 5;2.0:868092. doi: 10.3389/fnetp.2022.868092. PMID: 36926081.

7. Hoehn M, Aswendt M. Functional hyperconnectivity related to brain disease: maladaptive process or element of resilience?. *Neural Regeneration Research*. 2022 Nov 18;18.0(7.0):1489. doi: 10.4103/1673-5374.361541. PMID: 36571347.

8. Liu L, Xiang M, Cai X, Wu B, Chen C, Cai N, Ao D. Multi-omics analyses of gut microbiota via 16S rRNA gene sequencing, LC-MS/MS and diffusion tension imaging reveal aberrant microbiota-gut-brain axis in very low or extremely low birth weight infants with white matter injury. *BMC Microbiology*. 2023 Dec 6;23.0(1.0):387. doi: 10.1186/s12866-023-03103-5. PMID: 38057706.

9. Chen G, Du X, Cui J, Song J, Xiong M, Zeng X, Yang H, Xu K. Role of gut microbiota in ischemic stroke: A narrative review of human and animal studies. *Neuroprotection/Neuroprotection (Chichester, England. Print)*. 2024 May 16;2.0(2.0):120-136. doi: 10.1002/nep3.44. PMID: 41383698.

10. Small SL. Precision neurology. *Ageing Research Reviews*. 2024 Dec 8;104.0:102632. doi: 10.1016/j.arr.2024.102632. PMID: 39657848.

11. Iriah SC, Rodriguez N, Febo M, Morrissette M, Strandwitz P, Kulkarni P, Ferris CF. The microbiome's influence on the neurobiology of opioid addiction and brain connectivity. *Brain Research Bulletin*. 2024 Dec 6;220.0:111159. doi: 10.1016/j.brainresbull.2024.111159. PMID: 39645048.

12. Dai J, Yang J, Han S, Li N, Wang S, Xia S, Kim H, Jun Y, Lee S, Kitagawa Y, Xie F, Yang L, Shen S, Chen LL, Turner DP, Hodin RA, Martyn JAJ, Mao J, You Z. Deficiency of intestinal alkaline phosphatase affects behavior and microglia activity in mice. *Brain Behavior and Immunity*. 2025 Feb 19;126.0:297-310. doi: 10.1016/j.bbi.2025.02.006. PMID: 39984137.

13. Scharwächter L, Schmitt FJ, Pallast N, Fink GR, Aswendt M. Network analysis of neuroimaging in mice. *NeuroImage*. 2022 Mar 17;253.0:119110. doi: 10.1016/j.neuroimage.2022.119110. PMID: 35311664.

14. Van der Linden A, Hoehn M. Monitoring Neuronal Network Disturbances of Brain Diseases: A Preclinical MRI Approach in the Rodent Brain. *Frontiers in Cellular Neuroscience*. 2022 Jan 3;15.0:815552. doi: 10.3389/fncel.2021.815552. PMID: 35046778.

15. Yeo XY, Chae WR, Lee HU, Bae H, Pettersson S, Grandjean J, Han W, Jung S. Nuanced contribution of gut microbiome in the early brain development of mice. *Gut Microbes*. 2023 Nov 27;15.0(2.0):2283911. doi: 10.1080/19490976.2023.2283911. PMID: 38010368.

16. Thompson SL, Ellegood J, Bowdish DME, Lerch JP, Foster JA. Sex- and brain region-specific alterations in brain volume in germ-free mice. *iScience*. 2024 Nov 19;27.0(12.0):111429. doi: 10.1016/j.isci.2024.111429. PMID: 39735434.

17. MAO Z, Zhang J, Guo L, WANG X, ZHU Z, Miao M. Therapeutic approaches targeting the gut microbiota in ischemic stroke: current advances and future directions. *Bioscience of Microbiota Food and Health*. 2024 Jan 1;43.0(4.0):321-328. doi: 10.12938/bmfh.2024-022. PMID: 39364121.

18. Carloni S, Strati F. The Gut Vascular Barrier: Linking the Intestinal Environment with Blood Circulation. *The microbiomes of humans, animals, plants, and the environment*. 2025 Jan 1;:57-74. doi: 10.1007/978-3-031-84065-4_4.

19. Dimaoui A, Mekhlouf YE, Lamtai M, Ibouzine-Dine L, Marzouki ZE, Touil T, Hessni AE, Mesfioui A. Early-life gut microbiota depletion leads to persistent CNS myelin impairment in adult Swiss albino mice. *Neuroscience and Behavioral Physiology*. 2025 Oct 1;55.0(8.0):1374-1384. doi: 10.1007/s11055-025-01893-x.

20. Lee S, Dubrof S, Ahmed I, Zhao Q, Callaway TR, Lourenco JM, Park HJ. Maternal docosahexaenoic acid supplementation shapes offspring gut microbiota to modulate the gut-brain axis in a sow-piglet model. *Frontiers in Nutrition*. 2026 Apr 13;13.0:1776896. doi: 10.3389/fnut.2026.1776896. PMID: 42051340.

21. Kim J, Kim DW, Lee AC, Mason M, Jouroukhin Y, Woo H, Yolken RH, Pletnikov MV. Homeostatic regulation of neuronal excitability by probiotics in male germ-free mice. *Journal of Neuroscience Research*. 2021 Dec 21;100.0(2.0):444-460. doi: 10.1002/jnr.24999. PMID: 34935171.

22. Yeo XY, Chae WR, Lee HU, Bae H, Pettersson S, Grandjean J, Han W, Jung S. Maternal microbes and early brain development in mouse. *bioRxiv (Cold Spring Harbor Laboratory)*. 2022 Dec 20;. doi: 10.1101/2022.12.19.521137.

23. Iriah SC. The neurological and behavioral effects of opioids. 2023 Jan 1;. doi: 10.17760/d20620617.

24. Green M, Foster JA. Gut-brain communication through microbes. *Elsevier eBooks*. 2024 Jan 1;:29-66. doi: 10.1016/b978-0-323-90144-4.00002-1.

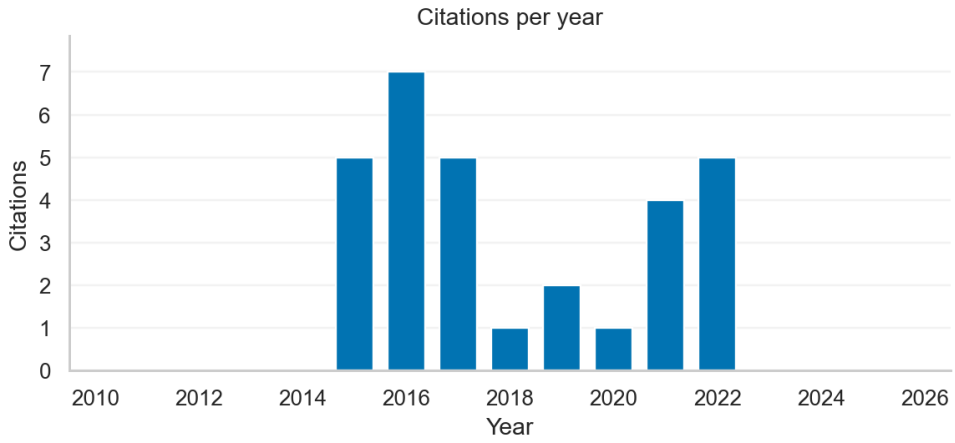
25. Kolobaric A, Ladouceur CD, Morris A, Methé BA, Jašarević E, Bylsma LM. Adolescent girls at familial risk for depression with more advanced adrenarche have altered gut microbiota. *bioRxiv (Cold Spring Harbor Laboratory)*. 2025 Aug 8;. doi: 10.1101/2025.08.07.669149.

26. Yener SM. The Gut-Brain Connection: How Diet Influences Mental Health. *Nutritional neurosciences*. 2025 Jan 1;:117-140. doi: 10.1007/978-981-95-1414-4_5.

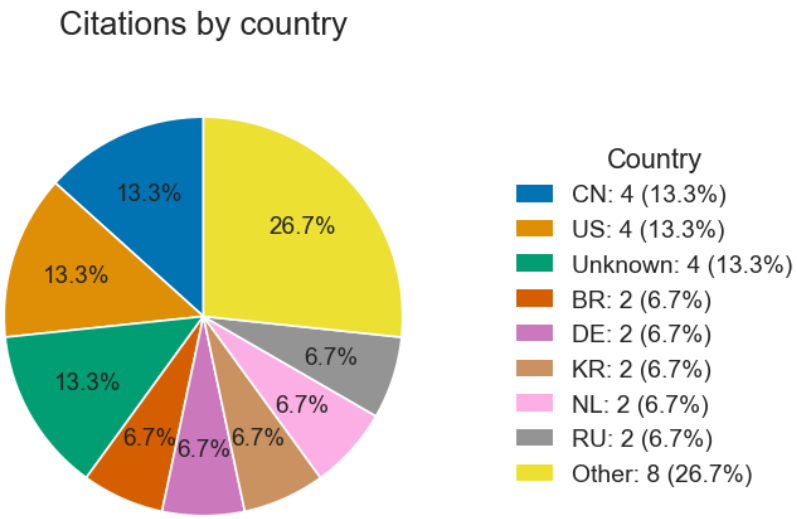
27. Teng L, Wu M, Song W, Wang Q, Luo X, Zhu L. Gut microbiota-mediated neuroinflammation following stroke: mechanisms and intervention strategies. *Journal of Translational Medicine*. 2026 Jul 30;24.0(1.0). doi: 10.1186/s12967-026-08694-x. PMID: 42595995.

Publication

Aswendt M, Adamczak JM, Tennstaedt A. A review of novel optical imaging strategies of the stroke pathology and stem cell therapy in stroke. *Frontiers in Cellular Neuroscience*. 2014 Aug 14;8:226. doi: 10.3389/fncel.2014.00226. PMID: 25177269.



Citation summary
Total citations: **30**
Average/year: **2.31**
Period: 2014-2026
SCImago cites/doc 2y (2025): **3.73**
SJR 2025: **1.43** (Q2)



Cited by

1. Kaskova ZM, Tsarkova AS, Yampolsky IV. 1001 lights: luciferins, luciferases, their mechanisms of action and applications in chemical analysis, biology and medicine. *Chemical Society Reviews*. 2016 Jan 1;45.0(21.0):6048-6077. doi: 10.1039/c6cs00296j. PMID: 27711774.

2. Parvez S, Kaushik M, Ali M, Alam MM, Ali J, Tabassum H, Kaushik P. Dodging blood brain barrier with “nano” warriors: Novel strategy against ischemic stroke. *Theranostics*. 2021 Dec 15;12.0(2.0):689-719. doi: 10.7150/thno.64806. PMID: 34976208.

3. Nierode GJ, Kwon PS, Dordick JS, Kwon S. Cell-Based Assay Design for High-Content Screening of Drug Candidates. *Journal of Microbiology and Biotechnology*. 2016 Feb 28;26.0(2.0):213-225. doi: 10.4014/jmb.1508.08007. PMID: 26428732.

4. Sarmento RM, Vasconcelos FF, Filho PPR, Wu W, de Albuquerque VHC. Automatic Neuroimage Processing and Analysis in Stroke-A Systematic Review. *IEEE Reviews in Biomedical Engineering*. 2019 Aug 23;13.0:130-155. doi: 10.1109/rbme.2019.2934500. PMID: 31449031.

5. Jurgielewicz P, Harmsen S, Wei EY, Bachmann MH, Ting R, Aras Ö. New imaging probes to track cell fate: reporter genes in stem cell research. *Cellular and Molecular Life Sciences*. 2017 Jul 3;74.0(24.0):4455-4469. doi: 10.1007/s00018-017-2584-z. PMID: 28674728.

6. Jendelová P, Kubinová Š, Sandvig I, Erceg S, Sandvig A, Syková E. Current developments in cell- and biomaterial-based approaches for stroke repair. *Expert Opinion on Biological Therapy*. 2015 Sep 30;16.0(1.0):43-56. doi: 10.1517/14712598.2016.1094457. PMID: 26419582.

7. Rizvi SFA, Zhang H. Emerging trends of receptor-mediated tumor targeting peptides: A review with perspective from molecular imaging modalities. *European Journal of Medicinal Chemistry*. 2021 May 15;221.0:113538. doi: 10.1016/j.ejmech.2021.113538. PMID: 34022717.

8. Kim JH, Kim SY, Kim B, Lee S, Cha SH, Lee D, Lee HJ. Prospects of Therapeutic Target and Directions for Ischemic Stroke. *Pharmaceuticals*. 2021 Apr 1;14.0(4.0):321. doi: 10.3390/ph14040321. PMID: 33916253.

9. Gavins FNE, Smith HK. Cell Tracking Technologies for Acute Ischemic Brain Injury. *Journal of Cerebral Blood Flow & Metabolism*. 2015 May 13;35.0(7.0):1090-1099. doi: 10.1038/jcbfm.2015.93. PMID: 25966948.

10. Pochechuev MS, Bilan DS, Fedotov IV, Kelmanson IV, Solotenkov MA, Stepanov EA, Kotova DA, Ivanova AD, Kostyuk AI, Raevskii RI, Александрович ЛА, Fedotov AB, Belousov VV, Zheltikov AM. Real-time fiber-optic recording of acute-ischemic-stroke signatures. *Journal of Biophotonics*. 2022 Jun 3;15.0(10.0):e202200050. doi: 10.1002/jbio.202200050. PMID: 35654757.

11. Tennstaedt A, Mastropietro A, Nelles M, Beyrau A, Hoehn M. In Vivo Fate Imaging of Intracerebral Stem Cell Grafts in Mouse Brain. *PLoS ONE*. 2015 Dec 7;10.0(12.0):e0144262. doi: 10.1371/journal.pone.0144262. PMID: 26641453.

12. Mukhtar R, Fazal MU, Saleem MS, Saleem S. Role of low-level laser therapy in post-herpetic neuralgia: a pilot study. *Lasers in Medical Science*. 2020 Jan 31;35.0(8.0):1759-1764. doi: 10.1007/s10103-020-02969-5. PMID: 32006261.

13. da Silva HR, Mamani JB, Nucci MP, Nucci LP, Kondo AT, Fantacini DMC, de Souza LEB, Picanço-Castro V, Covas DT, Kutner JM, Oliveira FA, Hamerschlak N, Gamarra LF. Triple-modal imaging of stem-cells labeled with multimodal nanoparticles, applied in a stroke model. *World Journal of Stem Cells*. 2019 Feb 26;11.0(2.0):100-123. doi: 10.4252/wjsc.v11.i2.100. PMID: 30842808.

14. Mezzanotte L, Iljas JD, Que I, Chan AB, Kaijzel EL, Hoeben RC, Löwik CW. Optimized Longitudinal Monitoring of Stem Cell Grafts in Mouse Brain Using a Novel Bioluminescent/Near Infrared Fluorescent Fusion Reporter. *Cell Transplantation*. 2017 Dec 1;26.0(12.0):1878-1889. doi: 10.1177/0963689717739718. PMID: 29390874.

15. Wang Y, Yu C, Li Y, Bao H, Li X, Fan H, Huang J, Zhang Z. In vivo MRI tracking and therapeutic efficacy of transplanted mesenchymal stem cells labeled with ferrimagnetic vortex iron oxide nanorings for liver fibrosis repair. *Nanoscale*. 2022 Jan 1;14.0(13.0):5227-5238. doi: 10.1039/d1nr08544a. PMID: 35315848.

16. Oh SS, Kim Y, Lee YB, Bae SK, Kim JS, An S, Choi J. Optical Modalities for Research, Diagnosis, and Treatment of Stroke and the Consequent Brain Injuries. *Applied Sciences*. 2022 Feb 11;12.0(4.0):1891. doi: 10.3390/app12041891.

17. Gao X, Wang Y, Hou H, Lyu Y, Wang H, Yao L, Zhang J, Cao F, Wang Y. In vivo bioluminescence imaging of hyperglycemia exacerbating stem cells on choroidal neovascularization in mice. *International Journal of Ophthalmology*. 2016 Apr 18;9.0(4.0):519-27. doi: 10.18240/ijo.2016.04.07. PMID: 27162722.

18. Moradpour S, Aliaghaei A, Bigdeli M. Effect of Sertoli Cell Transplant and Rapamycin Pretreatment on Middle Cerebral Artery Occlusion-Induced Brain Ischemia in a Rat Model. *Experimental and Clinical Transplantation*. 2021 Nov 1;19.0(11.0):1204-1211. doi: 10.6002/ect.2021.0198. PMID: 34812711.

19. Aswendt M, Collmann FM, Hoehn M. Neurobiological insights from bioluminescence imaging. *Oncotarget*. 2017 Aug 13;8.0(41.0):69198-69199. doi: 10.18632/oncotarget.20302. PMID: 29050195.

20. Cao J, Zhou M. In Vivo Fluorescence Reflectance Imaging with Subcutaneous Mouse Tumor Models. *Methods in molecular biology*. 2016 Jan 1;1444.0:27-35. doi: 10.1007/978-1-4939-3721-9_3. PMID: 27283414.

21. Boitet M, Grailhe R. Reporter Expression and Tissue Depth Quantification Using Bright and Broad-Range Spectrum Bioluminescence Probes. *Biosensors Journal*. 2017 Jan 1;6.0(2.0). doi: 10.4172/2090-4967.1000148.

22. Nowak B, Andrzejewska A, Rogujski P, Zawadzki M, Walczak P, Dorobek M, Łukomska B, Janowski M. Imaging of Stem Cell Therapy for Stroke and Beyond. *Springer eBooks*. 2022 Jan 1;:137-156. doi: 10.1007/978-981-16-8562-0_6.

23. Doeppner TR, Hermann DM. Editorial: Stem cells and progenitor cells in ischemic stroke-fashion or future?. *Frontiers in Cellular Neuroscience*. 2015 Aug 25;9.0:334. doi: 10.3389/fncel.2015.00334. PMID: 26379504.

24. Chandrasekharan P, Yang C. Nanobiomaterials for Molecular Imaging. 2017 Dec 1;:259-279. doi: 10.1002/9783527698646.ch11.

25. Aswendt M, Boehm-Sturm P, Hoehn M. Chapter 10 Neural Stem Cells. *Pan Stanford Publishing eBooks*. 2016 Oct 21;:283-310. doi: 10.1201/9781315364605-11.

26. Han H, Zhang W, Yuan L, Yan J, Wang W, Shi C, Li Y. Imaging and Tracking Transplanted Bone Marrow Stem Cells After Stroke. *Springer eBooks*. 2016 Dec 14;:311-332. doi: 10.1007/978-981-10-2929-5_14.

27. Thompson IJ. Manipulation of antigen presenting cells to enhance host immune responses to bacterial infection. *Imperial College Research Computing Service Data Repository*. 2015 Jun 1;. doi: 10.25560/40916.

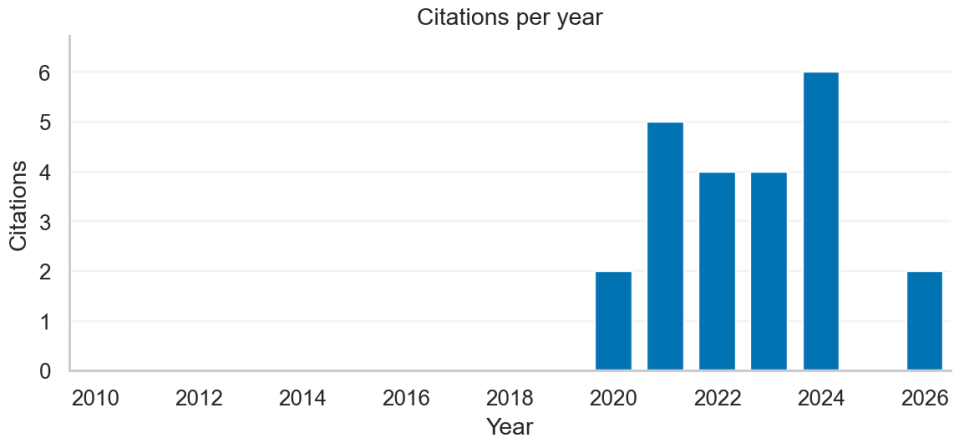
28. Kaur A. In Cellulo Studies. *Springer theses*. 2018 Jan 1;:129-147. doi: 10.1007/978-3-319-73405-7_6.

29. Lewis MA. Optical Imaging. *Imaging in medical diagnosis and therapy*. 2016 Feb 4;:141-163. doi: 10.1201/b19052-14.

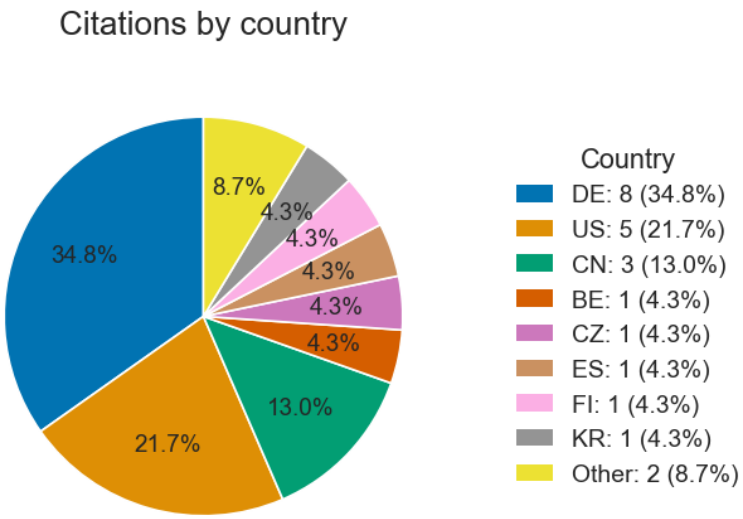
30. Lin C, Liu T, Reyes CJQ. In Vivo Fluorescence Imaging. *Encyclopedia of Analytical Chemistry*. 2022 Sep 26;:1-29. doi: 10.1002/9780470027318.a9777.

Publication

Pallast N, Wieters F, Nill M, Fink GR, Aswendt M. Graph theoretical quantification of white matter reorganization after cortical stroke in mice. *NeuroImage*. 2020 May 4;217:116873. doi: 10.1016/j.neuroimage.2020.116873. PMID: 32380139.



Citation summary
Total citations: **23**
Average/year: **3.29**
Period: 2020-2026
SCImago cites/doc 2y (2025): **5.31**
SJR 2025: **1.93** (Q1)



Cited by

1. Stokowska A, Aswendt M, Žucha D, Lohmann S, Wieters F, Suarez JM, Atkins AL, Li Y, Miteva M, Lewin J, Wiedermann D, Diedenhofen M, Naluai ÅT, Abaffy P, Valihrach L, Kubista M, Hoehn M, Pekny M, Pekna M. Complement C3a treatment accelerates recovery after stroke via modulation of astrocyte reactivity and cortical connectivity. *Journal of Clinical Investigation*. 2023 Mar 30;133(10.0). doi: 10.1172/jci162253. PMID: 36995772.

2. Latifi S, Carmichael ST. The emergence of multiscale connectomics-based approaches in stroke recovery. *Trends in Neurosciences*. 2024 Feb 23;47(4.0):303-318. doi: 10.1016/j.tins.2024.01.003. PMID: 38402008.

3. White JD, Arefin TM, Pugliese A, Lee CH, Gassen J, Zhang J, Kaffman A. Early life stress causes sex-specific changes in adult fronto-limbic connectivity that differentially drive learning. *eLife*. 2020 Dec 1;9. doi: 10.7554/elife.58301. PMID: 33259286.

4. Aswendt M, Green C, Sadler R, Llovera G, Dzikowski L, Heindl S, de Agüero MG, Diedenhofen M, Vogel SN, Wieters F, Wiedermann D, Liesz A, Hoehn M. The gut microbiota modulates brain network connectivity under physiological conditions and after acute brain ischemia. *iScience*. 2021 Sep 9;24(10.0):103095. doi: 10.1016/j.isci.2021.103095. PMID: 34622150.

5. Arefin TM, Lee C, White J, Zhang J, Kaffman A. Macroscopic Structural and Connectome Mapping of the Mouse Brain Using Diffusion Magnetic Resonance Imaging. *BIO-PROTOCOL*. 2021 Jan 1;11(22.0):e4221. doi: 10.21769/bioprotoc.4221. PMID: 34909442.

6. Islam R, White JD, Arefin TM, Mehta SS, Liu X, Polis B, Giuliano L, Ahmed S, Bowers CA, Zhang J, Kaffman A. Early adversity causes sex-specific deficits in perforant pathway connectivity and contextual memory in adolescent mice. *Biology of Sex Differences*. 2024 May 7;15(1.0):39. doi: 10.1186/s13293-024-00616-0. PMID: 38715106.

7. Li Z, Gao H, Zeng P, Jia Y, Kong X, Xu K, Bai R. Secondary Degeneration of White Matter After Focal Sensorimotor Cortical Ischemic Stroke in Rats. *Frontiers in Neuroscience*. 2021 Jan 18;14:611696. doi: 10.3389/fnins.2020.611696. PMID: 33536869.

8. Koukalova L, Chmelová M, Amlerova Z, Vargová L. Out of the core: the impact of focal ischemia in regions beyond the penumbra. *Frontiers in Cellular Neuroscience*. 2024 Mar 5;18:1336886. doi: 10.3389/fncel.2024.1336886. PMID: 38504666.

9. Wang L, Yang Z, Fudo S, Prasanna X, Yan Z, Vihinen H, Chen Y, Zhao Y, He X, Bu Q, Li H, Zhao Y, Jiang L, Qin F, Dai Y, Zhang N, Qin M, Kuang W, Zhao Y, Jokitalo E, Vattulainen I, Kajander T, Zhao H, Cen X. Membrane remodeling by FAM92A1 during brain development regulates neuronal morphology, synaptic function, and cognition. *Nature Communications*. 2024 Jul 23;15(1.0):6209. doi: 10.1038/s41467-024-50565-w. PMID: 39043703.

10. Scharwächter L, Schmitt FJ, Pallast N, Fink GR, Aswendt M. Network analysis of neuroimaging in mice. *NeuroImage*. 2022 Mar 17;253:119110. doi: 10.1016/j.neuroimage.2022.119110. PMID: 35311664.

11. Boboc IKŞ, Rotaru-Zăvăleanu AD, Călina DC, Albu CV, Cătălin B, Turcu-Ştiolică A. A Preclinical Systematic Review and Meta-Analysis of Behavior Testing in Mice Models of Ischemic Stroke. *Life*. 2023 Feb 17;13(2.0):567. doi: 10.3390/life13020567. PMID: 36836924.

12. Van der Linden A, Hoehn M. Monitoring Neuronal Network Disturbances of Brain Diseases: A Preclinical MRI Approach in the Rodent Brain. *Frontiers in Cellular Neuroscience*. 2022 Jan 3;15:815552. doi: 10.3389/fncel.2021.815552. PMID: 35046778.

13. Medaglia JD, Erickson BA, Pustina D, Kelkar AS, DeMarco AT, Dickens JV, Turkeltaub PE. Simulated Attack Reveals How Lesions Affect Network Properties in Poststroke Aphasia. *Journal of Neuroscience*. 2022 May 11;42(24.0):4913-4926. doi: 10.1523/jneurosci.1163-21.2022. PMID: 35545436.

14. Götz J, Wieters F, Fritz VJ, Käsgen O, Kalantari A, Fink GR, Aswendt M. Temporal and Spatial Gene Expression Profile of Stroke Recovery Genes in Mice. *Genes*. 2023 Feb 9;14(2.0):454. doi: 10.3390/genes14020454. PMID: 36833381.

15. Chen M, Wang Y, Li Z. Disrupted white matter structural networks in patients with acute ischemic stroke in the right basal ganglia. *Neuroscience*. 2024 Sep 26;568:68-75. doi: 10.1016/j.neuroscience.2024.08.003. PMID: 39341271.

16. Mahani FSN, Kalantari A, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic structural connectivity changes in cortical and cortico-striatal strokes in mice. *Neural Regeneration Research*. 2026 Feb 28;21(10.0):4945-4953. doi: 10.4103/nrr.nrr-d-25-00491. PMID: 41778731.

17. Aswendt M, Wieters F. Structural integrity and remodeling underlying functional recovery after stroke. *Neural Regeneration Research*. 2020 Dec 17;16(7.0):1423. doi: 10.4103/1673-5374.301004. PMID: 33318437.

18. Villa-González M, Pérez-Álvarez MJ. A 3R-Tau-mediated mechanism in oligodendrocytes: could it be the key for neuroprotection after stroke?. *Neural Regeneration Research*. 2021 Apr 23;16(12.0):2401-2402. doi: 10.4103/1673-5374.313027. PMID: 33907017.

19. Schlemm E, Jensen M, Kuceyeski AF, Jamison KW, Ingwersen T, Mayer C, Königsberg A, Boutitie F, Ebinger M, Endres M, Fiebach JB, Fiehler J, Galinović I, Lemmens R, Muir KW, Nighoghossian N, Pedraza S, Puig J, Simonsen CZ, Thijs V, Wouters A, Gerloff CP, Thomalla G, Cheng B. Early effect of thrombolysis on structural brain network organisation after anterior-circulation stroke in the randomized WAKE-UP trial. *Human Brain Mapping*. 2022 Sep 14;43(16.0):5053-5065. doi: 10.1002/hbm.26073. PMID: 36102287.

20. Mahani FSN, Kalantari A, Fink GR, Hoehn M, Aswendt M. A systematic review of the relationship between magnetic resonance imaging based resting-state and structural networks in the rodent brain. *Frontiers in Neuroscience*. 2023 Jul 24;17:1194630. doi: 10.3389/fnins.2023.1194630. PMID: 37554291.

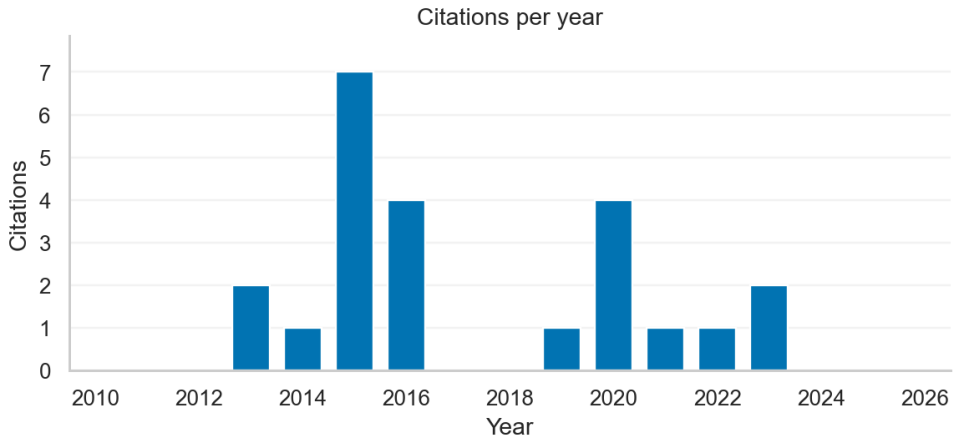
21. Niu B, Wu H, Li Y, Klugah-Brown B, Hanna GP, Yao Y, Jing J, Baig TI, Xia Y, Yao D, Biswal BB. Topological functional network analysis of cortical blood flow in hyperacute ischemic rats. *Brain Structure and Function*. 2024 Dec 26;230(1.0):20. doi: 10.1007/s00429-024-02864-7. PMID: 39724244.

22. Mahani F, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic Functional Connectivity Changes in Cortical and Cortico-Striatal Strokes in Mice. *CNS Neuroscience & Therapeutics*. 2026 Aug 1;32(8.0):e71114. doi: 10.1002/cns.71114. PMID: 42661330.

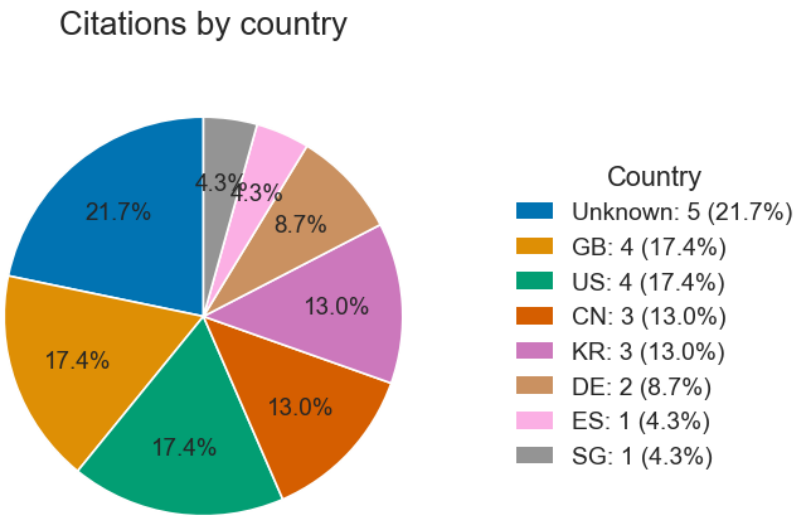
23. Im S, Suh J, Shim J, Baek H. Deterministic Tractography Analysis of Rat Brain Using SIGMA Atlas in 9.4T MRI. *Brain Sciences*. 2021 Dec 18;11(12.0):1656. doi: 10.3390/brainsci11121656. PMID: 34942958.

Publication

Tennstaedt A, Aswendt M, Adamczak JM, Hoehn M. Noninvasive Multimodal Imaging of Stem Cell Transplants in the Brain Using Bioluminescence Imaging and Magnetic Resonance Imaging. *Methods in molecular biology*. 2013 Jan 1;1052:153-166. doi: 10.1007/7651_2013_14. PMID: 23733537.



Citation summary
Total citations: **23**
Average/year: **1.64**
Period: 2013-2026
SCImago cites/doc 2y (2025): **1.07**
SJR 2025: **0.423** (Q4)



Cited by

1. Barrow M, Taylor A, Murray PA, Rosseinsky MJ, Adams DJ. Design considerations for the synthesis of polymer coated iron oxide nanoparticles for stem cell labelling and tracking using MRI. *Chemical Society Reviews*. 2015 Jan 1;44.0(19.0):6733-6748. doi: 10.1039/c5cs00331h. PMID: 26169237.

2. Tung JK, Berglund K, Gutekunst C, Hochgeschwender U, Gross RE. Bioluminescence imaging in live cells and animals. *Neurophotonics*. 2016 Apr 5;3.0(2.0):1-6. doi: 10.1117/1.nph.3.2.025001. PMID: 27226972.

3. Skidmore S, Barker RA. Challenges in the clinical advancement of cell therapies for Parkinson's disease. *Nature Biomedical Engineering*. 2023 Jan 12;7.0(4.0):370-386. doi: 10.1038/s41551-022-00987-y. PMID: 36635420.

4. Kang S, Lee S, Na JH, Yoon HI, Lee D, Koo H, Cho YW, Kim SH, Jeong SY, Kwon IC, Choi K, Kim K. Cell Labeling and Tracking Method without Distorted Signals by Phagocytosis of Macrophages. *Theranostics*. 2014 Jan 1;4.0(4.0):420-431. doi: 10.7150/thno.7265. PMID: 24578725.

5. Kim SM, Jeong CH, Woo JS, Ryu CH, Lee JH, Jeun S. In vivo near-infrared imaging for the tracking of systemically delivered mesenchymal stem cells: tropism for brain tumors and biodistribution. *International Journal of Nanomedicine*. 2015 Dec 1;11.0:13. doi: 10.2147/ijn.s97073. PMID: 26719691.

6. Aswendt M, Vogel S, Schäfer C, Jathoul AP, Pulé M, Hoehn M. Quantitative in vivo dual-color bioluminescence imaging in the mouse brain. *Neurophotonics*. 2019 May 7;6.0(2.0):1-11. doi: 10.1117/1.nph.6.2.025006. PMID: 31093514.

7. Jang SE, Qiu L, Chan LL, Tan E, Zeng L. Current Status of Stem Cell-Derived Therapies for Parkinson's Disease: From Cell Assessment and Imaging Modalities to Clinical Trials. *Frontiers in Neuroscience*. 2020 Oct 16;14.0:558532. doi: 10.3389/fnins.2020.558532. PMID: 33177975.

8. Barrow M, Taylor A, Carrión JG, Mandal P, Park BK, Poptani H, Murray PA, Rosseinsky MJ, Adams DJ. Co-precipitation of DEAE-dextran coated SPIONs: how synthesis conditions affect particle properties, stem cell labelling and MR contrast. *Contrast Media & Molecular Imaging*. 2016 Jun 30;11.0(5.0):362-370. doi: 10.1002/cmmi.1700. PMID: 27358113.

9. Chehade M, Srivastava AK, Bulte JWM. Co-Registration of Bioluminescence Tomography, Computed Tomography, and Magnetic Resonance Imaging for Multimodal In Vivo Stem Cell Tracking. *Tomography*. 2016 Jun 1;2.0(2.0):158-165. doi: 10.18383/j.tom.2016.00160. PMID: 27478872.

10. Fan Z, Cai H, Ge Z, Wang L, Zhang X, Li L, Zhai X. The Efficacy and Safety of Granulocyte Colony-Stimulating Factor for Patients with Stroke. *Journal of Stroke and Cerebrovascular Diseases*. 2015 Aug 1;24.0(8.0):1701-1708. doi: 10.1016/j.jstrokecerebrovasdis.2014.11.033. PMID: 26004861.

11. Raikwar SP, Thangavel R, Ahmed ME, Selvakumar GP, Kempuraj D, Wu K, Khan O, Bazley K, Bussinger B, Kukulka K, Zaheer SA, Iyer SS, Govindarajan R, Burton C, James D, Zaheer A. Real-Time Noninvasive Bioluminescence, Ultrasound and Photoacoustic Imaging in NFκB-RE-Luc Transgenic Mice Reveal Glia Maturation Factor-Mediated Immediate and Sustained Spatio-Temporal Activation of NFκB Signaling Post-Traumatic Brain Injury in a Gender-Specific Manner. *Cellular and Molecular Neurobiology*. 2020 Aug 12;41.0(8.0):1687-1706. doi: 10.1007/s10571-020-00937-9. PMID: 32785863.

12. Nguyen VP, Zhe J, Hu J, Ahmed U, Paulus YM. Molecular and cellular imaging of the eye. *Biomedical Optics Express*. 2023 Dec 11;15.0(1.0):360. doi: 10.1364/boe.502350. PMID: 38223186.

13. Ha S, Ahn S, Kim S, Joo Y, Chong YH, Suh Y, Chang K. In vivo imaging of human adipose-derived stem cells in Alzheimer's disease animal model. *Journal of Biomedical Optics*. 2013 Dec 2;19.0(5.0):1-5. doi: 10.1117/1.jbo.19.5.051206. PMID: 24297061.

14. Aswendt M, Henn N, Michalk S, Schneider G, Steiner M, Bissa U, Dose C, Hoehn M. Novel Bimodal Iron Oxide Particles for Efficient Tracking of Human Neural Stem Cells in Vivo. *Nanomedicine*. 2015 Aug 1;10.0(16.0):2499-2512. doi: 10.2217/nnm.15.94. PMID: 26296195.

15. Xia Y, Bao H, Huang J, Li X, Yu C, Zhang Z, Wang H. Near-infrared-persistent luminescence/bioluminescence imaging tracking of transplanted mesenchymal stem cells in pulmonary fibrosis. *Biomaterials Science*. 2020 Jan 1;8.0(11.0):3095-3105. doi: 10.1039/d0bm00063a. PMID: 32347847.

16. Imaging and Tracking Stem Cells. *Methods in molecular biology*. 2013 Jan 1;. doi: 10.1007/978-1-62703-559-0.

17. Martín A, Ramos-Cabrer P, Planas AM. Noninvasive Brain Imaging in Small Animal Stroke Models: MRI, PET, and SPECT. *Neuromethods*. 2016 Jan 1;:147-186. doi: 10.1007/978-1-4939-5620-3_11.

18. Meleshina AV, Cherkasova EI, Shirmanova MV, Khrapichev AA, Dudenkova VV, Zagaynova EV. Modern Techniques for Stem Cells in vivo Imaging (Review). *Современные технологии в медицине*. 2015 Dec 1;7.0(4.0):174-188. doi: 10.17691/stm2015.7.4.24.

19. Mousavinejad M, Skidmore S, Barone FG, Tyers P, Pisupati V, Poptani H, Plagge A, Barker R, Murray PA, Taylor A, Hill CJ. Assessing Human Embryonic Stem Cell-Derived Dopaminergic Neuron Progenitor Transplants Using Non-invasive Imaging Techniques. *Molecular Imaging and Biology*. 2020 May 6;22.0(5.0):1244-1254. doi: 10.1007/s11307-020-01499-4. PMID: 32378000.

20. Savitz SI, Parsha K. Enhancing Stroke Recovery with Cellular Therapies. *Stroke*. 2015 Aug 27;:981-991. doi: 10.1016/b978-0-323-29544-4.00060-8.

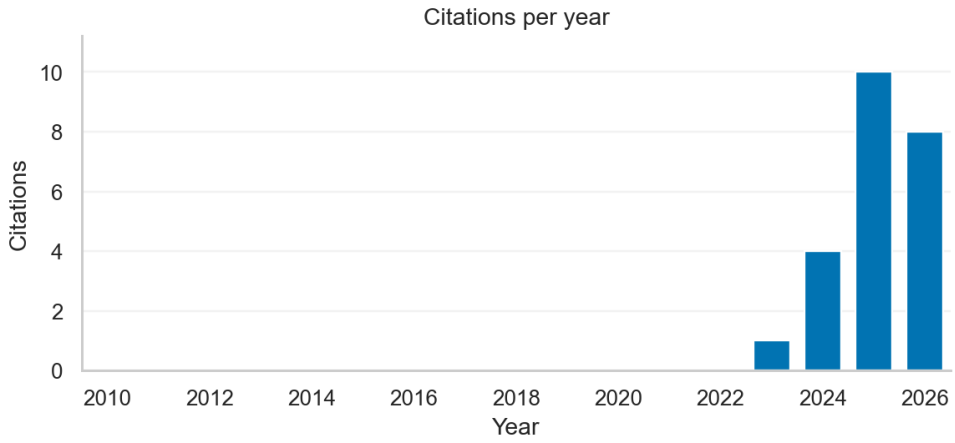
21. Li X, Yu C, Bao H, Chen Z, Liu X, Huang J, Zhang Z. CT/bioluminescence dual-modal imaging tracking of stem cells labeled with Au@PEI@PEG nanotracers and RfLuc in nintedanib-assisted pulmonary fibrosis therapy. *Nanomedicine Nanotechnology Biology and Medicine*. 2022 Jan 13;41.0:102517. doi: 10.1016/j.nano.2022.102517. PMID: 35032629.

22. Square P, Novgorod N, Federation R, State L. Modern Techniques for Stem Cells in vivo Imaging (Review). *CyberLeninK (CyberLeninka)*. 2015 Jan 1;.

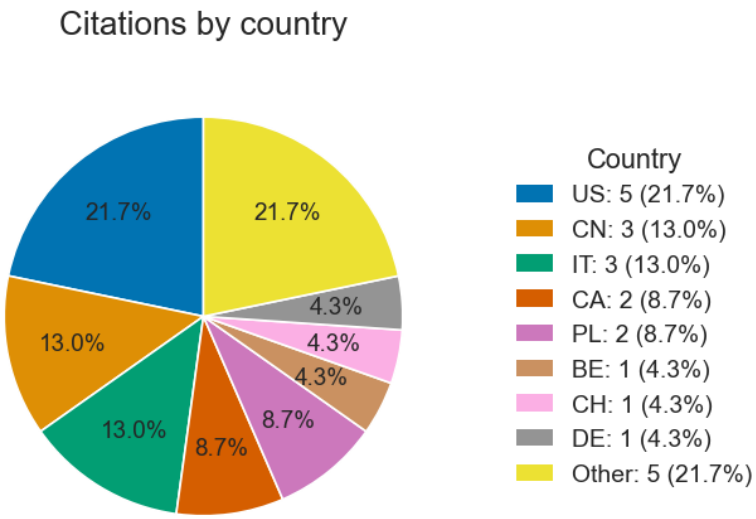
23. Satani N, Parsha K, Savitz SI. Enhancing Stroke Recovery With Cellular Therapies. *Stroke*. 2021 Apr 24;:900-911.e5. doi: 10.1016/b978-0-323-69424-7.00062-4.

Publication

Hoehn M, Aswendt M. Functional hyperconnectivity related to brain disease: maladaptive process or element of resilience?. *Neural Regeneration Research*. 2022 Nov 18;18(7):1489. doi: 10.4103/1673-5374.361541. PMID: 36571347.



Citation summary
Total citations: **23**
Average/year: **4.60**
Period: 2022-2026
SCImago cites/doc 2y (2025): **5.98**
SJR 2025: **1.43** (Q1)



Cited by

1. Arrigoni A, Previtali M, Bosticardo S, Pezzetti G, Poloni S, Capelli S, Napolitano A, Remuzzi A, Zangari R, Lorini FL, Sessa M, Daducci A, Caroli A, Gerevini S. Brain microstructure and connectivity in COVID-19 patients with olfactory or cognitive impairment. *NeuroImage Clinical*. 2024 Jan 1;43.0:103631. doi: 10.1016/j.nicl.2024.103631. PMID: 38878591.

2. Maas DA, Douw L. Multiscale network neuroscience in neuro-oncology: How tumors, brain networks, and behavior connect across scales. *Neuro-Oncology Practice*. 2023 Aug 22;10.0(6.0):506-517. doi: 10.1093/nop/npad044. PMID: 38026586.

3. Dark HE, Shafer AT, Cordon J, An Y, Lewis A, Moghekar A, Landman BA, Resnick SM, Walker KA. Association of Plasma Biomarkers of Alzheimer Disease and Neurodegeneration With Longitudinal Intra-Network Functional Brain Connectivity. *Neurology*. 2025 Jan 31;104.0(4.0):e210271. doi: 10.1212/wnl.0000000000210271. PMID: 39889254.

4. Pini L, Imbimbo BP, Griffa A, Allali G, Pomara N. Targeting brain connectivity in Alzheimer's disease with repurposed drugs. *Ageing Research Reviews*. 2025 Dec 19;114.0:103001. doi: 10.1016/j.arr.2025.103001. PMID: 41421724.

5. Zhang P, Zhou Y, Ni H, Huang Z, Tang C, Zhuge Q, Dong L, Zhang J. Altered functional connectivity of brainstem ARAS nuclei unveils the mechanisms of disorders of consciousness in sTBI: an exploratory study. *NeuroImage Clinical*. 2025 Jan 1;46.0:103787. doi: 10.1016/j.nicl.2025.103787. PMID: 40262479.

6. Chrobak AA, Bielak S, Nowaczek D, Żyrkowska A, Sobczak AM, Fąfrowicz M, Bryll A, Marek T, Dudek D, Siwek M. Divergent pattern of functional connectivity within the dorsal attention network differentiates schizophrenia and bipolar disorder patients. *Frontiers in Psychiatry*. 2024 Sep 19;15.0:1474313. doi: 10.3389/fpsyt.2024.1474313. PMID: 39364382.

7. Tremblay C, Pastor-Bernier A, Rheault F, Daneault V, Ayral V, Filiatrault M, Desaulniers L, Vo A, Gagnon J, Postuma RB, Dušek P, Mareček S, Varga Z, Klein JC, Hu MT, Lehéricy S, Arnulf I, Dodet P, Vidailhet M, Corvol J, Group FTIS, Rahayel S. Altered network efficiency in isolated REM sleep behavior disorder: A multicentric study. *Alzheimer s & Dementia*. 2025 Aug 1;21.0(8.0):e70574. doi: 10.1002/alz.70574. PMID: 40791089.

8. Malan NS, Baillet S, Tankha P, Roy M, Gopalakrishnan R. Altered cortical alpha modulations and connectivity in complex regional pain syndrome. *Journal of Pain*. 2025 Nov 8;39.0:105595. doi: 10.1016/j.jpain.2025.105595. PMID: 41207412.

9. Fiskum V, Winter-Hjelm N, Christiansen N, Sandvig A, Sandvig I. Microscale dysfunction and mesoscale compensation in degenerating neuronal networks. *Network Neuroscience*. 2026 Jan 1;10.0(3.0):594-612. doi: 10.1162/netn.a.552. PMID: 42529618.

10. Goldbeck SM, Costa DVDS, Yang SE, Whitt CC, Tora AE, Warren CA, Shin JH. Clostridioides difficile Infection in Aged Mice Decreases Memory Function, Which Can Be Protected with Alanyl-Glutamine Supplementation. *Journal of Nutrition*. 2025 Apr 11;155.0(6.0):1700-1709. doi: 10.1016/j.tjnut.2025.03.032. PMID: 40222582.

11. Martínez YE, Widmer M, Zimmermann J, Schönhammer JG, Luft AR, Jäncke L. Brain reorganization: altered functional connectivity in reward network after stroke. *NeuroImage Clinical*. 2025 Jan 1;48.0:103914. doi: 10.1016/j.nicl.2025.103914. PMID: 41289980.

12. Krukow P, Kopiś-Posiej N, Pablo VG, Rodríguez-González V, Gómez C, Poza J. Altered task-related brain network dynamics and performance consistency in a non-clinical group burdened with childhood trauma. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*. 2025 Dec 1;143.0:111571. doi: 10.1016/j.pnpbp.2025.111571. PMID: 41349875.

13. Mahani F, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic Functional Connectivity Changes in Cortical and Cortico-Striatal Strokes in Mice. *CNS Neuroscience & Therapeutics*. 2026 Aug 1;32.0(8.0):e71114. doi: 10.1002/cns.71114. PMID: 42661330.

14. Garmendia J, Labayru G, Jiménez-Marín A, Villanúa J, Cortés JM, de Munaín AL, Sistiaga A. Hyperconnectivity in resting-state fMRI as a marker of disease severity in Myotonic Dystrophy Type 1. *Journal of Neuromuscular Diseases*. 2025 Jan 1;12.0(1.0):90-101. doi: 10.1177/22143602241307197. PMID: 39973452.

15. Sastre-Yagüe D, Malerba SB, Rocchi F, Gini S, Mancini G, Stuefer A, Coletta L, Noei S, Markicevic M, Alvino FG, Zerbi V, Galbusera A, Mariani JRP, Panzeri S, Gozzi A. Cortical excitability inversely modulates fMRI connectivity via low-frequency neuronal coupling. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Mar 14;. doi: 10.64898/2026.03.12.710517.

16. Chen J, Liu T, Shi H. End-stage renal disease accompanied by mild cognitive impairment: A study and analysis of trimodal brain network fusion. *PLoS ONE*. 2024 Jun 13;19.0(6.0):e0305079. doi: 10.1371/journal.pone.0305079. PMID: 38870175.

17. Griffith O, Fornini R, Walter AE, Wilkes J, Bai X, Slobounov S. Comorbidity of concussion and depression alters brain functional connectivity in collegiate student-athletes. *Brain Research*. 2024 Aug 26;1845.0:149200. doi: 10.1016/j.brainres.2024.149200. PMID: 39197571.

18. Tremblay C, Pastor-Bernier A, Rheault F, Daneault V, Ayral V, Filiatrault M, Desaulniers L, Vo A, Gagnon J, Postuma RB, Dušek P, Mareček S, Varga Z, Klein JC, Hu MT, Lehéricy S, Arnulf I, Dodet P, Vidailhet M, Corvol J, Group IS, Rahayel S. Altered Network Efficiency in Isolated REM Sleep Behaviour Disorder: A Multicentric Study. *medRxiv*. 2025 Jul 10;. doi: 10.1101/2025.07.08.25331130.

19. Yang F, Liao W, Duan Y. Alterations of white matter connectivity in alopecia areata patients. *Brain Imaging and Behavior*. 2026 Mar 6;20.0(2.0). doi: 10.1007/s11682-026-01123-z. PMID: 41787052.

20. Spruyt L, Mlinarič T, Dusart N, Reinartz M, Van Hulle MM, Sunaert SG, Van Laere K, Dupont PJ, Vandenberghe RR. The electrophysiological basis of resting-state fMRI hyperconnectivity in early Alzheimer's disease. *Alzheimer s Research & Therapy*. 2026 Apr 6;18.0(1.0). doi: 10.1186/s13195-026-02003-w. PMID: 41943113.

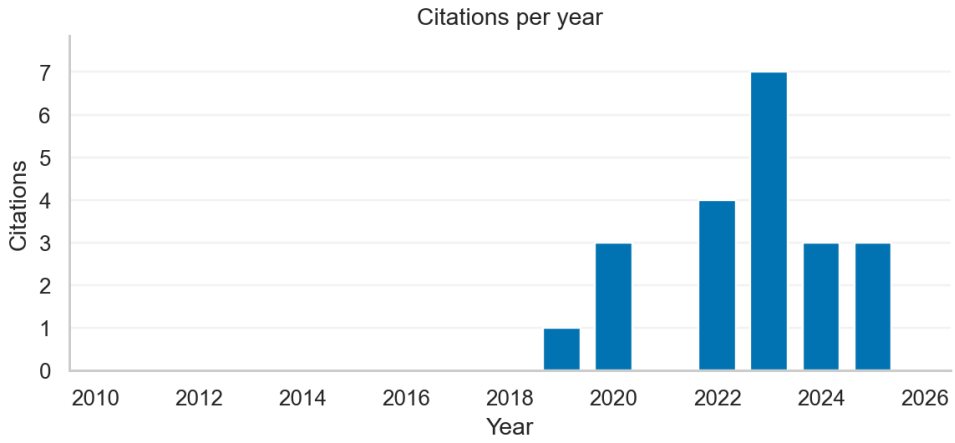
21. Rabut C, Shivaei S, Heiles B, Daghlian G, Shapiro MG. Trimodal ultrasound imaging of brain-tumor interaction. *iScience*. 2026 Sep 23;29.0(10.0):117572. doi: 10.1016/j.isci.2026.117572. PMID: 42828291.

22. Kim D, Kim S, Kim S, Kim E, Kim S, Kim S, Joo EY. Distinct thalamostriatal functional network dysregulation in comorbid insomnia and sleep apnea. *SLEEP*. 2026 Sep 25;. doi: 10.1093/sleep/zsag252. PMID: 42789003.

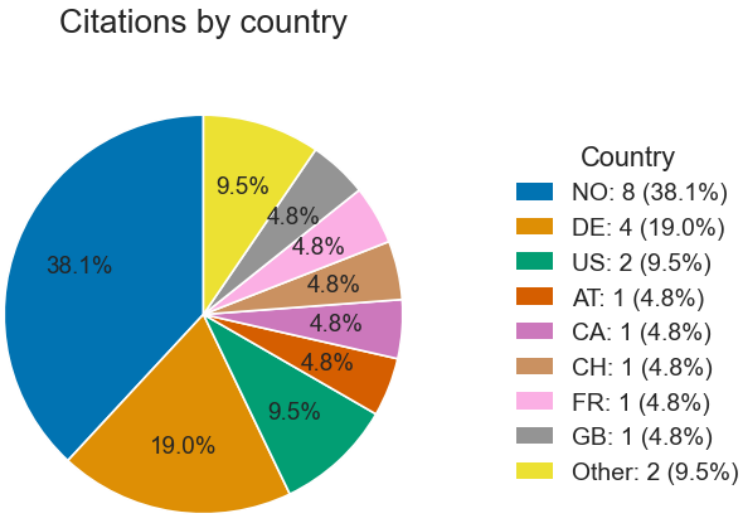
23. Pickering M, Morel V, Pereira B, Pickering GP, Chapurlat RD, Tournadre A, Chassain C, Sontheimer A. Osteocalcin and impaired cerebral resting-state functional connectivity in postmenopausal osteoporosis: a pilot case-control study. *Postgraduate Medicine*. 2026 Oct 14;:1-9. doi: 10.1080/00325481.2026.2739000. PMID: 42870572.

Publication

Pallast N, Wieters F, Fink GR, Aswendt M. Atlas-based imaging data analysis tool for quantitative mouse brain histology (AIDAhisto). *Journal of Neuroscience Methods*. 2019 Aug 12;326:108394. doi: 10.1016/j.jneumeth.2019.108394. PMID: 31415844.



Citation summary
Total citations: **21**
Average/year: **2.62**
Period: 2019-2026
SCImago cites/doc 2y (2025): **2.93**
SJR 2025: **0.788** (Q2)



Cited by

1. Yates SC, Groeneboom NE, Coello CS, Lichtenthaler SF, Kuhn P, Demuth H, Hartlage-Rübsamen M, Roßner S, Leergaard TB, Kreshuk A, Puchades MA, Bjaalie JG. QUINT: Workflow for Quantification and Spatial Analysis of Features in Histological Images From Rodent Brain. *Frontiers in Neuroinformatics*. 2019 Dec 2;13.0:75. doi: 10.3389/fninf.2019.00075. PMID: 31849633.

2. Kleven H, Bjerke IE, Clascá F, Groenewegen HJ, Bjaalie JG, Leergaard TB. Waxholm Space atlas of the rat brain: a 3D atlas supporting data analysis and integration. *Nature Methods*. 2023 Oct 2;20.0(11.0):1822-1829. doi: 10.1038/s41592-023-02034-3. PMID: 37783883.

3. Groeneboom NE, Yates SC, Puchades MA, Bjaalie JG. Nutil: A Pre- and Post-processing Toolbox for Histological Rodent Brain Section Images. *Frontiers in Neuroinformatics*. 2020 Aug 21;14.0:37. doi: 10.3389/fninf.2020.00037. PMID: 32973479.

4. Aswendt M, Pallast N, Wieters F, Baues M, Hoehn M, Fink GR. Lesion Size- and Location-Dependent Recruitment of Contralesional Thalamus and Motor Cortex Facilitates Recovery after Stroke in Mice. *Translational Stroke Research*. 2020 Mar 12;12.0(1.0):87-97. doi: 10.1007/s12975-020-00802-3. PMID: 32166716.

5. Pallast N, Wieters F, Nill M, Fink GR, Aswendt M. Graph theoretical quantification of white matter reorganization after cortical stroke in mice. *NeuroImage*. 2020 May 4;217.0:116873. doi: 10.1016/j.neuroimage.2020.116873. PMID: 32380139.

6. Lauridsen K, Ly A, Prévost ED, McNulty CJ, McGovern DJ, Tay JW, Dragavon JM, Root DH. A Semi-Automated Workflow for Brain Slice Histology Alignment, Registration, and Cell Quantification (SHARCQ). *eNeuro*. 2022 Mar 1;9.0(2.0):ENEURO.0483-21.2022. doi: 10.1523/eneuro.0483-21.2022. PMID: 35396257.

7. Sadeghi M, Ramos-Prats A, Neto PC, Castaldi F, Crowley D, Matulewicz P, Paradiso E, Freysinger W, Ferraguti F, Goebel G. Localization and Registration of 2D Histological Mouse Brain Images in 3D Atlas Space. *Neuroinformatics*. 2023 Jun 25;21.0(3.0):615-630. doi: 10.1007/s12021-023-09632-8. PMID: 37357231.

8. Puchades MA, Yates SC, Csucs G, Carey H, Balkir A, Leergaard TB, Bjaalie JG. Software and pipelines for registration and analyses of rodent brain image data in reference atlas space. *Frontiers in Neuroinformatics*. 2025 Sep 24;19.0:1629388. doi: 10.3389/fninf.2025.1629388. PMID: 41069395.

9. Bokiniec P, Whitmire CJ, Leva TM, Poulet JF. Brain-wide connectivity map of mouse thermosensory cortices. *Cerebral Cortex*. 2022 Sep 15;33.0(8.0):4870-4885. doi: 10.1093/cercor/bhac386. PMID: 36255325.

10. Bjerke IE, Yates SC, Carey H, Bjaalie JG, Leergaard TB. Scaling up cell-counting efforts in neuroscience through semi-automated methods. *iScience*. 2023 Aug 7;26.0(9.0):107562. doi: 10.1016/j.isci.2023.107562. PMID: 37636060.

11. Vercio LL, Green RM, Robertson S, Guo S, Dauter A, Marchini M, Vidal-García M, Zhao X, Mahika A, Marcucio R, Hallgrímsson B, Forkert ND. Segmentation of Tissues and Proliferating Cells in Light-Sheet Microscopy Images of Mouse Embryos Using Convolutional Neural Networks. *IEEE Access*. 2022 Jan 1;10.0:105084-105100. doi: 10.1109/access.2022.3210542. PMID: 36660260.

12. Kleven H, Reiten I, Blixhavn CH, Schlegel U, Øvsthus M, Papp EA, Puchades MA, Bjaalie JG, Leergaard TB, Bjerke IE. A neuroscientist's guide to using murine brain atlases for efficient analysis and transparent reporting. *Frontiers in Neuroinformatics*. 2023 Mar 9;17.0:1154080. doi: 10.3389/fninf.2023.1154080. PMID: 36970659.

13. Ostrova IV, Babkina AS, Lyubomudrov MA, Grechko AV, Golubev AM. Photochemicallly Induced Thrombosis as a Model of Ischemic Stroke. *General Reanimatology*. 2023 Jun 24;19.0(3.0):54-65. doi: 10.15360/1813-9779-2023-3-54-65.

14. Piluso S, Souedet N, Jan C, Hérard A, Clouchoux C, Delzescaux T. giRAff: an automated atlas segmentation tool adapted to single histological slices. *Frontiers in Neuroscience*. 2024 Jan 11;17.0:1230814. doi: 10.3389/fnins.2023.1230814. PMID: 38274499.

15. Chie JAKC, Persohn SA, Miner EW, Burton CP, Salama P, Territo PR. Total Variation Based 2D Image Registration of Post-Mortem Mouse Brain Images. *IEEE International Symposium on Biomedical Imaging (ISBI)*. 2024 May 27;:1-4. doi: 10.1109/isbi56570.2024.10635196.

16. Müller B. On the cusp of x-ray tomographic mapping of the human brain and its 10^11 cells. *Desy Publications Database (Deutsches Elektronen-Synchrotron DESY)*. 2024 Nov 25;:501. doi: 10.1117/12.3032298.

17. Damigos G, Zacharaki EI, Zerva N, Pavlopoulos A, Chatzikyrkou K, Koumenti A, Moustakas K, Pantos C, Mourouzis I, Lourbopoulos A. Machine learning based analysis of stroke lesions on mouse tissue sections. *Journal of Cerebral Blood Flow & Metabolism*. 2022 Feb 25;42.0(8.0):1463-1477. doi: 10.1177/0271678x221083387. PMID: 35209753.

18. Kleven H, Gillespie T, Zehl L, Dickscheid T, Bjaalie JG, Martone ME, Leergaard TB. AtOM, an ontology model to standardize use of brain atlases in tools, workflows, and data infrastructures. *Scientific Data*. 2023 Jul 26;10.0(1.0):486. doi: 10.1038/s41597-023-02389-4. PMID: 37495585.

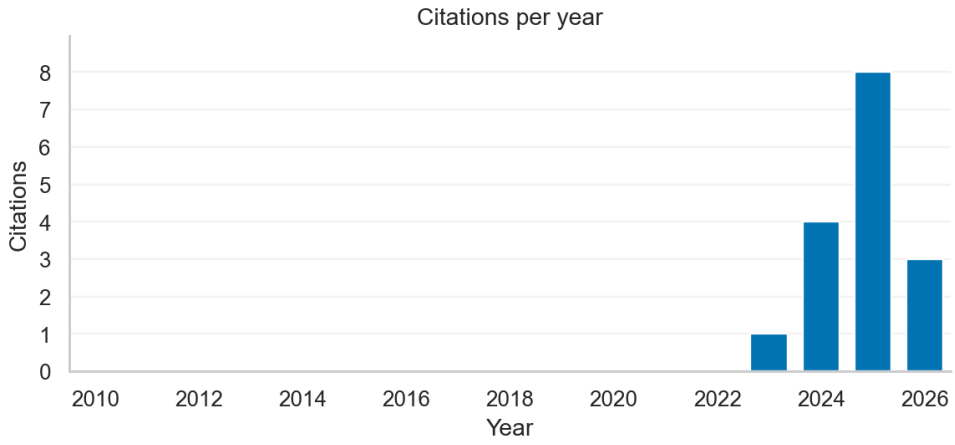
19. Kleven H, Gillespie T, Zehl L, Dickscheid T, Bjaalie JG, Martone ME, Leergaard TB. AtOM, an ontology model for standardizing use of brain atlases in tools, workflows, and data infrastructures. *bioRxiv (Cold Spring Harbor Laboratory)*. 2023 Jan 23;. doi: 10.1101/2023.01.22.525049.

20. Blaschke SJ, Backes H, Vlachakis S, Rautenberg N, Demir S, Wiedermann D, Aswendt M, Fink GR, Schroeter M, Rueger MA. Subacute cathodal transcranial direct current stimulation rescues secondary thalamic neurodegeneration after cortical stroke in mice. *Experimental Neurology*. 2025 Dec 23;398.0:115604. doi: 10.1016/j.expneurol.2025.115604. PMID: 41448520.

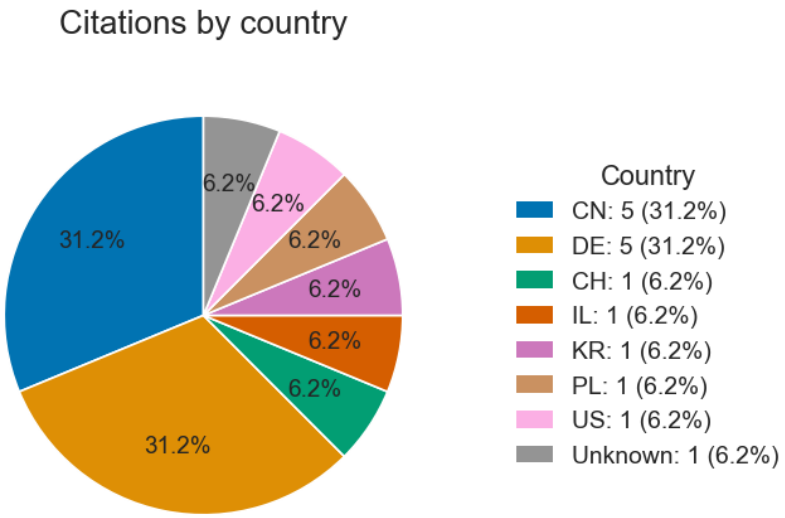
21. Năstase A, Yip PK, Uff CEG, O'Meara I, Barjat H. A computational workflow for microscopy-guided ion identification in clinical mass spectrometry imaging datasets. *bioRxiv (Cold Spring Harbor Laboratory)*. 2025 Dec 25;. doi: 10.64898/2025.12.25.694368.

Publication

Blaschke SJ, Vlachakis S, Pallast N, Walter HL, Volz LJ, Wiedermann D, Fink GR, Hoehn M, Aswendt M, Schroeter M, Rueger MA. Transcranial Direct Current Stimulation Reverses Stroke-Induced Network Alterations in Mice. *Stroke*. 2023 Jun 28;54(8):2145-2155. doi: 10.1161/strokeaha.123.042808. PMID: 37377015.



Citation summary
Total citations: **16**
Average/year: **4.00**
Period: 2023-2026
SCImago cites/doc 2y (2025): **6.75**
SJR 2025: **2.86** (Q1)



Cited by

1. Yan W, Lin Y, Chen Y, Wang Y, Wang J, Zhang M. Enhancing Neuroplasticity for Post-Stroke Motor Recovery: Mechanisms, Models, and Neurotechnology. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. 2025 Jan 1;33:1156-1168. doi: 10.1109/tnsre.2025.3551753. PMID: 40100694.

2. Antal A, Bjekić J, Ganho-Ávila A, Alekseichuk I, Assecondi S, Bergmann TO, Bikson M, Brunelin J, Brunoni AR, Charvet LE, Chen RE, Kadosh RC, Diedrich L, D'Urso G, Ferrucci R, Filipović SR, Fitzgerald PB, Flöel A, Fröhlich F, George MS, Hamilton RH, Haueisen J, Hallett M, Herrmann CS, Hummel FC, Jaberzadeh SJ, Langguth B, B ML, Lefaucheur J, Miniussi C, Moliadze V, Nikander M, Nikolin S, Nitsche MA, Opitz A, O'Shea J, Padberg F, Plewnia C, Priori A, Ramasawmy P, Razza LB, Rossi S, Rothwell JC, Rueger MA, Ruffini G, Sack AT, Salvador R, Schellhorn K, Schuhmann T, Shirota Y, Siebner HR, Thielscher A, Ugawa Y, Uusitalo S, Wexler A, Paulus W, Vanderhasselt M, Van Waes V, Wessel MJ, Wischnewski M, Baeken C, Ziemann U. Low intensity transcranial electric stimulation: Safety, ethical, legal regulatory and application guidelines (2017-2025: An update) - endorsed by the European Society for Brain Stimulation (ESBS) and by the International Federation for Clinical Neurophysiology (IFCN). *Clinical Neurophysiology*. 2025 Nov 23;184:2111436. doi: 10.1016/j.clinph.2025.2111436. PMID: 41622107.

3. Liu Y, Lin WC, Bai Z, Ge Y, Xiao Y, Zhu F, Zhang J, Wang B, Li Q, Gao L, Li Y, Shan C, Nagai J, Wu Y, Xu T, Gao Z. Lcn2 from neutrophil extracellular traps induces astrogliosis and post-stroke emotional disorders. *Neuron*. 2025 Oct 10;113(24.0):4199-4216.e8. doi: 10.1016/j.neuron.2025.09.018. PMID: 41075784.

4. Markowska A, Tarnacka B. Molecular Changes in the Ischemic Brain as Non-Invasive Brain Stimulation Targets-TMS and tDCS Mechanisms, Therapeutic Challenges, and Combination Therapies. *Biomedicines*. 2024 Jul 13;12(7.0):1560. doi: 10.3390/biomedicines12071560. PMID: 39062133.

5. Raghavan P. Top-Down and Bottom-Up Mechanisms of Motor Recovery Poststroke. *Physical Medicine and Rehabilitation Clinics of North America*. 2023 Aug 29;35(2.0):235-257. doi: 10.1016/j.pmr.2023.07.006. PMID: 38514216.

6. Chen Z, Huang X, Xia NL, Gu M, Xu J, Lu M, Chen H, Xiong C, Chen Y. Next-Generation Neurotechnologies Inspired by Motor Primitive Model for Restoring Human Natural Movement. *Research*. 2025 Jan 1;8:0942. doi: 10.34133/research.0942. PMID: 41235025.

7. Kong X, Lyu W, Lin X, Feng H, Xu L, Li C, Sun X, Lin C, Li J, Wei P. Transcranial direct current stimulation enhances the protective effect of isoflurane preconditioning on cerebral ischemia/reperfusion injury: A new mechanism associated with the nuclear protein Akirin2. *CNS Neuroscience & Therapeutics*. 2024 Sep 1;30(9.0):e70033. doi: 10.1111/cns.70033. PMID: 39267282.

8. Demir S, Fink GR, Rueger MA, Blaschke SJ. Effects of noninvasive brain stimulation on motor functions in animal models of ischemia and trauma in the central nervous system. *Neural Regeneration Research*. 2025 Jun 19;21(4.0):1264-1276. doi: 10.4103/nrr.nrr-d-24-01613. PMID: 40536917.

9. Lee HJ, Shin HK, Shin Y, Kim J, Choi BT. The Fundamental Mechanism of Transcranial Electrical Stimulation in Post-Stroke Rehabilitation. *Frontiers in Bioscience-Landmark*. 2026 Jan 21;31(1.0):46519. doi: 10.31083/fbl46519. PMID: 41609073.

10. Niu B, Wu H, Li Y, Klugah-Brown B, Hanna GP, Yao Y, Jing J, Baig TI, Xia Y, Yao D, Biswal BB. Topological functional network analysis of cortical blood flow in hyperacute ischemic rats. *Brain Structure and Function*. 2024 Dec 26;230(1.0):20. doi: 10.1007/s00429-024-02864-7. PMID: 39724244.

11. Mahani F, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic Functional Connectivity Changes in Cortical and Cortico-Striatal Strokes in Mice. *CNS Neuroscience & Therapeutics*. 2026 Aug 1;32(8.0):e71114. doi: 10.1002/cns.71114. PMID: 42661330.

12. Hensel L, Eickhoff SB, Grefkes C. Funktionelle Magnetresonanztomografie - Update. *Klinische Neurophysiologie*. 2024 Dec 1;55(4.0):236-246. doi: 10.1055/a-1974-6535.

13. Agrawal B, Feuermann Y, Panov J, Kaphzan H. From Immediate Impact to Enduring Change: A Transcriptomic Comparison of tDCS's Temporal Effects and Its Long-Term Equivalence with TMS. *International Journal of Molecular Sciences*. 2025 Sep 4;26(17.0):8634. doi: 10.3390/ijms26178634. PMID: 40943554.

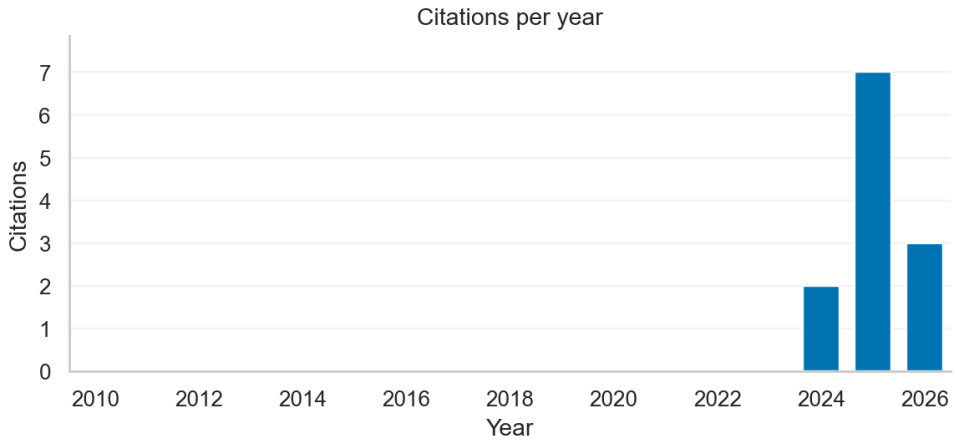
14. Rautenberg N, Blaschke SJ, Vlachakis S, Schroeter M, Fink GR, Rueger MA. Transcranial direct current stimulation (tDCS) in conscious mice: a refined method for repetitive applications without restraint. *Laboratory Animals*. 2025 Dec 14;60(2.0):157-170. doi: 10.1177/00236772251391978. PMID: 41392400.

15. Blaschke SJ, Backes H, Vlachakis S, Rautenberg N, Demir S, Wiedermann D, Aswendt M, Fink GR, Schroeter M, Rueger MA. Subacute cathodal transcranial direct current stimulation rescues secondary thalamic neurodegeneration after cortical stroke in mice. *Experimental Neurology*. 2025 Dec 23;398:115604. doi: 10.1016/j.expneurol.2025.115604. PMID: 41448520.

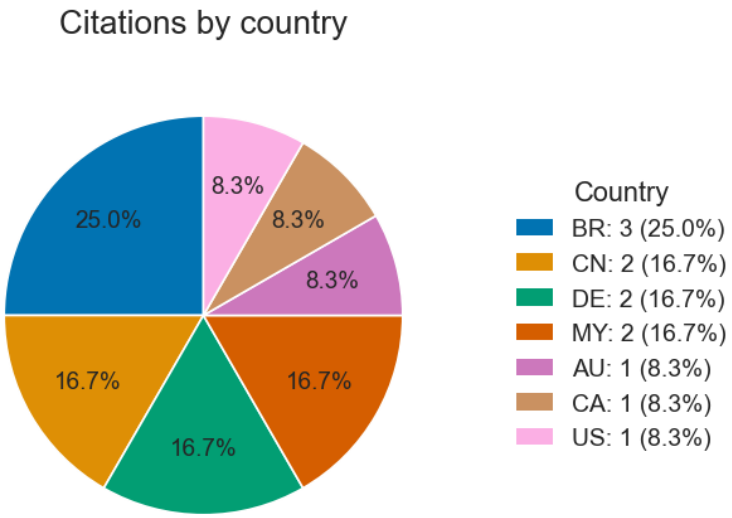
16. Xing S, Yuan Y, Ren P, Chen Z, Fu Y. Combined transcranial electrical stimulation (tES) and cognitive training (CT) for cognitive impairment: Evidence from clinical applications and basic research. *Behavioural Brain Research*. 2026 Feb 2;504:116078. doi: 10.1016/j.bbr.2026.116078. PMID: 41638258.

Publication

Paß T, Ricke KM, Hofmann P, Chowdhury R, Nie Y, Chinnery PF, Endepols H, Neumaier B, Carvalho AF, Rigoux L, Steculorum SM, Prudent J, Riemer T, Aswendt M, Liss B, Brachvogel B, Wiesner RJ. Preserved striatal innervation maintains motor function despite severe loss of nigral dopaminergic neurons. *Brain*. 2024 Apr 4;147(9):3189-3203. doi: 10.1093/brain/awae089. PMID: 38574200.



Citation summary
Total citations: **12**
Average/year: **4.00**
Period: 2024-2026
SCImago cites/doc 2y (2025): **9.66**
SJR 2025: **4.75** (Q1)



Cited by

1. Bulk J, Schmehr JN, Ackels T, Beleza RO, Carvalho AF, Gouveia A, Rigoux L, Hellier V, Cremer AL, Backes H, Schaefer AT, Steculorum SM. A food-sensitive olfactory circuit drives anticipatory satiety. *Nature Metabolism*. 2025 Jun 11;7.0(6.0):1246-1265. doi: 10.1038/s42255-025-01301-1. PMID: 40500386.

2. Wright EC, Scott E, Tian L. Applications of functional neurotransmitter release imaging with genetically encoded sensors in psychiatric research. *Neuropsychopharmacology*. 2024 Jun 28;50.0(1.0):269-273. doi: 10.1038/s41386-024-01903-5. PMID: 38942957.

3. Rocha GS, Freire MAM, Falcão D, Outeiro TF, Lima RR, Santos JRD. Neurodegeneration in Parkinson's disease: are we looking at the right spot?. *Molecular Brain*. 2025 Aug 1;18.0(1.0):68. doi: 10.1186/s13041-025-01218-6. PMID: 40751268.

4. Freire MAM, Rocha GS, Lemos NAM, Lima RR, Bittar S, Jenkins LB, Falcão D, Steinbusch HW, Santos JRD. Insights into Parkinson's Disease Pathology Focusing on Glial Response and Apoptosis in a Classic Rat Model of Dopaminergic Degeneration. *Neuroglia*. 2025 Sep 18;6.0(3.0):36. doi: 10.3390/neuroglia6030036.

5. Tang C, Yang J, Lei X, Zhang M, Chen Y, Peng X, He D. Unraveling the complexity of neurodegeneration: heterogeneous damage patterns of locus coeruleus and substantia nigra in Alzheimer's disease. *Alzheimer s & Dementia*. 2025 Sep 1;21.0(9.0):e70605. doi: 10.1002/alz.70605. PMID: 40891260.

6. Chen C, Zhao Y, Wang J, Pan D, Wang X, Xu Y, Yan J, Wang L, Yang X, Lu M, Liu G. Mitochondrial Dysfunction in Neurodegenerative Diseases. *MedComm*. 2025 Aug 27;6.0(9.0). doi: 10.1002/mco2.70326.

7. Bianchetti ME, Ferreira AFF, Britto LR. Inhibition of neuroinflammation by GIBH-130 (AD-16) reduces neurodegeneration, motor deficits, and proinflammatory cytokines in a hemiparkinsonian model. *Frontiers in Neuroanatomy*. 2024 Dec 16;18.0:1511951. doi: 10.3389/fnana.2024.1511951. PMID: 39736920.

8. Reimon LC, Gouveia A, Carvalho AF, Schmehr JN, Mehdi ME, Moreira-Soto RD, Ardanaz CG, Bulk J, Rigoux L, Klemm P, Cremer AL, Dethloff F, Hinze Y, Backes H, Giavalisco P, Steculorum SM. Fat sensory cues in early life program central response to food and obesity. *Nature Metabolism*. 2025 Dec 1;7.0(12.0):2451-2473. doi: 10.1038/s42255-025-01405-8. PMID: 41326798.

9. Trudeau L, Denis R. Anatomy of dopamine neuron projections: Axonal arbors, varicosities, and synapses. *Handbook of behavioral neuroscience*. 2025 Jan 1;:37-47. doi: 10.1016/b978-0-443-29867-7.00011-6.

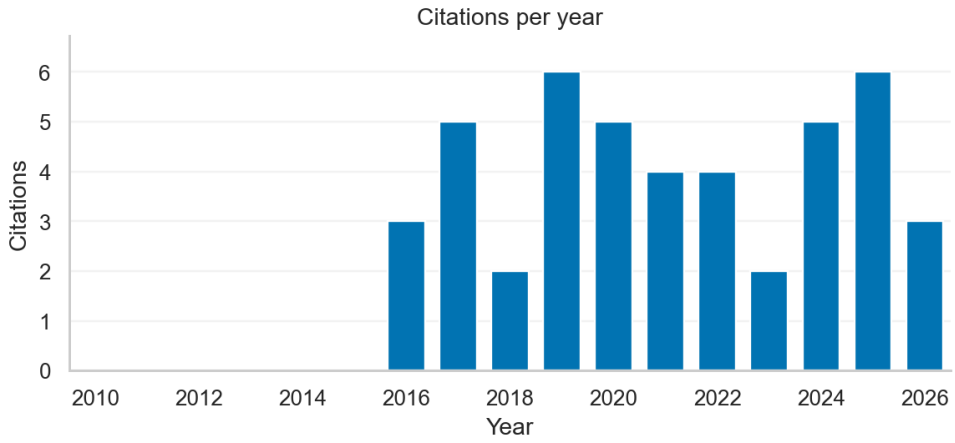
10. Edwards S, Loipersberger C, Arthur E, Collins-Praino LE, Corrigan F. Effects of pre-existing olfactory inflammation on Parkinson's disease related pathology following diffuse traumatic brain injury. *Experimental Neurology*. 2026 Jul 10;405.0:115915. doi: 10.1016/j.expneurol.2026.115915. PMID: 42431347.

11. Mohammad SI, Kareem A, Rekha MM, Vasudevan A, Jabir MS, Nayak PP, Abosaoda MK, Arora V, Nanda A, Hsu C. Mitochondria-centered metabolic microdomains: Regulatory roles in synaptic plasticity and circuit homeostasis. *Neurochemistry International*. 2026 Sep 15;200.0:106264. doi: 10.1016/j.neuint.2026.106264. PMID: 42744222.

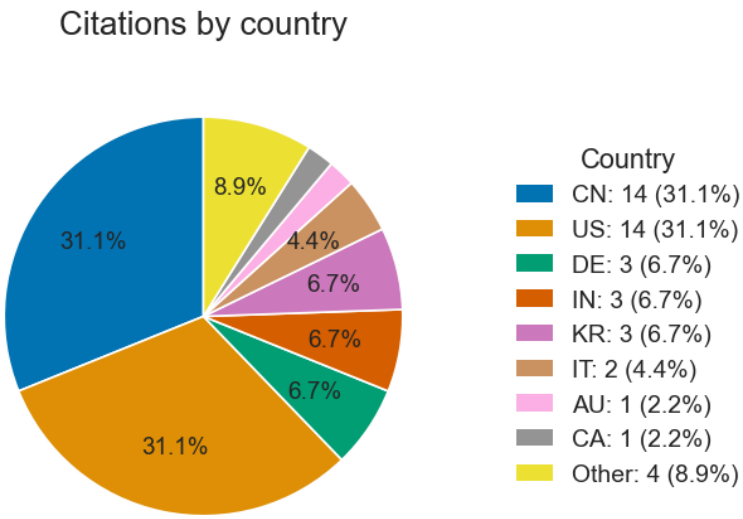
12. Ramli MDC, Abidin ZZ, Okunsai B, Umbar GSA, Mauta TE. Parkinson's disease pathogenesis: From molecular insights to novel therapies. *Elsevier eBooks*. 2026 Oct 2;:41-95. doi: 10.1016/b978-0-443-27702-3.00027-3.

Publication

Cheng M, Aswendt M, Steinberg GK. Optogenetic Approaches to Target Specific Neural Circuits in Post-stroke Recovery. *Neurotherapeutics*. 2015 Dec 23;13(2):325-340. doi: 10.1007/s13311-015-0411-5. PMID: 26701667.



Citation summary
Total citations: **45**
Average/year: **3.75**
Period: 2015-2026
SCImago cites/doc 2y (2025): **7.11**
SJR 2025: **2.38** (Q1)



Cited by

1. Boddington LJ, Reynolds JNJ. Targeting interhemispheric inhibition with neuromodulation to enhance stroke rehabilitation. *Brain stimulation*. 2017 Jan 11;10.0(2):214-222. doi: 10.1016/j.brs.2017.01.006. PMID: 28117178.

2. Su F, Xu W. Enhancing Brain Plasticity to Promote Stroke Recovery. *Frontiers in Neurology*. 2020 Oct 30;11.0:554089. doi: 10.3389/fneur.2020.554089. PMID: 33192987.

3. Tennant KA, Taylor SL, White ER, Brown CE. Optogenetic rewiring of thalamocortical circuits to restore function in the stroke injured brain. *Nature Communications*. 2017 Jun 23;8.0(1):15879. doi: 10.1038/ncomms15879. PMID: 28643802.

4. Cho M, Han J, Suh J, Kim JJ, Ryu JR, Min IS, Sang M, Lim S, Kim TS, Kim K, Kang K, Hwang K, Kim K, Hong E, Nam M, Kim J, Song YM, Lee GJ, Cho I, Yu KJ. Fully bioresorbable hybrid opto-electronic neural implant system for simultaneous electrophysiological recording and optogenetic stimulation. *Nature Communications*. 2024 Mar 6;15.0(1):2000. doi: 10.1038/s41467-024-45803-0. PMID: 38448437.

5. Cassidy JM, Mark JI, Cramer SC. Functional connectivity drives stroke recovery: shifting the paradigm from correlation to causation. *Brain*. 2021 Dec 17;145.0(4):1211-1228. doi: 10.1093/brain/awab469. PMID: 34932786.

6. Choy CS, Cloherty SL, Pirogova E, Fang Q. Virtual Reality Assisted Motor Imagery for Early Post-Stroke Recovery: A Review. *IEEE Reviews in Biomedical Engineering*. 2022 Apr 5;16.0:487-498. doi: 10.1109/rbme.2022.3165062. PMID: 35380970.

7. Sun H, Wang H, Wu C, Liu G, He M, Zhang H, Hou F, Liao H. Enhancing Neuron Activity Promotes Functional Recovery by Inhibiting Microglia-Mediated Synapse Elimination After Stroke. *Stroke*. 2025 Jan 8;56.0(2):505-516. doi: 10.1161/strokeaha.124.049265. PMID: 39772780.

8. Shah AM, Ishizaka S, Cheng M, Wang EH, Bautista AR, Levy SL, Smerin D, Sun G, Steinberg GK. Optogenetic neuronal stimulation of the lateral cerebellar nucleus promotes persistent functional recovery after stroke. *Scientific Reports*. 2017 Jun 1;7.0(1):46612. doi: 10.1038/srep46612. PMID: 28569261.

9. Adewole DO, Serruya MD, Harris JP, Burrell JC, Petrov D, Chen HI, Wolf JA, Cullen DK. The Evolution of Neuroprosthetic Interfaces. *Critical Reviews in Biomedical Engineering*. 2016 Jan 1;44.0(1-02):123-152. doi: 10.1615/critrevbiomedeng.2016017198. PMID: 27652455.

10. Lechtenberg KJ, Meyer ST, Doyle JB, Peterson TC, Buckwalter MS. Augmented β2-adrenergic signaling dampens the neuroinflammatory response following ischemic stroke and increases stroke size. *Journal of Neuroinflammation*. 2019 May 28;16.0(1):112. doi: 10.1186/s12974-019-1506-4. PMID: 31138227.

11. Mahmoudi P, Veladi H, Pakdel F. Optogenetics, Tools and Applications in Neurobiology. *Journal of Medical Signals & Sensors*. 2017 Jan 1;7.0(2):71. doi: 10.4103/2228-7477.205506. PMID: 28553579.

12. Zhang Q, Hu DX, He F, Li CY, Qi G, Cai H, Li TX, Ming J, Zhang P, Chen X, Tian B. Locus coeruleus-CA1 projections are involved in chronic depressive stress-induced hippocampal vulnerability to transient global ischaemia. *Nature Communications*. 2019 Jul 3;10.0(1):2942. doi: 10.1038/s41467-019-10795-9. PMID: 31270312.

13. Montagni E, Resta F, Mascaro ALA, Pavone FS. Optogenetics in Brain Research: From a Strategy to Investigate Physiological Function to a Therapeutic Tool. *Photonics*. 2019 Aug 20;6.0(3):92. doi: 10.3390/photonics6030092.

14. Zeiger W, Marosi M, Saggi SS, Noble N, Samad I, Portera-Cailliau C. Barrel cortex plasticity after photothrombotic stroke involves potentiating responses of pre-existing circuits but not functional remapping to new circuits. *Nature Communications*. 2021 Jun 25;12.0(1):3972. doi: 10.1038/s41467-021-24211-8. PMID: 34172735.

15. Conti E, Scaglione A, de Vito G, Calugi F, Pasquini M, Pizzorusso T, Micera S, Mascaro ALA, Pavone FS. Combining Optogenetic Stimulation and Motor Training Improves Functional Recovery and Perilesional Cortical Activity. *Neurorehabilitation and neural repair*. 2021 Nov 11;36.0(2):107-118. doi: 10.1177/15459683211056656. PMID: 34761714.

16. Lu C, Wu X, Ma H, Wang Q, Wang Y, Luo Y, Li C, Xu H. Optogenetic Stimulation Enhanced Neuronal Plasticities in Motor Recovery after Ischemic Stroke. *Neural Plasticity*. 2019 Mar 14;2019.0:1-9. doi: 10.1155/2019/5271573. PMID: 31007684.

17. Vahdat S, Pendharkar AV, Chiang T, Harvey SS, □□□□, Cao Z, Kim AY, Choy M, Chen H, Lee HJ, Cheng M, Lee JH, Steinberg GK. Brain-wide neural dynamics of poststroke recovery induced by optogenetic stimulation. *Science Advances*. 2021 Aug 11;7.0(33). doi: 10.1126/sciadv.abd9465. PMID: 34380610.

18. Yang C, Zhang T, Huang K, Xiong M, Liu H, Wang P, Zhang Y. Increased both cortical activation and functional connectivity after transcranial direct current stimulation in patients with post-stroke: A functional near-infrared spectroscopy study. *Frontiers in Psychiatry*. 2022 Dec 7;13.0:1046849. doi: 10.3389/fpsyt.2022.1046849. PMID: 36569623.

19. Shin SS, Pelled G. Novel Neuromodulation Techniques to Assess Interhemispheric Communication in Neural Injury and Neurodegenerative Diseases. *Frontiers in Neural Circuits*. 2017 Mar 8;11.0:15. doi: 10.3389/fncir.2017.00015. PMID: 28337129.

20. Hyung S, Park J, Jung K. Application of optogenetic glial cells to neuron-glial communication. *Frontiers in Cellular Neuroscience*. 2023 Oct 5;17.0:1249043. doi: 10.3389/fncel.2023.1249043. PMID: 37868193.

21. Prescott K, Münch AE, Brahms E, Weigel MK, Inoue K, Buckwalter MS, Liddelow SA, Peterson TC. Blocking of microglia-astrocyte proinflammatory signaling is beneficial following stroke. *Frontiers in Molecular Neuroscience*. 2024 Jan 3;16.0:1305949. doi: 10.3389/fnmol.2023.1305949. PMID: 38240014.

22. Kale MB, Wankhede NL, Bishoyi AK, Ballal S, Kalia R, Arya R, Kumar S, Khalid M, Gulati M, Umare MD, Taksande BG, Upaganlawar AB, Umekar MJ, Kopalli SR, Fareed M, Koppula S. Emerging biophysical techniques for probing synaptic transmission in neurodegenerative disorders. *Neuroscience*. 2024 Nov 26;565.0:63-79. doi: 10.1016/j.neuroscience.2024.11.055. PMID: 39608699.

23. Singh AK, Dhalayat K, Dhobale SV, Ghosh B, Datta A, Borah A, Bhattacharya P. Unravelling the brain resilience following stroke: From injury to rewiring of the brain through pathway activation, drug targets, and therapeutic interventions. *Ageing Research Reviews*. 2025 May 21;109.0:102780. doi: 10.1016/j.arr.2025.102780. PMID: 40409413.

24. Kim RG, Cho J, Park J, Kim YR, Lee MC, Kim H. Neuron type-specific optogenetic stimulation for differential stroke recovery in chronic capsular infarct. *Experimental & Molecular Medicine*. 2024 Jun 2;56.0(6):1439-1449. doi: 10.1038/s12276-024-01253-8. PMID: 38825647.

25. Bo B, Li Y, Li W, Wang Y, Tong S. Optogenetic translocation of protons out of penumbral neurons is protective in a rodent model of focal cerebral ischemia. *Brain stimulation*. 2020 Mar 19;13.0(3):881-890. doi: 10.1016/j.brs.2020.03.008. PMID: 32289721.

26. Peterson TC, Lechtenberg KJ, Piening B, Lucas TA, Wei E, Chaïb H, Dowdell AK, Snyder M, Buckwalter MS. Obesity Drives Delayed Infarct Expansion, Inflammation, and Distinct Gene Networks in a Mouse Stroke Model. *Translational Stroke Research*. 2020 Jun 25;12.0(2):331-346. doi: 10.1007/s12975-020-00826-9. PMID: 32588199.

27. Geng Y, Li Z, Zhu J, Du C, Yuan F, Cai X, Ali A, Yang J, Tang C, Cong Z, Ma C. Advances in Optogenetics Applications for Central Nervous System Injuries. *Journal of Neurotrauma*. 2022 Oct 28;40.0(13-14):1297-1316. doi: 10.1089/neu.2022.0290. PMID: 36305381.

28. Duan Y, Hua X, Zheng M, Wu J, Xing X, Li Y, Xu J. Corticocortical paired associative stimulation for treating motor dysfunction after stroke: study protocol for a randomised sham-controlled double-blind clinical trial. *BMJ Open*. 2022 Jan 1;12.0(1):e053991. doi: 10.1136/bmjopen-2021-053991. PMID: 35027421.

29. Li S, Xing X, Hua X, Zhang Y, Wu J, Shan C, Wang H, Zheng M, Xu J. Electroacupuncture modulates abnormal brain connectivity after ischemia reperfusion injury in rats: A graph theory-based approach. *Brain and Behavior*. 2024 May 1;14.0(5):e3504. doi: 10.1002/brb3.3504. PMID: 38698583.

30. Mannel A, Khunt D, Agrawal V, Schelling K, Calderón E, la Fougère CJ, Vega SC. Optimizing stroke lesion segmentation: A dual-approach using Gaussian mixture models and nnU-Net. *Computers in Biology and Medicine*. 2025 May 2;192.0(Pt A):110221. doi: 10.1016/j.compbiomed.2025.110221. PMID: 40318493.

31. Chu C, Chen Z, Liang C, Wu P, Guo W, Chen B, Huang C, Li S, Chang C, Liang Y, Kan S, Lo Y, Huang W, Chen Y, Chen S. A Closed-Loop-Capable Neural Interface Platform for Deep Brain Modulation via Integrated Non-Viral Gene Delivery, NIR Optogenetics, and Electrophysiological Recording. *Advanced Science*. 2025 Nov 19;13.0(7):e15060. doi: 10.1002/advs.202515060. PMID: 41255225.

32. Pendharkar AV, Levy SL, Ho AL, Sussman ES, Cheng M, Steinberg GK. Optogenetic modulation in stroke recovery. *Neurosurgical FOCUS*. 2016 May 1;40.0(5):E6. doi: 10.3171/2016.2.focus163. PMID: 27132527.

33. Bo B, Li Y, Li W, Wang Y, Tong S. Optogenetic Excitation of Ipsilesional Sensorimotor Neurons is Protective in Acute Ischemic Stroke: A Laser Speckle Imaging Study. *IEEE Transactions on Biomedical Engineering*. 2018 Oct 4;66.0(5):1372-1379. doi: 10.1109/tbme.2018.2872965. PMID: 30281433.

34. Bo B, Li Y, Li W, Wang Y, Tong S. Neurovascular Coupling Impairment in Acute Ischemic Stroke by Optogenetics and Optical Brain Imaging. *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society*. 2020 Jul 1;2020.0:3727-3730. doi: 10.1109/embc44109.2020.9176641. PMID: 33018811.

35. Li K, Liang H, Qiu J, Zhang X, Cai B, Wang D, Zhang D, Lin B, Han H, Yang G, Zhu Z. Reveal the mechanism of brain function with fluorescence microscopy at single-cell resolution: from neural decoding to encoding. *Journal of NeuroEngineering and Rehabilitation*. 2025 May 27;22.0(1):118. doi: 10.1186/s12984-025-01655-3. PMID: 40426214.

36. Wang A, Li J, Liu H, Yue D. Physiological Regulatory Mechanisms of Neurovascular Coupling and the Role of Its Dysfunction in Neurological Diseases. *Journal of Integrative Neuroscience*. 2026 Feb 26;25.0(2):42796. doi: 10.31083/jin42796. PMID: 41762050.

37. Kunze T, Haueisen J, Knösche TR. Emergence of cognitive priming and structure building from the hierarchical interaction of canonical microcircuit models. *Biological Cybernetics*. 2019 Feb 14;113.0(3):273-291. doi: 10.1007/s00422-019-00792-y. PMID: 30767085.

38. Somuncu O, Berns HM, Sanchez JG. New Pioneers of Optogenetics in Neuroscience. *Advances in experimental medicine and biology*. 2019 Jan 1;1288.0:47-60. doi: 10.1007/5584_2019_473. PMID: 31983055.

39. Carmel JB, Willis DE. Neural Circuits Catch Fire. *Neurotherapeutics*. 2016 Mar 18;13.0(2):261-263. doi: 10.1007/s13311-016-0428-4. PMID: 26984475.

40. Aswendt M, Green C, Hoehn M. Plastic Network Changes During Brain Disease. *Handbook of behavioral neuroscience*. 2018 Jan 1;:415-424. doi: 10.1016/b978-0-12-812028-6.00022-7.

41. Conti E, Mascaro ALA, Scaglione A, de Vito G, Calugi F, Pasquini M, Pizzorusso T, Micera S, Pavone FS. Restoration of motor-evoked cortical activity is a distinguishing feature of the most effective rehabilitation therapy after stroke. *bioRxiv (Cold Spring Harbor Laboratory)*. 2020 Mar 6;. doi: 10.1101/2020.03.05.974972.

42. Prescott K, Münch AE, Brahms E, Weigel MK, Inoue K, Buckwalter MS, Liddelow SA, Peterson TC. Blocking Formation of Neurotoxic Reactive Astrocytes is Beneficial Following Stroke. *bioRxiv (Cold Spring Harbor Laboratory)*. 2023 Oct 16;. doi: 10.1101/2023.10.11.561918. PMID: 37905154.

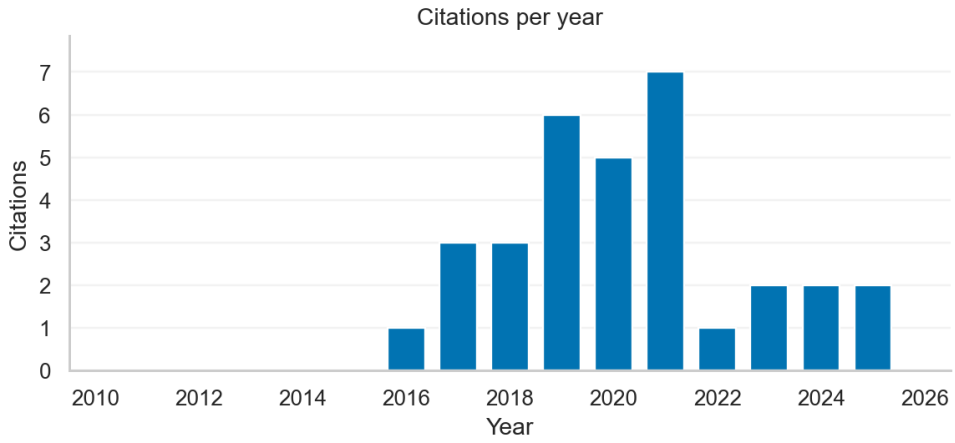
43. Trisal A, Singh AK. CLARITY in Alzheimer's Research: Merging Tissue Transparency with Next-Gen Neurotechnologies. *Journal of Neuroscience Methods*. 2025 Aug 13;423.0:110554. doi: 10.1016/j.jneumeth.2025.110554. PMID: 40816537.

44. Xing X, Wu J, Ma J, Zheng M, Hua X, Xu J, Wang Y. Altered resting-state network effects in patients after stroke following cortico-cortical target transcranial magnetic stimulation. *Brain Communications*. 2026 Jan 1;8.0(4):fcag305. doi: 10.1093/braincomms/fcag305. PMID: 42614885.

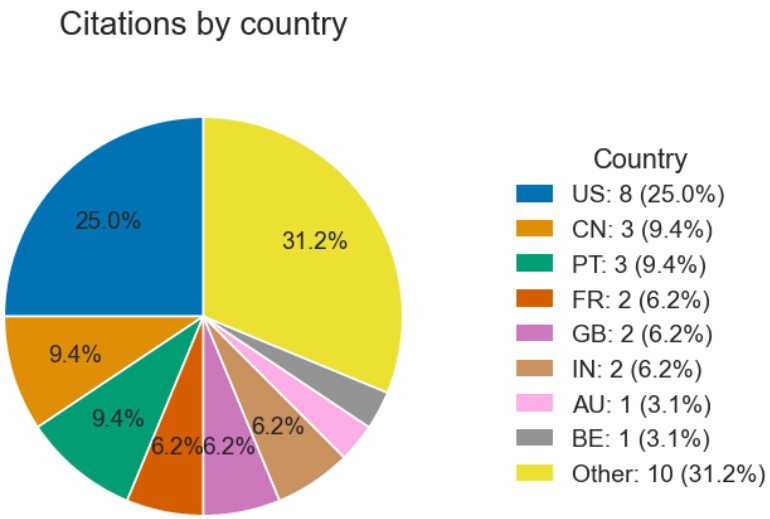
45. Ikramuddin S, Huang Z, Zhu L, Hu X, Zheng H, Feng W. Current and emerging neuromodulation techniques for motor recovery after ischemic stroke: a 2026 update. *Expert Review of Neurotherapeutics*. 2026 Oct 13;:1-15. doi: 10.1080/14737175.2026.2744756. PMID: 42867124.

Publication

Aswendt M, Schwarz MK, Abdelmoula WM, Dijkstra J, Dedeurwaerdere S. Whole-Brain Microscopy Meets In Vivo Neuroimaging: Techniques, Benefits, and Limitations. *Molecular Imaging and Biology*. 2016 Sep 2;19(1):1-9. doi: 10.1007/s11307-016-0988-z. PMID: 27590493.



Citation summary
Total citations: **32**
Average/year: **2.91**
Period: 2016-2026
SCImago cites/doc 2y (2025): **2.79**
SJR 2025: **0.74** (Q2)



Cited by

1. Gupta CN, Turner JA, Calhoun VD. Source-based morphometry: a decade of covarying structural brain patterns. *Brain Structure and Function*. 2019 Nov 7;224.0(9.0):3031-3044. doi: 10.1007/s00429-019-01969-8. PMID: 31701266.

2. Mi X, Chen A, Duarte D, Carey EM, Taylor CR, Braaker PN, Bright M, Almeida R, Lim J, Ruetten VMS, Wang Y, Wang M, Zhang W, Zheng WH, Reitman ME, Huang Y, Wang X, Li L, Deng H, Shi S, Poskanzer KE, Lyons DA, Nimmerjahn A, Ahrens MB, Yu G. Fast, accurate, and versatile data analysis platform for the quantification of molecular spatiotemporal signals. *Cell*. 2025 Apr 8;188.0(10.0):2794-2809.e21. doi: 10.1016/j.cell.2025.03.012. PMID: 38766026.

3. Leuze CW, Aswendt M, Ferenczi E, Liu CW, Hsueh B, Goubran M, Tian Q, Steinberg GK, Zeineh M, Deisseroth KA, McNab JA. The separate effects of lipids and proteins on brain MRI contrast revealed through tissue clearing. *NeuroImage*. 2017 Apr 12;156.0:412-422. doi: 10.1016/j.neuroimage.2017.04.021. PMID: 28411157.

4. Laviña B. Brain Vascular Imaging Techniques. *International Journal of Molecular Sciences*. 2016 Dec 30;18.0(1.0):70. doi: 10.3390/ijms18010070. PMID: 28042833.

5. Lindsey BW, Douek AM, Loosli F, Kaslin J. A Whole Brain Staining, Embedding, and Clearing Pipeline for Adult Zebrafish to Visualize Cell Proliferation and Morphology in 3-Dimensions. *Frontiers in Neuroscience*. 2018 Jan 17;11.0:750. doi: 10.3389/fnins.2017.00750. PMID: 29386991.

6. Leuze CW, Goubran M, Baraković M, Aswendt M, Tian Q, Hsueh B, Crow A, Weber EMM, Steinberg GK, Zeineh M, Plowey ED, Daducci A, Innocenti G, Thiran J, Deisseroth KA, McNab JA. Comparison of diffusion MRI and CLARITY fiber orientation estimates in both gray and white matter regions of human and primate brain. *NeuroImage*. 2020 Dec 29;228.0:117692. doi: 10.1016/j.neuroimage.2020.117692. PMID: 33385546.

7. Carter LM, Poty S, Sharma SK, Lewis JS. Preclinical optimization of antibody-based radiopharmaceuticals for cancer imaging and radionuclide therapy-Model, vector, and radionuclide selection. *Journal of Labelled Compounds and Radiopharmaceuticals*. 2018 Feb 7;61.0(9.0):611-635. doi: 10.1002/jlcr.3612. PMID: 29412489.

8. Foster DS, Nguyen AT, Chinta MS, Salhotra A, Jones RE, Mascharak S, Titan AL, Ransom RC, da Silva O, Foley E, Briger E, Longaker MT. A Clearing Technique to Enhance Endogenous Fluorophores in Skin and Soft Tissue. *Scientific Reports*. 2019 Oct 31;9.0(1.0):15791. doi: 10.1038/s41598-019-50359-x. PMID: 31673001.

9. Czéh B, Simon M. Benefits of animal models to understand the pathophysiology of depressive disorders. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*. 2020 Jul 28;106.0:110049. doi: 10.1016/j.pnpbp.2020.110049. PMID: 32735913.

10. Son J, Park SJ, Ha T, Lee S, Cho H, Choi J. Electrophysiological Monitoring of Neurochemical-Based Neural Signal Transmission in a Human Brain-Spinal Cord Assembloid. *ACS Sensors*. 2022 Jan 19;7.0(2.0):409-414. doi: 10.1021/acssensors.1c02279. PMID: 35044765.

11. Xue J, Liu Y, Ji T, Zhao A, Liang Y, Deng H, Wang Q, Zhang Y, Yang L, Yang A. Comparison between technetium-99m methoxyisobutylisonitrile scintigraphy and ultrasound in the diagnosis of parathyroid adenoma and parathyroid hyperplasia. *Nuclear Medicine Communications*. 2018 Sep 26;39.0(12.0):1129-1137. doi: 10.1097/mnm.0000000000000921. PMID: 30239472.

12. Carneiro IC, Carvalho SD, Henrique R, Oliveira LMC, Tuchin VV. Moving tissue spectral window to the deep-ultraviolet via optical clearing. *Journal of Biophotonics*. 2019 Aug 29;12.0(12.0):e201900181. doi: 10.1002/jbio.201900181. PMID: 31465137.

13. Naqvi SMZA, Zhang Y, Tahir MN, Ullah Z, Ahmed S, Wu J, Raghavan V, Abdulraheem MI, Ping JF, Hu X, Hu J, Hu J. Advanced strategies of the in-vivo plant hormone detection. *TrAC Trends in Analytical Chemistry*. 2023 Jul 7;166.0:117186. doi: 10.1016/j.trac.2023.117186.

14. Wykes RC, Khoo HM, Caciagli L, Blumenfeld H, Golshani P, Kapur J, Stern JM, Bernasconi N, Dedeurwaerdere S, Bernasconi N. WONOEP appraisal: Network concept from an imaging perspective. *Epilepsia*. 2019 Jun 9;60.0(7.0):1293-1305. doi: 10.1111/epi.16067. PMID: 31179547.

15. Zheng F, Zhou J, Ouyang Z, Zhang J, Wang X, Muyldermans S, Van Ginderachter JA, Devoogdt N, Wen Y, Schoonooghe S, Raes G. Development and Characterization of Nanobodies Targeting the Kupffer Cell. *Frontiers in Immunology*. 2021 Feb 22;12.0:641819. doi: 10.3389/fimmu.2021.641819. PMID: 33692811.

16. Ahmad IN, Kushwaha P, Usmani SA, Tiwari A. Polymeric Micelles: Revolutionizing Cancer Therapeutics for Enhanced Efficacy. *BioNanoScience*. 2025 Jan 16;15.0(1.0). doi: 10.1007/s12668-025-01803-y.

17. Silva PHI, Pépin M, Figurek A, Gutiérrez-Jiménez E, Bobot M, Iervolino A, Mattace-Raso FU, Hoorn EJ, Bailey MA, Hénaut L, Nielsen R, Frische S, Trepiccione F, Hafez G, ALTUNKAYNAK-CAMCA HO, Endlich N, Unwin RJ, Capasso G, Pešić V, Massy ZA, Wagner CA. Animal models to study cognitive impairment of chronic kidney disease. *American Journal of Physiology-Renal Physiology*. 2024 Apr 18;326.0(6.0):F894-F916. doi: 10.1152/ajprenal.00338.2023. PMID: 38634137.

18. Yamada H, Makino S, Okunaga I, Miyake T, Nonaka K, Trejo JAO, Tominaga T, Empitu MA, Kadariswantiningsih IN, Kerever A, Komiya A, Ichikawa T, Arikawa-Hirasawa E, Yanagita M, Asanuma K. Beyond 2D: A scalable and highly sensitive method for a comprehensive 3D analysis of kidney biopsy tissue. *PNAS Nexus*. 2023 Dec 21;3.0(1.0):pgad433. doi: 10.1093/pnasnexus/pgad433. PMID: 38193136.

19. Chourrout M, Rositi H, Ong E, Hubert V, Paccalet A, Foucault L, Autret A, C BF, Olivier C, Bolbos RI, Peyrin F, Da Silva CC, Meyronet D, Raineteau O, Elleaume H, Brun E, Chauveau F, Wiart M. Brain virtual histology with X-ray phase-contrast tomography Part I: whole-brain myelin mapping in white-matter injury models. *Biomedical Optics Express*. 2021 Nov 5;13.0(3.0):1620. doi: 10.1364/boe.438832. PMID: 35415001.

20. Jones EE, Quiason CM, Dale S, Shahidi-Latham SK. Feasibility Assessment of a MALDI FTICR Imaging Approach for the 3D Reconstruction of a Mouse Lung. *Journal of the American Society for Mass Spectrometry*. 2017 Apr 11;28.0(8.0):1709-1715. doi: 10.1007/s13361-017-1658-3. PMID: 28401432.

21. Carbó-Carreté M, Cañete-Massé C, Peró M, Guàrdia-Olmos J. Using fMRI to Assess Brain Activity in People With Down Syndrome: A Systematic Review. *Frontiers in Human Neuroscience*. 2020 Apr 27;14.0:147. doi: 10.3389/fnhum.2020.00147. PMID: 32395104.

22. Fogli B, Corthout N, Kerstens A, Bosse F, Klimaschewski LP, Munck S, Schweigreiter R. Imaging axon regeneration within synthetic nerve conduits. *Scientific Reports*. 2019 Jul 12;9.0(1.0):10095. doi: 10.1038/s41598-019-46579-w. PMID: 31300753.

23. Lu W, Pei Z, Hu W, Tan C, Tong X, Feng Y, Sun X. Recent progress in optical clearing of eye tissues. *Experimental Eye Research*. 2021 Oct 15;212.0:108796. doi: 10.1016/j.exer.2021.108796. PMID: 34662543.

24. Porter DD, Morton PD. Clearing techniques for visualizing the nervous system in development, injury, and disease. *Journal of Neuroscience Methods*. 2020 Jan 13;334.0:108594. doi: 10.1016/j.jneumeth.2020.108594. PMID: 31945400.

25. Gomes N, Tuchin VV, Oliveira LMC. Refractive Index Matching Efficiency in Colorectal Mucosa Treated With Glycerol. *IEEE Journal of Selected Topics in Quantum Electronics*. 2021 Jan 13;27.0(4.0):1-8. doi: 10.1109/jstqe.2021.3050208.

26. Oliveira LMC, Tuchin VV. Tissue Optics. *SpringerBriefs in physics*. 2019 Jan 1;:1-15. doi: 10.1007/978-3-030-33055-2_1.

27. McDonald MW, Jeffers MS, Filadelfi M, Vicencio A, Heidenreich G, Wu J, Silasi G. Localizing Microemboli within the Rodent Brain through Block-Face Imaging and Atlas Registration. *eNeuro*. 2021 Jul 1;8.0(4.0):ENEURO.0216-21.2021. doi: 10.1523/eneuro.0216-21.2021. PMID: 34272259.

28. Martínez-Lorenzana G, Gamal-Eltrabily M, Tello-García IA, Martínez-Torres A, Becerra-González M, González-Hernández A, Condés-Lara M. CLARITY with neuronal tracing and immunofluorescence to study the

somatosensory system in rats. *Journal of Neuroscience Methods*. 2020 Dec 24;350.0:109048. doi: 10.1016/j.jneumeth.2020.109048. PMID: 33359224.

29. Amar SK, Anand U, Verma A. Cerebrovascular Imaging in a Rodent Stroke Model. *Springer eBooks*. 2021 Jan 1;:97-105. doi: 10.1007/978-981-33-6679-4_7.

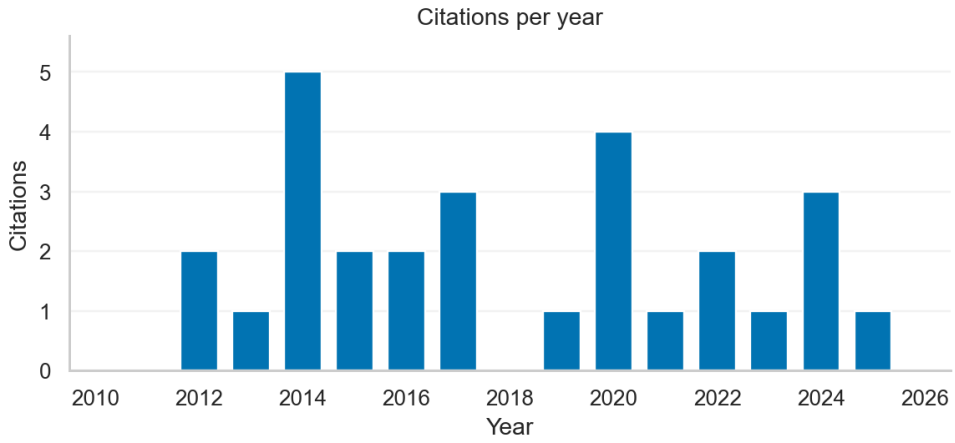
30. Chourrout M, Rositi H, Ong E, Hubert V, Paccalet A, Foucault L, Autret A, C BF, Olivier C, Bolbos RI, Peyrin F, Da Silva CC, Meyronet D, Raineteau O, Elleaume H, Brun E, Chauveau F, Wiart M. Brain virtual histology with X-ray phase-contrast tomography Part I: whole-brain myelin mapping in white-matter injury models. *bioRxiv (Cold Spring Harbor Laboratory)*. 2021 Mar 25;. doi: 10.1101/2021.03.24.436852.

31. Fay MG, Lang PJ, Denu DS, O'Connor N, Haydock BW, Blaisdell JO, Roussel N, Wilson AL, Aronson SR, Angstman PJ, Gong C, Butola T, Devinsky O, Basu J, Tomer R, Glaser JR. ClearScope: a fully integrated light sheet theta microscope for sub-cellular resolution imaging without lateral size constraints. *bioRxiv (Cold Spring Harbor Laboratory)*. 2024 Aug 19;. doi: 10.1101/2024.08.15.608141. PMID: 39229056.

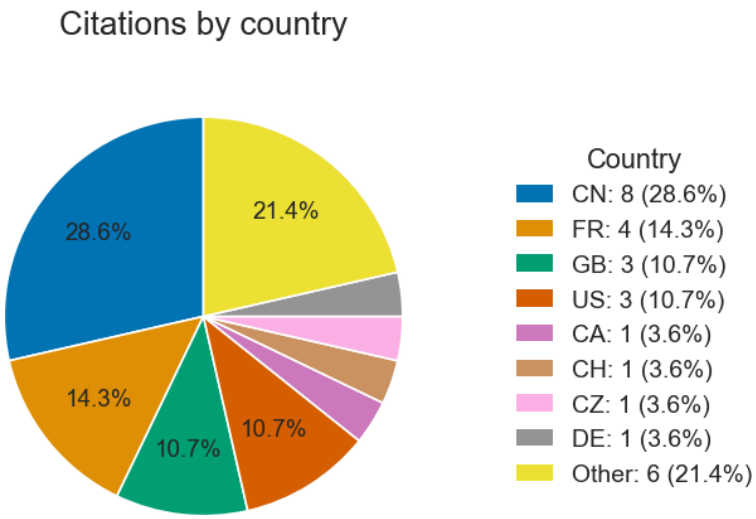
32. Henein MY. Atherosclerosis and Vascular Imaging 2016. *MDPI eBooks*. 2017 Jun 16;. doi: 10.3390/books978-3-03842-433-8.

Publication

Napolitano R, Pariani GM, Fedeli F, Baranyai Z, Aswendt M, Aime S, Gianolio E. Synthesis and Relaxometric Characterization of a MRI Gd-Based Probe Responsive to Glutamic Acid Decarboxylase Enzymatic Activity. *Journal of Medicinal Chemistry*. 2013 Mar 8;56(6):2466-2477. doi: 10.1021/jm301831f. PMID: 23469759.



Citation summary
Total citations: **28**
Average/year: **2.00**
Period: 2013-2026
SCImago cites/doc 2y (2025): **6.84**
SJR 2025: **1.73** (Q1)



Cited by

1. Hingorani DV, Bernstein AS, Pagel MD. A review of responsive MRI contrast agents: 2005-2014. *Contrast Media & Molecular Imaging*. 2014 Oct 29;10.0(4.0):245-265. doi: 10.1002/cmmi.1629. PMID: 25355685.

2. Davies G, Kramberger I, Davis JJ. Environmentally responsive MRI contrast agents. *Chemical Communications*. 2013 Jan 1;49.0(84.0):9704. doi: 10.1039/c3cc44268c. PMID: 24040650.

3. Botár R, Molnár E, Trencsényi G, Kiss J, Kálmán FK, Tircsó G. Stable and Inert Mn(II)-Based and pH-Responsive Contrast Agents. *Journal of the American Chemical Society*. 2020 Jan 13;142.0(4.0):1662-1666. doi: 10.1021/jacs.9b09407. PMID: 31927927.

4. Baranyai Z, Tircsó G, Rösch F. The Use of the Macrocyclic Chelator DOTA in Radiochemical Separations. *European Journal of Inorganic Chemistry*. 2019 Oct 30;2020.0(1.0):36-56. doi: 10.1002/ejic.201900706.

5. Meng Q, Wu M, Shang Z, Zhang Z, Zhang R. Responsive gadolinium(III) complex-based small molecule magnetic resonance imaging probes: Design, mechanism and application. *Coordination Chemistry Reviews*. 2022 Jan 25;457.0:214398. doi: 10.1016/j.ccr.2021.214398.

6. Yue P, Nagendraraj T, Wang G, Jin Z, Angelovski G. The role of responsive MRI probes in the past and the future of molecular imaging. *Chemical Science*. 2024 Jan 1;15.0(48.0):20122-20154. doi: 10.1039/d4sc04849k. PMID: 39611034.

7. Lacerda S, Tóth É. Lanthanide Complexes in Molecular Magnetic Resonance Imaging and Theranostics. *ChemMedChem*. 2017 May 4;12.0(12.0):883-894. doi: 10.1002/cmdc.201700210. PMID: 28470934.

8. Lv X, Feng L, Ai C, Hou J, Wang P, Zou L, Cheng J, Ge G, Cui J, Yang L. A Practical and High-Affinity Fluorescent Probe for Uridine Diphosphate Glucuronosyltransferase 1A1: A Good Surrogate for Bilirubin. *Journal of Medicinal Chemistry*. 2017 Nov 10;60.0(23.0):9664-9675. doi: 10.1021/acs.jmedchem.7b01097. PMID: 29125289.

9. Hingorani DV, Yoo B, Bernstein AS, Pagel MD. Detecting Enzyme Activities with Exogenous MRI Contrast Agents. *Chemistry - A European Journal*. 2014 Jul 2;20.0(32.0):9840-9850. doi: 10.1002/chem.201402474. PMID: 24990812.

10. Du J, Li W, Tan H. Three-layer medical image fusion with tensor-based features. *Information Sciences*. 2020 Mar 21;525.0:93-108. doi: 10.1016/j.ins.2020.03.051.

11. Liu M, Gao J, Zhang Y, Zhou X, Wang Y, Wu L, Tian Z, Tang J. Recent advances in bioresponsive macrocyclic gadolinium(iii) complexes for MR imaging and therapy. *Dalton Transactions*. 2025 Jan 1;54.0(17.0):6741-6777. doi: 10.1039/d5dt00191a. PMID: 40085150.

12. Kitos AA, Castañeda R, Comeau ZJ, Mavragani N, Calvert ND, Kirby A, Martínez-Santiesteban F, Pallister PJ, Scholl TJ, Murugesu M, Shuhendler AJ, Brusso JL. Cluster-based redox-responsive super-atomic MRI contrast agents. *Chem*. 2024 Oct 21;11.0(3.0):102330. doi: 10.1016/j.chempr.2024.09.029.

13. Aswendt M, Gianolio E, Pariani GM, Napolitano R, Fedeli F, Himmelreich U, Aime S, Hoehn M. In vivo imaging of inhibitory, GABAergic neurons by MRI. *NeuroImage*. 2012 May 17;62.0(3.0):1685-1693. doi: 10.1016/j.neuroimage.2012.05.029. PMID: 22609794.

14. Bonnet CS, Tóth É. Molecular Magnetic Resonance Imaging Probes Based on Ln3+ Complexes. *Advances in inorganic chemistry*. 2015 Nov 6;:43-96. doi: 10.1016/bs.adioch.2015.09.002.

15. Shen X, Pan Y, Liang G. Development of Macrocyclic Mn(II)−Bispyridine Complexes as pH-Responsive Magnetic Resonance Imaging Contrast Agents. *European Journal of Inorganic Chemistry*. 2023 Feb 21;26.0(12.0). doi: 10.1002/ejic.202200786.

16. Wang L, Wang L, Sun XJ, Fu L, Wang X, Wang X, Chen L, Huang YL. Detection of uridine diphosphate glucuronosyltransferase 1A1 for pancreatic cancer imaging and treatment via a “turn-on” fluorescent probe. *The Analyst*. 2024 Jan 1;149.0(10.0):2877-2886. doi: 10.1039/d4an00035h. PMID: 38567989.

17. Buzhor MG, Avram L, Frish L, Cohen Y, Amir RJ. Fluorinated smart micelles as enzyme-responsive probes for 19 F-magnetic resonance. *Journal of Materials Chemistry B*. 2015 Dec 15;4.0(18.0):3037-3042. doi: 10.1039/c5tb02445e. PMID: 32263042.

18. Bernsen MR, Ruggiero A, van Straten M, Kotek G, Haeck JC, Wielopolski PA, Krestin GP. Computed Tomography and Magnetic Resonance Imaging. *Recent results in cancer research/Recent Results in Cancer Research*. 2012 Oct 17;187.0:3-63. doi: 10.1007/978-3-642-10853-2_1. PMID: 23179877.

19. Hermann P, Kotek J, Kubíček V. Ten-Membered Rings or Lager With One or More Nitrogen Atoms. *Elsevier eBooks*. 2021 Sep 23;:591-683. doi: 10.1016/b978-0-12-818655-8.00128-1.

20. Bernsen MR, van Straten M, Kotek G, Warnert EAH, Haeck JC, Ruggiero A, Wielopolski PA, Krestin GP. Computed Tomography and Magnetic Resonance Imaging. *Recent results in cancer research/Recent Results in Cancer Research*. 2020 Jan 1;:31-110. doi: 10.1007/978-3-030-42618-7_2.

21. Brethon A, Bouix-Peter C, Clary L, Fournier J, Harris CS, Lardy C, Roche D, Rodeschini V, Talano S. An expedient synthesis of non-racemic N-alkylated pyrrolidin-2,5-diones and piperidin-2,6-diones as peptidomimetics. *Tetrahedron Letters*. 2016 Nov 21;57.0(52.0):5924-5927. doi: 10.1016/j.tetlet.2016.11.083.

22. Schroeter A, Rudin M, Gianolio E, Viale A, Castelli DD, Aime S, Hövener J, Bastiaansen J, Comment A, Düwel S, Ardenkjær-Larsen JH, Becker M. MRI. *Springer eBooks*. 2017 Jan 1;:227-324. doi: 10.1007/978-3-319-42202-2_13.

23. Sun C, Lin H, Gao J. Recent advances in biological-environment-responsive smart MRI contrast agents. *Scientia Sinica Chimica*. 2016 Dec 20;47.0(2.0):119-132. doi: 10.1360/n032016-00134.

24. Stasiuk GJ. Metal Complexes as T1 MRI Contrast Agents. *Elsevier eBooks*. 2020 Apr 27;:741-770. doi: 10.1016/b978-0-12-409547-2.14931-5.

25. Calatrava-Pérez E, Surender E, Truman LK, Máille GMÓ, Henwood AF, Scanlan EM, Gunnlaugsson T. Targeting Supramolecular Imaging Agents for a Wide Range of Applications. *Royal Society of Chemistry eBooks*. 2022 Apr 1;:1-42. doi: 10.1039/9781782624028-00001.

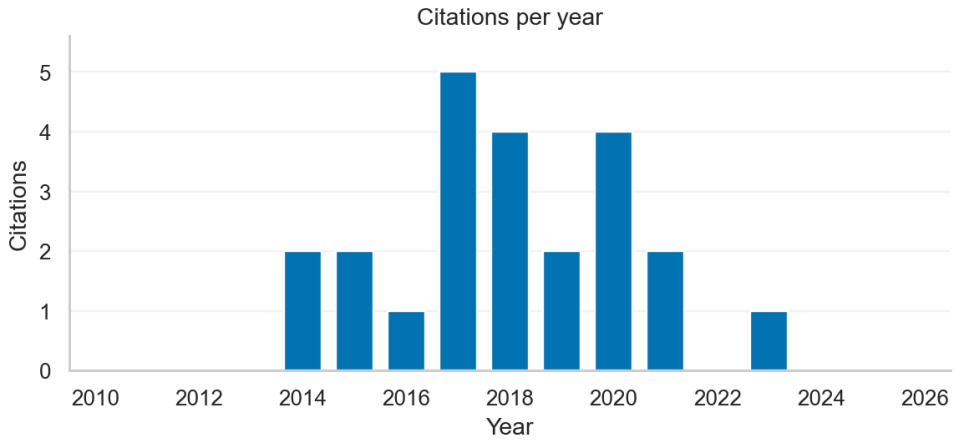
26. Nelson K, Winter PM, Shokeen M, Wang S, Berezin MY. Nanoparticles for Bioimaging. 2014 Sep 25;:151-192. doi: 10.1002/9781118873151.ch6.

27. Bonnet CS, Tóth É. MRI Contrast Agents. *HAL (Le Centre pour la Communication Scientifique Directe)*. 2014 May 2;:321-354. doi: 10.1002/9781118697191.ch12.

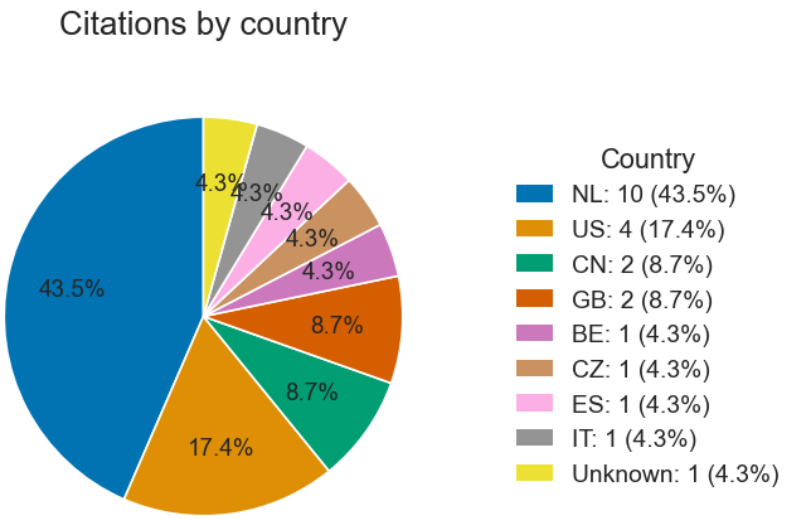
28. Sowers MA. Advancements in branched bottlebrush polymers for responsive, targeted imaging. *DSpace@MIT (Massachusetts Institute of Technology)*. 2014 Jan 1;.

Publication

Srinivas M, Boehm-Sturm P, Aswendt M, Pracht E, Figdor CG, de Vries IJ, Hoehn M. In vivo 19F MRI for Cell Tracking. *Journal of Visualized Experiments*. 2013 Nov 25;(81):e50802. doi: 10.3791/50802. PMID: 24299964.



Citation summary
Total citations: **23**
Average/year: **1.64**
Period: 2013-2026
SCImago cites/doc 2y (2025): **1.06**
SJR 2025: **0.409** (Q2)



Cited by

1. Swider E, Koshkina O, Tel J, Cruz LJ, de Vries IJM, Srinivas M. Customizing poly(lactic-co-glycolic acid) particles for biomedical applications. *Acta Biomaterialia*. 2018 Apr 10;73:38-51. doi: 10.1016/j.actbio.2018.04.006. PMID: 29653217.

2. Ottoboni L, von Wunster B, Martino G. Therapeutic Plasticity of Neural Stem Cells. *Frontiers in Neurology*. 2020 Mar 20;11:148. doi: 10.3389/fneur.2020.00148. PMID: 32265815.

3. Krekorian M, Fruhwirth GO, Srinivas M, Figdor CG, Heskamp S, Witney TH, Aarntzen EH. Imaging of T-cells and their responses during anti-cancer immunotherapy. *Theranostics*. 2019 Jan 1;9(25.0):7924-7947. doi: 10.7150/thno.37924. PMID: 31656546.

4. Staal AHJ, Becker K, Tagit O, van Riessen NK, Koshkina O, Veltien AA, Bouvain P, Cortenbach KR, Scheenen TWJ, Flögel U, Temme S, Srinivas M. In vivo clearance of 19F MRI imaging nanocarriers is strongly influenced by nanoparticle ultrastructure. *Biomaterials*. 2020 Aug 12;261:120307. doi: 10.1016/j.biomaterials.2020.120307. PMID: 32927288.

5. Zheng Y, Huang J, Zhu T, Li R, Wang Z, Ma F, Zhu J. Stem Cell Tracking Technologies for Neurological Regenerative Medicine Purposes. *Stem Cells International*. 2017 Jan 1;2017:1-9. doi: 10.1155/2017/2934149. PMID: 29138636.

6. Liu Z, Li Z. Molecular Imaging in Tracking Tumor-Specific Cytotoxic T Lymphocytes (CTLs). *Theranostics*. 2014 Jan 1;4(10.0):990-1001. doi: 10.7150/thno.9268. PMID: 25157278.

7. Scarfe L, Brillant N, Kumar JD, Ali N, Alrumayh A, Amali MO, Barbellion S, Jones V, Niemeijer M, Potdevin S, Roussignol G, Vaganov A, Barbaric I, Barrow M, Burton NC, Connell JJ, Dazzi F, Edsbagge J, French NS, Holder JC, Hutchinson C, Jones D, Kalber TL, Lovatt CA, Lythgoe MF, Patel S, Patrick PS, Piner J, Reinhardt J, Ricci E, Sidaway JE, Stacey GN, Lewis PJS, Sullivan GJ, Taylor A, Wilm B, Poptani H, Murray PA, Goldring CE, Park BK. Preclinical imaging methods for assessing the safety and efficacy of regenerative medicine therapies. *npj Regenerative Medicine*. 2017 Oct 11;2(1.0):28. doi:

10.1038/s41536-017-0029-9. PMID: 29302362.

8. Fernández-Susavila H, Bugallo-Casal A, Castillo J, Campos F. Adult Stem Cells and Induced Pluripotent Stem Cells for Stroke Treatment. *Frontiers in Neurology*. 2019 Aug 28;10:908. doi: 10.3389/fneur.2019.00908. PMID: 31555195.

9. Iafrate M, Fruhwirth GO. How Non-invasive in vivo Cell Tracking Supports the Development and Translation of Cancer Immunotherapies. *Frontiers in Physiology*. 2020 Apr 3;11:154. doi: 10.3389/fphys.2020.00154. PMID: 32327996.

10. Swider E, Daoudi K, Staal AHJ, Koshkina O, van Riessen NK, van Dinther EAW, de Vries IJM, de Korte CL, Srinivas M. Clinically-Applicable Perfluorocarbon-Loaded Nanoparticles For In vivo Photoacoustic, 19F Magnetic Resonance And Fluorescent Imaging. *Nanotheranostics*. 2018 Jan 1;2(3.0):258-268. doi: 10.7150/ntno.26208. PMID: 29868350.

11. Bernsen MR, Guenoun J, van Tiel ST, Krestin GP. Nanoparticles and clinically applicable cell tracking. *British Journal of Radiology*. 2015 Aug 7;88(1054.0):20150375. doi: 10.1259/bjr.20150375. PMID: 26248872.

12. Gálisová A, Herynek V, Swider E, Sticová E, Pátíková A, Košinová L, Kříž J, Hájek M, Srinivas M, Jirák D. A Trimodal Imaging Platform for Tracking Viable Transplanted Pancreatic Islets In Vivo: F-19 MR, Fluorescence, and Bioluminescence Imaging. *Molecular Imaging and Biology*. 2018 Aug 30;21(3.0):454-464. doi: 10.1007/s11307-018-1270-3. PMID: 30167995.

13. Liang S, Dresselaers T, Louchami K, Zhu C, Liu Y, Himmelreich U. Comparison of different compressed sensing algorithms for low SNR 19F MRI applications-Imaging of transplanted pancreatic islets and cells labeled with perfluorocarbons. *NMR in Biomedicine*. 2017 Aug 25;30(11.0). doi: 10.1002/nbm.3776. PMID: 28841762.

14. Maxouri O, Bodalal Z, Daal MRR, Rostami S, Rodriguez I, Akkari L, Srinivas M, Bernards R, Beets-Tan RGH. How to 19F MRI: applications, technique, and getting started. *BJR|Open*. 2023 Sep 25;5(1.0):20230019. doi: 10.1259/bjro.20230019. PMID: 37953866.

15. Bartusik-Aebisher D, Aebisher D. 19F applications in drug development and imaging - a review. *Biomedicine & Pharmacotherapy*. 2014 Jul 1;68(6.0):813-817. doi: 10.1016/j.biopha.2014.07.012. PMID: 25107839.

16. Constantinides C, Maguire ML, Stork L, Swider E, Srinivas M, Carr CA, Schneider JE. Temporal accumulation and localization of isoflurane in the C57BL/6 mouse and assessment of its potential contamination in 19F MRI with perfluoro-crown-ether-labeled cardiac progenitor cells at 9.4 Tesla. *Journal of Magnetic Resonance Imaging*. 2016 Dec 19;45(6.0):1659-1667. doi: 10.1002/jmri.25564. PMID: 27990708.

17. Xiao Z, Puré E. Imaging of T-cell Responses in the Context of Cancer Immunotherapy. *Cancer Immunology Research*. 2021 May 1;9(5.0):490-502. doi: 10.1158/2326-6066.cir-20-0678. PMID: 33941536.

18. Zheng L, Wang Y, Yang B, Zhang B, Wu Y. Islet Transplantation Imaging in vivo. *Diabetes Metabolic Syndrome and Obesity*. 2020 Sep 1;Volume 13:3301-3311. doi: 10.2147/dmso.s263253. PMID: 33061492.

19. Mastrogiacomo S, Dou W, Koshkina O, Boerman OC, Jansen JA, Heerschap A, Srinivas M, Walboomers XF. Perfluorocarbon/Gold Loading for Noninvasive in Vivo Assessment of Bone Fillers Using 19F Magnetic Resonance Imaging and Computed Tomography. *ACS Applied Materials & Interfaces*. 2017 Jun 21;9(27.0):22149-22159. doi: 10.1021/acsami.7b04075. PMID: 28635249.

20. Kadyk LC, DeWitt ND, Gomperts BN. Proceedings: Regenerative Medicine for Lung Diseases: A CIRM Workshop Report. *Stem Cells Translational Medicine*. 2017 Aug 9;6(10.0):1823-1828. doi: 10.1002/sctm.17-0047. PMID: 28791807.

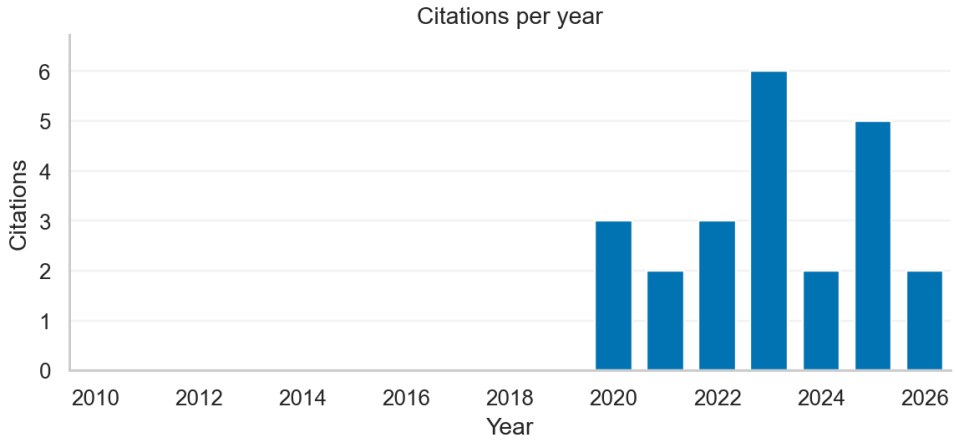
21. Staal X, Koshkina O, Srinivas M. In vivo 19-fluorine magnetic resonance imaging. *Elsevier eBooks*. 2018 Sep 28;:397-424. doi: 10.1016/b978-0-12-812733-9.00011-8.

22. Staal AHJ, Veltien AA, Srinivas M, Scheenen TWJ. 19F MRI Imaging Strategies to Reduce Isoflurane Artifacts in In Vivo Images. *Molecular Imaging and Biology*. 2021 Oct 20;24(1.0):71-81. doi: 10.1007/s11307-021-01653-6. PMID: 34671928.

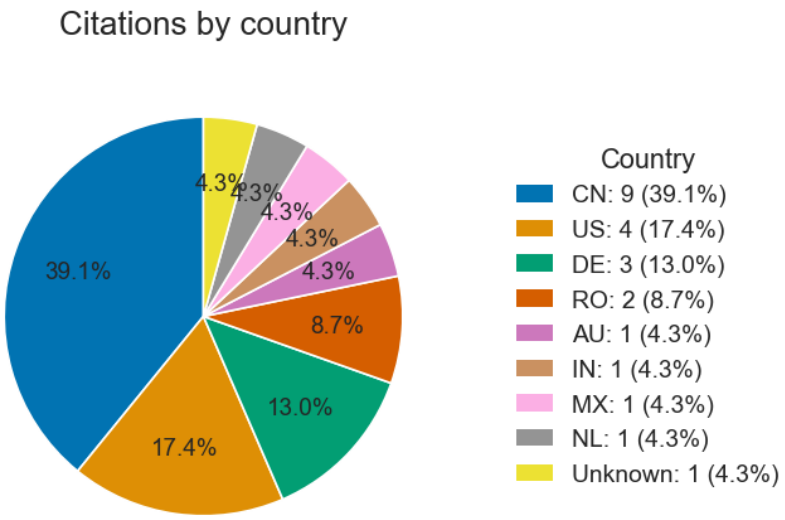
23. Bernsen MR, Guenoun J, van Tiel ST, Krestin GP, Mc E. NANOPARTICLES FOR DIAGNOSTIC IMAGING AND RADIOTHERAPY SPECIAL FEATURE: REVIEW ARTICLE Nanoparticles and clinically applicable cell tracking. 2015 Jan 1;.

Publication

Glover JC, Aswendt M, Boulland J, Lojk J, Stamenković S, Anđjus PR, Fiori F, Hoehn M, Mitrečić D, Pavlin M, Cavalli S, Frati C, Quaini F, (BIONECA) OBOTECA1. In vivo Cell Tracking Using Non-invasive Imaging of Iron Oxide-Based Particles with Particular Relevance for Stem Cell-Based Treatments of Neurological and Cardiac Disease. *Molecular Imaging and Biology*. 2019 Dec 4;22(6):1469-1488. doi: 10.1007/s11307-019-01440-4. PMID: 31802361.



Citation summary
Total citations: **23**
Average/year: **2.88**
Period: 2019-2026
SCImago cites/doc 2y (2025): **2.79**
SJR 2025: **0.74** (Q2)



Cited by

1. Rowe SP, Pomper MG. Molecular imaging in oncology: Current impact and future directions. *CA A Cancer Journal for Clinicians*. 2021 Dec 13;72(4.0):333-352. doi: 10.3322/caac.21713. PMID: 34902160.

2. Channa GM, Hassan D, Zeeshan M, Ruiz-Pulido G, Ebrahimi N, Mobashar A, Pourmadadi M, Rahdar A, Sargazi S, Fathi-karkan S, Medina DI, Diez-Pascual AM. Advances in nanotechnology versus stem cell therapy for the theranostics of Huntington's disease. *Journal of Drug Delivery Science and Technology*. 2023 Jul 19;87:104774. doi: 10.1016/j.jddst.2023.104774.

3. Zhao W, Li C, Chang J, Zhou H, Wang D, Sun J, Liu T, Peng H, Wang Q, Li Y, Whittaker AK. Advances and prospects of RAFT polymerization-derived nanomaterials in MRI-assisted biomedical applications. *Progress in Polymer Science*. 2023 Sep 18;146:101739. doi: 10.1016/j.progpolymsci.2023.101739.

4. Nigam S, Gjelaj E, Wang R, Wei G, Wang P. Machine Learning and Deep Learning Applications in Magnetic Particle Imaging. *Journal of Magnetic Resonance Imaging*. 2024 Feb 15;61(1.0):42-51. doi: 10.1002/jmri.29294. PMID: 38358090.

5. Gong Z, He Y, Zhou M, Xin H, Pan M, Fiaz M, Liu H, Yan F. Ultrasound imaging tracking of mesenchymal stem cells intracellularly labeled with biosynthetic gas vesicles for treatment of rheumatoid arthritis. *Theranostics*. 2022 Jan 1;12(5.0):2370-2382. doi: 10.7150/thno.66905. PMID: 35265215.

6. Schäfer R, Schwab MG, Siegel G, von Ameln-Mayerhofer A, Buadze M, Lourhmati A, Wendel H, Kluba T, Krueger MA, Calaminus C, Scheer E, Dominici M, Grisendi G, Doeppner TR, Schlechter J, Finzel AK, Gross D, Klaffschenkel RA, Gehring FK, Spohn G, Kretschmer A, Bieback K, Krämer-Albers E, Barth K, Eckert A, Elser S, Schmehl J, Claussen CD, Seifried E,

Hermann DM, Northoff H, Danielyan L. Modulating endothelial adhesion and migration impacts stem cell therapies efficacy. *EBioMedicine*. 2020 Sep 14;60:102987. doi: 10.1016/j.ebiom.2020.102987. PMID: 32942121.

7. Yan X, Li S, Yan H, Yu C, Liu F. IONPs-Based Medical Imaging in Cancer Care: Moving Beyond Traditional Diagnosis and Therapeutic Assessment. *International Journal of Nanomedicine*. 2023 Mar 31;Volume 18:1741-1763. doi: 10.2147/ijn.s399047. PMID: 37034271.

8. Yang X, Tian D, He W, Lv W, Fan J, Li H, Jin W, Meng X. Cellular and molecular imaging for stem cell tracking in neurological diseases. *Stroke and Vascular Neurology*. 2020 Oct 29;6(1.0):121-127. doi: 10.1136/svn-2020-000408. PMID: 33122254.

9. Smeets LCC, Sengun E, Trayford C, van Cranenbroek B, de Jonge MI, Dallaglio K, Hütten MC, Schoberer M, Ophelders DRMG, Wolfs TGAM, Molen RG, van Rijt S. Gold Mesoporous Silica-Coated Nanoparticles for Quantifying and Qualifying Mesenchymal Stem Cell Distribution; a Proof-of-Concept Study in Large Animals. *ACS Applied Bio Materials*. 2025 Feb 3;8(2.0):1511-1523. doi: 10.1021/acsabm.4c01714. PMID: 39900538.

10. Braidy N, Wen WX, Bongers A, Sachdev PS. Applications of magnetic particle imaging in the dementias. *Current Opinion in Psychiatry*. 2020 Nov 13;34(2.0):186-192. doi: 10.1097/yco.0000000000000668. PMID: 33196564.

11. Oberländer J, Ayerbe R, Cabellos J, Marques RDC, Li B, Günday-Türeli N, Türeli AE, Ofir R, Ish-Shalom E, Mailänder V. Higher Loading of Gold Nanoparticles in PAD Mesenchymal-like Stromal Cells Leads to a Decreased Exocytosis. *Cells*. 2022 Jul 28;11(15.0):2323. doi: 10.3390/cells11152323. PMID: 35954168.

12. Li X, Qin S, Liao X, Yang L, Liu A. Magnetic nanoparticles influence the biological function of mesenchymal stem cells. *Scientific Reports*. 2025 Jul 30;15(1.0):27862. doi: 10.1038/s41598-025-13083-3. PMID: 40739120.

13. Xue R, Liu Z, Liu L, Sun S, Gong Z. Ultrasound Imaging of Macrophages Intracellularly Labelled with Biosynthetic Gas Vesicles. *Molecular Imaging and Biology*. 2024 Aug 28;26(5.0):761-767. doi: 10.1007/s11307-024-01946-6. PMID: 39198330.

14. Xia C, Wang J, Bu R, Hu X. Ultrasonically targeted destruction of SDF1 nanobubbles carrying mesenchymal stem cells improves renal tubulointerstitial fibrosis. *International Journal of Pharmaceutics*. 2025 Dec 29;690:126542. doi: 10.1016/j.ijpharm.2025.126542. PMID: 41475473.

15. Nekolla SG, Saraste A, Sager HB, Makowski MR, Schwaiger M. Congestive Heart Failure. *Molecular Imaging*. 2021 Jan 1;:1167-1191. doi: 10.1016/b978-0-12-816386-3.00050-8.

16. Herea D, Labusca LS, Lupu N, Chiriac H. Magnetic particles for drug delivery. *Elsevier eBooks*. 2023 Jan 1;:259-304. doi: 10.1016/b978-0-12-823294-1.00002-6.

17. Ma X, Henry K, Healy KM, Luo TD, Smith TL, Li Z. In Vivo Tracking of Amniotic Fluid Derived Stem Cells on Acellular Nerve Graft. *The Journal Of Hand Surgery*. 2025 Mar 22;51(1.0):103.e1-103.e8. doi: 10.1016/j.jhsa.2025.02.012. PMID: 40117437.

18. Shaikh S. Molecular Imaging and Stem Cell Imaging. *Springer eBooks*. 2022 Jan 1;:237-256. doi: 10.1007/978-981-16-9535-3_20.

19. Iron imaging in neuroinflammation. *Advances in magnetic resonance technology and applications*. 2023 Jan 1;:51-78. doi: 10.1016/b978-0-323-91771-1.00013-7.

20. Labusca LS, Herea D, Chiriac H, Lupu N. Magnetic sensors for regenerative medicine. *Elsevier eBooks*. 2023 Jan 1;:401-433. doi: 10.1016/b978-0-12-823294-1.00012-9.

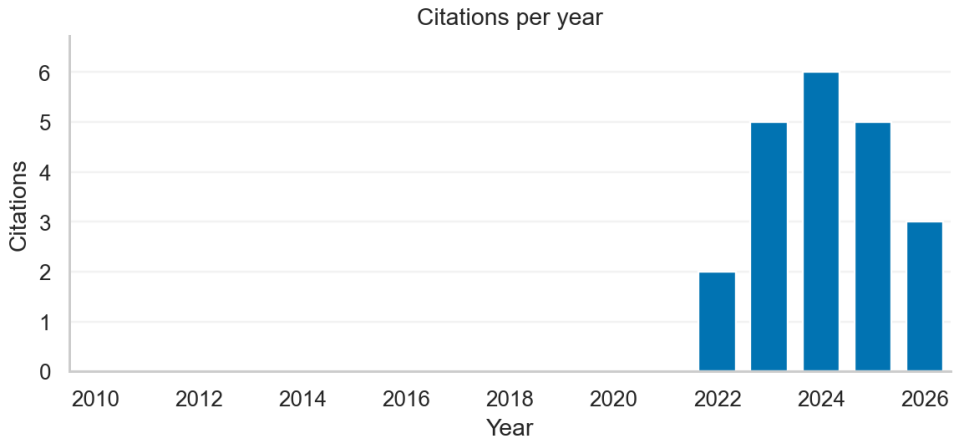
21. Cruz KA, Zhang R, Ma X. MRI Tracking of MPIO-Labeled Amniotic Fluid-Derived Stem Cells on an Acellular Nerve Graft in a Rat Model. *Methods in molecular biology*. 2025 Jan 1;. doi: 10.1007/7651_2025_686. PMID: 41288959.

22. Zhang P, Lin Z, Cheng X, Zhao X, Tang H, Jiao M, Jing L, Luo X. Advancing traumatic brain injury diagnosis through nanomaterial-based imaging technologies. *Nano Today*. 2026 Mar 21;69:103028. doi: 10.1016/j.nantod.2026.103028.

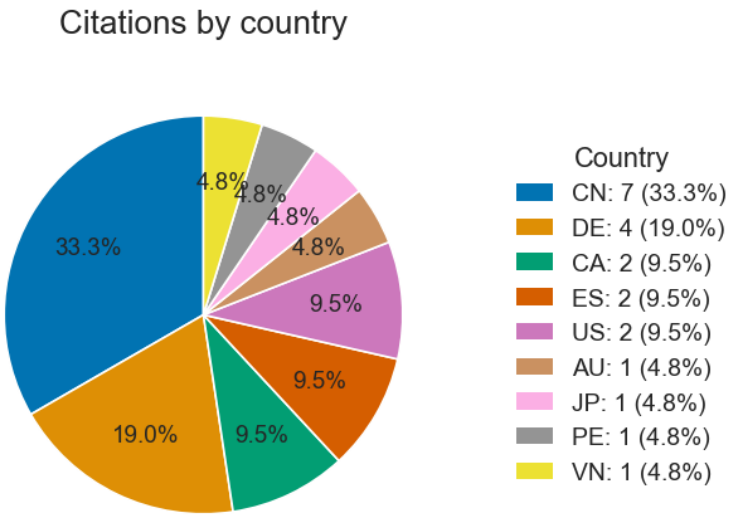
23. Sun Z, Zhao J, Li J, Ma X, Wang Z, Wang Y, Y L, Bai Y, Wei D, Jiang W. Biomaterial platforms for enhanced stem cell therapy in ischemic stroke. *Biomaterials*. 2026 Jul 11;336:124427. doi: 10.1016/j.biomaterials.2026.124427. PMID: 42468381.

Publication

Scharwächter L, Schmitt FJ, Pallast N, Fink GR, Aswendt M. Network analysis of neuroimaging in mice. *NeuroImage*. 2022 Mar 17;253:119110. doi: 10.1016/j.neuroimage.2022.119110. PMID: 35311664.



Citation summary
Total citations: **21**
Average/year: **4.20**
Period: 2022-2026
SCImago cites/doc 2y (2025): **5.31**
SJR 2025: **1.93** (Q1)



Cited by

1. Hoehn M, Aswendt M. Functional hyperconnectivity related to brain disease: maladaptive process or element of resilience?. *Neural Regeneration Research*. 2022 Nov 18;18.0(7.0):1489. doi: 10.4103/1673-5374.361541. PMID: 36571347.

2. Stapleton MC, Koch S, Cortes DRE, Wyman SK, Schwab KE, Mueller SG, McKennan CG, Boehm-Sturm P, Wu YL. Apolipoprotein-E deficiency leads to brain network alteration characterized by diffusion MRI and graph theory. *Frontiers in Neuroscience*. 2023 Nov 21;17.0:1183312. doi: 10.3389/fnins.2023.1183312. PMID: 38075287.

3. Mellado S, Touahri N, Montagud-Romero S, Perpiñá-Clérigues C, Garcia-García F, Moreno-Manzano V, Guerri C, Rodríguez-Arias M, Pascual M. Role of miRNAs from mesenchymal stem cell-derived extracellular vesicles in neuroinflammation and behavioral impairments induced by chronic alcohol consumption in female mice. *Neural Regeneration Research*. 2025 Jun 19;21.0(6.0):2370-2379. doi: 10.4103/nrr.nrr-d-24-01260. PMID: 40537146.

4. Lee BH, McGovern AJ, Lieblich SE, Mitchell SV, Yin J, Epp JR, Galea LA. Sex-specific factors and APOEε4 genotype alter functional connectivity at middle age. *Psychoneuroendocrinology*. 2025 Oct 24;183.0:107655. doi: 10.1016/j.psyneuen.2025.107655. PMID: 41160972.

5. Kimura I, Senoo A, Abo M. Changes in Structural Neural Networks in the Recovery Process of Motor Paralysis after Stroke. *Brain Sciences*. 2024 Feb 21;14.0(3.0):197. doi: 10.3390/brainsci14030197. PMID: 38539586.

6. Sun D, Shaik NEK, Unnithan RR, French CR. Hippocampal cognitive and relational map paradigms explored by multisensory encoding recording with wide-field calcium imaging. *iScience*. 2023 Nov 30;27.0(1.0):108603. doi: 10.1016/j.isci.2023.108603. PMID: 38094852.

7. Sun Q, Hu Y, Deng S, Xiong Y, Huang Z. A visualization pipeline for in vivo two-photon volumetric astrocytic calcium imaging. *Journal of Biomedical Research*. 2022 Jan 1;36.0(5.0):358. doi: 10.7555/jbr.36.20220099. PMID: 36130733.

8. Fuentes-Mendoza JM, Fernanda-Castillo M, Valdez-Esqueda F, Garza-Boullosa R, Gama-Reyes EG, Arreola MF, Concepción-Zavaleta MJ. Determining factors of neuroplasticity in patients with brain tumors: impact of connectome and artificial intelligence. A narrative review. *Expert Review of Anticancer Therapy*. 2025 Oct 31;26.0(2.0):179-189. doi: 10.1080/14737140.2025.2583399. PMID: 41169209.

9. Mahani FSN, Kalantari A, Fink GR, Hoehn M, Aswendt M. A systematic review of the relationship between magnetic resonance imaging based resting-state and structural networks in the rodent brain. *Frontiers in Neuroscience*. 2023 Jul 24;17.0:1194630. doi: 10.3389/fnins.2023.1194630. PMID: 37554291.

10. Niu B, Wu H, Li Y, Klugah-Brown B, Hanna GP, Yao Y, Jing J, Baig TI, Xia Y, Yao D, Biswal BB. Topological functional network analysis of cortical blood flow in hyperacute ischemic rats. *Brain Structure and Function*. 2024 Dec 26;230.0(1.0):20. doi: 10.1007/s00429-024-02864-7. PMID: 39724244.

11. Wang H, Zhao H, Li C, Dong J, Zhao J, Yue H, Lai Y, Zhao L, Wang H, Zhang J, Xu X, Yao B, Zhou H, Nie B, Du X, Peng R. Disrupted Topological Organization of Brain Network in Rats with Spatial Memory Impairments Induced by Acute Microwave Radiation. *Brain Sciences*. 2023 Jun 28;13.0(7.0):1006. doi: 10.3390/brainsci13071006. PMID: 37508937.

12. Jin S, Li J, Wang J. Brain connectome from neuronal morphology. *Network Neuroscience*. 2025 Jan 1;9.0(3.0):913-937. doi: 10.1162/netn_a_00458. PMID: 41132688.

13. Wang J, Jin S, Li J. Brain connectome from neuronal morphology. *Research Square*. 2024 Feb 27;. doi: 10.21203/rs.3.rs-3913903/v1.

14. Wendlandt TA, Wenk P, Henschke JU, Michalek A, Noesselt T, Pakan JM, Budinger E. Neural networks representing temporal expectation in mice. *bioRxiv (Cold Spring Harbor Laboratory)*. 2024 Apr 26;. doi: 10.1101/2024.04.25.589683.

15. Altahini S, Doering J, Kuchling J, Backhaus H, Kreye J, Backhaus RG, Finke C, Prüß H, Stroh A. Maternofetal transfer of human NMDAR-antibodies leads to cortical network defect in adolescence. *bioRxiv (Cold Spring Harbor Laboratory)*. 2023 Dec 1;. doi: 10.1101/2023.11.30.569370.

16. Lv K, Zhang C, Liu B, Yang A, Luan J, Hu P, Yao Z, Liu J, Ma G. White matter structural changes before and after microvascular decompression for hemifacial spasm. *Brain Structure and Function*. 2024 Mar 19;229.0(4.0):959-970. doi: 10.1007/s00429-023-02741-9. PMID: 38502329.

17. Mellado S, Touahri N, Montagud-Romero S, Perpiñá-Clérigues C, Garcia-García F, Moreno-Manzano V, Guerri C, Rodríguez-Arias M, Pascual M. Unveiling the role of miRNAs from MSC-EVs in neuroinflammation and behavioral impairments induced by chronic alcohol consumption. *bioRxiv (Cold Spring Harbor Laboratory)*. 2024 Oct 3;. doi: 10.1101/2024.10.03.616225.

18. Hoang NT, Dilhani N, Walpola T, Kulathilake CD, Masahiro A, Senoo A. Age-Related Distinct Patterns of Global and Language Network Nodes in Fluent and Non-Fluent Aphasia: A Graph Theory Analysis of Diffusion Tensor Imaging Tractography. *Investigative Magnetic Resonance Imaging*. 2025 Jan 1;29.0(4.0):201. doi: 10.13104/imri.2025.0027.

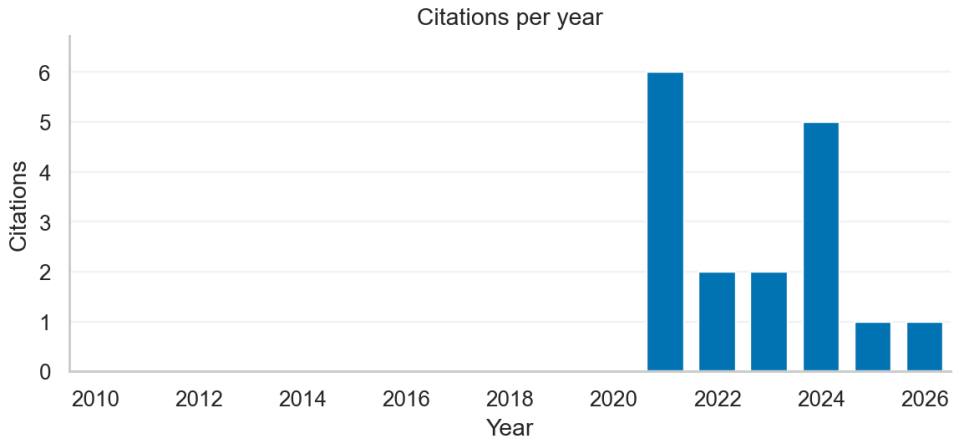
19. West D, Coulson NW, Cortes DRE, Schwab KE, Becker-Szurszewski T, Hartwick S, Stapleton MC, Saladino GM, Lo CW, Patterson CM, Subramanian S, Rajan DS, Wu YL. High-Definition Brain Network (HDBN) Delineation of CDKL5 Deficiency Disorder (CDD) in Genetically Engineered Mice. *Biomolecules*. 2026 Apr 28;16.0(5.0):652. doi: 10.3390/biom16050652. PMID: 42194004.

20. Yazdani Z, Bélanger E, Moreaud M, Llinares J, Allard A, Marquet PP, Desrosiers P. Graph-based characterization of in vitro neuronal network maturation using machine learning and digital holographic microscopy. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Jun 23;. doi: 10.64898/2026.06.18.732973.

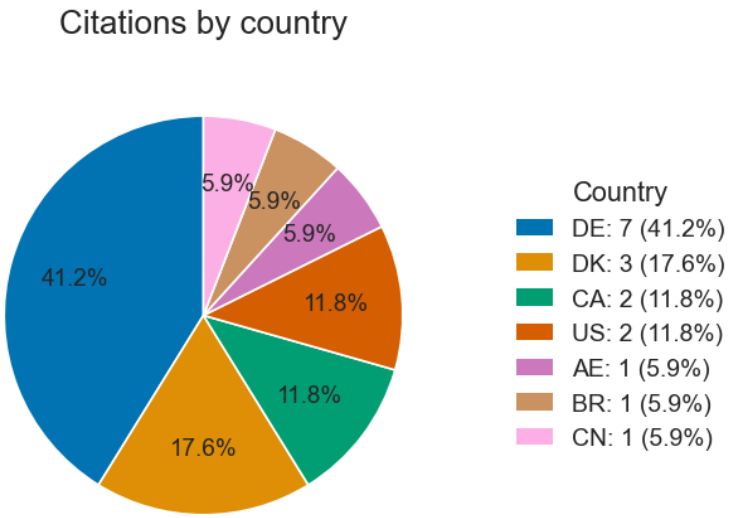
21. Da Wang, Li H, Zeng Y, Gao J, Xu M, Bo B, Pei M, Liang Z, Zhou N, Thompson GJ. Macroscopic cerebral energy efficiency corresponds to neuron reorganization in awake and anesthetized mice. *Frontiers in Neuroscience*. 2026 Oct 9;20.0. doi: 10.3389/fnins.2026.1949911.

Publication

Apetz N, Kordys E, Simon M, Mang B, Aswendt M, Wiedermann D, Neumaier B, Drzezga AE, Timmermann L, Endepols H. Effects of subthalamic deep brain stimulation on striatal metabolic connectivity in a rat hemiparkinsonian model. *Disease Models & Mechanisms*. 2019 May 1;12(5). doi: 10.1242/dmm.039065. PMID: 31064773.



Citation summary
Total citations: **17**
Average/year: **2.12**
Period: 2019-2026
SCImago cites/doc 2y (2025): **2.61**
SJR 2025: **1.31** (Q1)



Cited by

1. Miterko LN, Lin T, Zhou J, van der Heijden ME, Beckinghausen J, White JJ, Sillitoe RV. Neuromodulation of the cerebellum rescues movement in a mouse model of ataxia. *Nature Communications*. 2021 Feb 26;12.0(1.0):1295. doi: 10.1038/s41467-021-21417-8. PMID: 33637754.

2. Zhang KK, Matin R, Gorodetsky C, Ibrahim GM, Gouveia FV. Systematic review of rodent studies of deep brain stimulation for the treatment of neurological, developmental and neuropsychiatric disorders. *Translational Psychiatry*. 2024 Apr 11;14.0(1.0):186. doi: 10.1038/s41398-023-02727-5. PMID: 38605027.

3. Casado-Sainz A, Gudmundsen F, Bærentzen SL, Lange D, Ringsted A, Tejada IM, Medina S, Lee H, Svarer C, Keller SH, Schain M, Kjærby C, Fisher PM, Cumming PD, Palner M. Dorsal striatal dopamine induces fronto-cortical hypoactivity and attenuates anxiety and compulsive behaviors in rats. *Neuropsychopharmacology*. 2021 Nov 1;47.0(2.0):454-464. doi: 10.1038/s41386-021-01207-y. PMID: 34725486.

4. Knörr S, Musacchio T, Paulat R, Matthies C, Endres H, Wenger N, Harms C, Ip CW. Experimental deep brain stimulation in rodent models of movement disorders. *Experimental Neurology*. 2021 Nov 15;348.0:113926. doi: 10.1016/j.expneurol.2021.113926. PMID: 34793784.

5. Grotemeyer A, Petschner T, Peach RL, Hoehl D, Knauer T, Thomas U, Endres H, Blum R, Sendtner M, Volkmann J, Ip CW. Standardized wireless deep brain stimulation system for mice. *npj Parkinson s Disease*. 2024 Aug 14;10.0(1.0):153. doi: 10.1038/s41531-024-00767-2. PMID: 39143106.

6. Williams D. Why so slow? Models of parkinsonian bradykinesia. *Nature reviews. Neuroscience*. 2024 Jun 27;25.0(8.0):573-586. doi: 10.1038/s41583-024-00830-0. PMID: 38937655.

7. Ruiz MCM, Guimarães RP, Mortari MR. Parkinson’s disease rodent models: Are they suitable for DBS research?. *Journal of Neuroscience Methods*. 2022 Aug 5;380.0:109687. doi: 10.1016/j.jneumeth.2022.109687. PMID: 35940355.

8. Andree A, Li N, Butenko K, Kober M, Chen JZ, Higuchi T, Fauser M, Storch A, Ip CW, Kühn AA, Horn A, van Rienen U. Deep brain stimulation electrode modeling in rats. *Experimental Neurology*. 2022 Jan 10;350.0:113978. doi: 10.1016/j.expneurol.2022.113978. PMID: 35026227.

9. Real CC, Binda KH, Thomsen MB, Lillethorup TP, Brooks DJ, Landau AM. Selecting the Best Animal Model of Parkinson’s Disease for Your Research Purpose: Insight from in vivo PET Imaging Studies. *Current Neuropharmacology*. 2023 Feb 17;21.0(5.0):1241-1272. doi: 10.2174/1570159x21666230216101659. PMID: 36797611.

10. Endepols H, Zlatopolskiy BD, Zischler J, Alavinejad N, Apetz N, Vus S, Drzezga AE, Neumaier B. Imaging of cerebral tryptophan metabolism using 7-[18F]FTrp-PET in a unilateral Parkinsonian rat model. *NeuroImage*. 2021 Dec 20;247.0:118842. doi: 10.1016/j.neuroimage.2021.118842. PMID: 34942366.

11. Wu Z, Ren Z, Gao R, Sun K, Sun F, Liu T, Zheng S, Wang W, Zhang G. Impact of subthalamic nucleus deep brain stimulation at different frequencies on neurogenesis in a rat model of Parkinson's disease. *Heliyon*. 2024 May 1;10.0(10.0):e30730. doi: 10.1016/j.heliyon.2024.e30730. PMID: 38784548.

12. Endepols H, Anglada-Huguet M, Mandelkow E, Neumaier B, Mandelkow E, Drzezga AE. Fragmentation of functional resting state brain networks in a transgenic mouse model of tau pathology: A metabolic connectivity study using [18F]FDG-PET. *Experimental Neurology*. 2023 Dec 3;372.0:114632. doi: 10.1016/j.expneurol.2023.114632. PMID: 38052272.

13. Endepols H, Apetz N, Vieth L, Lesser C, Schulte-Holtey L, Neumaier B, Drzezga AE. Cerebellar Metabolic Connectivity during Treadmill Walking before and after Unilateral Dopamine Depletion in Rats. *International Journal of Molecular Sciences*. 2024 Aug 7;25.0(16.0):8617. doi: 10.3390/ijms25168617. PMID: 39201305.

14. Apetz N, Paralikar KJ, Neumaier B, Drzezga AE, Wiedermann D, Iyer R, Munns G, Scott ER, Timmermann L, Endepols H. Towards chronic deep brain stimulation in freely moving hemiparkinsonian rats: applicability and functionality of a fully implantable stimulation system. *Journal of Neural Engineering*. 2021 Feb 19;18.0(3.0):036018. doi: 10.1088/1741-2552/abe806. PMID: 33607640.

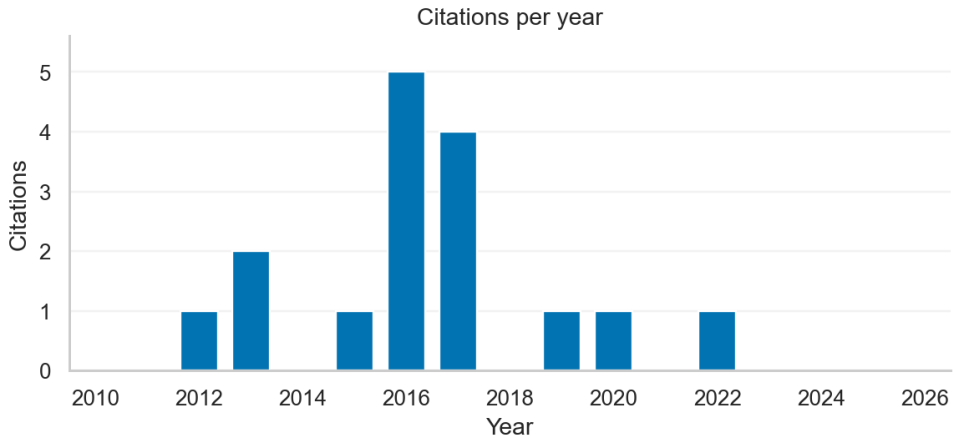
15. Casado-Sainz A, Gudmundsen F, Bærentzen SL, Lange D, Ringsted A, Tejada IM, Medina S, Lee H, Svarer C, Keller SH, Schain M, Kjærby C, Fisher PM, Cumming PD, Palner M. Dorsal striatal dopamine induces fronto-cortical hypoactivity and implies reduced anxiety and compulsive behaviors in rats. *bioRxiv (Cold Spring Harbor Laboratory)*. 2021 Feb 11;. doi: 10.1101/2021.02.11.430770.

16. Miterko LN, Peacoe LE, Duraine L, Zuo Z, Sillitoe RV. Cerebellar deep brain stimulation rescues Purkinje cell mitochondrial density in a genetic mouse model of cerebellar ataxia. *Brain Research Bulletin*. 2025 Dec 24;234.0:111704. doi: 10.1016/j.brainresbull.2025.111704. PMID: 41453498.

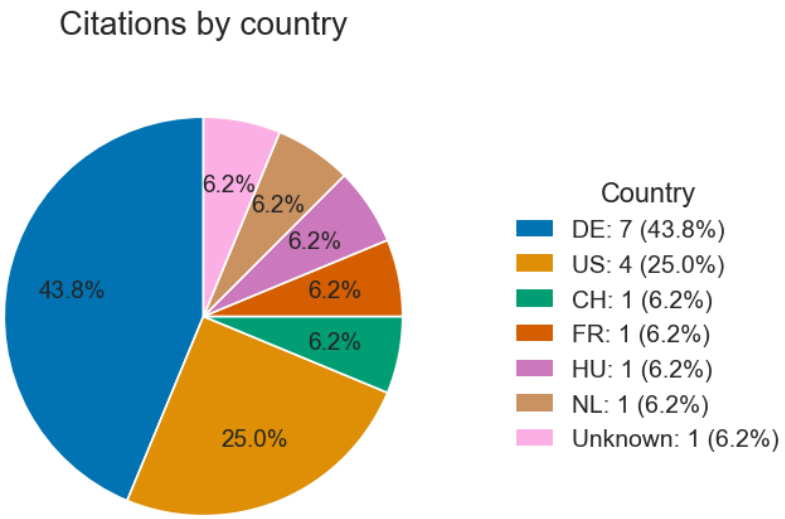
17. Campos ACP, Pagano RL, Hamani C. Neurorestorative and neuroimmune effects of deep brain stimulation. *Elsevier eBooks*. 2026 Jan 1;:55-73. doi: 10.1016/b978-0-323-99742-3.00003-5.

Publication

Aswendt M, Gianolio E, Pariani GM, Napolitano R, Fedeli F, Himmelreich U, Aime S, Hoehn M. In vivo imaging of inhibitory, GABAergic neurons by MRI. *NeuroImage*. 2012 May 17;62(3):1685-1693. doi: 10.1016/j.neuroimage.2012.05.029. PMID: 22609794.



Citation summary
Total citations: **16**
Average/year: **1.07**
Period: 2012-2026
SCImago cites/doc 2y (2025): **5.31**
SJR 2025: **1.93** (Q1)



Cited by

1. Sedgwick AC, Brewster JT, Harvey P, Iovan DA, Smith GST, He X, Tian H, Sessler JL, James TD. Metal-based imaging agents: progress towards interrogating neurodegenerative disease. *Chemical Society Reviews*. 2020 Jan 1;49.0(10.0):2886-2915. doi: 10.1039/c8cs00986d. PMID: 32226991.

2. Taj SH, Kho W, Aswendt M, Collmann FM, Green C, Adamczak JM, Tennstaedt A, Hoehn M. Dynamic Modulation of Microglia/Macrophage Polarization by miR-124 after Focal Cerebral Ischemia. *Journal of Neuroimmune Pharmacology*. 2016 Aug 18;11.0(4.0):733-748. doi: 10.1007/s11481-016-9700-y. PMID: 27539642.

3. Lacerda S, Tóth É. Lanthanide Complexes in Molecular Magnetic Resonance Imaging and Theranostics. *ChemMedChem*. 2017 May 4;12.0(12.0):883-894. doi: 10.1002/cmdc.201700210. PMID: 28470934.

4. Pallast N, Diedenhofen M, Blaschke SJ, Wieters F, Wiedermann D, Hoehn M, Fink GR, Aswendt M. Processing Pipeline for Atlas-Based Imaging Data Analysis of Structural and Functional Mouse Brain MRI (AIDAmri). *Frontiers in Neuroinformatics*. 2019 Jun 4;13.0:42. doi: 10.3389/fninf.2019.00042. PMID: 31231202.

5. Boehm-Sturm P, Aswendt M, Minassian A, Michalk S, Mengler L, Adamczak JM, Mezzanotte L, Löwik CW, Hoehn M. A multi-modality platform to image stem cell graft survival in the naïve and stroke-damaged mouse brain. *Biomaterials*. 2013 Dec 16;35.0(7.0):2218-2226. doi: 10.1016/j.biomaterials.2013.11.085. PMID: 24355489.

6. Tennstaedt A, Aswendt M, Adamczak JM, Collienne U, Selt M, Schneider G, Henn N, Schaefer C, Lagouge M, Wiedermann D, Kloppenburg P, Hoehn M. Human neural stem cell intracerebral grafts show spontaneous early neuronal differentiation after several weeks. *Biomaterials*. 2015 Jan 12;44.0:143-154. doi: 10.1016/j.biomaterials.2014.12.038. PMID: 25617134.

7. Cruz LJ, Que I, Aswendt M, Chan AB, Hoehn M, Löwik CW. Targeted nanoparticles for the non-invasive detection of traumatic brain injury by optical imaging and fluorine magnetic resonance imaging. *Nano Research*. 2016 Apr 14;9.0(5.0):1276-1289. doi: 10.1007/s12274-016-1023-z.

8. Uselman TW, Medina CS, Gray HB, Jacobs RE, Bearer EL. Longitudinal manganese-enhanced magnetic resonance imaging of neural projections and activity. *NMR in Biomedicine*. 2022 Mar 6;35.0(6.0):e4675. doi: 10.1002/nbm.4675. PMID: 35253280.

9. Napolitano R, Pariani GM, Fedeli F, Baranyai Z, Aswendt M, Aime S, Gianolio E. Synthesis and Relaxometric Characterization of a MRI Gd-Based Probe Responsive to Glutamic Acid Decarboxylase Enzymatic Activity. *Journal of Medicinal Chemistry*. 2013 Mar 8;56.0(6.0):2466-2477. doi: 10.1021/jm301831f. PMID: 23469759.

10. Vogel SN, Aswendt M, Nelles M, Henn N, Schneider G, Hoehn M. Initial graft size and not the innate immune response limit survival of engrafted neural stem cells. *Journal of Tissue Engineering and Regenerative Medicine*. 2017 Jun 9;12.0(3.0):784-793. doi: 10.1002/term.2497. PMID: 28599089.

11. Selt M, Tennstaedt A, Beyrau A, Nelles M, Schneider G, Löwik CW, Hoehn M. In Vivo Non-Invasive Tracking of Macrophage Recruitment to Experimental Stroke. *PLoS ONE*. 2016 Jun 24;11.0(6.0):e0156626. doi: 10.1371/journal.pone.0156626. PMID: 27341631.

12. Adamczak JM, Hoehn M. In vivo imaging of cell transplants in experimental ischemia. *Progress in brain research*. 2012 Jan 1;201.0:55-78. doi: 10.1016/b978-0-444-59544-7.00004-4. PMID: 23186710.

13. Schroeter A, Rudin M, Gianolio E, Viale A, Castelli DD, Aime S, Hövener J, Bastiaansen J, Comment A, Düwel S, Ardenkjær-Larsen JH, Becker M. MRI. *Springer eBooks*. 2017 Jan 1;:227-324. doi: 10.1007/978-3-319-42202-2_13.

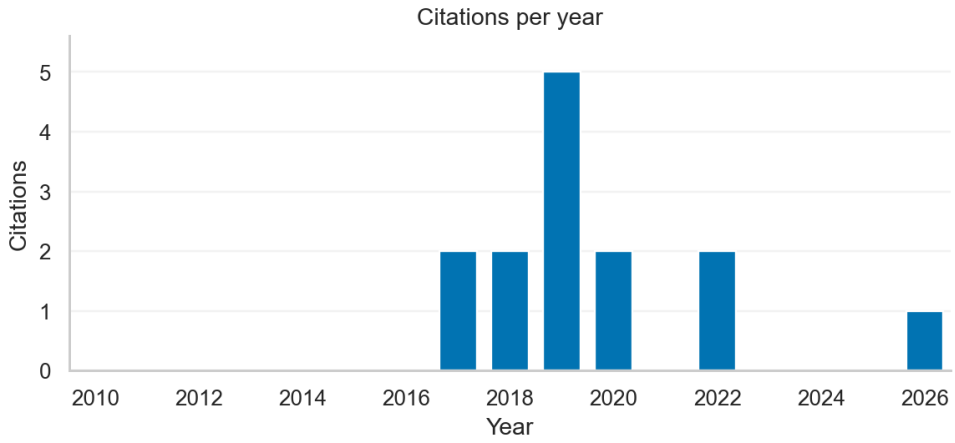
14. Modo M. Gaining Mechanistic Insights into Cell Therapy Using Magnetic Resonance Imaging. *Current Stem Cell Reports*. 2016 Jul 7;2.0(3.0):221-227. doi: 10.1007/s40778-016-0059-z.

15. Chen B, Liebman C, Rabbani P, Cho M. Stem Cell Tissue Engineering and Regenerative Medicine. 2017 Feb 8;:1-20. doi: 10.1002/9781119193272.ch1.

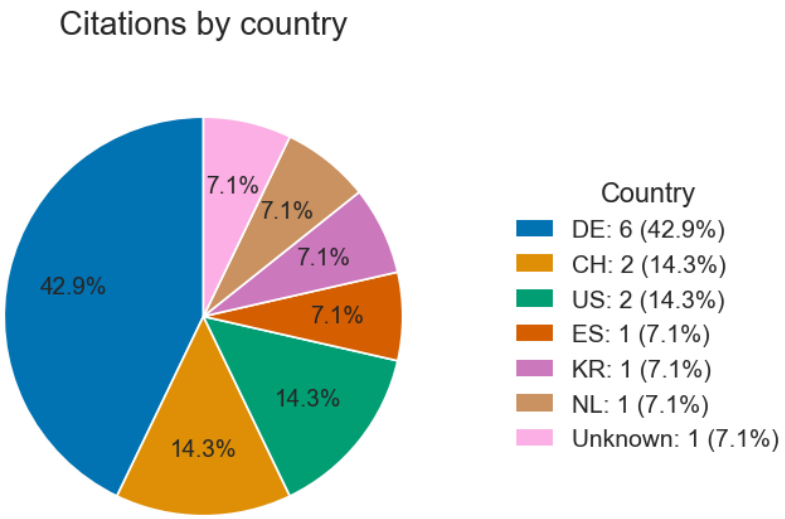
16. Aswendt M, Boehm-Sturm P, Hoehn M. Chapter 10 Neural Stem Cells. *Pan Stanford Publishing eBooks*. 2016 Oct 21;:283-310. doi: 10.1201/9781315364605-11.

Publication

Vogel SN, Aswendt M, Nelles M, Henn N, Schneider G, Hoehn M. Initial graft size and not the innate immune response limit survival of engrafted neural stem cells. *Journal of Tissue Engineering and Regenerative Medicine*. 2017 Jun 9;12(3):784-793. doi: 10.1002/term.2497. PMID: 28599089.



Citation summary
Total citations: **14**
Average/year: **1.40**
Period: 2017-2026
SCImago cites/doc 2y (2025): **1.76**
SJR 2025: **0.62** (Q2)



Cited by

1. Yun S, Shin T, Lee J, Cho MH, Kim I, Kim J, Jung K, Lee I, Cheon J, Park KI. Design of Magnetically Labeled Cells (Mag-Cells) for in Vivo Control of Stem Cell Migration and Differentiation. *Nano Letters*. 2018 Feb 2;18.0(2.0):838-845. doi: 10.1021/acs.nanolett.7b04089. PMID: 29393650.

2. Collmann FM, Pijnenburg R, Taj SH, Minassian A, Folz-Donahue K, Kukat C, Aswendt M, Hoehn M. Individual in vivo Profiles of Microglia Polarization After Stroke, Represented by the Genes iNOS and Ym1. *Frontiers in Immunology*. 2019 Jun 4;10.0:1236. doi: 10.3389/fimmu.2019.01236. PMID: 31214190.

3. Zenchak JR, Palmateer BL, Dorka N, Brown TM, Wagner L, Medendorp WE, Petersen ED, Prakash M, Hochgeschwender U. Bioluminescence-driven optogenetic activation of transplanted neural precursor cells improves motor deficits in a Parkinson's disease mouse model. *Journal of Neuroscience Research*. 2018 Mar 25;98.0(3.0):458-468. doi: 10.1002/jnr.24237. PMID: 29577367.

4. Aswendt M, Vogel S, Schäfer C, Jathoul AP, Pulé M, Hoehn M. Quantitative in vivo dual-color bioluminescence imaging in the mouse brain. *Neurophotonics*. 2019 May 7;6.0(2.0):1-11. doi: 10.1117/1.nph.6.2.025006. PMID: 31093514.

5. Minassian A, Green C, Diedenhofen M, Vogel SN, Heß S, Stoeber M, Radmilović MD, Wiedermann D, Kloppenburg P, Hoehn M. Human Neural Stem Cell Induced Functional Network Stabilization After Cortical Stroke: A Longitudinal Resting-State fMRI Study in Mice. *Frontiers in Cellular Neuroscience*. 2020 Apr 7;14.0:86. doi: 10.3389/fncel.2020.00086. PMID: 32317940.

6. Weber R, Bodenmann C, Uhr D, Zürcher KJ, Wanner D, Generali M, Nitsch RM, Rust R, Tackenberg C. Intracerebral Transplantation and In Vivo Bioluminescence Tracking of Human Neural Progenitor Cells in the Mouse Brain. *Journal of Visualized Experiments*. 2022 Jan 27;(179.0). doi: 10.3791/63102. PMID: 35156654.

7. Weber R, Bodenmann C, Uhr D, Zürcher KJ, Wanner D, Generali M, Nitsch RM, Rust R, Tackenberg C. Intracerebral Transplantation and In Vivo Bioluminescence Tracking of Human Neural Progenitor Cells in the Mouse Brain. *Journal of Visualized Experiments*. 2022 Jan 27;(179.0). doi: 10.3791/63102-v.

8. Vogel SN, Schäfer C, Heß S, Folz-Donahue K, Nelles M, Minassian A, Schwarz MK, Kukat C, Ehrlich M, Zaehres H, Kloppenburg P, Hoehn M, Aswendt M. The in vivo timeline of differentiation of engrafted human neural progenitor cells. *Stem Cell Research*. 2019 Mar 25;37.0:101429. doi: 10.1016/j.scr.2019.101429. PMID: 30933718.

9. Green C, Minassian A, Vogel SN, Diedenhofen M, Wiedermann D, Hoehn M. Persistent Quantitative Vitality of Stem Cell Graft Is Necessary for Stabilization of Functional Brain Networks After Stroke. *Frontiers in Neurology*. 2019 Apr 5;10.0:335. doi: 10.3389/fneur.2019.00335. PMID: 31024429.

10. Madsen SD, Giler MK, Bunnell BA, O'Connor KC. Illuminating the Regenerative Properties of Stem Cells In Vivo with Bioluminescence Imaging. *Biotechnology Journal*. 2020 Oct 22;16.0(3.0):e2000248. doi: 10.1002/biot.202000248. PMID: 33089922.

11. Aswendt M, Vogel S, Hoehn M. In vivo bioluminescence imaging to elucidate stem cell graft differentiation. *Neural Regeneration Research*. 2019 Sep 16;15.0(1.0):61. doi: 10.4103/1673-5374.264449. PMID: 31535647.

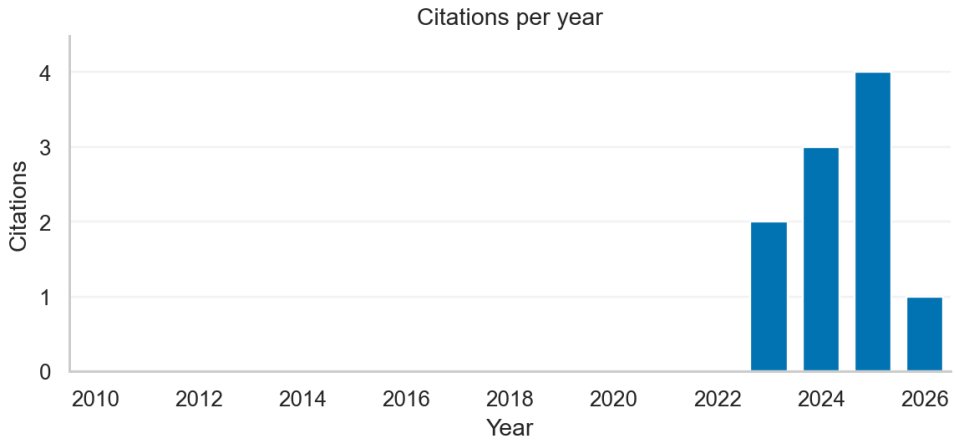
12. Aswendt M, Collmann FM, Hoehn M. Neurobiological insights from bioluminescence imaging. *Oncotarget*. 2017 Aug 13;8.0(41.0):69198-69199. doi: 10.18632/oncotarget.20302. PMID: 29050195.

13. Untitled. *Oncotarget*. 2017 Sep 19;8.0(41.0). doi: 10.18632/oncotarget.v8i41.

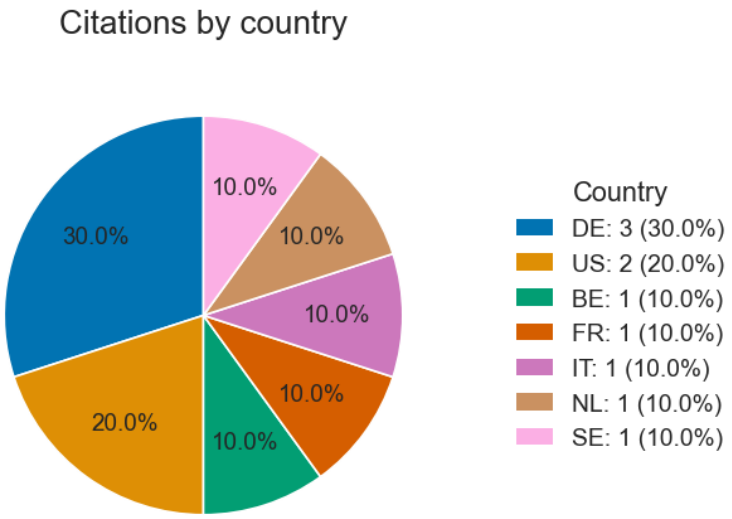
14. Rodriguez BP, Manero-Roig I, Salvador-Moya J, Basanta-Torres R, Martín-Aragón D, Hernández-Sánchez S, Luzuriaga J, Martín A, Lampin-Saint-Amaux A, Lanore F, Unda FJ, Ibarretxe G, Pineda JR. Human dental pulp stem cells grafted into C57BL/6J hippocampus differentiate towards immature neuronal like cells displaying action potential firing activity. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Jun 22;. doi: 10.64898/2026.06.16.732586.

Publication

Tavares A, Mezzanotte L, McDougald W, Bernsen MR, Vanhove C, Aswendt M, Ielacqua GD, Gremse F, MORAN CM, Warnock G, Kuntner C, Huisman MC. Community Survey Results Show that Standardisation of Preclinical Imaging Techniques Remains a Challenge. *Molecular Imaging and Biology*. 2022 Dec 8;25(3):560-568. doi: 10.1007/s11307-022-01790-6. PMID: 36482032.



Citation summary
Total citations: **10**
Average/year: **2.00**
Period: 2022-2026
SCImago cites/doc 2y (2025): **2.79**
SJR 2025: **0.74** (Q2)



Cited by

1. Keizers B, Stroet MCM, Ali M, Floru S, Saliën J, Mezzanotte L, Delikatny EJ, Gibbs SL, Giuliani S, Gioux S, Ingelberts H, Kruijff S, Ntziachristos V, LaRochelle EPM, Rogalla S, Rosenthal EL, Samkoe KS, Tichauer KM, Vahrmeijer AL, Witjes MJH, Voskuil FJ, Gorpas DS, Hernot S, van der Zaag PJ. Reflect: reporting guidelines for preclinical, translational and clinical fluorescence molecular imaging studies. *npj Imaging*. 2025 Nov 25;3.0(1):62. doi: 10.1038/s44303-025-00125-8. PMID: 41291176.

2. Vanhove C, Koole MJB, Costa PF, Schottelius M, Mannheim JG, Kuntner C, Warnock G, McDougald W, Tavares A, Bernsen MR. Preclinical SPECT and PET: Joint EANM and ESMI procedure guideline for implementing an efficient quality control programme. *European Journal of Nuclear Medicine and Molecular Imaging*. 2024 Jul 15;51.0(13):3822-3839. doi: 10.1007/s00259-024-06824-5. PMID: 39008066.

3. Mahani FSN, Kalantari A, Fink GR, Hoehn M, Aswendt M. A systematic review of the relationship between magnetic resonance imaging based resting-state and structural networks in the rodent brain. *Frontiers in Neuroscience*. 2023 Jul 24;17.0:1194630. doi: 10.3389/fnins.2023.1194630. PMID: 37554291.

4. Kalantari A, Shahbazi M, Schneider M, Raikes AC, Frazão VV, Bhattrai A, Carnevale L, Diao Y, Franx BAA, Gammaraccio F, Goncalves L, Lee SD, van Leeuwen EM, Michalek A, Mueller SG, Rivera-Olvera A, Padró D, Selim MK, van der Toorn A, Varriano F, Vrooman RM, Wenk P, Albers HE, Boehm-Sturm P, Budinger E, Canals S, De Santis S, Brinton RD, Dijkhuizen RM, Eixarch E, Forloni G, Grandjean J, Hekmatyar K, Jacobs RE, Jelescu I, Kurniawan ND, Lembo G, Longo DL, Maria NSS, Micotti E, Muñoz-Moreno E, Ramos-Cabrer P, Reichardt W, Sòria G, Ielacqua GD, Aswendt M. Automated quality control of small animal MR neuroimaging data. *Imaging Neuroscience*. 2024 Jan 1;2.0:1-23. doi: 10.1162/imag_a_00317. PMID: 40212822.

5. Mannheim JG, Lan W, Conti M, Siedler F, Krueger MA, Herfert K, la Fougère CJ, Schmidt FP. Feasibility of in vivo small animal imaging using a clinical total-body PET/CT system. *EJNMMI Physics*. 2025 Jul 23;12.0(1):71. doi: 10.1186/s40658-025-00782-z. PMID: 40699299.

6. Deyhle RT, Bernhardsson C, Olsson LE, Sydoff M. Development of a method for evaluation of absorbed doses in preclinical in vivo µCT. *Radiation Protection Dosimetry*. 2025 Mar 18;201.0(13-14):868-876. doi: 10.1093/rpd/ncaf023. PMID: 40875276.

7. Petit-Demoulière B, Huzard D. Data welfare is animal welfare: Building a WellFAIR research ecosystem. *Neuroscience Applied*. 2026 Jan 1;5.0:106998. doi: 10.1016/j.nsa.2026.106998. PMID: 41909773.

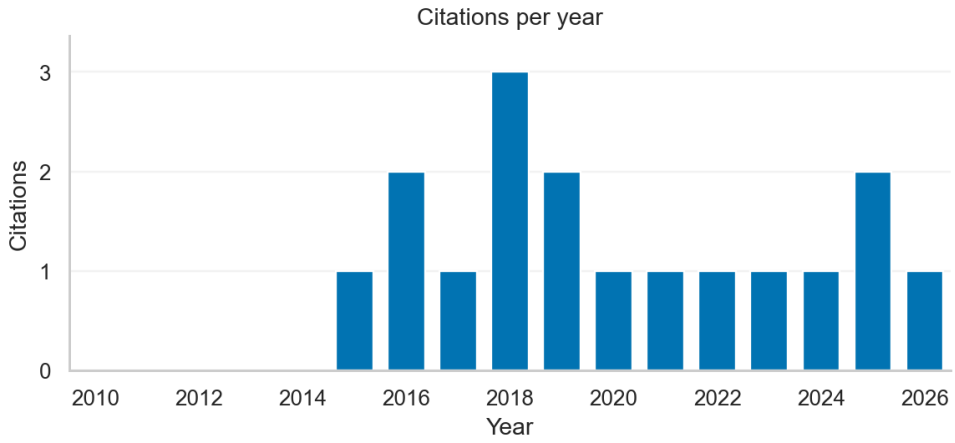
8. Gaudio H, Padmanabhan V, Landis WP, Silva LEV, Slovis J, Starr JP, Weeks MK, Widmann NJ, Forti RMM, Laurent GH, Ranieri NR, Mi F, Degani RE, Hallowell T, Delso N, Calkins H, Dobrzynski C, Haddad S, Kao S, Hwang M, Shi L, Baker WB, Tsui FR, Morgan RW, Kilbaugh TJ, Ko TS. A novel translational bioinformatics framework for facilitating multimodal data analyses in preclinical models of neurological injury. *Scientific Reports*. 2024 Dec 28;14.0(1):30710. doi: 10.1038/s41598-024-79973-0. PMID: 39730412.

9. Gaudio H, Padmanabhan V, Landis WP, Silva LEV, Slovis J, Starr JP, Weeks MK, Widmann NJ, Forti RMM, Laurent GH, Ranieri NR, Mi F, Degani RE, Hallowell T, Delso N, Calkins H, Dobrzynski C, Haddad S, Kao S, Hwang M, Shi L, Baker WB, Tsui FR, Morgan RW, Kilbaugh TJ, Ko TS. A Template for Translational Bioinformatics: Facilitating Multimodal Data Analyses in Preclinical Models of Neurological Injury. *bioRxiv (Cold Spring Harbor Laboratory)*. 2023 Jul 19;. doi: 10.1101/2023.07.17.547582. PMID: 37503137.

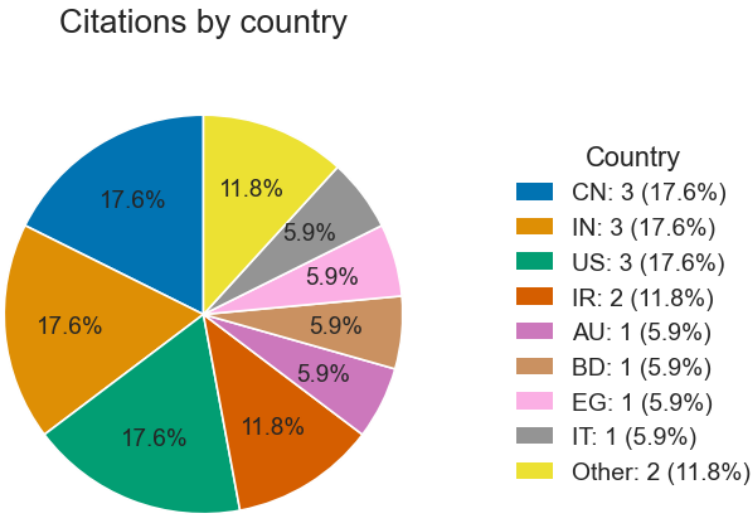
10. Pirozzi MA, Prinster A. Strategies for Standardization in Hybrid Imaging. *Springer eBooks*. 2025 Jan 1;:133-146. doi: 10.1007/978-3-031-86228-1_11.

Publication

Aswendt M, Henn N, Michalk S, Schneider G, Steiner M, Bissa U, Dose C, Hoehn M. Novel Bimodal Iron Oxide Particles for Efficient Tracking of Human Neural Stem Cells in Vivo. *Nanomedicine*. 2015 Aug 1;10(16):2499-2512. doi: 10.2217/nnm.15.94. PMID: 26296195.



Citation summary
Total citations: **17**
Average/year: **1.42**
Period: 2015-2026
SCImago cites/doc 2y (2025): **3.96**
SJR 2025: **0.681** (Q1)



Cited by

1. Pandey PK, Sharma AK, Gupta UD. Blood brain barrier: An overview on strategies in drug delivery, realisticin vitromodeling andin vivolive tracking. *Tissue Barriers*. 2015 Dec 15;4.0(1.0):e1129476. doi: 10.1080/21688370.2015.1129476. PMID: 27141418.

2. Zhang B, Yan W, Zhu Y, Yang W, Le W, Chen B, Zhu R, Cheng L. Nanomaterials in Neural-Stem-Cell-Mediated Regenerative Medicine: Imaging and Treatment of Neurological Diseases. *Advanced Materials*. 2018 Mar 15;30.0(17.0):e1705694. doi: 10.1002/adma.201705694. PMID: 29543350.

3. Pham BTT, Colvin EK, Pham NT, Kim B, Fuller E, Moon EA, Barbey R, Yuen S, Rickman BH, Bryce NS, Bickley SA, Tanudji MR, Jones S, Howell VM, Hawkett BS. Biodistribution and Clearance of Stable Superparamagnetic Maghemite Iron Oxide Nanoparticles in Mice Following Intraperitoneal Administration. *International Journal of Molecular Sciences*. 2018 Jan 10;19.0(1.0):205. doi: 10.3390/ijms19010205. PMID: 29320407.

4. D'Urso O, Drago F. Pharmacological significance of extra-oral taste receptors. *European Journal of Pharmacology*. 2021 Sep 5;910.0:174480. doi: 10.1016/j.ejphar.2021.174480. PMID: 34496302.

5. Asil SM, Ahlawat J, Barroso GG, Narayan M. Application of Nanotechnology in Stem-Cell-Based Therapy of Neurodegenerative Diseases. *Applied Sciences*. 2020 Jul 15;10.0(14.0):4852. doi: 10.3390/app10144852.

6. Madhavan L, Umashankar A, Corenblum MJ, Ray S, Yoshimaru ES, Trouard TP, Valdez M. Effects of the iron oxide nanoparticle Molday ION Rhodamine B on the viability and regenerative function of neural stem cells: relevance to clinical translation. *International Journal of Nanomedicine*. 2016 Apr 1;11.0:1731. doi: 10.2147/ijn.s102006. PMID: 27175074.

7. Senousy MA, Eid AH, Zaki MB, Abd-Elmawla MA, Ghaiad HR, El-Shiekh RA, Ahmed S, Khalifa HO. An integrative neuropharmacological review of Huntington’s disease challenges and the role of novel formulations in addressing pharmacological-pharmaceutical limitations. *Frontiers in Pharmacology*. 2026 May 7;17.0:1794983. doi: 10.3389/fphar.2026.1794983. PMID: 42181874.

8. Chen D, Monteiro-Riviere NA, Zhang LW. Intracellular imaging of quantum dots, gold, and iron oxide nanoparticles with associated endocytic pathways. *Wiley Interdisciplinary Reviews Nanomedicine and Nanobiotechnology*. 2016 Jul 15;9.0(2.0). doi: 10.1002/wnan.1419. PMID: 27418010.

9. Glover JC, Aswendt M, Boulland J, Lojk J, Stamenković S, Anđjus PR, Fiori F, Hoehn M, Mitrečić D, Pavlin M, Cavalli S, Frati C, Quaini F, (BIONECA) OBOTECA1. In vivo Cell Tracking Using Non-invasive Imaging of Iron Oxide-Based Particles with Particular Relevance for Stem Cell-Based Treatments of Neurological and Cardiac Disease. *Molecular Imaging and Biology*. 2019 Dec 4;22.0(6.0):1469-1488. doi: 10.1007/s11307-019-01440-4. PMID: 31802361.

10. Shahrukh M, Jiba U, Khan MA, Amin S, Ahmad FJ, Hasan N. Nanotechnology-driven approaches for the management of Huntington's disease. *Journal of Drug Delivery Science and Technology*. 2025 Jul 23;113.0:107318. doi: 10.1016/j.jddst.2025.107318.

11. Mallett CL, Shuboni-Mulligan DD, Shapiro EM. Tracking Neural Progenitor Cell Migration in the Rodent Brain Using Magnetic Resonance Imaging. *Frontiers in Neuroscience*. 2019 Jan 11;12.0:995. doi: 10.3389/fnins.2018.00995. PMID: 30686969.

12. Alizadeh R, Asghari A, Taghizadeh-Hesary F, Moradi S, Farhadi MR, Mehdizadeh M, Simorgh S, Nourazarian A, Shademan B, Susanabadi A, Kamrava SK. Intranasal delivery of stem cells labeled by nanoparticles in neurodegenerative disorders: Challenges and opportunities. *Wiley Interdisciplinary Reviews Nanomedicine and Nanobiotechnology*. 2023 Jul 6;15.0(6.0):e1915. doi: 10.1002/wnan.1915. PMID: 37414546.

13. Soleymani-Goloujeh M, Hosseini S, Eslaminejad MB. Advanced Nanotechnology Approaches as Emerging Tools in Cellular-Based Technologies. *Advances in experimental medicine and biology*. 2022 Jan 1;1409.0:127-144. doi: 10.1007/5584_2022_725. PMID: 35816248.

14. Deng S, Li Y, Zhao G. Imaging Gliomas with Nanoparticle-Labeled Stem Cells. *Chinese Medical Journal*. 2018 Mar 8;131.0(6.0):721-730. doi: 10.4103/0366-6999.226900. PMID: 29521296.

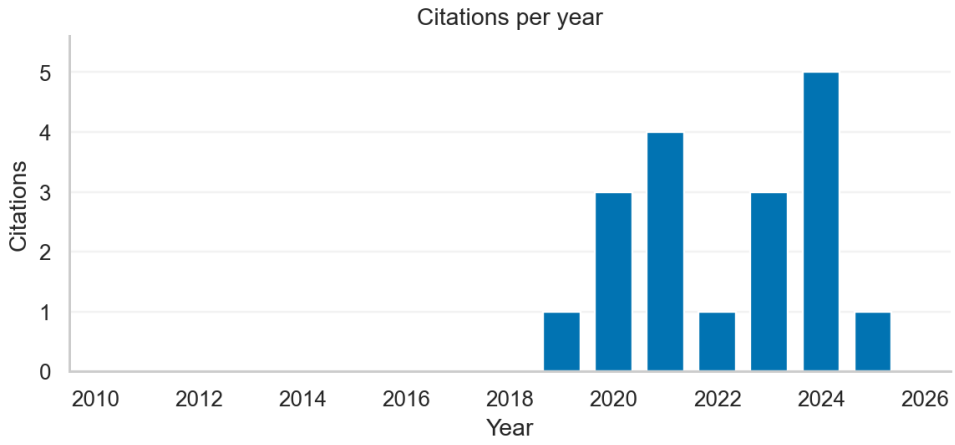
15. Begh MZA, Zehravi M, Bhuiyan MAK, Molla MR, Raman K, Emran TB, Ullah H, Ahmad IS, Osman H, Khandaker MU. Recent advances in stem cell approaches to neurodegeneration: A comprehensive review with mechanistic insight. *Pathology - Research and Practice*. 2025 May 14;271.0:156013. doi: 10.1016/j.prp.2025.156013. PMID: 40381433.

16. Wiraja C. Non-invasive nanosensor monitoring of mRNA expression for regenerative medicine application. *DR-NTU (Nanyang Technological University)*. 2017 Jan 1;. doi: 10.32657/10356/72855.

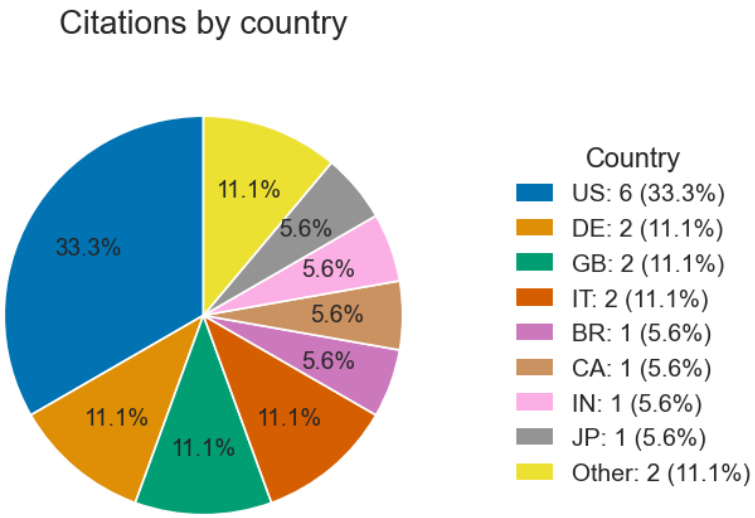
17. Sharma M, Gupta T, Gupta N, Rawat R, Badoniya N, Pandey E, Sharma S. Targeted Therapy for Neurological Disorders Using Nanomaterial-Based Drug Delivery Vehicle. *Smart nanomaterials technology*. 2024 Jan 1;:143-170. doi: 10.1007/978-981-97-0308-1_7.

Publication

Vogel SN, Schäfer C, Heß S, Folz-Donahue K, Nelles M, Minassian A, Schwarz MK, Kukat C, Ehrlich M, Zaehres H, Kloppenburg P, Hoehn M, Aswendt M. The in vivo timeline of differentiation of engrafted human neural progenitor cells. *Stem Cell Research*. 2019 Mar 25;37:101429. doi: 10.1016/j.scr.2019.101429. PMID: 30933718.



Citation summary
Total citations: **18**
Average/year: **2.25**
Period: 2019-2026
SCImago cites/doc 2y (2025): **0.63**
SJR 2025: **0.333** (Q3)



Cited by

1. Jin L, Tai Y, Nam J. Piezoelectric silk fibroin nanofibers: Structural optimization to enhance piezoelectricity and biostability for neural tissue engineering. *Nano Energy*. 2024 Oct 17;132.0:110367. doi: 10.1016/j.nanoen.2024.110367.

2. Palma-Tortosa S, Martín BC, Kokaia Z, Tornero D. Neuronal Replacement in Stem Cell Therapy for Stroke: Filling the Gap. *Frontiers in Cell and Developmental Biology*. 2021 Apr 6;9.0:662636. doi: 10.3389/fcell.2021.662636. PMID: 33889578.

3. Salwa, Kumar L. Engrafted stem cell therapy for Alzheimer's disease: A promising treatment strategy with clinical outcome. *Journal of Controlled Release*. 2021 Sep 9;338.0:837-857. doi: 10.1016/j.jconrel.2021.09.007. PMID: 34509587.

4. Kwokdinata C, Ramanujam V, Chen J, Oliveira PF, Nai MH, Chooi WH, Lim CT, Ng S, David L, Chew SY. Encapsulation of Human Spinal Cord Progenitor Cells in Hyaluronan-Gelatin Hydrogel for Spinal Cord Injury Treatment. *ACS Applied Materials & Interfaces*. 2023 Sep 26;15.0(44.0):50679-50692. doi: 10.1021/acsami.3c07419. PMID: 37751213.

5. Rodrigue B, Sajish M, Salmaso N, Aguilar-Valles A. Orchestrating Neural Development Through mRNA Translation Regulation. *Journal of Neurochemistry*. 2025 Jun 1;169.0(6.0):e70128. doi: 10.1111/jnc.70128. PMID: 40509688.

6. Galichet C, Clayton RW, Lovell-Badge RH. Novel Tools and Investigative Approaches for the Study of Oligodendrocyte Precursor Cells (NG2-Glia) in CNS Development and Disease. *Frontiers in Cellular Neuroscience*. 2021 Apr 29;15.0:673132. doi: 10.3389/fncel.2021.673132. PMID: 33994951.

7. Minassian A, Green C, Diedenhofen M, Vogel SN, Heß S, Stoeber M, Radmilović MD, Wiedermann D, Kloppenburg P, Hoehn M. Human Neural Stem Cell Induced Functional Network Stabilization After Cortical Stroke: A Longitudinal Resting-State fMRI Study in Mice. *Frontiers in Cellular Neuroscience*. 2020 Apr 7;14.0:86. doi: 10.3389/fncel.2020.00086.

PMID: 32317940.

8. Lara RP, Goding JA, Green RA. Adaptive biomimicry: design of neural interfaces with enhanced biointegration. *Current Opinion in Biotechnology*. 2021 Oct 26;72.0:62-68. doi: 10.1016/j.copbio.2021.10.004. PMID: 34715548.

9. Paranjape N, Lin YT, Flores-Ramirez Q, Sarin V, Johnson AB, Chu JSF, Paredes MF, Wiita AP. A CRISPR-engineered isogenic model of the 22q11.2 A-B syndromic deletion. *Scientific Reports*. 2023 May 11;13.0(1.0):7689. doi: 10.1038/s41598-023-34325-2. PMID: 37169815.

10. Maalouf K, Vaine CA, Frederick DM, Yoshinaga A, Obuchi W, Mahjoum S, Nieland L, Ali JA, Bragg DC, Breakefield XO, Breyne K. Tracking human neurologic disease status in mouse brain/plasma using reporter-tagged, EV-associated biomarkers. *Molecular Therapy*. 2023 May 17;31.0(7.0):2206-2219. doi: 10.1016/j.ymthe.2023.05.011. PMID: 37198883.

11. Madsen SD, Giler MK, Bunnell BA, O'Connor KC. Illuminating the Regenerative Properties of Stem Cells In Vivo with Bioluminescence Imaging. *Biotechnology Journal*. 2020 Oct 22;16.0(3.0):e2000248. doi: 10.1002/biot.202000248. PMID: 33089922.

12. Yamashita N, Morita M, Yokota H, Mimori-Kiyosue Y. Digital Spindle: A New Way to Explore Mitotic Functions by Whole Cell Data Collection and a Computational Approach. *Cells*. 2020 May 19;9.0(5.0):1255. doi: 10.3390/cells9051255. PMID: 32438637.

13. Calhoun C, Kan S, Stover AE, Harb JF, Monuki ES, Wang R, Schwartz PH. Human iPSC-derived neural stem cells engraft and improve pathophysiology of MPS I mice. *Molecular Therapy - Methods & Clinical Development*. 2024 Nov 5;32.0(4.0):101367. doi: 10.1016/j.omtm.2024.101367. PMID: 39764351.

14. Aswendt M, Vogel S, Hoehn M. In vivo bioluminescence imaging to elucidate stem cell graft differentiation. *Neural Regeneration Research*. 2019 Sep 16;15.0(1.0):61. doi: 10.4103/1673-5374.264449. PMID: 31535647.

15. Merighi F, De Vincentiis S, Onorati M, Raffa V. Long-term Mouse Spinal Cord Organotypic Slice Culture as a Platform for Validating Cell Transplantation in Spinal Cord Injury. *Journal of Visualized Experiments*. 2024 Apr 12;(206.0). doi: 10.3791/66704. PMID: 38682940.

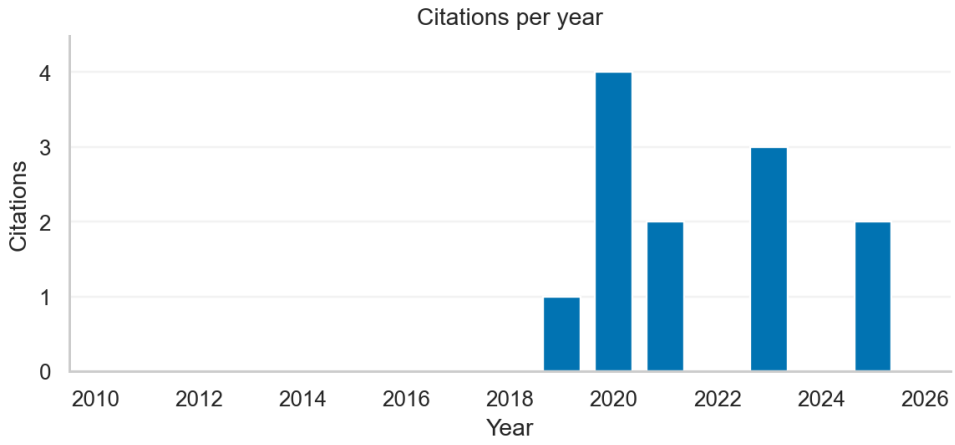
16. da Costa NMM, Caetano HIP, Aguiar LM, Parisi L, Ghezzi B, Elviri L, Zuardi LR, de Oliveira PT, Palioto DB. The Influence of Physiological Blood Clot on Osteoblastic Cell Response to a Chitosan-Based 3D Scaffold-A Pilot Investigation. *Biomimetics*. 2024 Dec 21;9.0(12.0):782. doi: 10.3390/biomimetics9120782. PMID: 39727786.

17. Paranjape N, Lin YT, Flores-Ramirez Q, Sarin V, Johnson AB, Chu JSF, Paredes MF, Wiita AP. A CRISPR-engineered Isogenic Model Reveals Altered Neuronal Phenotypes of the 22q11.2 A-B Syndromic Deletion. *bioRxiv (Cold Spring Harbor Laboratory)*. 2022 Jun 24;. doi: 10.1101/2022.06.22.497212.

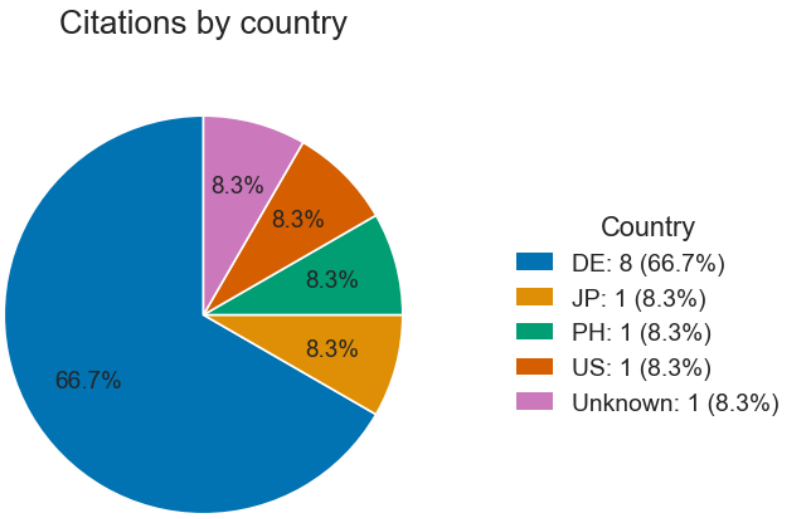
18. Merighi F, De Vincentiis S, Onorati M, Raffa V. JoVE Video Dataset. 2024 Apr 12;. doi: 10.3791/66704-v.

Publication

Pallast N, Wieters F, Nill M, Fink GR, Aswendt M. Cloud-based relational database for multimodal animal data. *Database*. 2018 Jan 1;2018. doi: 10.1093/database/bay124. PMID: 30576483.



Citation summary
Total citations: **12**
Average/year: **1.33**
Period: 2018-2026
SCImago cites/doc 2y (2025): **2.33**
SJR 2025: **0.943** (Q1)



Cited by

1. Aswendt M, Wilhelmsson U, Wieters F, Stokowska A, Schmitt FJ, Pallast N, de Pablo Y, Mohammed L, Hoehn M, Pekna M, Pekny M. Reactive astrocytes prevent maladaptive plasticity after ischemic stroke. *Progress in Neurobiology*. 2021 Dec 15;209.0:102199. doi: 10.1016/j.pneurobio.2021.102199. PMID: 34921928.

2. Aswendt M, Pallast N, Wieters F, Baues M, Hoehn M, Fink GR. Lesion Size- and Location-Dependent Recruitment of Contralesional Thalamus and Motor Cortex Facilitates Recovery after Stroke in Mice. *Translational Stroke Research*. 2020 Mar 12;12.0(1.0):87-97. doi: 10.1007/s12975-020-00802-3. PMID: 32166716.

3. Pallast N, Diedenhofen M, Blaschke SJ, Wieters F, Wiedermann D, Hoehn M, Fink GR, Aswendt M. Processing Pipeline for Atlas-Based Imaging Data Analysis of Structural and Functional Mouse Brain MRI (AIDAmri). *Frontiers in Neuroinformatics*. 2019 Jun 4;13.0:42. doi: 10.3389/fninf.2019.00042. PMID: 31231202.

4. Pallast N, Wieters F, Nill M, Fink GR, Aswendt M. Graph theoretical quantification of white matter reorganization after cortical stroke in mice. *NeuroImage*. 2020 May 4;217.0:116873. doi: 10.1016/j.neuroimage.2020.116873. PMID: 32380139.

5. Ohta T, Hananoe A, Fukushima-Nomura A, Ashizaki K, Sekita A, Seita J, Kawakami E, Sakurada K, Amagai M, Koseki H, Kawasaki H. Best practices for multimodal clinical data management and integration: An atopic dermatitis research case. *Allergology International*. 2023 Dec 14;73.0(2.0):255-263. doi: 10.1016/j.alit.2023.11.006. PMID: 38102028.

6. Götz J, Wieters F, Fritz VJ, Käsgen O, Kalantari A, Fink GR, Aswendt M. Temporal and Spatial Gene Expression Profile of Stroke Recovery Genes in Mice. *Genes*. 2023 Feb 9;14.0(2.0):454. doi: 10.3390/genes14020454. PMID: 36833381.

7. Kalantari A, Hambrock C, Grefkes C, Fink GR, Aswendt M. Proportional recovery in mice with cortical stroke. *Experimental Neurology*. 2025 Feb 4;386.0:115180. doi: 10.1016/j.expneurol.2025.115180. PMID: 39914643.

8. Aswendt M, Wieters F. Structural integrity and remodeling underlying functional recovery after stroke. *Neural Regeneration Research*. 2020 Dec 17;16.0(7.0):1423. doi: 10.4103/1673-5374.301004. PMID: 33318437.

9. Renschen A, Matsunaga A, Oksenberg JR, Santaniello A, Didonna A. TopoDB: a novel multifunctional management system for laboratory animal colonies. *Database*. 2020 Jan 1;2020.0. doi: 10.1093/database/baaa098. PMID: 33206961.

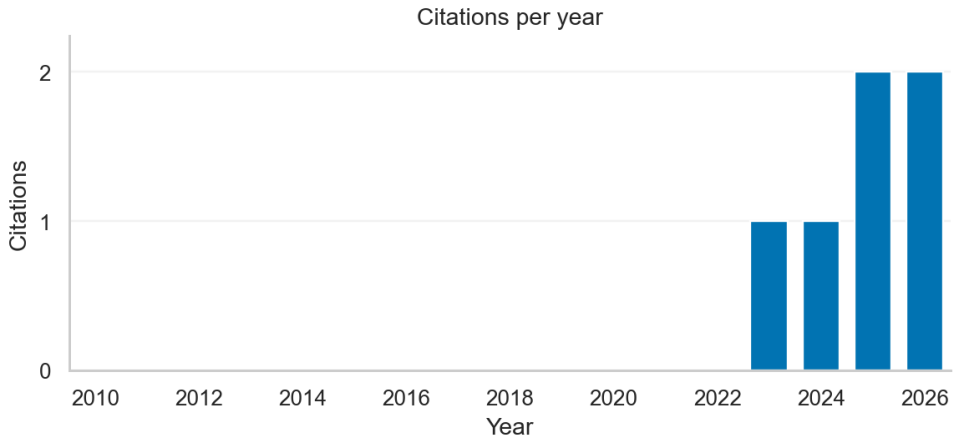
10. Kalantari A, Szczepanik M, Heunis S, Mönch C, Hanke M, Wächtler T, Aswendt M. How to establish and maintain a multimodal animal research dataset using DataLad. *Scientific Data*. 2023 Jun 5;10.0(1.0):357. doi: 10.1038/s41597-023-02242-8. PMID: 37277500.

11. Picker TS. Digitalization in Laboratories of the Pharmaceutical Industry. *Methods and principles in medicinal chemistry*. 2021 Sep 24;:397-420. doi: 10.1002/9783527823048.ch8.

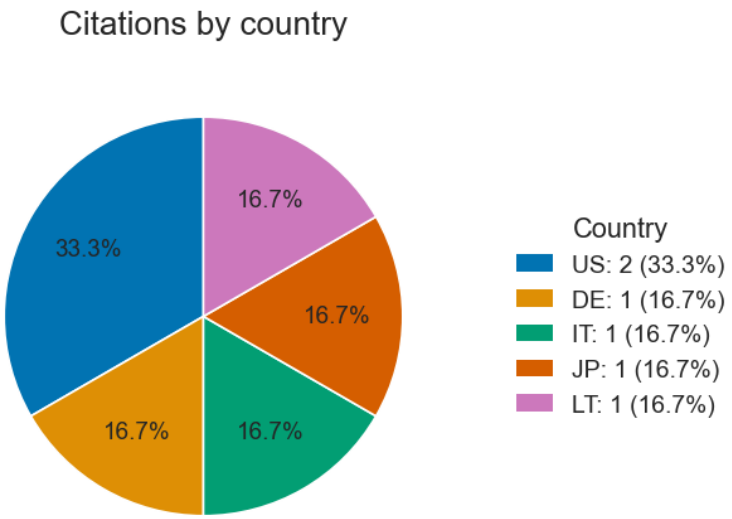
12. Andres PRO, Repalam MCR. Provincial animal health monitoring system and geospatial analysis in the Province of Laguna. *AIP conference proceedings*. 2025 Jan 1;3287.0:030001. doi: 10.1063/5.0262098.

Publication

Götz J, Wieters F, Fritz VJ, Käsgen O, Kalantari A, Fink GR, Aswendt M. Temporal and Spatial Gene Expression Profile of Stroke Recovery Genes in Mice. *Genes*. 2023 Feb 9;14(2):454. doi: 10.3390/genes14020454. PMID: 36833381.



Citation summary
Total citations: **6**
Average/year: **1.50**
Period: 2023-2026
SCImago cites/doc 2y (2025): **3.32**
SJR 2025: **0.823** (Q2)



Cited by

1. Irisa K, Shichita T. Neural repair mechanisms after ischemic stroke. *Inflammation and Regeneration*. 2025 Mar 17;45.0(1.0):7. doi: 10.1186/s41232-025-00372-7. PMID: 40098163.

2. Valančienė J, Melaika K, Šliachtenko A, Šiaurytė K, Ekkert A, Jatužis D. Stroke genetics and how it Informs novel drug discovery. *Expert Opinion on Drug Discovery*. 2024 Mar 4;19.0(5.0):553-564. doi: 10.1080/17460441.2024.2324916. PMID: 38494780.

3. Cimmino G, Conte S, Palumbo D, Sperlongano S, Torella M, Della Corte A, Golino P. The Novel Role of Noncoding RNAs in Modulating Platelet Function: Implications in Activation and Aggregation. *International Journal of Molecular Sciences*. 2023 Apr 21;24.0(8.0):7650. doi: 10.3390/ijms24087650. PMID: 37108819.

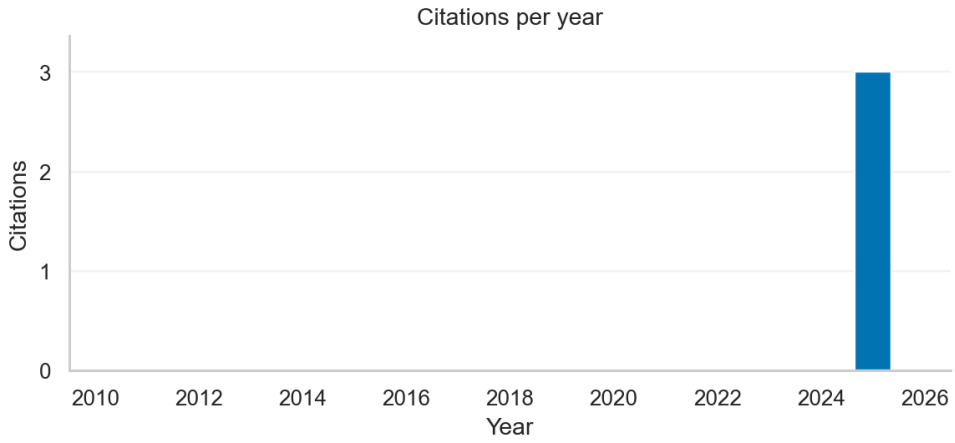
4. Kalantari A, Hambrock C, Grefkes C, Fink GR, Aswendt M. Proportional recovery in mice with cortical stroke. *Experimental Neurology*. 2025 Feb 4;386.0:115180. doi: 10.1016/j.expneurol.2025.115180. PMID: 39914643.

5. Betz D, Alers VA, Kenwood MR, Zuurbier KR, Coimbra R, Rhoton P, Plautz EJ, Douglas PM, Ramirez DMO, Stowe AM, Goldberg MP. Shared Transcriptomic Signatures in Perilesional and Contralesional Cortex. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Feb 4;. doi: 10.64898/2026.02.02.703306. PMID: 41676687.

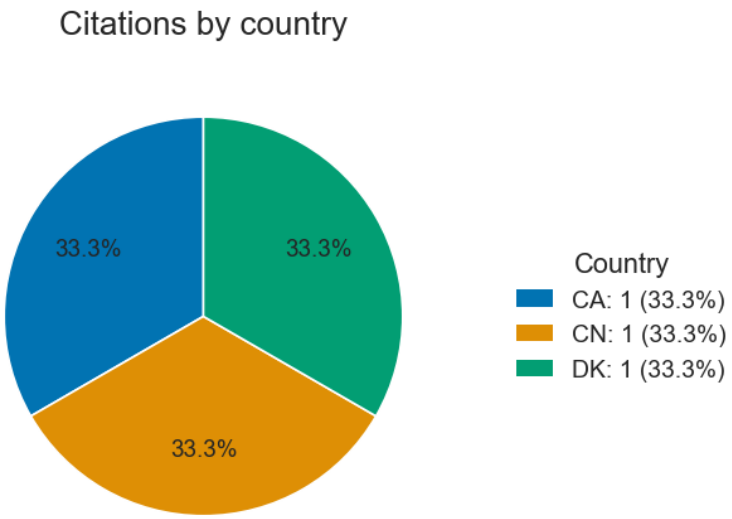
6. Betz D, Alers VA, Kenwood MR, Zuurbier KR, Coimbra R, Rhoton P, Plautz EJ, Douglas PM, Ramirez DMO, Stowe AM, Goldberg MP. Shared transcriptomic signatures in perilesional and contralesional cortex after ischemic stroke. *Journal of Neuroinflammation*. 2026 Jun 15;23.0(1.0). doi: 10.1186/s12974-026-03885-1. PMID: 42298604.

Publication

Kalantari A, Hambrock C, Grefkes C, Fink GR, Aswendt M. Proportional recovery in mice with cortical stroke. *Experimental Neurology*. 2025 Feb 4;386:115180. doi: 10.1016/j.expneurol.2025.115180. PMID: 39914643.



Citation summary
Total citations: **3**
Average/year: **1.50**
Period: 2025-2026
SCImago cites/doc 2y (2025): **5.03**
SJR 2025: **1.39** (Q1)



Cited by

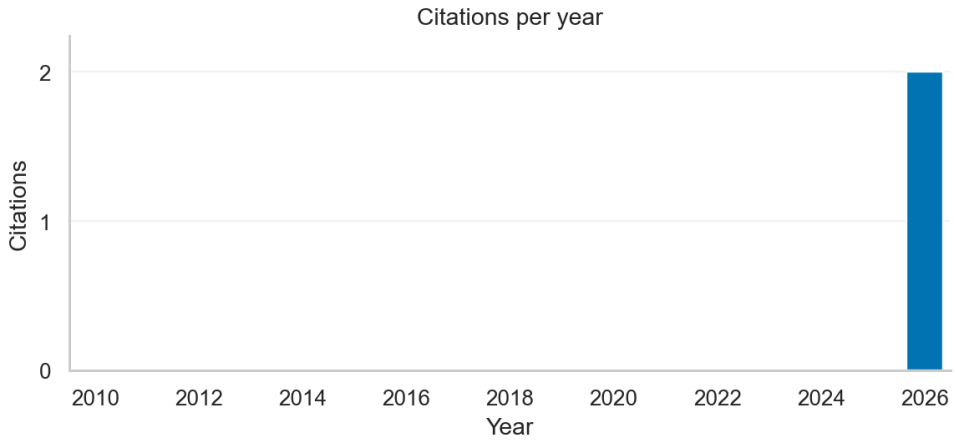
1. Haarslev C, Bradshaw KP, Corradini S, Raffaele S, Fumagalli M, Buckwalter MS, Lambertsen KL. Harnessing anti-TNFR1 therapy has a limited impact on inflammation and motor function after stroke in obese male mice. *Brain Research*. 2025 Aug 18;1865:149888. doi: 10.1016/j.brainres.2025.149888. PMID: 40829684.

2. Cai Y, Duan C, Wen S, Ling Z, Huang G, Zou J, Zeng Q. Progression of cognitive and functional decline before and after stroke: evidence from four prospective cohort studies across 23 countries. *Journal of Neurology*. 2025 Nov 20;272(12.0):779. doi: 10.1007/s00415-025-13511-4. PMID: 41264137.

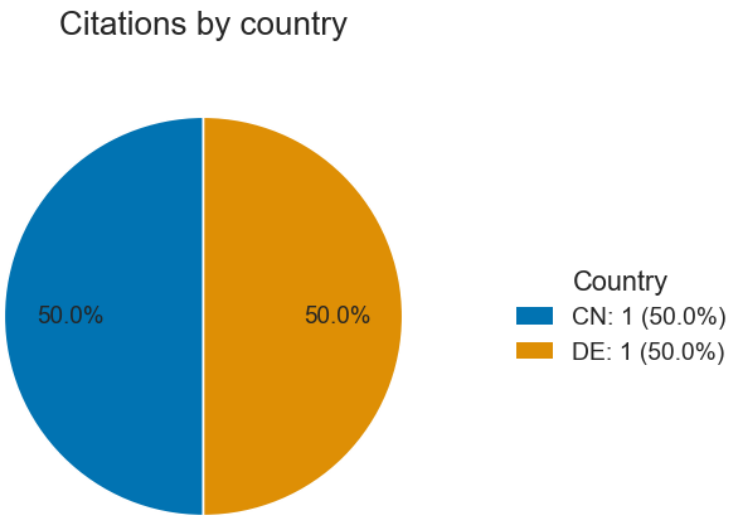
3. Van Sprengel M, Butterworth J, Reeson PL, Brown CE. Retrograde and anterograde trans-synaptic viral tracing of neuronal connections reveals local and distant effects of ischemic stroke on dendritic spines. *Journal of Cerebral Blood Flow & Metabolism*. 2025 May 25;45(10.0):1891-1904. doi: 10.1177/0271678x251345360. PMID: 40415372.

Publication

Mahani FSN, Kalantari A, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic structural connectivity changes in cortical and cortico-striatal strokes in mice. *Neural Regeneration Research*. 2026 Feb 28;21(10):4945-4953. doi: 10.4103/nrr.nrr-d-25-00491. PMID: 41778731.



Citation summary
Total citations: **2**
Average/year: **2.00**
Period: 2026-2026
SCImago cites/doc 2y (2025): **5.98**
SJR 2025: **1.43** (Q1)



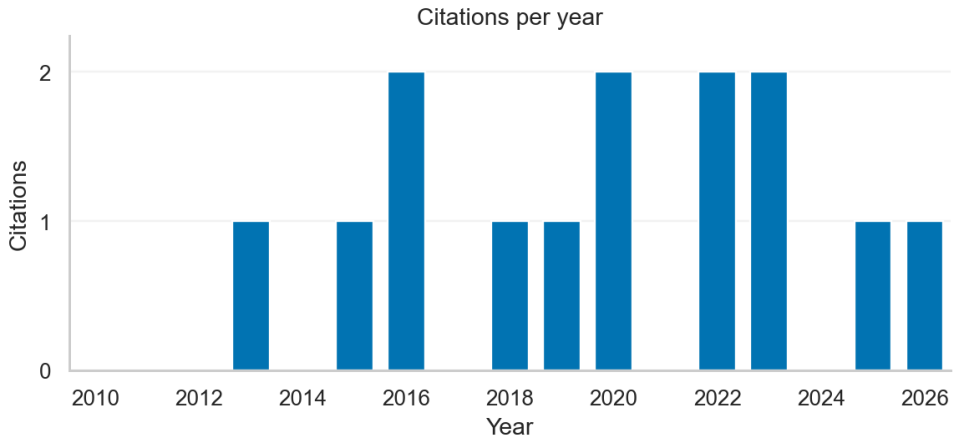
Cited by

1. Mahani F, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic Functional Connectivity Changes in Cortical and Cortico-Striatal Strokes in Mice. *CNS Neuroscience & Therapeutics*. 2026 Aug 1;32(8.0):e71114. doi: 10.1002/cns.71114. PMID: 42661330.

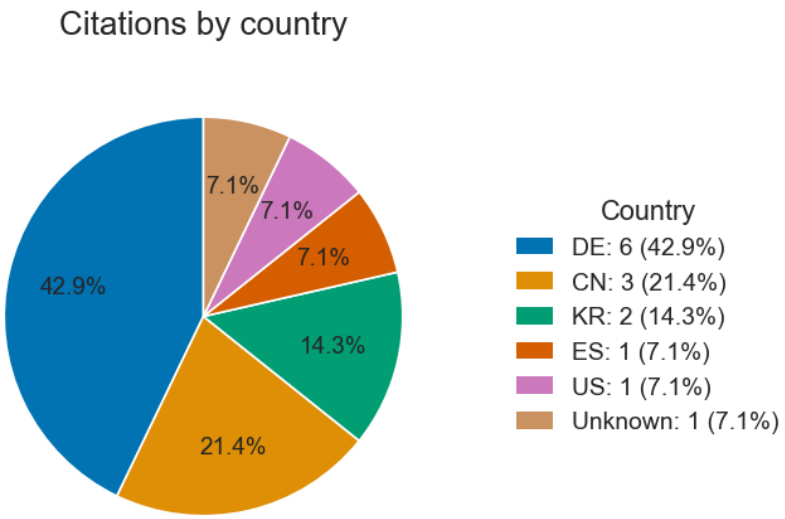
2. Song M, Bu N, Wei L, Chen J. Microglial handling of myelin debris and remyelination after ischaemic stroke: recognition, intracellular processing, lipid fate, and oligodendroglial repair. *Frontiers in Cellular Neuroscience*. 2026 Sep 15;20:1945478. doi: 10.3389/fncel.2026.1945478. PMID: 42812911.

Publication

Hoehn M, Aswendt M. Structure-function relationship of cerebral networks in experimental neuroscience: Contribution of magnetic resonance imaging. *Experimental Neurology*. 2012 May 1;242:65-73. doi: 10.1016/j.expneurol.2012.04.014. PMID: 22572591.



Citation summary
Total citations: **14**
Average/year: **0.93**
Period: 2012-2026
SCImago cites/doc 2y (2025): **5.03**
SJR 2025: **1.39** (Q1)



Cited by

1. Wankhede NL, Koppula S, Umare PD, Umare MD, Lokhande SK, Taksande BG, Upaganlawar AB, Umekar MJ, Kopalli SR, Kale MB. Neuroimaging biomarkers: a critical review of their role in neurological disease diagnosis and management. *Brain Disorders*. 2025 Apr 21;18.0:100222. doi: 10.1016/j.dscb.2025.100222.

2. Pallast N, Diedenhofen M, Blaschke SJ, Wieters F, Wiedermann D, Hoehn M, Fink GR, Aswendt M. Processing Pipeline for Atlas-Based Imaging Data Analysis of Structural and Functional Mouse Brain MRI (AIDAmri). *Frontiers in Neuroinformatics*. 2019 Jun 4;13.0:42. doi: 10.3389/fninf.2019.00042. PMID: 31231202.

3. Zheng N, Li M, Wu Y, Kaewborisuth C, Li Z, Gui Z, Wu J, Cai A, Lin K, Su K, Xiang H, Tian X, Manyande A, Xu F, Wang J. A novel technology for in vivo detection of cell type-specific neural connection with AQP1-encoding rAAV2-retro vector and metal-free MRI. *NeuroImage*. 2022 Jun 19;258.0:119402. doi: 10.1016/j.neuroimage.2022.119402. PMID: 35732245.

4. Cheng M, Aswendt M, Steinberg GK. Optogenetic Approaches to Target Specific Neural Circuits in Post-stroke Recovery. *Neurotherapeutics*. 2015 Dec 23;13.0(2.0):325-340. doi: 10.1007/s13311-015-0411-5. PMID: 26701667.

5. Aswendt M, Schwarz MK, Abdelmoula WM, Dijkstra J, Dedeurwaerdere S. Whole-Brain Microscopy Meets In Vivo Neuroimaging: Techniques, Benefits, and Limitations. *Molecular Imaging and Biology*. 2016 Sep 2;19.0(1.0):1-9. doi: 10.1007/s11307-016-0988-z. PMID: 27590493.

6. Yang X, Kendrick KM, Wu Q, Chen T, Lama S, Cheng B, Li S, Huang X, Gong Q. Structural and Functional Connectivity Changes in the Brain Associated with Shyness but Not with Social Anxiety. *PLoS ONE*. 2013 May 10;8.0(5.0):e63151. doi: 10.1371/journal.pone.0063151. PMID: 23675458.

7. Scharwächter L, Schmitt FJ, Pallast N, Fink GR, Aswendt M. Network analysis of neuroimaging in mice. *NeuroImage*. 2022 Mar 17;253.0:119110. doi: 10.1016/j.neuroimage.2022.119110. PMID: 35311664.

8. Carbó-Carreté M, Cañete-Massé C, Peró M, Guàrdia-Olmos J. Using fMRI to Assess Brain Activity in People With Down Syndrome: A Systematic Review. *Frontiers in Human Neuroscience*. 2020 Apr 27;14.0:147. doi: 10.3389/fnhum.2020.00147. PMID: 32395104.

9. Aswendt M, Wieters F. Structural integrity and remodeling underlying functional recovery after stroke. *Neural Regeneration Research*. 2020 Dec 17;16.0(7.0):1423. doi: 10.4103/1673-5374.301004. PMID: 33318437.

10. Mahani FSN, Kalantari A, Fink GR, Hoehn M, Aswendt M. A systematic review of the relationship between magnetic resonance imaging based resting-state and structural networks in the rodent brain. *Frontiers in Neuroscience*. 2023 Jul 24;17.0:1194630. doi: 10.3389/fnins.2023.1194630. PMID: 37554291.

11. Aswendt M, Green C, Hoehn M. Plastic Network Changes During Brain Disease. *Handbook of behavioral neuroscience*. 2018 Jan 1;:415-424. doi: 10.1016/b978-0-12-812028-6.00022-7.

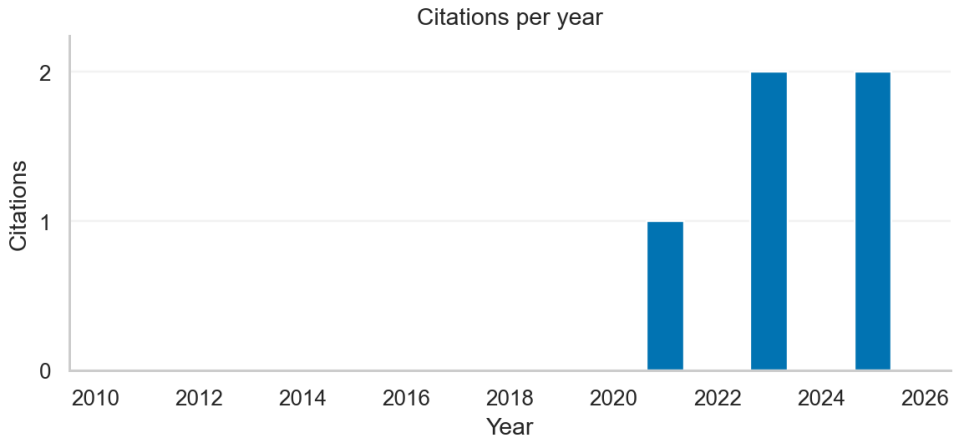
12. Kale MB, Wankhede NL, Taksande BG, Upaganlawar A, Umekar MJ, Koppula S, Kopalli SR. Imaging-Based Biomarkers in Neurological Diseases-A Critical Review. *Preprints.org*. 2023 Sep 7;. doi: 10.20944/preprints202309.0536.v1.

13. Aswendt M, Boehm-Sturm P, Hoehn M. Chapter 10 Neural Stem Cells. *Pan Stanford Publishing eBooks*. 2016 Oct 21;:283-310. doi: 10.1201/9781315364605-11.

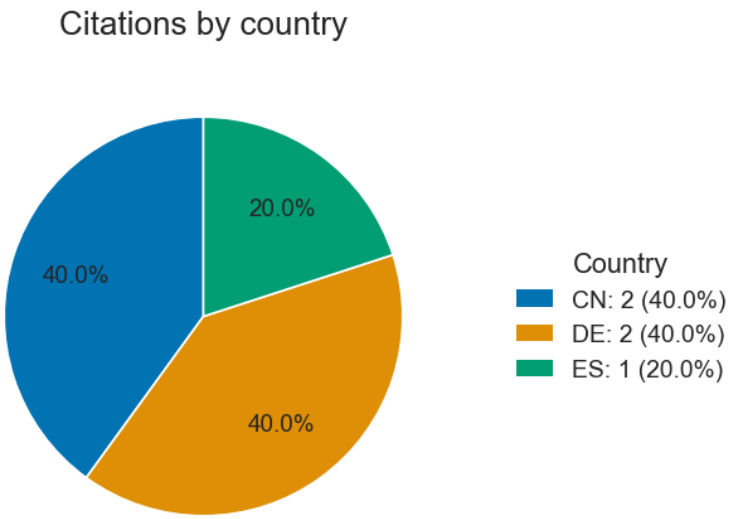
14. Zou Y, Lai X, Liu Q, Zhang H, Li H, Li W, He D, Yao L, Yang X. Transcranial magnetic stimulation combined with functional near-infrared spectroscopy to elucidate the neurophysiological mechanisms of post-stroke hemiplegia: a systematic review. *Frontiers in Neurology*. 2026 Jun 10;17.0:1824157. doi: 10.3389/fneur.2026.1824157. PMID: 42358925.

Publication

Aswendt M, Wieters F. Structural integrity and remodeling underlying functional recovery after stroke. *Neural Regeneration Research*. 2020 Dec 17;16(7):1423. doi: 10.4103/1673-5374.301004. PMID: 33318437.



Citation summary
Total citations: **5**
Average/year: **0.71**
Period: 2020-2026
SCImago cites/doc 2y (2025): **5.98**
SJR 2025: **1.43** (Q1)



Cited by

1. Song J, Zeng J, Chen X, Wang J, Zhang Y, Gao Y, Wang R, Jiang N, Lin Y, Li R. Anti-neuroinflammatory agent rhein lysinate-based self-assembled injectable hydrogel loaded with ZL006 for promoting post-stroke functional recovery. *Biomaterials*. 2025 Jan 23;318:123124. doi: 10.1016/j.biomaterials.2025.123124. PMID: 39884131.

2. Rogalewski A, Schäbitz W. Stroke recovery enhancing therapies: lessons from recent clinical trials. *Neural Regeneration Research*. 2021 Aug 29;17(4.0):717. doi: 10.4103/1673-5374.314287. PMID: 34472456.

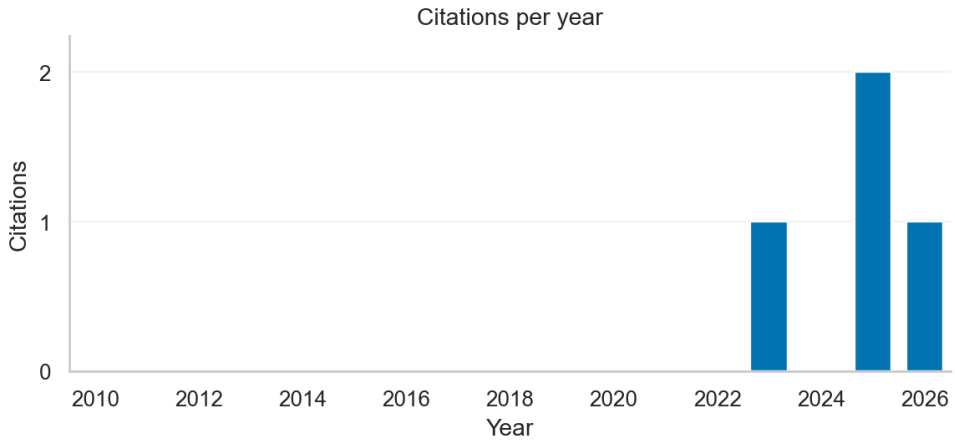
3. Román EP, Martos IC, Núñez JM, Valenza-Peña G, Santiago MG, Valenza MC. Effectiveness of self-management interventions during the peri-hospitalization period in patients with stroke: A systematic review and meta-analysis. *Clinical Rehabilitation*. 2023 Aug 7;38(1.0):34-46. doi: 10.1177/02692155231193563. PMID: 37551095.

4. Rogalewski A, Schabitz WR. Therapien zur Verbesserung der Schlaganfall-Regeneration - Ergebnisse klinischer Studien im wissenschaftlichen Kontext. *Fortschritte der Neurologie · Psychiatrie*. 2023 Dec 1;91(12.0):516-522. doi: 10.1055/a-2181-1026. PMID: 38081165.

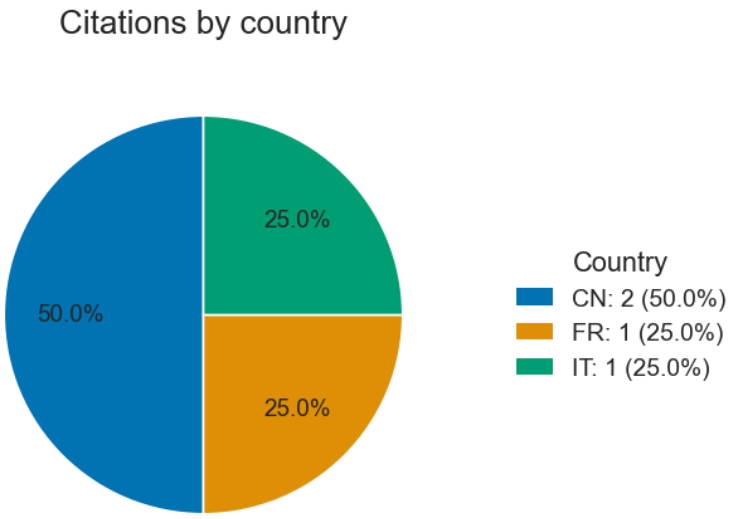
5. Zhong Y, Mi P, Qin X. Advances in the Application of rTMS in the Rehabilitation of Post-Stroke Swallowing Dysfunction. *Journal of Biosciences and Medicines*. 2025 Jan 1;13(5.0):231-240. doi: 10.4236/jbm.2025.135018.

Publication

Wieters F, Gruhn M, Büschges A, Fink GR, Aswendt M. Terminal H-reflex Measurements in Mice. *Journal of Visualized Experiments*. 2022 Jun 16;(184). doi: 10.3791/63304. PMID: 35781283.



Citation summary
Total citations: **4**
Average/year: **0.80**
Period: 2022-2026
SCImago cites/doc 2y (2025): **1.06**
SJR 2025: **0.409** (Q2)



Cited by

1. Marchioretti C, Zanetti G, Pirazzini M, Gherardi G, Nogara L, Andreotti R, Martini PGV, Marcucci L, Canato M, Nath SR, Zuccaro E, Chivet M, Mammucari C, Pacifici M, Raffaello A, Rizzuto R, Mattarei A, Desbats MA, Salviati L, Megighian A, Sorarú G, Pegoraro E, Belluzzi E, Pozzuoli A, Biz C, Ruggieri P, Romualdi C, Lieberman AP, Babu GJ, Sandri M, Blaauw B, Basso M, Pennuto M. Defective excitation-contraction coupling and mitochondrial respiration precede mitochondrial Ca2+ accumulation in spinobulbar muscular atrophy skeletal muscle. *Nature Communications*. 2023 Feb 6;14(1):602. doi: 10.1038/s41467-023-36185-w. PMID: 36746942.

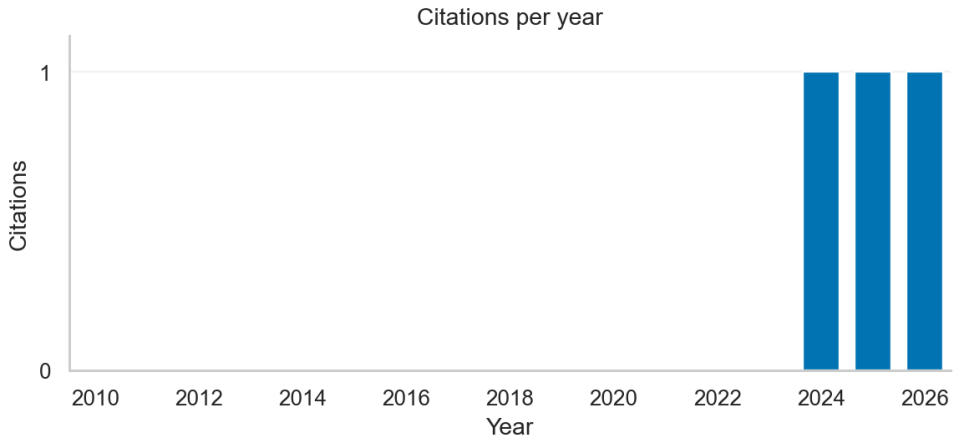
2. Gong H, Zhang Z, Duan Z, Mao X, Wu Y, Rao J, Du X. Mechanisms of Different Motor Neurons in the Occurrence of Spasticity After Spinal Cord Injury: A Narrative Review. *International Journal of Molecular Sciences*. 2025 May 28;26(11):5162. doi: 10.3390/ijms26115162. PMID: 40507976.

3. Zhang Q, Wang J, Ma L, Ma L, Zhong J, Li F, Xu Q, Ma L, Ma L, Xiu J, Wang X. Motion-Style Scalp Acupuncture Ameliorates Post-Stroke Muscle Spasticity Through Cerebral Blood Flow Augmentation and 5-HT 2A R-Mediated Spinal KCC2 Reactivation. *Brain and Behavior*. 2025 Nov 26;15(12):e71064. doi: 10.1002/brb3.71064. PMID: 41299887.

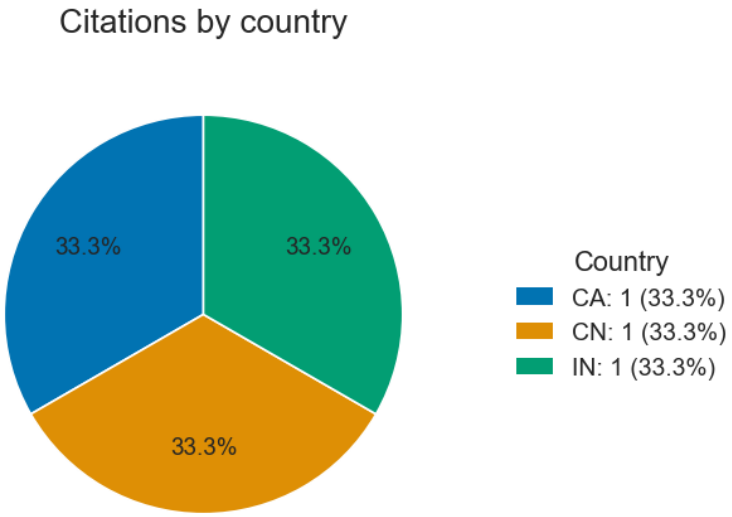
4. Ertlen C, Bonnet M, Seblani M, Coyle T, Brezun J, Serratrice N, Decherchi P, Marqueste T. Treadmill Training Improves Functional Recovery After Injection of Adipose-Derived Stromal Vascular Fraction in Rats With a Spinal Cord Injury. *Journal of Neuroscience Research*. 2026 Sep 1;104(9):e70158. doi: 10.1002/jnr.70158. PMID: 42764209.

Publication

Mahani FSN, Kalantari A, Fink GR, Hoehn M, Aswendt M. A systematic review of the relationship between magnetic resonance imaging based resting-state and structural networks in the rodent brain. *Frontiers in Neuroscience*. 2023 Jul 24;17:1194630. doi: 10.3389/fnins.2023.1194630. PMID: 37554291.



Citation summary
Total citations: **3**
Average/year: **0.75**
Period: 2023-2026
SCImago cites/doc 2y (2025): **3.73**
SJR 2025: **1.07** (Q2)



Cited by

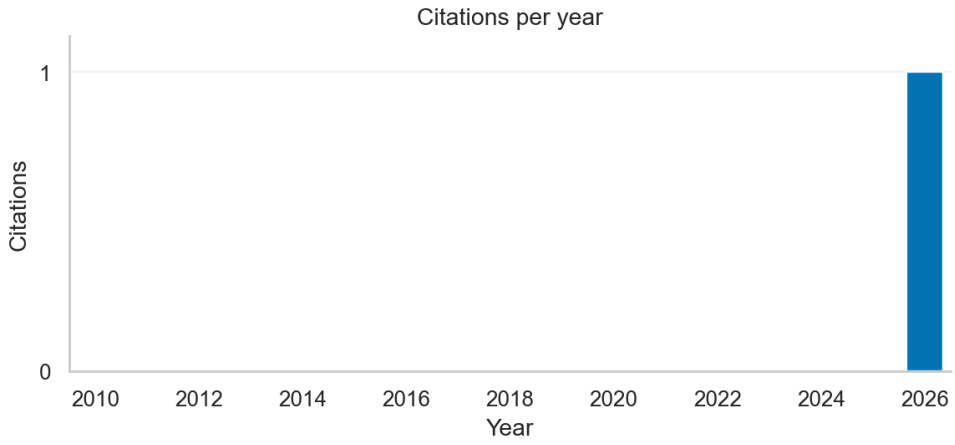
1. Gaudreault F, Desjardins M. Microvascular structure variability explains variance in fMRI functional connectivity. *Brain Structure and Function*. 2025 Feb 8;230.0(2.0):39. doi: 10.1007/s00429-025-02899-4. PMID: 39921726.

2. Prajapati S, Bhagawati R, Alam I, Jebakumari SA. Unraveling Disease-Specific Diffusion Patterns in Human Brain Using Diffusion Tensor Image Analysis. *Lecture notes in electrical engineering*. 2024 Oct 19;:14-20. doi: 10.1007/978-981-97-8043-3_3.

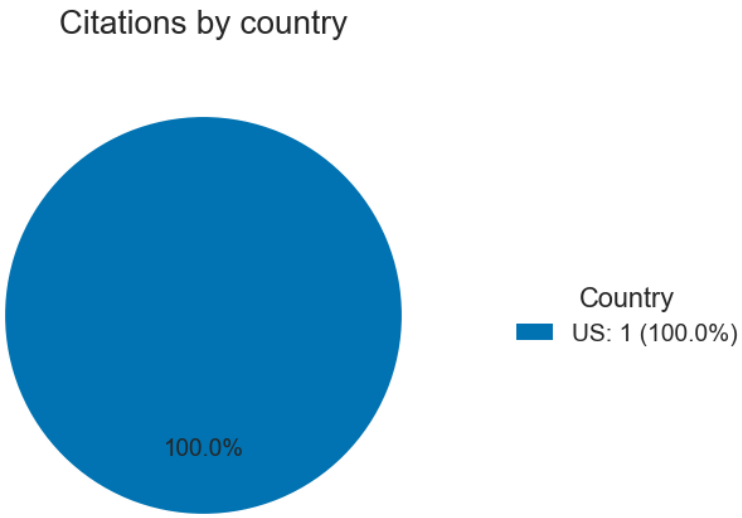
3. Han S, Liu Z, Fang J, Wen Z, Yuan X, Zhao F, Si T, Manyande A, Li B, Wang J, Tian X. Hypoactivity of the prelimbic cortex projecting to the lateral entorhinal cortex contributes to neuropathic pain-induced object recognition memory impairment in mice. *Neurotherapeutics*. 2026 Jan 1;23.0(1.0):e00842. doi: 10.1016/j.neurot.2026.e00842. PMID: 41620328.

Publication

Kalantari A, Shahbazi M, Schneider M, Raikes AC, Frazão VV, Bhattrai A, Carnevale L, Diao Y, Franx BAA, Gammaraccio F, Goncalves L, Lee SD, van Leeuwen EM, Michalek A, Mueller SG, Rivera-Olvera A, Padró D, Selim MK, van der Toorn A, Varriano F, Vrooman RM, Wenk P, Albers HE, Boehm-Sturm P, Budinger E, Canals S, De Santis S, Brinton RD, Dijkhuizen RM, Eixarch E, Forloni G, Grandjean J, Hekmatyar K, Jacobs RE, Jelescu I, Kurniawan ND, Lembo G, Longo DL, Maria NSS, Micotti E, Muñoz-Moreno E, Ramos-Cabrer P, Reichardt W, Sòria G, Ielacqua GD, Aswendt M. Automated quality control of small animal MR neuroimaging data. *Imaging Neuroscience*. 2024 Jan 1;2:1-23. doi: 10.1162/imag_a_00317. PMID: 40212822.



Citation summary
Total citations: **1**
Average/year: **0.33**
Period: 2024-2026
SCImago cites/doc 2y (2025): **2.37**
SJR 2025: **0.791** (Q2)

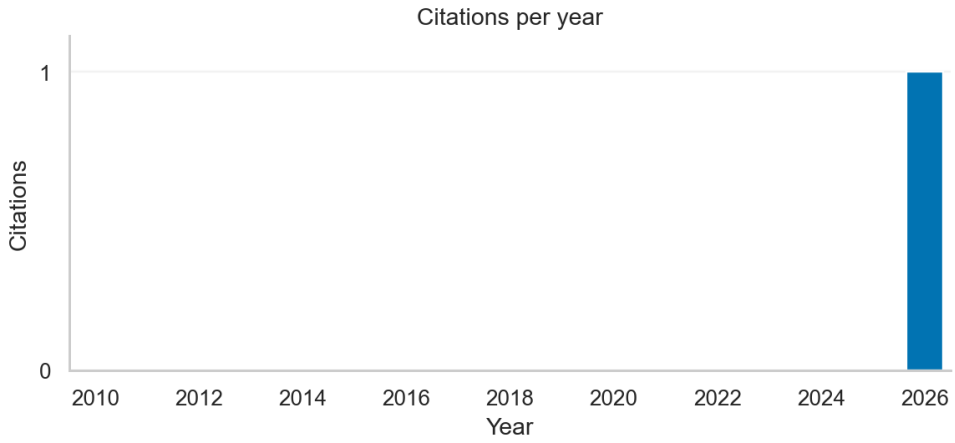


Cited by

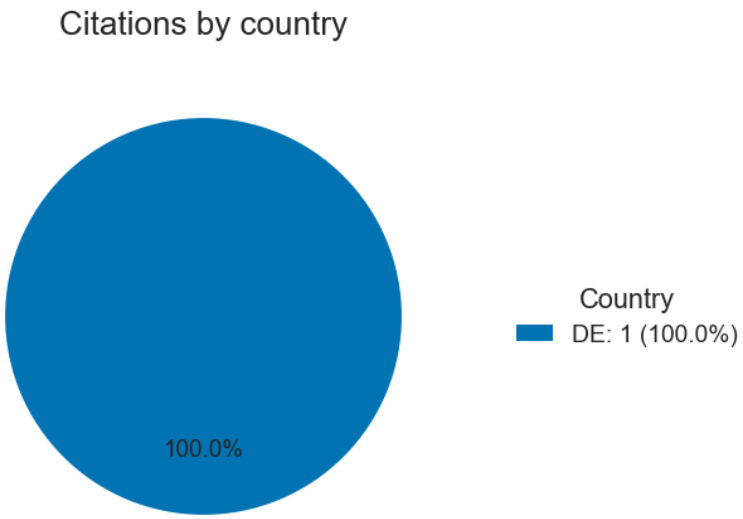
1. Raikes AC, Bhattrai A, Wang T, Wiegand JL, Brinton RD. Humanized APOE mouse brain volume increases over age irrespective of sex and APOE genotype: implications for translational validity to the human. *Frontiers in Neuroscience*. 2026 Jul 14;20:1843319. doi: 10.3389/fnins.2026.1843319. PMID: 41280128.

Publication

Mahani F, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic Functional Connectivity Changes in Cortical and Cortico-Striatal Strokes in Mice. *CNS Neuroscience & Therapeutics*. 2026 Aug 1;32(8):e71114. doi: 10.1002/cns.71114. PMID: 42661330.



Citation summary
Total citations: **1**
Average/year: **1.00**
Period: 2026-2026
SCImago cites/doc 2y (2025): **6.46**
SJR 2025: **1.74** (Q1)

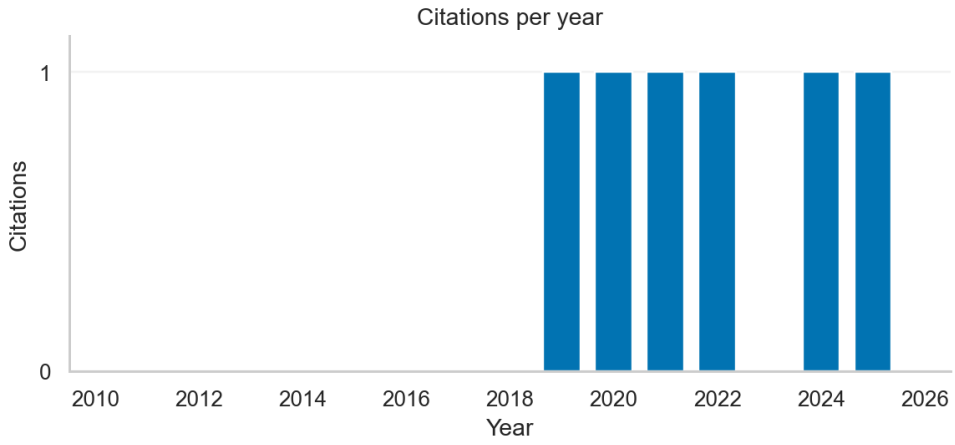


Cited by

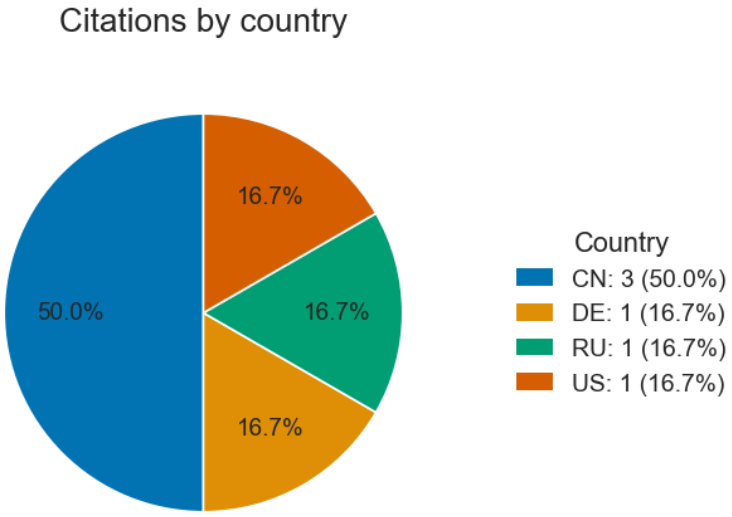
1. Mahani F, Hoehn M, Fink GR, Aswendt M. Structure-function coupling reveals hemisphere-specific network reorganization after stroke. *bioRxiv (Cold Spring Harbor Laboratory)*. 2026 Sep 24;. doi: 10.64898/2026.09.18.752042.

Publication

Aswendt M, Vogel S, Hoehn M. In vivo bioluminescence imaging to elucidate stem cell graft differentiation. *Neural Regeneration Research*. 2019 Sep 16;15(1):61. doi: 10.4103/1673-5374.264449. PMID: 31535647.



Citation summary
Total citations: **6**
Average/year: **0.75**
Period: 2019-2026
SCImago cites/doc 2y (2025): **5.98**
SJR 2025: **1.43** (Q1)



Cited by

1. Hu X, Li Z, Zhang D, Yang Y, Fu S, Zhang Z, Yang R, Xiong K. A systematic summary of survival and death signalling during the life of hair follicle stem cells. *Stem Cell Research & Therapy*. 2021 Aug 11;12.0(1.0):453. doi: 10.1186/s13287-021-02527-y. PMID: 34380571.

2. Aswendt M, Vogel S, Schäfer C, Jathoul AP, Pulé M, Hoehn M. Quantitative in vivo dual-color bioluminescence imaging in the mouse brain. *Neurophotonics*. 2019 May 7;6.0(2.0):1-11. doi: 10.1117/1.nph.6.2.025006. PMID: 31093514.

3. Xia Y, Bao H, Huang J, Li X, Yu C, Zhang Z, Wang H. Near-infrared-persistent luminescence/bioluminescence imaging tracking of transplanted mesenchymal stem cells in pulmonary fibrosis. *Biomaterials Science*. 2020 Jan 1;8.0(11.0):3095-3105. doi: 10.1039/d0bm00063a. PMID: 32347847.

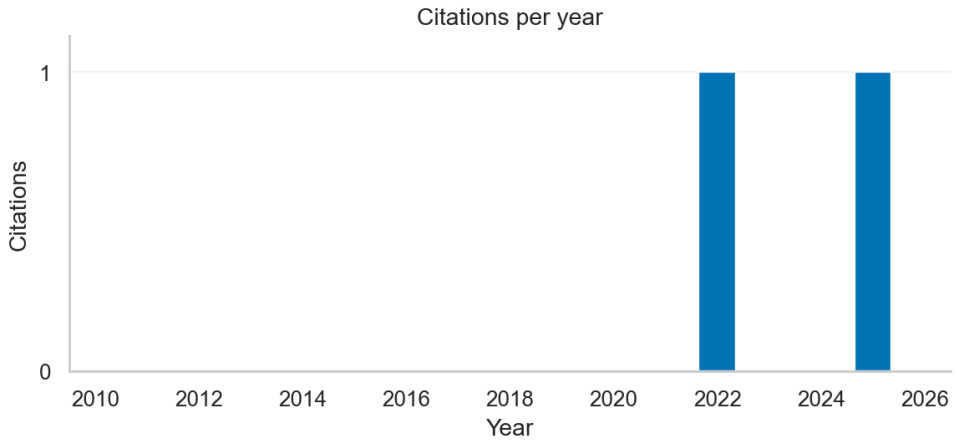
4. Modo M, Didwischus N, Kisel AA. An overview of noninvasive imaging strategies in neural engineering. *Elsevier eBooks*. 2024 Sep 20;:301-350. doi: 10.1016/b978-0-323-95730-4.00001-9.

5. Li X, Yu C, Bao H, Chen Z, Liu X, Huang J, Zhang Z. CT/bioluminescence dual-modal imaging tracking of stem cells labeled with Au@PEI@PEG nanotracers and RfLuc in nintedanib-assisted pulmonary fibrosis therapy. *Nanomedicine Nanotechnology Biology and Medicine*. 2022 Jan 13;41.0:102517. doi: 10.1016/j.nano.2022.102517. PMID: 35032629.

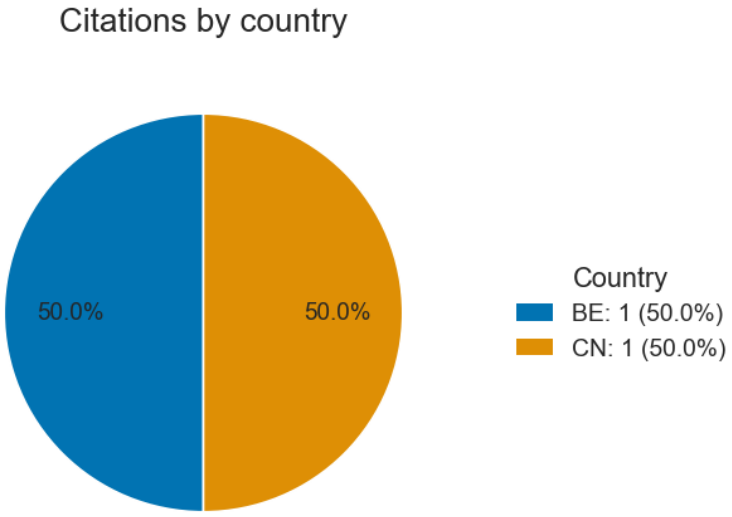
6. Esimbekova EN, Kratasyuk VA, Rozanova NA, Lonshakova-Mukina VI, Torgashina IG, Komleva YK, Saridis MR, Korsakova SA, Yurchenko SO, Salmina AB. Current and prospective trends in the application of bioluminescent analysis in experimental brain studies. *Talanta*. 2025 Jun 24;296.0:128510. doi: 10.1016/j.talanta.2025.128510. PMID: 40582108.

Publication

Aswendt M, Green C, Hoehn M. Plastic Network Changes During Brain Disease. *Handbook of behavioral neuroscience*. 2018 Jan 1;:415-424. doi: 10.1016/b978-0-12-812028-6.00022-7.



Citation summary
Total citations: **2**
Average/year: **0.22**
Period: 2018-2026
SCImago cites/doc 2y (2025): **0**
SJR 2025: **0.133** (Q4)



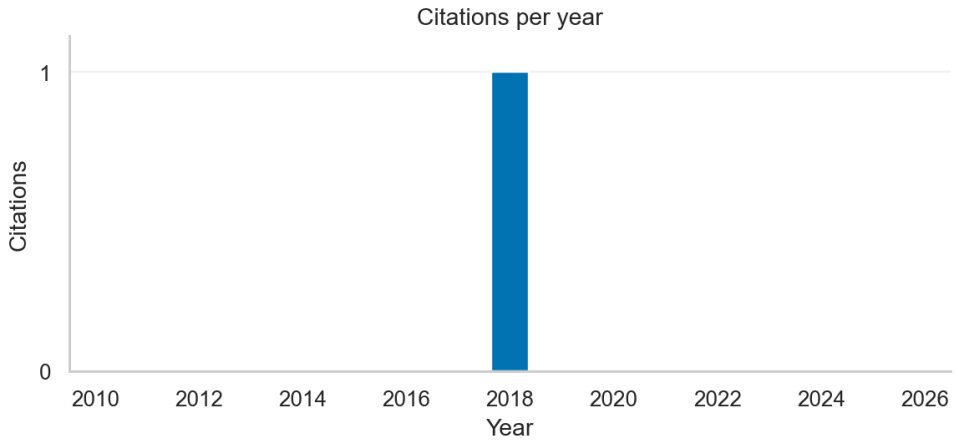
Cited by

1. Li N, He N, Li M, Xu L. From neural decoding to design intervention: A systematic review of human emotional responses to architectural spaces through neuroimaging techniques. *Wellbeing Space and Society*. 2025 Oct 4;9:100305. doi: 10.1016/j.wss.2025.100305.

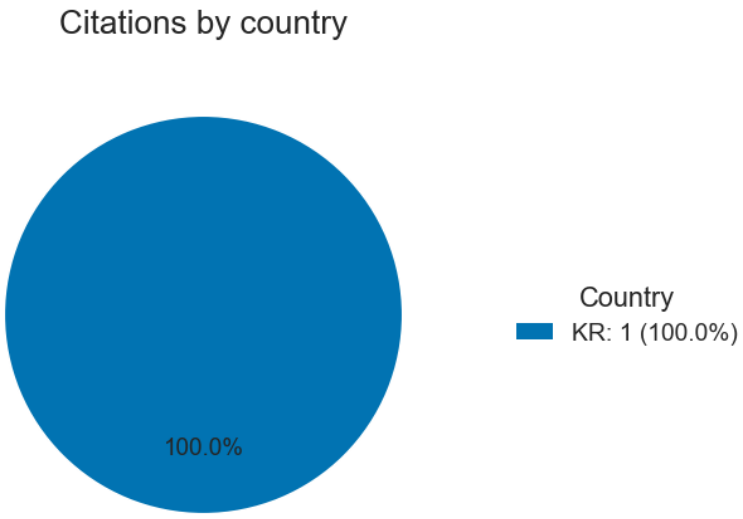
2. Van der Linden A, Hoehn M. Monitoring Neuronal Network Disturbances of Brain Diseases: A Preclinical MRI Approach in the Rodent Brain. *Frontiers in Cellular Neuroscience*. 2022 Jan 3;15:815552. doi: 10.3389/fncel.2021.815552. PMID: 35046778.

Publication

Aswendt M, Collmann FM, Hoehn M. Neurobiological insights from bioluminescence imaging. *Oncotarget*. 2017 Aug 13;8(41):69198-69199. doi: 10.18632/oncotarget.20302. PMID: 29050195.



Citation summary
Total citations: **1**
Average/year: **0.10**
Period: 2017-2026
SCImago cites/doc 2y (2025): **1.89**
SJR 2025: **0.828** (Q2)

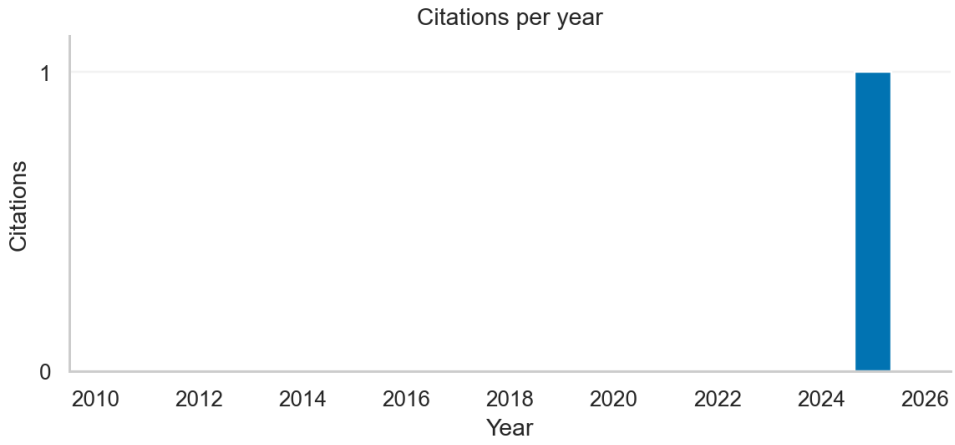


Cited by

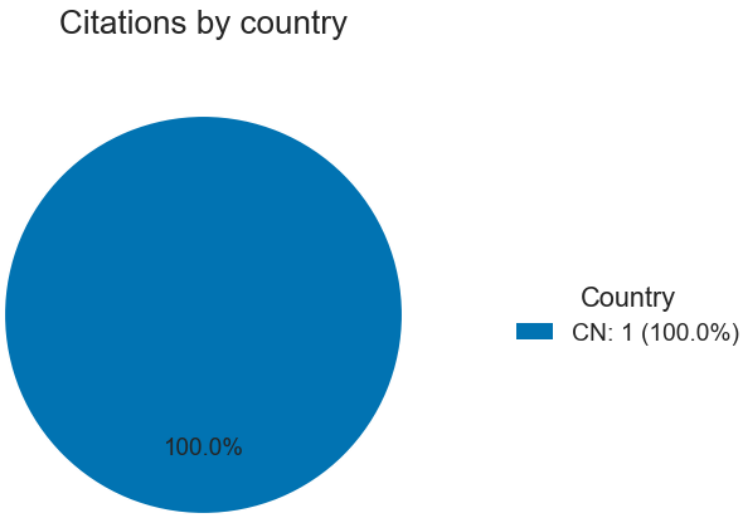
1. Heo H, Jo J, Jung JI, Han Y, Lee S, Kim S, Kwon S, Kim K, Hwang BJ, Kee Y, Lee BD, Kang D, Her S. Improved dynamic monitoring of transcriptional activity during longitudinal analysis in the mouse brain. *Biology Open*. 2018 Jan 1;8(1). doi: 10.1242/bio.037168. PMID: 30341106.

Publication

Wieters F, Gruhn M, Büschges A, Fink GR, Aswendt M. Terminal H-reflex Measurements in Mice. *Journal of Visualized Experiments*. 2022 Jun 16;(184). doi: 10.3791/63304-v.



Citation summary
Total citations: **1**
Average/year: **0.20**
Period: 2022-2026
SCImago cites/doc 2y (2025): **1.06**
SJR 2025: **0.409** (Q2)

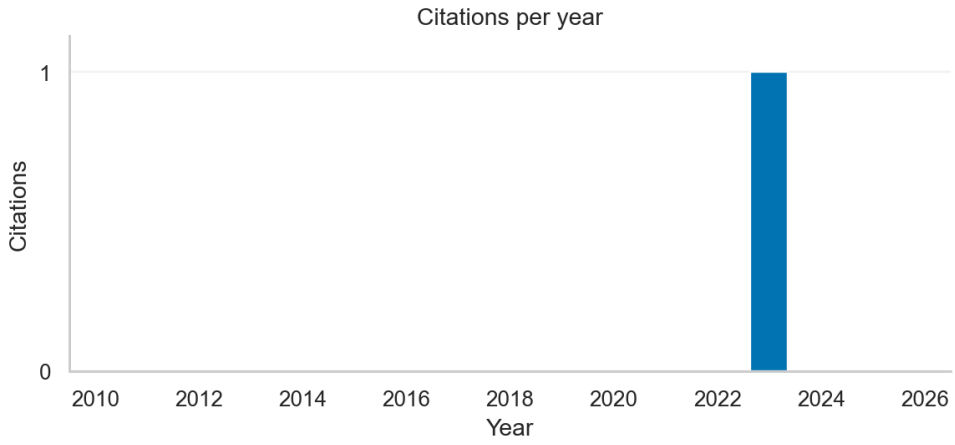


Cited by

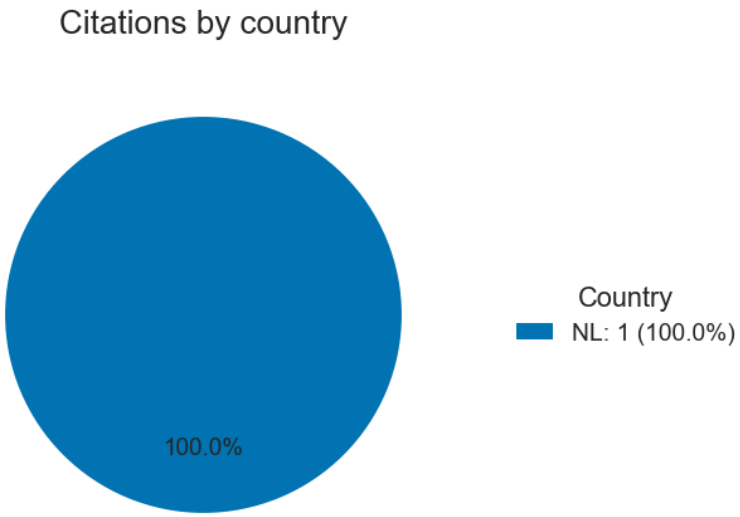
1. Gong H, Zhang Z, Duan Z, Mao X, Wu Y, Rao J, Du X. Mechanisms of Different Motor Neurons in the Occurrence of Spasticity After Spinal Cord Injury: A Narrative Review. *International Journal of Molecular Sciences*. 2025 May 28;26(11):5162. doi: 10.3390/ijms26115162. PMID: 40507976.

Publication

Srinivas M, Boehm-Sturm P, Aswendt M, Pracht E, Figdor CG, de Vries IJ, Hoehn M. In vivo 19F MRI for Cell Tracking. *Journal of Visualized Experiments*. 2013 Nov 25;(81). doi: 10.3791/50802-v.



Citation summary
Total citations: **1**
Average/year: **0.07**
Period: 2013-2026
SCImago cites/doc 2y (2025): **1.06**
SJR 2025: **0.409** (Q2)

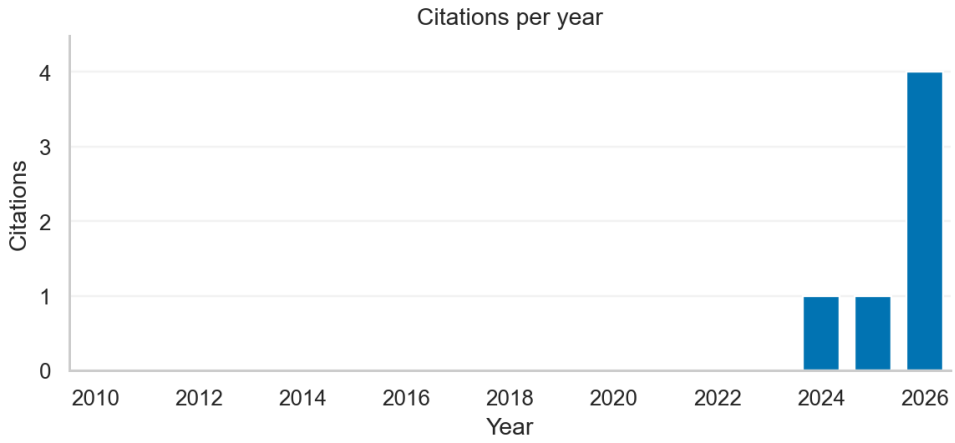


Cited by

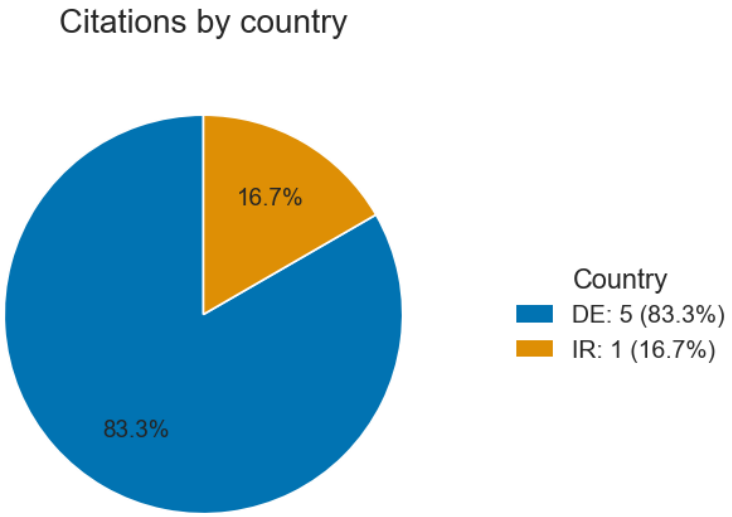
1. Maxouri O, Bodalal Z, Daal MRR, Rostami S, Rodriguez I, Akkari L, Srinivas M, Bernards R, Beets-Tan RGH. How to 19F MRI: applications, technique, and getting started. *BJR|Open*. 2023 Sep 25;5(1):20230019. doi: 10.1259/bjro.20230019. PMID: 37953866.

Publication

Kalantari A, Szczepanik M, Heunis S, Mönch C, Hanke M, Wächtler T, Aswendt M. How to establish and maintain a multimodal animal research dataset using DataLad. *Scientific Data*. 2023 Jun 5;10(1):357. doi: 10.1038/s41597-023-02242-8. PMID: 37277500.



Citation summary
Total citations: **6**
Average/year: **1.50**
Period: 2023-2026
SCImago cites/doc 2y (2025): **6.78**
SJR 2025: **1.91** (Q1)



Cited by

1. Kalantari A, Hambrock C, Grefkes C, Fink GR, Aswendt M. Proportional recovery in mice with cortical stroke. *Experimental Neurology*. 2025 Feb 4;386:115180. doi: 10.1016/j.expneurol.2025.115180. PMID: 39914643.

2. Mahani FSN, Kalantari A, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic structural connectivity changes in cortical and cortico-striatal strokes in mice. *Neural Regeneration Research*. 2026 Feb 28;21(10.0):4945-4953. doi: 10.4103/nrr.nrr-d-25-00491. PMID: 41778731.

3. Kalantari A, Shahbazi M, Schneider M, Raikes AC, Frazão VV, Bhattrai A, Carnevale L, Diao Y, Franx BAA, Gammaraccio F, Goncalves L, Lee SD, van Leeuwen EM, Michalek A, Mueller SG, Rivera-Olvera A, Padró D, Selim MK, van der Toorn A, Varriano F, Vrooman RM, Wenk P, Albers HE, Boehm-Sturm P, Budinger E, Canals S, De Santis S, Brinton RD, Dijkhuizen RM, Eixarch E, Forloni G, Grandjean J, Hekmatyar K, Jacobs RE, Jelescu I, Kurniawan ND, Lembo G, Longo DL, Maria NSS, Micotti E, Muñoz-Moreno E, Ramos-Cabrer P, Reichardt W, Sòria G, Ielacqua GD, Aswendt M. Automated quality control of small animal MR neuroimaging data. *Imaging Neuroscience*. 2024 Jan 1;2:1-23. doi: 10.1162/imag_a_00317. PMID: 40212822.

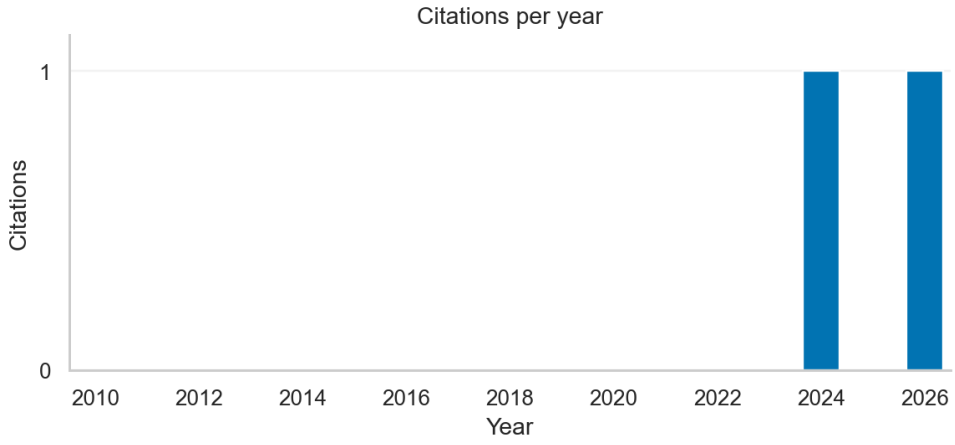
4. Mahani F, Diedenhofen M, Green C, Wiedermann D, Fink GR, Hoehn M, Aswendt M. Dynamic Functional Connectivity Changes in Cortical and Cortico-Striatal Strokes in Mice. *CNS Neuroscience & Therapeutics*. 2026 Aug 1;32(8.0):e71114. doi: 10.1002/cns.71114. PMID: 42661330.

5. Mirhosseini SS, Naseri H, Siahlou B, Arasi MP, Eslami SM, Safaei AA. FAIR m-BIDS: Advancing brain data utilization through multimodal and FAIR principles. *Scientific Data*. 2026 Feb 27;13(1.0). doi: 10.1038/s41597-026-06790-7. PMID: 41760682.

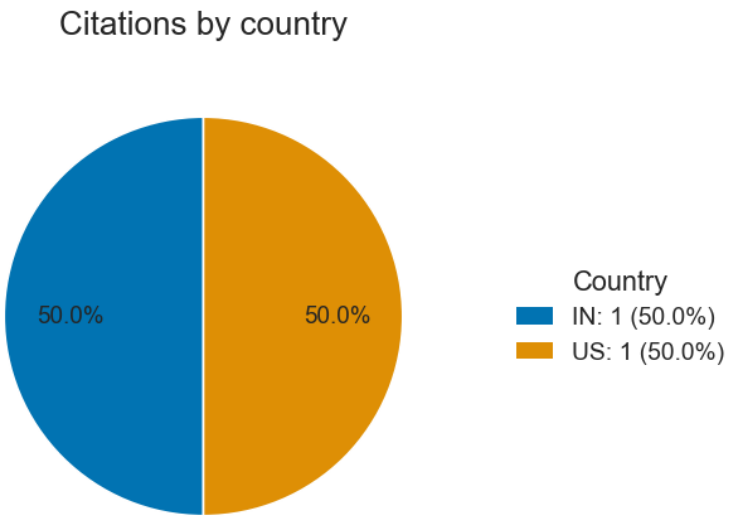
6. Elwes MR, Jaberansary M, Tang FK, Aswendt M, Beyan O, Kutafina E. TrainTracks - federated learning for reproducible research on sensitive medical data. *BMC Medical Informatics and Decision Making*. 2026 May 21;26(1.0). doi: 10.1186/s12911-026-03553-7. PMID: 42169119.

Publication

Paß T, Ricke KM, Hofmann P, Chowdhury R, Nie Y, Chinnery PF, Endepols H, Neumaier B, Carvalho AF, Rigoux L, Steculorum SM, Prudent J, Riemer T, Aswendt M, Brachvogel B, Wiesner RJ. Preserved striatal innervation and motor function despite severe loss of nigral dopamine neurons following mitochondrial dysfunction induced by mtDNA mutations. *bioRxiv (Cold Spring Harbor Laboratory)*. 2023 Jul 8;. doi: 10.1101/2023.07.07.547791.



Citation summary
Total citations: **2**
Average/year: **0.50**
Period: 2023-2026



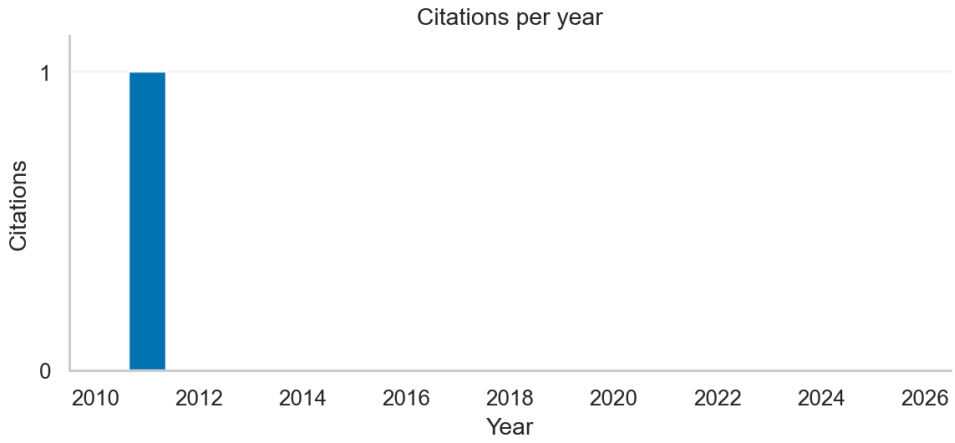
Cited by

1. Wright EC, Scott E, Tian L. Applications of functional neurotransmitter release imaging with genetically encoded sensors in psychiatric research. *Neuropsychopharmacology*. 2024 Jun 28;50(1):269-273. doi: 10.1038/s41386-024-01903-5. PMID: 38942957.

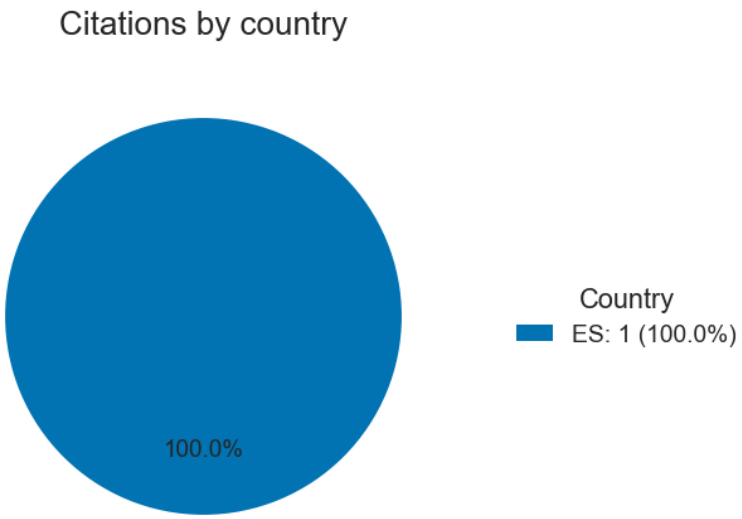
2. Dwivedi V, Thanekar R, Khare K, Taksande BG, Umekar MJ, Mangrulkar SV. Mitochondrial dysfunction and cellular senescence in ageing sarcopenia with current insights and targeted strategies. *Molecular Biology Reports*. 2026 Feb 27;53(1). doi: 10.1007/s11033-026-11609-0. PMID: 41758483.

Publication

Aswendt M, Gianolio E, Pariani P, Napolitano R, Fedeli F, Himmelreich U, Hoehn M, Aime S. Imaging GABAergic neurons by MRI: Use of a responsive contrast agent for glutamic acid decarboxylase. 2011 Jun 1;6:31.



Citation summary
Total citations: **1**
Average/year: **0.06**
Period: 2011-2026



Cited by

1. Ramos-Cabrer P, Hoehn M. MRI Stem Cell Tracking for Therapy in Experimental Cerebral Ischemia. *Translational Stroke Research*. 2011 Oct 26;3(1):22-35. doi: 10.1007/s12975-011-0111-3. PMID: 24323753.